



University of Groningen
Master's Thesis

**From Rotations to Spin Contrast:
Semiclassical and Quantum Dynamics of
Spin-Embedded Nanorotors in Stern–Gerlach
Interferometry**

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Abstract

Quantum-gravity-induced entanglement of masses (QGEM) proposes to test the possible quantum nature of gravity by determining whether it can mediate entanglement between mesoscopic objects prepared in spatial superpositions.

Motivated by this protocol, this thesis develops a self-contained framework for the translational, rotational and spin dynamics of an axially symmetric diamagnetic diamondnanorotor with an embedded spin-1 defect, dictating the nitrogen-vacancy centre as its principal physical realisation.

Beginning with rotation theory and rigid-body mechanics, we derive the kinematics, canonical momenta and three-dimensional semiclassical Hamiltonian of the system. We treat the centre of mass and orientation classically and, the internal spin, quantum mechanically. Then, through controlled approximations, the resulting spin-conditioned forces and torques are reduced.

The equations are verified and solved numerically using symbolic checks, fourth-order Runge–Kutta propagation, parameter scans and Monte Carlo sampling to compare the model hierarchy and examine trajectories, gyroscopic stabilisation, interferometric closure and robustness.

Finally, fluctuations around the semiclassical branches are quantised to determine the final mechanical wave-packet overlap and spin contrast, including finite-temperature effects.

The resulting framework connects the mathematical structure of rotations to the mechanical evolution and quantum closure of spin-based nanorotor interferometry relevant to QGEM.

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“What I cannot create, I do not understand.”

— Richard P. Feynman

Notation

We maintain a similar notation to the core material [1, 2].

Symbols

For matrices, vectors, and related objects such as frame origins, we use the following notation (N denotes the number of dimensions, usually $N = 3$):

- $\mathbf{A}$ – (uppercase bold) matrices;
- $\mathbf{a}$ – (lowercase bold) vectors ($1 \times N$ arrays);
- $\vec{\mathbf{a}}$ – (lowercase bold with arrow) frame origins ($N \times N$ matrices);
- $|\mathbf{A}| := \det(\mathbf{A})$;
- $\|\mathbf{r}\| := \sqrt{\sum_i r_i^2}$;
- $\overrightarrow{OP_k(t)}$, $\overrightarrow{CP_k(t)}$, and $\overrightarrow{CP_k}$ – if neither, one, or both of the points defining a vector are time-dependent, we may leave this dependence implicit when it is clear from context. This keeps the notation compact. When we do show the dependence explicitly, we do not write it on both points inside the vector, e.g. $\overrightarrow{C(t)P_k(t)}$, as in [1].
- s_α , c_α – sinus and cosinus of some angle α

When we say that ”matrices $\mathbf{A}$ and $\mathbf{B}$ are orthogonal”, we mean that each matrix is orthogonal individually, not that they are orthogonal to each other.

Abbreviations and Special Terms

- r.m. – rotation matrix. Many sources refer to p.r.m.’s simply as r.m.’s, for conciseness. We will do the same, since we will often make general statements that hold regardless of the r.m.’s handedness;
- p.r.m. – proper rotation matrix;
- imp.r.m. – improper rotation matrix;
- $\mathbf{1}$ – identity matrix of the context-required dimension (usually 3×3);
- r.ax. – rotation axis (the axis around which an r.m. rotates an object);
- f.o.r. – frame of reference;
- c.o.m. – center of mass;
- compatible – commuting (for operators).

Comments

Text with a vertical line on the left side marks comments that may be skipped if they are obvious to the reader. They give extra explanation or a different view of the surrounding material.

Products

We clarify what the different matrix and vector products represent. This mainly serves as a sanity check, since conventions differ across scientific fields.

Vector Vector Products

- dyadic product

$$\mathbf{a} \otimes \mathbf{b} = \mathbf{a}\mathbf{b}^T = \begin{pmatrix} a_1b_1 & a_1b_2 & a_1b_3 \\ a_2b_1 & a_2b_2 & a_2b_3 \\ a_3b_1 & a_3b_2 & a_3b_3 \end{pmatrix} \quad \Bigg| \quad C_{ij} = a_ib_j \quad (1)$$

- dot product

$$\mathbf{a} \cdot \mathbf{b} = \mathbf{a}^T \mathbf{b} = \sum_{i=0}^3 a_ib_i \quad (2)$$

- cross product

It is only defined in $N = 3$ and $N = 7$ dimensions (for reasons outside the scope of this thesis [3]), so no general formula for $\mathbf{c}_i$ is given.

$$\mathbf{a} \times \mathbf{b} = \begin{pmatrix} a_2b_3 - a_3b_2 \\ a_3b_1 - a_1b_3 \\ a_1b_2 - a_2b_1 \end{pmatrix} \quad (3)$$

- Haramard product

$$\mathbf{a} \circ \mathbf{b} = \begin{pmatrix} a_1b_1 \\ a_2b_2 \\ a_3b_3 \end{pmatrix} \quad (4)$$

Matrix Matrix Products

- Hadamard product

$$\mathbf{A} \otimes \mathbf{B} = \begin{pmatrix} \mathbf{A}_{11}\mathbf{B}_{11} & \mathbf{A}_{12}\mathbf{B}_{12} & \mathbf{A}_{13}\mathbf{B}_{13} \\ \mathbf{A}_{21}\mathbf{B}_{21} & \mathbf{A}_{22}\mathbf{B}_{22} & \mathbf{A}_{23}\mathbf{B}_{23} \\ \mathbf{A}_{31}\mathbf{B}_{31} & \mathbf{A}_{32}\mathbf{B}_{32} & \mathbf{A}_{33}\mathbf{B}_{33} \end{pmatrix} \quad \Bigg| \quad C_{ij} = \mathbf{A}_{ij}\mathbf{B}_{ij} \quad (5)$$

- dot product

$$\mathbf{A}\mathbf{B} = \begin{pmatrix} \mathbf{A}_{11}\mathbf{B}_{11} + \mathbf{A}_{12}\mathbf{B}_{21} + \mathbf{A}_{13}\mathbf{B}_{31} & \dots \\ \dots & \dots \end{pmatrix} \quad \Bigg| \quad C_{ij} = \sum_{k=1}^N \mathbf{A}_{ik}\mathbf{B}_{kj} \quad (6)$$

Matrix Vector Products

- dot product

$$\mathbf{A}\mathbf{a} = \begin{pmatrix} \mathbf{A}_{11}\mathbf{a}_1 + \mathbf{A}_{12}\mathbf{a}_2 + \mathbf{A}_{13}\mathbf{a}_3 \\ \mathbf{A}_{21}\mathbf{a}_1 + \mathbf{A}_{22}\mathbf{a}_2 + \mathbf{A}_{23}\mathbf{a}_3 \\ \mathbf{A}_{31}\mathbf{a}_1 + \mathbf{A}_{32}\mathbf{a}_2 + \mathbf{A}_{33}\mathbf{a}_3 \end{pmatrix} \quad \Bigg| \quad \mathbf{c}_i = \sum_{k=1}^N \mathbf{A}_{ik}\mathbf{a}_k \quad (7)$$

Units

We use natural units:

$$c = \hbar = G = k_B = 1, \quad (8)$$

where c is the speed of light, $\hbar$ is the Planck constant, G is the gravitational constant and k_B is the Boltzmann constant.

Commutators

$$[A, B]_{(0)} = A \quad \Bigg| \quad [A, B]_{(k+1)} = [[A, B]_{(k)}, B] \quad (9)$$

1 Introduction

1.1 Quantum gravity as a low-energy experimental question

Whether the gravitational interaction is fundamentally quantum is one of the few questions in quantum gravity that can be investigated without first selecting a particular microscopic theory of spacetime.

While both general relativity and quantum theory are extraordinarily successful within their domains, their joint empirical regime remains largely unexplored and no unifying theory has yet prevailed. Since quantum gravity’s direct signatures are expected to be extremely weak, QGEM aims to test its nature through more accessible controlled quantum systems rather than Planck-scale dynamics or the direct detection of individual gravitons [4].

In Feynman’s 1957 Chapel Hill discussion, he considered a mass placed in a spatial superposition and asked whether its gravitational field must too enter a corresponding superposition, thereby requiring gravity itself to be treated quantum mechanically [4, 5]. The modern quantum-information formulation was introduced independently in 2017 by Bose, Mazumdar and collaborators, and by Marletto and Vedral [5, 6]: the quantum-gravity-induced entanglement of masses (QGEM) protocol.

In it, the branches of the spatial superpositions of two mesoscopic masses interact gravitationally and acquire branch-dependent relative phases. After recombination, these phases are mapped onto measurable spin correlations, from which the entanglement can be witnessed. Under the protocol’s locality assumptions, if nongravitational interactions are sufficiently suppressed, the mediator cannot be represented as a classical channel (i.e., only carries classical information), but rather as a quantum bus (i.e., it must be capable of transmitting quantum information) [5, 6]. While this would not uniquely identify a theory of quantum gravity, nor constitute a direct graviton detection, it would however show that gravity can mediate entanglement in a regime otherwise well described by Newtonian gravity.

While the concept is simple, the experimental requirements are severe. An observable gravitational phase favours heavier masses, larger differences between the branch separations and longer interaction times. Each one, however, makes coherent preparation and recombination vastly more difficult. Moreover, an experiment must control environmental decoherence, magnetic noise and electromagnetic interactions, including Casimir and induced-dipole effects, which can mask or imitate the desired signal [7, 8, 9].

This thesis therefore addresses the set of mechanical prerequisites on which its observation depends: the controlled creation and high-fidelity closure of a spin-dependent interferometer for a massive nanorotor.

1.2 Spin-controlled massive superpositions

Before a spatial superposition can be generated, the nanorotor must first be levitated at a stable equilibrium against gravity. An inhomogeneous magnetic field pushes diamagnetic particles towards regions of weaker field, thus allowing for magneto-gravitational trapping without the requirement that the particle has to carry a permanent charge. The explicit trap is reconstructed from the leading quadrupole, hexapole and octopole components of the field employed in [10], where diamond nanocrystals were levitated and feedback-cooled in high vacuum.

This static trap levitation field determines the equilibrium position and the small oscillations around it and must remain present throughout the interferometric sequence. By contrast, the superimposed Stern–Gerlach field supplies the controllable spin-dependent gradient which is switched between the separation, free-evolution, and recombination stages. It is therefore the entire field $\mathbf{B}^{\text{sf}}(\mathbf{r}, t) = \mathbf{B}_{\text{tr}}(\mathbf{r}) + \mathbf{B}_{\text{SG}}(\mathbf{r}, t)$, that affects its quantization direction through the internal spin Hamiltonian.

The levitation and spin dynamics must therefore be modelled consistently, without double-counting the trapping field or imposing an unphysical one-dimensional field geometry. This is especially important when the embedded spin (i.e., the defect) is displaced from the centre of mass (c.o.m.).

Through its nitrogen-vacancy (NV) centre, diamond provides both the diamagnetic response required for levitation and a controllable internal spin [10, 11]. Its negatively charged ground state forms an

optically addressable spin triplet that can be initialised, manipulated and read out, while its zero-field splitting defines a preferred axis fixed inside the crystal [11].

Once levitated, the spin superposition is subjected to the time-dependent Stern–Gerlach field, during which a microwave spin inversion redirects and helps recombines the branches. We depict schematically the superposition of the two conditional mechanical states, $|\psi_{i,+}(t)\rangle$ and $|\psi_{i,-}(t)\rangle$, and the final (t_f) branch overlap using the

$$|\Psi_i(t)\rangle = \frac{1}{\sqrt{2}}(|+1\rangle_i|\psi_{i,+}(t)\rangle + |-1\rangle_i|\psi_{i,-}(t)\rangle) \quad \Bigg| \quad C_i(t_f) = |\langle\psi_{i,-}(t_f)|\psi_{i,+}(t_f)\rangle|, \quad (10)$$

where $C_i = 1$ corresponds to perfect recombination, while $C_i < 1$ indicates residual mechanical distinguishability and reduced spin contrast. At t_{flip} , the microwave pulse ideally maps $|+1\rangle_i \leftrightarrow |-1\rangle_i$ without instantaneously changing the mechanical states,

$$|\Psi_i(t_{\text{flip}}^+)\rangle = \frac{1}{\sqrt{2}}(|-1\rangle_i|\psi_{i,+}(t_{\text{flip}})\rangle + |+1\rangle_i|\psi_{i,-}(t_{\text{flip}})\rangle), \quad (11)$$

leaving the two wave packets to evolve under the opposite spin-conditioned forces.

The QGEM signal is ultimately inferred from correlations between the two spins, which, in order to be reliable, must be facilitated by a sufficiently large spin coherence (i.e., large overlap $C_i(t_f)$), which suffers from any residual translational or rotational mismatch.

1.3 Rotational closure and gyroscopic stabilisation

Once the spin Hamiltonian is treated as orientation-dependent, the point mass approximation of the nanoparticle no longer suffices. Furthermore, the centre-of-mass position, orientation in $SO(3)$ and angular momentum may evolve differently along the two spin-conditioned arms.

For interference, closing the centre-of-mass trajectory in position and momentum is necessary, but not sufficient. At the classical level, the orientations and angular momenta must also match, while the corresponding quantum states must overlap in their widths and correlations. The rotational motion and its coupling to embedded spins has therefore become a functional part of nanorotor interferometry [12, 13].

For a massive object, even a small final mismatch can leave the two branches distinguishable and thereby heavily suppress the spin contrast. This sensitivity of Stern–Gerlach interference is historically known as the Humpty–Dumpty problem [14, 15, 16], since all the king’s horses and all the king’s men couldn’t put the nanorotor together again.

Rotational dynamics further worsen this problem, since the interferometer’s branches may remain distinguishable through libration, precession or rotation about the body axis [17]. This rotational mismatch is the central physical problem considered in this thesis.

Without an initial stabilizing rotation, the aligned configuration is stable for the $m_s = -1$ branch but unstable for $m_s = +1$ one [18] and using $m_s = 0$ and $m_s = -1$ weakens the spin-dependent force, reducing the attainable Stern–Gerlach separation. A proposed solution is a sufficiently rapid initial rotation (approximately) along the nanorotor’s common symmetry and embedded-spin axis [18]. The resulting gyroscopic restoring effect suppresses the libration instability recovering the largest available separation.

The main paper we base this thesis on, [19], develops this mechanism for cylindrical nanorotors and studies its mass and geometrical dependence. We reconstruct this proposal within a self-consistent framework, rederiving its mechanics, stating each assumption and approximation explicitly, and testing the reduced descriptions against the complete model.

1.4 Scope and methodological strategy

The aim of this thesis is therefore to understand how the nanorotor’s translational, rotational and spin dynamics combine, which approximations may be used consistently and when the two spin-conditioned branches close both classically and quantum mechanically.

Great emphasis was put on the derivational first two technical chapters. Since the embedded spin is fixed within a rotating body, the later dynamics directly depend on the definitions of rotations, reference

frames and angular momenta. An inconsistent convention at this stage would propagate into the Hamiltonian and ultimately into the predicted forces, torques and interferometric closure, which is why we enforce a consistent notation, rather than a sum of independently imported final formulas. Wormer’s treatment of the classical and quantum rigid rotor [1] provides the mathematical foundation, from which we introduce intermediate argumentations, derivations and adaptations to the present problem.

With this framework, we describe the nanorotor semiclassically: its c.o.m. and orientation follow classical trajectories, while the embedded spin receives a quantum mechanical treatment. Because the spin energy depends on both the defect’s position and orientation relative to the magnetic field, each spin branch experiences a different mechanical potential, thus different forces and torques.

The complete three-dimensional description is retained as the parent model, while simpler centred, aligned, one-dimensional and small-angle descriptions are derived from it only after their assumptions have been clearly stated, justified and quantified. This allows us to determine when and to what degree a reduced model remains physically justified.

A common numerical treatment for the analytical models follows. We reconstruct the trap and interferometric protocol, compare the complete and reduced dynamics, identify gyroscopically stable operating regions and test their sensitivity to preparation and control errors. Symbolic checks, fourth-order Runge–Kutta propagation, parameter scans and Monte Carlo sampling facilitate these comparisons.

Finally, classical branch trajectories agreement does not by itself guarantee a large spin contrast. The mechanical fluctuations around the two semiclassical branches are therefore quantized and propagated to determine their final wave-packet overlap $C_i(t_f)$. Finite-temperature corrections and direct wave-function propagation are then used to examine effects not fully captured by the simplest harmonic treatment. This completes the connection between the nanorotor’s classical motion and the quantum mechanical closure required for the QGEM spin readout.

1.5 Research objectives

The work is organised around five objectives.

1. Derive a consistent description of rotations in the laboratory, body and spin frames, together with the rigid-body kinematics and Hamiltonian structure required for the nanorotor.
2. Construct a complete three-dimensional semiclassical spin–rotor Hamiltonian, derive its branch-dependent forces and torques and obtain the reduced models through controlled approximations.
3. Determine the stability, spin-gap, adiabaticity, symmetry and resonance conditions for the gyroscopically stabilised Stern–Gerlach protocol, together with their dependence on the mass, shape and defect geometry.
4. Implement a numerical treatment for the exact and reduced equations, simulate the dynamics and compare the trajectories, closure, parameter scaling and robustness.
5. Quantize the fluctuations around the conditional trajectories and determine how translational and rotational mismatch, wave-packet deformation, nonlinear libration and finite temperature affect the final overlap and spin contrast.

Together, these objectives address the Humpty Dumpty question of whether a massive spin-embedded nanorotor can be separated and subsequently recombined without leaving a massive observable record of the followed spin path in its mechanical state.

2 Mathematical framework: rotations

We begin with real rotations in Euclidean three-space (as transformations of vectors), reinterpret them as elements of $SO(3)$, and finally as unitary transformations acting on quantum operators.

2.1 Rotation matrices

2.1.1 Basic structure

An r.m. $\mathbf{R}$ is an $N \times N$ linear map which acts on N -component column vectors, *rotating and reflecting them about the chosen coordinate origin* $\mathbf{0}$. Therefore, *it cannot translate them* (i.e., preserves the origin):

$$\mathbf{r} \mapsto \mathbf{R}\mathbf{r} \quad | \quad \mathbf{0} \mapsto \mathbf{0} \quad | \quad \text{for } N = 3: \quad \mathbf{R} = \begin{pmatrix} R_{11} & R_{12} & R_{13} \\ R_{21} & R_{22} & R_{23} \\ R_{31} & R_{32} & R_{33} \end{pmatrix} = (\mathbf{r}_1 \ \mathbf{r}_2 \ \mathbf{r}_3) = \begin{pmatrix} r_{11} & r_{21} & r_{31} \\ r_{12} & r_{22} & r_{32} \\ r_{13} & r_{23} & r_{33} \end{pmatrix}. \quad (12)$$

2.1.2 Unitarity/orthogonality

A complex matrix $\mathbf{U}$ is unitarity if its columns form an orthonormal set under the Hermitian inner product:

$$\mathbf{u}_i^\dagger \mathbf{u}_j = \delta_{ij} \Leftrightarrow \mathbf{U}^\dagger = \mathbf{U}^{-1} \Rightarrow |\det(\mathbf{U})| = 1. \quad (13)$$

For a real matrix this becomes orthogonality:

$$\mathbf{r}_i \cdot \mathbf{r}_j = \delta_{ij} \Leftrightarrow \mathbf{R}^T = \mathbf{R}^{-1} \Rightarrow \det(\mathbf{R}) = \begin{cases} +1 & \mathbf{R} \text{ right handed} \\ -1 & \mathbf{R} \text{ left handed} \end{cases}. \quad (14)$$

Being orthogonal, $\mathbf{R}$ preserves the length of the vectors acts on:

$$\|\mathbf{R}\mathbf{v}\| = \sqrt{(\mathbf{R}\mathbf{v}) \cdot (\mathbf{R}\mathbf{v})} = \sqrt{\mathbf{v}^T \mathbf{R}^T \mathbf{R} \mathbf{v}} = \sqrt{\mathbf{v}^T \mathbf{v}} = \|\mathbf{v}\|. \quad (15)$$

2.1.3 Proper/improper

Theorem 1. *The matrix handedness determines its reflectivity.* Let $\mathbf{R} \in O(3)$ and $\mathbf{r}_i$ its columns:

$$\mathbf{r}_i \times \mathbf{r}_j = \det(\mathbf{R}) \sum_{k=1}^3 \varepsilon_{ijk} \mathbf{r}_k \quad \left| \quad \det(\mathbf{R}) = \begin{cases} +1 & \text{p.r.m. (does not reflect)} \\ -1 & \text{imp.r.m. (does reflect)} \end{cases} \right. \quad (16)$$

The proof is given in **Appendix A.1**.

Lemma 1. *Cross product under orthogonal maps.* Let $\mathbf{A} \in O(3)$. Then, for any $\mathbf{u}, \mathbf{v} \in \mathbb{R}^3$,

$$\mathbf{A}(\mathbf{u} \times \mathbf{v}) = \det(\mathbf{A})(\mathbf{A}\mathbf{u}) \times (\mathbf{A}\mathbf{v}). \quad (17)$$

Proof Using **Thm. 1** for the orthogonal frame $\mathbf{A}\hat{\mathbf{e}}_i$, and using $\mathbf{u} = u_i \hat{\mathbf{e}}_i$ and $\mathbf{v} = v_j \hat{\mathbf{e}}_j$, we get:

$$(\mathbf{A}\mathbf{u}) \times (\mathbf{A}\mathbf{v}) = u_i v_j (\mathbf{A}\hat{\mathbf{e}}_i) \times (\mathbf{A}\hat{\mathbf{e}}_j) = \det(\mathbf{A}) u_i v_j \varepsilon_{ijk} \mathbf{A}\hat{\mathbf{e}}_k = \det(\mathbf{A}) \mathbf{A}(\mathbf{u} \times \mathbf{v}),$$

where we remind the reader of the Einstein notation used. Since $\det(\mathbf{A}) = \pm 1$, this is equivalent to

$$\mathbf{A}(\mathbf{u} \times \mathbf{v}) = \det(\mathbf{A})(\mathbf{A}\mathbf{u}) \times (\mathbf{A}\mathbf{v}).$$

□

2.1.4 Parametrization

It is often useful to factor a p.r.m. into elementary rotations. Since $SO(3)$ is three-dimensional, three real parameters suffice locally. The word "locally" is important: Euler-angle parametrizations cover $SO(3)$, but they are not globally one-to-one because of the usual coordinate singularities.

To visualize such a rotation, we use **Fig. 1**. In it, a frame of reference is rotated in three steps in the zxz convention, with angles α , β and γ : e.g., for the x axis, $\hat{\mathbf{e}}_x \xrightarrow{\alpha} \hat{N}_x \xrightarrow{\beta} \hat{M}_x \xrightarrow{\gamma} \hat{n}_x$.

Theorem 2. Euler-angle coverage of $SO(3)$. Using the Euclidean r.m.s,

$$\mathbf{R}_x(\theta) := \begin{pmatrix} 1 & 0 & 0 \\ 0 & c_\theta & -s_\theta \\ 0 & s_\theta & c_\theta \end{pmatrix} \quad \mathbf{R}_y(\theta) := \begin{pmatrix} c_\theta & 0 & s_\theta \\ 0 & 1 & 0 \\ -s_\theta & 0 & c_\theta \end{pmatrix} \quad \mathbf{R}_z(\theta) := \begin{pmatrix} c_\theta & -s_\theta & 0 \\ s_\theta & c_\theta & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad \text{for } \theta = \{\alpha, \beta, \gamma\}, \quad (18)$$

any p.r.m. can be factorized in some parameterization, such as:

$$\mathbf{R}_{zyx} = \mathbf{R}_z(\alpha)\mathbf{R}_y(\beta)\mathbf{R}_x(\gamma) = \begin{pmatrix} c_\alpha c_\beta - s_\alpha c_\gamma + s_\beta s_\gamma c_\alpha & s_\alpha s_\gamma + s_\beta c_\alpha c_\gamma \\ s_\alpha c_\beta & s_\alpha s_\beta s_\gamma + c_\alpha c_\gamma & s_\alpha s_\beta c_\gamma - s_\gamma c_\alpha \\ -s_\beta & s_\gamma c_\beta & c_\beta c_\gamma \end{pmatrix} \quad \text{Tait-Bryan} \quad (19)$$

$$\mathbf{R}_{zxx} = \mathbf{R}_z(\alpha)\mathbf{R}_x(\beta)\mathbf{R}_z(\gamma) = \begin{pmatrix} c_\alpha c_\gamma - s_\alpha c_\beta s_\gamma - c_\alpha s_\gamma - s_\alpha c_\beta c_\gamma & s_\alpha s_\beta \\ s_\alpha c_\gamma + c_\alpha c_\beta s_\gamma - s_\alpha s_\gamma + c_\alpha c_\beta c_\gamma & -c_\alpha s_\beta \\ s_\beta s_\gamma & s_\beta c_\gamma & c_\beta \end{pmatrix} \quad \text{Euler.} \quad (20)$$

While Eqs. 19, 20 follow trivially, the coverage statement proof and the reconstruction of the angles are given in **Appendix A.2**. In its making, we also used the following Lemma:

Lemma 2. Each row and column of $\mathbf{R} \in SO(3)$, has unit norm.

Proof For any $\mathbf{R} \in SO(3)$, where we denote by $\mathbf{r}_i$ its columns and by $\mathbf{t}_i$ its rows, we have:

$$\mathbf{R}^T \mathbf{R} = \mathbf{1} \Rightarrow \|\mathbf{r}_i\| = 1 \quad \bigg| \quad \mathbf{R} \mathbf{R}^T = \mathbf{1} \Rightarrow \|\mathbf{t}_i\| = 1. \quad (21)$$

□

As stated, the preceding parametrizations only cover p.r.m.s. An imp.r.m. $\mathbf{S}$ can, however, always be obtained from the product of a fixed, conventional imp.r.m. $\boldsymbol{\sigma}$ and a p.r.m.:

$$\mathbf{S} = \boldsymbol{\sigma} \mathbf{R}, \quad \det \boldsymbol{\sigma} = -1, \quad \mathbf{R} = \boldsymbol{\sigma}^T \mathbf{S} \in SO(3). \quad (22)$$

2.1.5 The invariant vector of an r.m.

A non-orthogonal matrix $\mathbf{R}$ does not preserve length (Eq. 15): it will stretch/compress the object and may also rotate, reflect it: that is, it has no eigenvectors. when applied on some, the line spanned by the resulting vector is preserved, but if the eigenvalue is $\neq 1$, the vector is stretched/compressed. Euler's theorem below makes this statement precise.

Theorem 3. Euler's axis theorem for proper rotations. Every non-identity p.r.m. ($\mathbf{R} \in SO(3) \setminus \{1\}$) has a unique invariant axis. Equivalently, it has exactly two unit eigenvectors $\pm \mathbf{n}$ with eigenvalue +1:

$$\mathbf{R} \mathbf{n} = \mathbf{n}. \quad (23)$$

If $\mathbf{R}$ fixes two linearly independent vectors, then $\mathbf{R} = \mathbf{1}$.

Extension 3.1. Every non-(-1) imp.r.m. $\mathbf{R} \in O(3) \setminus (SO(3) \cup \{-1\})$ has a unique reversed axis. Equivalently, it has exactly two unit eigenvectors $\pm \mathbf{n}$ with eigenvalue -1:

$$\mathbf{R} \mathbf{n} = -\mathbf{n}. \quad (24)$$

If $\mathbf{R}$ reverses two linearly independent vectors, then $\mathbf{R} = -\mathbf{1}$.

The proof is given in **Appendix A.3**.

The invariant unit vector of a p.r.m. ($\hat{\mathbf{n}}$), together with the rotation angle around itself (φ), define an r.m. through its rotation axis (r.ax.) as $\mathbf{R} = \mathbf{R}(\hat{\mathbf{n}}, \varphi) = \mathbf{R}_{\hat{\mathbf{n}}}(\varphi)$. Thus, for any arbitrary vector $\mathbf{r}$, we have:

$$\mathbf{r} = \mathbf{r}_{\parallel} + \mathbf{r}_{\perp} = (\mathbf{r} \cdot \hat{\mathbf{n}}) \hat{\mathbf{n}} + \mathbf{x}_{\perp} \quad \bigg| \quad \mathbf{x}_{\perp} = \mathbf{r} - (\mathbf{r} \cdot \hat{\mathbf{n}}) \hat{\mathbf{n}} \quad \bigg| \quad \mathbf{y}_{\perp} = \hat{\mathbf{n}} \times \mathbf{r} \quad \bigg| \quad \mathbf{x}'_{\perp} = \cos(\varphi) \mathbf{x}_{\perp} + \sin(\varphi) \mathbf{y}_{\perp}. \quad (25)$$

Therefore, rotating $\mathbf{r}$ around $\hat{\mathbf{n}}$ ignores $\mathbf{r}_{\parallel}$, but rotates $\mathbf{x}_{\perp}$ in the perpendicular plane.

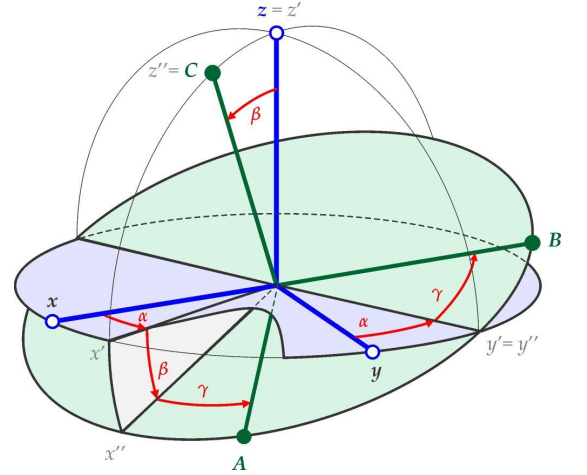


Figure 1: F.o.r. rotation using the Euler (zxz) parametrization as a product of rotations starting from the fixed frame $(\hat{x}, \hat{y}, \hat{z})$ to the fully zxz rotated final one $(\hat{n}_x, \hat{n}_y, \hat{n}_z) = (\hat{n}_x, \hat{n}_y, \hat{n}_z)$. We denote the between steps using: $(\hat{x}', \hat{y}', \hat{z}') = (\hat{N}_x, \hat{N}_y, \hat{N}_z)$ and $(\hat{x}'', \hat{y}'', \hat{z}'') = (\hat{M}_x, \hat{M}_y, \hat{M}_z)$.

We express this effect as **Rodrigues' rotation formula**:

$$\begin{aligned}
\mathbf{R}(\hat{\mathbf{n}}, \varphi) \mathbf{r} &= \cos(\varphi) [\mathbf{r} - (\mathbf{r} \cdot \hat{\mathbf{n}}) \hat{\mathbf{n}}] + \sin(\varphi) \hat{\mathbf{n}} \times \mathbf{r} + (\mathbf{r} \cdot \hat{\mathbf{n}}) \hat{\mathbf{n}} \\
&= \mathbf{r} \cos(\varphi) + (\mathbf{r} \cdot \hat{\mathbf{n}}) \hat{\mathbf{n}} [1 - \cos(\varphi)] + \hat{\mathbf{n}} \times \mathbf{r} \sin(\varphi) \\
&= [\mathbf{1} + \sin(\varphi) \mathbf{N} + (1 - \cos(\varphi)) \mathbf{N}^2] \mathbf{r},
\end{aligned} \tag{26}$$

where in the last line we made use of **Eqs. 3, 1**

$$\hat{\mathbf{n}} \times \mathbf{r} = \begin{pmatrix} 0 & -n_3 & n_2 \\ n_3 & 0 & -n_1 \\ -n_2 & n_1 & 0 \end{pmatrix} \mathbf{r} := \mathbf{N} \mathbf{r} \quad \left| \quad (\mathbf{r} \cdot \hat{\mathbf{n}}) \hat{\mathbf{n}} = \hat{\mathbf{n}} \otimes \hat{\mathbf{n}} \mathbf{r}, \right. \tag{27}$$

which, combined with $\hat{\mathbf{n}}$'s orthonormality ($\sum_{i=1}^3 n_i^2 = 1$), give

$$\mathbf{N}^T \mathbf{N} = -\mathbf{N}^2 = (\mathbf{n}^T \mathbf{n}) \mathbf{1} - \mathbf{n} \mathbf{n}^T = \mathbf{1} - \mathbf{n} \mathbf{n}^T = \mathbf{1} - \mathbf{n} \otimes \mathbf{n}, \tag{28}$$

and the rest is a matter of rearranging the terms.

Corollary 1. *The principal rotation angle $\varphi \in [0, \pi]$ of a p.r.m. is determined by its trace:*

$$\text{Tr}(\mathbf{R}) = 1 + 2 \cos(\varphi) \quad \Rightarrow \quad \varphi = \arccos \left(\frac{\text{Tr}(\mathbf{R}) - 1}{2} \right) \tag{29}$$

and its orientation is contained in the antisymmetric part when restricting to $\varphi \in (0, \pi)$:

$$\mathbf{R} - \mathbf{R}^T = 2 \sin \varphi \mathbf{N} \quad \Rightarrow \quad \begin{pmatrix} 0 & -n_3 & n_2 \\ n_3 & 0 & -n_1 \\ -n_2 & n_1 & 0 \end{pmatrix} = \frac{\mathbf{R} - \mathbf{R}^T}{2 \sin \varphi} \quad \Rightarrow \quad \hat{\mathbf{n}} = \frac{1}{2 \sin \varphi} \begin{pmatrix} R_{32} - R_{23} \\ R_{13} - R_{31} \\ R_{21} - R_{12} \end{pmatrix}. \tag{30}$$

Equivalently, the pairs $(\hat{\mathbf{n}}, \varphi)$ and $(-\hat{\mathbf{n}}, -\varphi)$ describe the same rotation. The trace fixes only the principal angle, while the antisymmetric part fixes the corresponding oriented axis.

For $\varphi \in \{0, \pi\}$, the antisymmetric part vanishes. For $\varphi = 0$, $\mathbf{R} = \mathbf{1}$ and there is no unique axis; for $\varphi = \pi$, the axis must instead be found as the eigenvector with eigenvalue $+1$.

The importance of rewriting $\mathbf{R}$ in terms of an antisymmetric matrix $\mathbf{N}$ stems from $SO(3)$'s connection to its Lie algebra in **Sec. 2.2.1**.

2.1.6 $\mathbf{R}(\hat{\mathbf{n}} \rightarrow \mathbf{A}\hat{\mathbf{n}})$

Similarly to how $\mathbf{R}(\hat{\mathbf{n}}, \varphi) \hat{\mathbf{n}} = \hat{\mathbf{n}}$, we note that applying a length-preserving transformation $\mathbf{A} \in O(3)$ on the r.ax. $\hat{\mathbf{n}}$ creates a vector $\mathbf{A}\hat{\mathbf{n}}$ invariant to a new r.m. $\mathbf{A} \mathbf{R} \mathbf{A}^T$ (which maintains its p./imp. classification):

$$\mathbf{A}\hat{\mathbf{n}} = \mathbf{A} \mathbf{R}(\hat{\mathbf{n}}, \varphi) \mathbf{A}^T (\mathbf{A}\hat{\mathbf{n}}) \quad | \quad \mathbf{A} \mathbf{N}(\mathbf{n}) \mathbf{A}^T = \det(\mathbf{A}) \mathbf{N}(\mathbf{A}\mathbf{n}), \tag{31}$$

where in the r.h.s. equation we used **Lem. 1**, reminding ourselves that $\mathbf{N} = \hat{\mathbf{n}} \times \mathbf{r}$, thus showing that φ must now be handled with the handedness of $\mathbf{A}$. We underline this effect by applying Rodrigues' formula:

$$\mathbf{A} \mathbf{R}(\mathbf{n}, \varphi) \mathbf{A}^T = \mathbf{R}(\mathbf{A}\mathbf{n}, \det(\mathbf{A})\varphi) = \mathbf{R}(\det(\mathbf{A})\mathbf{A}\mathbf{n}, \varphi). \tag{32}$$

2.1.7 Euler Parametrization: stacking

Eq.32 allows rewriting the Euler parametrization in a different light: instead of a series of rotations in a fixed frame, one can have rotations in the object's frame:

$$\hat{\mathbf{e}}_{y'} = \hat{\mathbf{e}}_y \quad | \quad \hat{\mathbf{e}}_{z'} = \hat{\mathbf{e}}_z \quad | \quad \hat{\mathbf{e}}_{x'} = \mathbf{R}_z(\alpha) \hat{\mathbf{e}}_x \quad | \quad \hat{\mathbf{e}}_{z''} = \mathbf{R}_z(\alpha) \mathbf{R}_x(\beta) \hat{\mathbf{e}}_z. \tag{33}$$

$$\begin{aligned}
\mathbf{R}_z(\alpha) \mathbf{R}_x(\beta) \mathbf{R}_z(\gamma) &= [\mathbf{R}_z(\alpha) \mathbf{R}_x(\beta) \mathbf{R}_z(\gamma)] \left[\mathbf{R}_x(\beta)^T [\mathbf{R}_z(\alpha)^T \mathbf{R}_z(\alpha)] \mathbf{R}_x(\beta) \right] [\mathbf{R}_z(\alpha)^T \mathbf{R}_z(\alpha)] \\
&= [\mathbf{R}_z(\alpha) \mathbf{R}_x(\beta) \mathbf{R}_z(\gamma)] [\mathbf{R}_z(\alpha) \mathbf{R}_x(\beta)]^T \mathbf{R}_z(\alpha) \mathbf{R}_x(\beta) \mathbf{R}_z(\alpha)^T \mathbf{R}_z(\alpha) \\
&= \left[[\mathbf{R}_z(\alpha) \mathbf{R}_x(\beta)] \mathbf{R}_z(\gamma) [\mathbf{R}_z(\alpha) \mathbf{R}_x(\beta)]^T \right] \left[\mathbf{R}_z(\alpha) \mathbf{R}_x(\beta) \mathbf{R}_z(\alpha)^T \right] [\mathbf{R}_z(\alpha)] \\
&= \mathbf{R}_{z''}(\gamma) \mathbf{R}_{x'}(\beta) \mathbf{R}_z(\alpha)
\end{aligned} \tag{34}$$

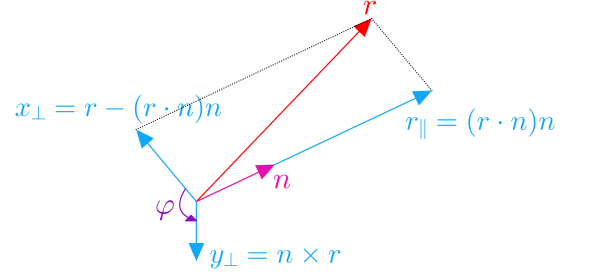


Figure 2: Axis-angle decomposition of $\mathbf{r}$ with respect to the rotation axis $\hat{\mathbf{n}}$. The parallel component $\mathbf{r}_{\parallel}$ is invariant, while $\mathbf{x}_{\perp}$ rotates by φ in the perpendicular plane spanned by $\mathbf{x}_{\perp}$ and $\mathbf{y}_{\perp}$.

2.2 Rotations in Group Theory: $SO(3)$ and $so(3)$

2.2.1 Lie Theory: definitions

P.r.m.s are elements of $SO(3)$ and imp.r.m.s of $O(3) \setminus SO(3)$. Let us now summarize these groups:

- O – orthogonal $\mathbf{R}^T \mathbf{R} = \mathbf{1}$, leading to $\det(\mathbf{R}) = \pm 1$;
- S – special, meaning determinant $\det(\mathbf{R}) = +1$ (i.e., proper);
- 3 – the matrices act on a three-dimensional real vector space;
- Together, they form a Lie group under ordinary matrix multiplication.

Definition of a Lie group: It is a group (i.e., closed, associative, has an identity element and is invertible) whose elements can be smoothly parametrized. Thus, it is a smooth manifold, with smooth multiplication and smooth inversion.

It is convenient to represent an n -dimensional Lie group G (elements $g(\varphi)$, $\varphi = (\varphi_1, \dots, \varphi_n)$) by $d \times d$ matrices $\mathbf{D}(\varphi) = \mathbf{D}(g(\varphi))$ which respect $\mathbf{D}(g_1 g_2) = \mathbf{D}(g_1) \mathbf{D}(g_2)$ and preserve the identity $\mathbf{D}(e) = \mathbf{1}$.

For $n=3$ rotation angles/axis, $d=3$ in the fundamental rep (the most "basic" and "natural" representation of the group). The terms in **Eq.18** are the basis of the fundamental rep (with $\mathbf{R}_{xyz/zxz}(\varphi=0) = \mathbf{1}$).¹

Definition of path connectedness and simple connectedness: A space is **path connected** if any two of its points can be joined by a continuous path. It is **simply connected** if, in addition, every closed loop can be continuously shrunk to a point inside the same space.

$SO(3)$ is not simply connected. Every rotation can be represented by an axis $\hat{n}$ and an angle $0 \leq \varphi \leq \pi$: topologically $SO(3)$ is a solid ball of radius π , the identity rotation at its center. However, opposite points on the boundary are identified: $\mathbf{R}_{\hat{n}}(\pi) = \mathbf{R}_{-\hat{n}}(\pi)$. Therefore, a continuous 2π rotation starts at the center, reaches the angle π boundary, re-enters through the identified $-\pi$ boundary, and returns to the center.

This passage cannot be removed by continuously deforming the loop while keeping it closed; only two such passages can cancel. Therefore only a 4π loop is shrinkable to the identity loop: $\pi_1(SO(3)) \cong \mathbb{Z}_2$.²

Definition of a Lie algebra: It is a vector space equipped with a Lie bracket (i.e., a bilinear, antisymmetric product satisfying the Jacobi identity), which, for a matrix Lie group G , is defined as the tangent space at the identity element $\mathbf{1}$ (T_1):

$$\mathfrak{g} = T_1 G = \left\{ \mathbf{A} = \left. \frac{d\mathbf{G}(t)}{dt} \right|_{t=0} \mid \mathbf{G}(t) \in G, \mathbf{G}(0) = \mathbf{1} \right\} \quad \left| \quad [\mathbf{A}_i, \mathbf{A}_j] = \mathbf{A}_i \mathbf{A}_j - \mathbf{A}_j \mathbf{A}_i. \quad (35)$$

Since every $\mathbf{R}(\varphi) \in SO(3)$ satisfies $\mathbf{R}(\varphi)^T \mathbf{R}(\varphi) = \mathbf{1}$, we infer about $so(3)$ by differentiating at $t=0$:

$$\left. \frac{d}{dt} [\mathbf{R}(t)^T \mathbf{R}(t)] \right|_{t=0} = \mathbf{A}^T + \mathbf{A} = 0, \quad \text{for } \mathbf{A} = \dot{\mathbf{R}}(t)|_{t=0} \quad \Rightarrow \quad T_1 SO(3) = so(3) \subseteq \left\{ \mathbf{A} \in M_3(\mathbb{R}) \mid \mathbf{A}^T = -\mathbf{A} \right\}. \quad (36)$$

We proved that all terms of $so(3)$ are real antisymmetric 3×3 matrices, but to test the reverse, we ask: *How do we go back and forth between the Lie group and its algebra?*

2.2.2 Tangent space at identity / exponential map

Let $D_{\text{exp}}(\boldsymbol{\xi})$ denote a representation of a Lie group element in exponential coordinates near the identity and $\boldsymbol{\xi}$ (not an Euler-angle triple) labelling one Lie-algebra element (then exponentiated as a single object):

$$X = \xi_n T_n = \sum_n \xi_n T_n. \quad (37)$$

In an infinitesimal neighbourhood of the identity, the Taylor expansion begins as

$$D_{\text{exp}}(d\boldsymbol{\xi}) = \mathbf{1} + d\xi_n T_n + \mathcal{O}(d\xi^2) \quad \left| \quad D_{\text{exp}}(\mathbf{0}) = \mathbf{1} \quad \left| \quad T_n := \left. \frac{\partial D_{\text{exp}}}{\partial \xi_n} \right|_{\boldsymbol{\xi}=\mathbf{0}}. \quad (38)$$

¹In **Eq.18** we used the same φ for all 3 matrices, so we can omit the index n in favor of the used parameter.

² $\pi_1(X) = \{\text{closed loops in } X \text{ based at a point}\} / \text{continuous deformation} - \text{closed loops in } X \text{ where loops count as the same if one can be continuously deformed into the other.}$

For several independent directions, the higher-order terms in the Taylor expansion around identity should not only be written only as $(d\varphi_n)^k T_n^k$, since mixed factors appear:

$$\frac{1}{2}(d\varphi_n T_n)^2 = \frac{1}{2}d\varphi_m d\varphi_n T_m T_n. \quad (39)$$

The matrices/operators T_n are the generators of the representation. If the coordinates ξ_n are not superfluous, then the T_n are linearly independent and span the tangent space at the identity. This is a local statement: global uniqueness of φ_n is not required; Euler angles, for example, are not globally unique.

For $SO(3)$, the vector $\boldsymbol{\xi}$ should not be confused with Euler angles such as $\boldsymbol{\varphi} = (\alpha, \beta, \gamma)$, which instead define product coordinates. Instead, it is the axis–angle vector:

$$\boldsymbol{\xi} = \varphi \hat{\mathbf{n}} \quad \Rightarrow \quad \|\boldsymbol{\xi}\| - \text{rotation angle} \quad \& \quad \hat{\mathbf{n}} - \text{rotation axis}. \quad (40)$$

$\mathbf{1}$ denotes the identity on the space on which the representation acts: in the real vector representation it is the 3×3 identity matrix $\mathbf{1}_3$, while in a quantum/unitary representation it is the identity operator $\hat{\mathbf{1}}_{\mathcal{H}}$ on the Hilbert space.

For the real $SO(3)$ matrix representation and, respectively, the quantum/unitary one we thus have:

$$D_{\text{exp}} = \mathbf{R} \quad \Bigg| \quad T_n = \mathbf{L}_n \quad \Bigg| \quad \mathbf{R}(d\boldsymbol{\xi}) = \mathbf{1}_3 + d\xi_n \mathbf{L}_n + \mathcal{O}(d\xi^2) \quad (41)$$

$$D_{\text{exp}} = \hat{U} \quad \Bigg| \quad T_n = -i\hat{L}_n \quad \Bigg| \quad \hat{U}(d\boldsymbol{\xi}) = \hat{\mathbf{1}}_{\mathcal{H}} - id\xi_n \hat{L}_n + \mathcal{O}(d\xi^2). \quad (42)$$

The factor $-i$ belongs only to the unitary convention. It appears because the actual Lie-algebra element generating a unitary curve must be anti-Hermitian, while the angular-momentum operator $\hat{L}_n$ is chosen Hermitian. This will be clarified in **Sec. 2.2.3**.

By making a large number of N such infinitesimal rotations $(d\xi)^k = (\xi/N)^k$ and using only the leading order $\mathcal{O}(d\xi)$, the full Taylor expansion is obtained, also known as **the exponential map**:

$$D(\boldsymbol{\xi}) = \lim_{N \rightarrow \infty} (\mathbf{1} + d\xi T_n)^N = \lim_{N \rightarrow \infty} \left(\mathbf{1} + \frac{\xi_n T_n}{N} \right)^N = \exp(\xi_n T_n) \quad (43)$$

$$\mathbf{R}(\boldsymbol{\xi}) = e^{\xi_n \mathbf{L}_n} \quad \Bigg| \quad \hat{U}(\boldsymbol{\xi}) = e^{-i\xi_n \hat{L}_n}, \quad (44)$$

where higher-order terms in the expansion become infinitesimal in the limit around $\mathbf{1}$.

Here, we re-emphasize the difference between $\boldsymbol{\xi}$ and $\boldsymbol{\varphi}$: the index n follows the Euler summation convention for the exponential coordinates ξ_n , but Euler angles φ_n are product coordinates which follow the Baker–Campbell–Hausdorff (BCH) formula:

$$e^A e^B = \exp \left(A + B + \frac{1}{2}[A, B] + \frac{1}{12}[A, [A, B]] + \frac{1}{12}[B, [B, A]] + \dots \right). \quad (45)$$

That is, $\boldsymbol{\xi}$ are the resulting coordinates from the exponential product of the $\boldsymbol{\varphi}$ ones that result from the BCH formula, with the requirement here is that the commutator of two generators returns a generator, preserving the map. In **Sec. 2.2.3** where we address this by introducing the **Lie bracket**.

Despite this, there is one important nuance: the exponential map is local by construction, and for a general Lie group it need not be globally surjective. For $SO(3)$, however, it is surjective for a concrete geometric reason: by Euler’s axis theorem (**Thm. 3**), every p.r.m. is a rotation about some axis by some angle. Therefore, every $\mathbf{R} \in SO(3)$ can always be written as an exponential, although not uniquely, as follows from the usual axis–angle identifications, such as $\varphi \sim \varphi + 2\pi$ and $(\hat{\mathbf{n}}, \varphi) \sim (-\hat{\mathbf{n}}, -\varphi)$.

Theorem 4. *Every element of the identity component of a Lie group can be written as a finite product of exponentials of elements of its Lie algebra.*

We return to **Eq. 36** to prove the stronger statement of equality. Let $\mathbf{A} \in \mathfrak{so}_3$. Then $\text{Tr } \mathbf{A} = 0$, and

$$\mathbf{R}(t) = e^{t\mathbf{A}} \Rightarrow \mathbf{R}(t)^T \mathbf{R}(t) = e^{t\mathbf{A}^T} e^{t\mathbf{A}} = e^{-t\mathbf{A}} e^{t\mathbf{A}} = \mathbf{1} \quad \Bigg| \quad \det \mathbf{R}(t) = \det(e^{t\mathbf{A}}) = e^{t \text{Tr } \mathbf{A}} = 1 \quad \Bigg| \quad \left. \frac{d}{dt} e^{t\mathbf{A}} \right|_{t=0} = \mathbf{A}. \quad (46)$$

Therefore $\mathbf{R}(t) \in SO(3)$ and every real antisymmetric 3×3 matrix is tangent to $SO(3)$ at the identity:

$$\mathfrak{so}(3) = T_1 SO(3) = \left\{ \mathbf{A} \in M_3(\mathbb{R}) \mid \mathbf{A}^T = -\mathbf{A} \right\}. \quad (47)$$

A convenient basis is obtained by differentiating the elementary rotations:

$$\mathbf{L}_1 = \left. \frac{\partial \mathbf{R}_x(\varphi)}{\partial \varphi} \right|_{\varphi=0} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix}, \quad \mathbf{L}_2 = \left. \frac{\partial \mathbf{R}_y(\varphi)}{\partial \varphi} \right|_{\varphi=0} = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{pmatrix}, \quad \mathbf{L}_3 = \left. \frac{\partial \mathbf{R}_z(\varphi)}{\partial \varphi} \right|_{\varphi=0} = \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad (48)$$

Every $\mathbf{A} \in so(3)$ can therefore be written as

$$\mathbf{A} = a_n \mathbf{L}_n, \quad (\mathbf{L}_n)_{ij} = -\varepsilon_{nij}. \quad (49)$$

Thus, (x, y, z) and $(1, 2, 3)$ are used interchangeably, with $(x \leftrightarrow 1)$, $(y \leftrightarrow 2)$, and $(z \leftrightarrow 3)$.

2.2.3 Hermitian vs antisymmetric generators

The factor $-i$ in the unitary form of **Eq. 43** marks the change from the real orthogonal matrix rep of $SO(3)$ to the quantum/unitary one. In the former, the generators are themselves Lie-algebra elements:

$$\mathbf{R}(\boldsymbol{\xi}) = e^{\boldsymbol{\xi}_n \mathbf{L}_n} \quad \Bigg| \quad \mathbf{L}_n^T = -\mathbf{L}_n. \quad (50)$$

In the quantum/unitary rep, the angular-momentum operators $\hat{L}_n$ are chosen Hermitian, while the actual Lie-algebra elements in the exponent, $-i\hat{L}_n$, are anti-Hermitian:

$$\hat{U}(\boldsymbol{\xi}) = e^{-i\boldsymbol{\xi}_n \hat{L}_n} \quad \Bigg| \quad \hat{L}_n^\dagger = \hat{L}_n, \quad (-i\hat{L}_n)^\dagger = -(-i\hat{L}_n). \quad (51)$$

The reason is physical. **Quantum rotations are generated by angular-momentum operators, and angular momentum is an observable.** Thus, $\hat{L}_n$ are chosen Hermitian so that their expectation values and eigenvalues are real.³ Mathematically, however, a unitary curve $\hat{U}(t) = e^{t\hat{X}}$ is generated by an anti-Hermitian element $\hat{X}^\dagger = -\hat{X}$. The factor $-i$ reconciles these two facts by setting $\hat{X} = -i\hat{L}$.

The **Lie bracket** must close in whichever convention is being used. For real antisymmetric generators, the commutator is again real antisymmetric. For Hermitian quantum generators, the commutator is anti-Hermitian, so writing it again in a Hermitian basis requires a factor of i :

$$\begin{aligned} [\mathbf{L}_i, \mathbf{L}_j]^T = -[\mathbf{L}_i, \mathbf{L}_j] &\Rightarrow [\mathbf{L}_i, \mathbf{L}_j] = c_{ij}^k \mathbf{L}_k, \\ [\hat{L}_i, \hat{L}_j]^\dagger = -[\hat{L}_i, \hat{L}_j] &\Rightarrow [\hat{L}_i, \hat{L}_j] = i c_{ij}^k \hat{L}_k. \end{aligned} \quad (52)$$

The structure constants c_{ij}^k encode the Lie algebra itself; Hermiticity or antisymmetry only reflects the representation convention. For the $SO(3)$ basis in **Eq. 48**, using **Eq. 49** gives

$$[\mathbf{L}_i, \mathbf{L}_j] = \varepsilon_{ijk} \mathbf{L}_k. \quad (53)$$

The quantum convention is obtained by representing the same real Lie-algebra element as an anti-Hermitian operator, thus preserving the Lie bracket, i.e., the algebra:

$$\rho(\mathbf{L}_i) = -i\hat{L}_i \quad (54)$$

$$\rho([\mathbf{L}_i, \mathbf{L}_j]) = [\rho(\mathbf{L}_i), \rho(\mathbf{L}_j)] \Rightarrow \varepsilon_{ijk}(-i\hat{L}_k) = [-i\hat{L}_i, -i\hat{L}_j] = -[\hat{L}_i, \hat{L}_j] \Rightarrow [\hat{L}_i, \hat{L}_j] = i\varepsilon_{ijk} \hat{L}_k. \quad (55)$$

2.2.4 $O(n)$

As discussed above, the exponential map does not rely on simple connectedness: for a matrix Lie group, every Lie-algebra element $\mathbf{A}$ generates a continuous one-parameter curve $t \mapsto e^{t\mathbf{A}}$ with $e^{0\mathbf{A}} = \mathbf{1}$ which reaches every point of the group connected to the identity and none other.

For $O(n)$, the identity component is $SO(n)$:

$$O(n) = SO(n) \sqcup \boldsymbol{\sigma} SO(n) \quad \Bigg| \quad \det(\boldsymbol{\sigma}) = -1 \quad \Bigg| \quad O(n)_0 = SO(n), \quad (56)$$

where $\boldsymbol{\sigma} SO(n)$ is the disconnected component containing the imp.r.m.s. *It is not a subgroup*: it does not contain the identity, and the product of two imp.r.m.s has determinant $+1$. Equivalently, after choosing one fixed reflection $\boldsymbol{\sigma}$ with $\boldsymbol{\sigma}^2 = \mathbf{1}$, the determinant sign gives the split structure

$$O(n) \cong SO(n) \rtimes \mathbb{Z}_2. \quad (57)$$

³Since we use natural units (**Eq.8**), $\hbar=1$. Otherwise one writes $e^{-i\varphi_n \hat{J}_n/\hbar}$, with $\hat{J}_n$ the angular-momentum operator.

At the infinitesimal level, $O(n)$ and $SO(n)$ have the same Lie algebra, because the tangent space at the identity only sees the identity component:

$$o(n) = T_1 O(n) = T_1 SO(n) = so(n) = \left\{ \mathbf{A} \in M_n(\mathbb{R}) \mid \mathbf{A}^T = -\mathbf{A} \right\}. \quad (58)$$

Moreover, no real exponential can have determinant -1 . For any $\mathbf{B} \in M_n(\mathbb{R})$

$$\det(e^{\mathbf{B}}) = e^{\text{Tr} \mathbf{B}} > 0. \quad (59)$$

Therefore, exponentiating real antisymmetric generators gives elements of $SO(n)$, not imp.r.m.s.

Thus an imp.r.m. must be obtained by multiplying a p.r.m. by a fixed imp.r.m., as in **Eq. 56**.⁴ Any argument relying on the real exponential map therefore applies to the proper component $SO(n)$.

2.3 Adjoint actions: rotations of vector-like objects

The previous subsection fixed the Lie algebra of rotations and the required real/unitary conventions. We now show how rotations act on vector-like objects.

There are two parallel statements. In the real rep, rotations act by conjugation on antisymmetric matrices. In the quantum rep, rotations act by unitary conjugation on vector operators.

2.3.1 The vector form of an antisymmetric matrix

For every vector $\mathbf{a} \in \mathbb{R}^3$, its cross-product matrix is defined by

$$\mathbf{A} = [\mathbf{a}]_{\times} := a_i \mathbf{L}_i = \begin{pmatrix} 0 & -a_3 & a_2 \\ a_3 & 0 & -a_1 \\ -a_2 & a_1 & 0 \end{pmatrix} \longleftrightarrow \mathbf{a} = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix} \quad \left| \quad [\mathbf{a}]_{\times} \mathbf{b} = \mathbf{a} \times \mathbf{b}, \quad (60)$$

$$[\cdot]_{\times} : \mathbb{R}^3 \rightarrow so(3), \quad \mathbf{a} \mapsto [\mathbf{a}]_{\times}. \quad (61)$$

This map is therefore a linear isomorphism: it is the bridge between the vector form of Rodrigues' formula (**Eq. 26**) and the Lie-algebra form of rotations. Analogue to **Eq. 28**, but without the unitarity requirement,

$$\mathbf{A}^T \mathbf{A} = -\mathbf{A}^2 = (\mathbf{a}^T \mathbf{a}) \mathbf{1} - \mathbf{a} \mathbf{a}^T = (\mathbf{a}^T \mathbf{a}) \mathbf{1} - \mathbf{a} \otimes \mathbf{a}, \quad (62)$$

Lemma 3. Adjoint action on $so(3)$. Let $\mathbf{B} \in O(3)$. Then, the adjoint action of $\mathbf{B}$ on $so(3)$ is:

$$\text{Ad}_{\mathbf{B}}[\mathbf{a}]_{\times} = \mathbf{B}[\mathbf{a}]_{\times} \mathbf{B}^T = \det(\mathbf{B})[\mathbf{B}\mathbf{a}]_{\times} \longleftrightarrow \det(\mathbf{B})\mathbf{B}\mathbf{a}. \quad (63)$$

For $\mathbf{B} \in SO(3)$, $\det(\mathbf{B}) = +1$, so the three independent entries of an antisymmetric matrix transform like an ordinary vector. For $\mathbf{B} \in O(3) \setminus SO(3)$, the extra sign shows that they transform as an axial vector.

Proof For any $\mathbf{x} \in \mathbb{R}^3$, by making use of **Lem. 1**, we get:

$$\mathbf{B}[\mathbf{a}]_{\times} \mathbf{B}^T \mathbf{x} = \mathbf{B}(\mathbf{a} \times \mathbf{B}^T \mathbf{x}) = \det(\mathbf{B})(\mathbf{B}\mathbf{a}) \times (\mathbf{B}\mathbf{B}^T \mathbf{x}) = \det(\mathbf{B})(\mathbf{B}\mathbf{a}) \times \mathbf{x} = \det(\mathbf{B})[\mathbf{B}\mathbf{a}]_{\times} \mathbf{x}. \quad (64)$$

□

2.3.2 Vector operators

In quantum mechanics, the rotated object is usually not part of $M_3(\mathbb{R})$, but an operator acting on a Hilbert space. We need a criterion for when a triple of operators behaves like a vector under rotations.

A quantum vector operator is an ordered triple of operators whose components mix among themselves under infinitesimal rotations in the same way as the components of an ordinary vector:

$$\hat{\mathbf{V}} = \left(\hat{V}_1 \ \hat{V}_2 \ \hat{V}_3 \right)^T \quad \left| \quad [\hat{L}_i, \hat{V}_j] = i\varepsilon_{ijk} \hat{V}_k, \quad (65)$$

making, in particular, $\hat{\mathbf{L}}$ itself a vector operator. This is the infinitesimal form of the statement that $\hat{\mathbf{V}}$ rotates as a vector around the r.ax. specified by some classical unit vector $\hat{\mathbf{n}}$ with components n_j :

$$\|\hat{\mathbf{n}}\| = 1 \quad \left| \quad \hat{G} = n_j \hat{L}_j = \hat{\mathbf{n}} \cdot \hat{\mathbf{L}} \quad \left| \quad \hat{U}(\hat{\mathbf{n}}, \varphi) = e^{-i\varphi \hat{G}}. \quad (66)$$

⁴Some imp.r.m.s can be written as exponentials after allowing complex logarithms. Then the curve $e^{t\mathbf{A}}$ does not stay inside the real group $O(n)$, making such a complex $\mathbf{A}$ not a generator of $O(n)$ as a real Lie group.

Then, for a vector operator, the adjoint action creates a small rotation around the i^{th} axis as:

$$\text{Ad}_{\hat{U}_i^\dagger} \hat{\mathbf{V}} = e^{id\varphi \hat{L}_i} \hat{\mathbf{V}} e^{-id\varphi \hat{L}_i} = \hat{\mathbf{V}} + id\varphi [\hat{L}_i, \hat{\mathbf{V}}] + \mathcal{O}(d\varphi^2) \quad \left| \quad [\hat{L}_i, \hat{\mathbf{V}}] := \begin{pmatrix} [\hat{L}_i, \hat{V}_1] \\ [\hat{L}_i, \hat{V}_2] \\ [\hat{L}_i, \hat{V}_3] \end{pmatrix} \right. \quad (67)$$

$$(\text{Ad}_{\hat{U}_i^\dagger} \hat{\mathbf{V}})_j = \text{Ad}_{\hat{U}_i^\dagger} \hat{V}_j = e^{id\varphi \hat{L}_i} \hat{V}_j e^{-id\varphi \hat{L}_i} = \hat{V}_j + id\varphi [\hat{L}_i, \hat{V}_j] + \mathcal{O}(d\varphi^2) = \hat{V}_j - d\varphi \varepsilon_{ijk} \hat{V}_k + \mathcal{O}(d\varphi^2).$$

The unitary operator $\hat{U}(\hat{\mathbf{n}}, \varphi)$ rotates quantum states and its Hermitian transpose adjoint action represents the induced rotation of the operator components: the usual real r.m. acting on $\hat{\mathbf{V}}$.

Theorem 5. Unitary rotations of vector operators. For every vector operator $\hat{\mathbf{V}}$ satisfying **Eq. 65**,

$$\text{Ad}_{\hat{U}^\dagger} \hat{\mathbf{V}} = \hat{U}(\hat{\mathbf{n}}, \varphi)^\dagger \hat{\mathbf{V}} \hat{U}(\hat{\mathbf{n}}, \varphi) = \mathbf{R}(\hat{\mathbf{n}}, \varphi) \hat{\mathbf{V}}. \quad (68)$$

For half-integer spin reps, the unitary operators form a rep of the $SU(2)$ cover rather than a faithful one of $SO(3)$. This does not change **Eq. 68** as the central sign cancels in the adjoint action

$$\text{Ad}_{\hat{U}^\dagger} \hat{\mathbf{V}} = (-\hat{U})^\dagger \hat{\mathbf{V}} (-\hat{U}) = \hat{U}^\dagger \hat{\mathbf{V}} \hat{U}. \quad (69)$$

Proof Define

$$\begin{aligned} \hat{\mathbf{F}}(\varphi) = e^{i\varphi \hat{G}} \hat{\mathbf{V}} e^{-i\varphi \hat{G}} &\Rightarrow \frac{d\hat{\mathbf{F}}_i}{d\varphi} = ie^{i\varphi \hat{G}} [\hat{G}, \hat{V}_i] e^{-i\varphi \hat{G}} = in_j e^{i\varphi \hat{G}} [\hat{L}_j, \hat{V}_i] e^{-i\varphi \hat{G}} = in_j e^{i\varphi \hat{G}} (i\varepsilon_{jik} \hat{V}_k) e^{-i\varphi \hat{G}} \\ &= -n_j \varepsilon_{jik} (e^{i\varphi \hat{G}} \hat{V}_k e^{-i\varphi \hat{G}}) = n_j \varepsilon_{ijk} \hat{F}_k = (\hat{\mathbf{n}} \times \hat{\mathbf{F}})_i. \end{aligned} \quad (70)$$

Thus, in vector form, this is the same differential equation solved by an ordinary vector rotated around $\hat{\mathbf{n}}$

$$\frac{d\hat{\mathbf{F}}}{d\varphi} = \mathbf{N}(\hat{\mathbf{n}}) \hat{\mathbf{F}}, \quad \hat{\mathbf{F}}(0) = \hat{\mathbf{V}}, \quad \mathbf{N}(\hat{\mathbf{n}}) \mathbf{x} = \hat{\mathbf{n}} \times \mathbf{x} \quad \Rightarrow \quad \hat{\mathbf{F}}(\varphi) = e^{\varphi \mathbf{N}(\hat{\mathbf{n}})} \hat{\mathbf{V}} = \mathbf{R}(\hat{\mathbf{n}}, \varphi) \hat{\mathbf{V}}, \quad (71)$$

where the last equality is Rodrigues' formula from **Eq. 26**. $\square$

An equivalent Taylor-series proof is given in **Appendix A.4**, where the arising of the trigonometric terms in Rodrigues' formula is explicitly shown.

Corollary 2. A rotation of an operator around one of the coordinate axes can be expressed in terms of the elementary p.r.m.s introduced in **Eq. 18**:

$$\hat{U}_i(\varphi_i) = e^{-i\varphi_i \hat{L}_i} \Rightarrow \hat{U}_i(\varphi_i)^\dagger \hat{\mathbf{V}} \hat{U}_i(\varphi_i) = \mathbf{R}_i(\varphi_i) \hat{\mathbf{V}}. \quad (72)$$

A general p.r.m. can be built from elementary rotations. The same idea applies to the unitary operators generating quantum rotations, but the order must be handled carefully, as rotations about different axes do not commute. Using the same zxz convention as **Eq. 20**, define

$$\hat{U}_{zxz}(\alpha, \beta, \gamma) = \hat{U}_\alpha \hat{U}_\beta \hat{U}_\gamma = e^{-i\alpha \hat{L}_3} e^{-i\beta \hat{L}_1} e^{-i\gamma \hat{L}_3} \quad \left| \quad \hat{U}_{zxz}(\alpha, \beta, \gamma)^\dagger = \hat{U}_\gamma^\dagger \hat{U}_\beta^\dagger \hat{U}_\alpha^\dagger = e^{i\gamma \hat{L}_3} e^{i\beta \hat{L}_1} e^{i\alpha \hat{L}_3}. \quad (73)$$

Extension 5.1. In the adjoint action on an operator $\hat{U}$, the rotations nest so that the resulting component transformation appears in the same order as the corresponding real p.r.m.⁵:

$$\text{Ad}_{\hat{U}_{zxz}^\dagger} \hat{\mathbf{V}} = \hat{U}_{zxz}^\dagger \hat{\mathbf{V}} \hat{U}_{zxz} = \mathbf{R}_z(\gamma) \mathbf{R}_x(\beta) \mathbf{R}_z(\alpha) \hat{\mathbf{V}}. \quad (75)$$

Proof Making use of **Cor. 2** and that the numerical matrices $\mathbf{R}_i$ act only on the component index of $\hat{\mathbf{V}}$, thus commuting with the Hilbert-space operators $\hat{U}_i$, we directly obtain:

$$\begin{aligned} \text{Ad}_{\hat{U}_{zxz}^\dagger} \hat{\mathbf{V}} &= \hat{U}_{zxz}^\dagger \hat{\mathbf{V}} \hat{U}_{zxz} = \hat{U}_\gamma^\dagger \hat{U}_\beta^\dagger (\hat{U}_\alpha^\dagger \hat{\mathbf{V}} \hat{U}_\alpha) \hat{U}_\beta \hat{U}_\gamma = \hat{U}_\gamma^\dagger \hat{U}_\beta^\dagger (\mathbf{R}_z(\alpha) \hat{\mathbf{V}}) \hat{U}_\beta \hat{U}_\gamma \\ &= \hat{U}_\gamma^\dagger (\mathbf{R}_z(\alpha) \mathbf{R}_y(\beta) \hat{\mathbf{V}}) \hat{U}_\gamma = \mathbf{R}_z(\alpha) \mathbf{R}_y(\beta) \mathbf{R}_z(\gamma) \hat{\mathbf{V}}. \end{aligned} \quad (76)$$

⁵Similarly, for the zxz convention, we obtain:

$$\hat{U}_{zyx}(\alpha, \beta, \gamma) = e^{-i\alpha \hat{L}_3} e^{-i\beta \hat{L}_2} e^{-i\gamma \hat{L}_1} \quad \left| \quad \text{Ad}_{\hat{U}_{zyx}^\dagger} \hat{\mathbf{V}} = \hat{U}_{zyx}^\dagger \hat{\mathbf{V}} \hat{U}_{zyx} = \mathbf{R}_z(\alpha) \mathbf{R}_y(\beta) \mathbf{R}_x(\gamma) \hat{\mathbf{V}}. \quad (74)$$

□

The non-abelian point is not that an Euler product can never be written as a single exponential, but that it cannot be written by naively adding the angles in the exponent for generic $\{\alpha, \beta, \gamma\}$:

$$e^{-i\alpha\hat{L}_3}e^{-i\beta\hat{L}_1}e^{-i\gamma\hat{L}_3} \neq e^{-i[(\alpha+\gamma)\hat{L}_3+\beta\hat{L}_1]} \quad (77)$$

For the real $SO(3)$ rep, every p.r.m. still has an effective axis-angle representative

$$\mathbf{R}_z(\alpha)\mathbf{R}_x(\beta)\mathbf{R}_z(\gamma) = e^{\Phi[\hat{\mathbf{n}}_{\text{eff}}] \times}, \quad (78)$$

where the BCH formula explains why $\hat{\mathbf{n}}_{\text{eff}}$ and Φ are not obtained by component-wise angle addition.

If the unitary rep integrates to $SO(3)$, applying the rep gives the matching operator statement:

$$e^{-i\alpha\hat{L}_3}e^{-i\beta\hat{L}_1}e^{-i\gamma\hat{L}_3} = e^{-i\Phi\hat{\mathbf{n}}_{\text{eff}} \cdot \hat{\mathbf{L}}}. \quad (79)$$

For spinorial reps, the integrated group is the $SU(2)$ cover and the same single-exponential statement still holds, but the global identifications differ (e.g., a 2π rotation need not act as the identity). However, as we saw in **Eq. 69**, this global caveat does not affect the local commutator algebra.

2.4 $so(3)$ and $SO(3)$ products

We now collect the product identities required for comparing vector-like objects attached to different rotations or f.o.r.s. While the dot product only measures relative orientation, thus only depending on a relative orthogonal transformation, the cross product also produces a vector: after accounting for the relative orientation the result must also be oriented in space and handedness must be keep track of.

Lemma 4. Products in the $so(3) \simeq \mathbb{R}^3$ identification. *The trace of the product of two antisymmetric matrices $\mathbf{A}$ and $\mathbf{B}$ is twice the Euclidean dot product of their associated vectors $\mathbf{a}$ and $\mathbf{b}$:*

$$\mathbf{A} = [\mathbf{a}]_{\times} \quad \Big| \quad \mathbf{B} = [\mathbf{b}]_{\times} \quad \Rightarrow \quad \mathbf{a} \cdot \mathbf{b} = -\frac{1}{2} \text{Tr}(\mathbf{AB}) \quad \Big| \quad [\mathbf{a}]_{\times} \mathbf{b} = \mathbf{a} \times \mathbf{b} = \mathbf{Ab} = -\mathbf{Ba}. \quad (80)$$

Proof The cross-product identity follows directly from the definition of the cross-product matrix:

$$\mathbf{Ab} = [\mathbf{a}]_{\times} \mathbf{b} = \mathbf{a} \times \mathbf{b}, \quad \mathbf{Ba} = [\mathbf{b}]_{\times} \mathbf{a} = \mathbf{b} \times \mathbf{a} = -\mathbf{a} \times \mathbf{b}. \quad (81)$$

For the trace identity, we simply apply $\mathbf{AB}$ to an arbitrary vector $\mathbf{x}$ and use the vector triple product (BAC-CAB) identity,

$$\mathbf{ABx} = \mathbf{a} \times (\mathbf{b} \times \mathbf{x}) = \mathbf{b}(\mathbf{a} \cdot \mathbf{x}) - \mathbf{x}(\mathbf{a} \cdot \mathbf{b}) \quad \Rightarrow \quad \mathbf{AB} = \mathbf{ba}^T - (\mathbf{a} \cdot \mathbf{b})\mathbf{1}. \quad (82)$$

Taking the trace gives

$$\text{Tr}(\mathbf{AB}) = \text{Tr}(\mathbf{ba}^T) - 3(\mathbf{a} \cdot \mathbf{b}) = \mathbf{a} \cdot \mathbf{b} - 3\mathbf{a} \cdot \mathbf{b} = -2\mathbf{a} \cdot \mathbf{b}. \quad (83)$$

□

Lemma 5. Dot products after different orthogonal maps. *The dot product of two vectors $\mathbf{a}$ and $\mathbf{b}$, transformed by r.m.s $\mathbf{C}, \mathbf{D} \in O(3)$, respectively, is equivalent to the dot product of modified vectors:*

$$(\mathbf{Ca}) \cdot (\mathbf{Db}) = \mathbf{a} \cdot (\mathbf{C}^T \mathbf{Db}) = (\mathbf{D}^T \mathbf{Ca}) \cdot \mathbf{b}. \quad (84)$$

The relative orientation is carried by $\mathbf{C}^T \mathbf{D}$ or, equivalently, by $\mathbf{D}^T \mathbf{C}$.

The proof follows directly from the Euclidean inner product.

The cross product also depends on the relative orientation: after forming it in a chosen reference orientation, the result must rotated back with a handedness-accounting factor.

Lemma 6. Cross products after different orthogonal maps. *The cross product of two vectors $\mathbf{a}$ and $\mathbf{b}$, transformed by r.m.s $\mathbf{C}, \mathbf{D} \in O(3)$, respectively, can be equated using reference transformations.*

Using either $\mathbf{C}$ or $\mathbf{D}$ as the reference transformation, both vectors in the cross product have the same outer transformation that respectively matches them:

$$(\mathbf{Ca}) \times (\mathbf{Db}) = \det(\mathbf{C})\mathbf{C} [\mathbf{a} \times (\mathbf{C}^T \mathbf{Db})] = \det(\mathbf{D})\mathbf{D} [(\mathbf{D}^T \mathbf{Ca}) \times \mathbf{b}]. \quad (85)$$

Proof Set $\mathbf{b}'' = \mathbf{C}^T \mathbf{D} \mathbf{b}$. Then $\mathbf{D} \mathbf{b} = \mathbf{C} \mathbf{b}''$, and by using **Lem. 1**, we directly obtain

$$(\mathbf{C} \mathbf{a}) \times (\mathbf{D} \mathbf{b}) = (\mathbf{C} \mathbf{a}) \times (\mathbf{C} \mathbf{b}'') = \det(\mathbf{C}) \mathbf{C} (\mathbf{a} \times \mathbf{b}'') = \det(\mathbf{C}) \mathbf{C} \left[\mathbf{a} \times (\mathbf{C}^T \mathbf{D} \mathbf{b}) \right], \quad (86)$$

where the second form in **Eq. 85** follows in the same way after setting $\mathbf{a}'' = \mathbf{D}^T \mathbf{C} \mathbf{a}$. $\square$

To keep the $so(3) \simeq \mathbb{R}^3$ identification explicit, **Eq. 85** can also be written as

$$(\mathbf{C} \mathbf{a}) \times (\mathbf{D} \mathbf{b}) = \det(\mathbf{C}) \mathbf{C} [\mathbf{a}]_{\times} (\mathbf{C}^T \mathbf{D} \mathbf{b}). \quad (87)$$

If $\mathbf{C} = \mathbf{D}$ is a p.r.m., the relative orthogonal transformation and the determinant disappear:

$$(\mathbf{C} \mathbf{a}) \times (\mathbf{C} \mathbf{b}) = \mathbf{C} (\mathbf{a} \times \mathbf{b}), \quad (88)$$

where for $\mathbf{C}$ -imp.r.m., an extra minus sign appears as the same handedness effect encountered in **Lem. 3**.

3 Kinematics of a rigid body

A rigid body has both a spatial position and an orientation: we describe the first by its c.o.m. and the second by a time-dependent element r.m. In this section we connect this geometric description to the kinetic energy, canonical momenta and equations of motion (e.o.m.) later used for the nanorotor.

3.1 Rigid configurations and moving frames

Consider n point masses m_k moving in 3D Euclidean space and denote their positions at time t by $P_k(t)$. We choose a fixed right-handed orthonormal laboratory f.o.r. with origin O and basis triple $\vec{e}_O$, so that

$$\vec{OP}_k(t) = \vec{e}_O \mathbf{s}_k(t). \quad (89)$$

The c.o.m. $C(t)$ and total mass m are then

$$\vec{OC}(t) = \vec{e}_O \mathbf{c}(t) \quad (90)$$

$$\mathbf{c}(t) := \frac{1}{m} \sum_{k=1}^n m_k \mathbf{s}_k(t) \quad \left| \quad m := \sum_{k=1}^n m_k. \quad (91)$$

We also introduce the lab frame translated to the c.o.m., denoted by $\vec{e}_C$, and write the relative position of the k^{th} mass as

$$\vec{CP}_k(t) = \vec{e}_C \mathbf{r}_k(t), \quad \sum_{k=1}^n m_k \mathbf{r}_k(t) = \mathbf{0}. \quad (92)$$

A 3×3 matrix can encode the orientation of a f.o.r., but not also its location. To do both, one needs the affine/homogeneous form

$$\begin{pmatrix} \vec{e}_O^T \mathbf{t}_O \\ \mathbf{0}^T & 1 \end{pmatrix}, \quad (93)$$

where $\mathbf{t}_O$ is the coordinate column of the frame origin O . $\vec{e}_O$ thus represents the frame orientation and its location is represented by the index O . Just like r.m.s, f.o.r.s affect vector products only through their rotational part, but they must still survive outside any product, to keep specifying the coordinate reference.

Theorem 6. Rigidity through the Gram matrix. A system of point fixed masses is rigid iff the Gram matrix of its c.o.m.-relative vectors (i.e. the result of each individual product) is time-independent:

$$\vec{CP}_k(t) \cdot \vec{CP}_l(t) = \mathbf{r}_k(t)^T \mathbf{r}_l(t) = \rho_{kl} \quad \left| \quad \dot{\rho}_{kl} = 0, \quad k, l = 1, \dots, n. \quad (94)$$

Rigidity therefore means that the shape is fixed up to a global rotation.

The proof is given in **Appendix A.5**.

We attach a right-handed orthonormal f.o.r. $\vec{f}_C(t)$ (centered at $C(t)$) such that all mass coordinates are constant in this body frame as it may rotate around its c.o.m. With our convention that basis triples multiply component columns from the left, the body frame is related to the translated lab frame by

$$\vec{f}_C(t) = \vec{e}_C \mathbf{F}(t) \quad \left| \quad \vec{e}_C = \vec{f}_C(t) \mathbf{F}(t)^T \quad \left| \quad \mathbf{F}(t) \in SO(3). \quad (95)$$

If $\mathbf{r}_k^0 = \mathbf{r}_k(t=0)$ denotes the constant body-frame components of $\vec{CP}_k$, then

$$\vec{CP}_k(t) = \vec{f}_C(t) \mathbf{r}_k^0 = \vec{e}_C \mathbf{F}(t) \mathbf{r}_k^0 = \vec{e}_C \mathbf{r}_k(t) \quad \left| \quad \mathbf{r}_k(t) = \mathbf{F}(t) \mathbf{r}_k^0 \quad \left| \quad \mathbf{s}_k(t) = \mathbf{c}(t) + \mathbf{r}_k(t) = \mathbf{c}(t) + \mathbf{F}(t) \mathbf{r}_k^0, \quad (96)$$

fixing $\mathbf{r}_k^0$ in the body frame, while $\mathbf{r}_k(t)$ is generally time-dependent in the translated lab frame. This is the rigid body's kinematic content: three translational d.o.f. in $\mathbf{c}(t)$ and three rotational d.o.f. in $\mathbf{F}(t)$.

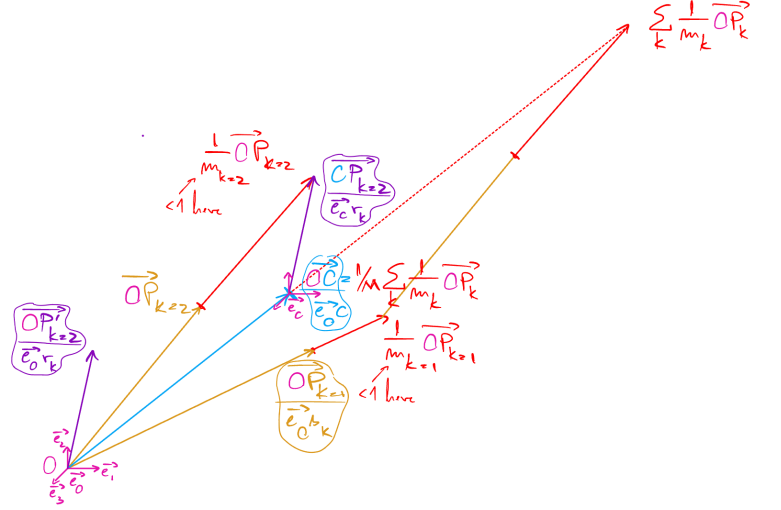


Figure 3: Schematic of the different vectors composing our system of coordinates, the c.o.m.s, f.o.r.s and the different relationships between them. In this visualization we used masses with amplitudes < 1 in order to get $\frac{1}{m_k} \|\vec{OP}_k\|^2 > \|\vec{OP}_k\|^2$ so the former is more easily visible.

3.2 Angular velocity and the zxz Euler parametrization

Since $\mathbf{F}(t) \in SO(3)$, both $\dot{\mathbf{F}}\mathbf{F}^T$ and $\mathbf{F}^T\dot{\mathbf{F}}$ are antisymmetric:

$$\mathbf{0} = \dot{\mathbf{I}} = \frac{d}{dt}[\mathbf{F}\mathbf{F}^T] = \dot{\mathbf{F}}\mathbf{F}^T + \mathbf{F}\dot{\mathbf{F}}^T \Rightarrow \dot{\mathbf{F}}\mathbf{F}^T = -[\dot{\mathbf{F}}\mathbf{F}^T]^T, \quad (97)$$

and we therefore define the space-fixed (sf) and body-fixed (bf) angular-velocity matrices as

$$\boldsymbol{\Omega}^{\text{sf}} := \dot{\mathbf{F}}\mathbf{F}^T \quad \Big| \quad \boldsymbol{\Omega}^{\text{bf}} := \mathbf{F}^T\dot{\mathbf{F}} \quad \Big| \quad \boldsymbol{\Omega}^{\text{sf}}, \boldsymbol{\Omega}^{\text{bf}} \in so(3) \quad \Rightarrow \quad \boldsymbol{\Omega}^{\text{bf}} = \mathbf{F}^T\boldsymbol{\Omega}^{\text{sf}}\mathbf{F} = \text{Ad}_{\mathbf{F}^T} \boldsymbol{\Omega}^{\text{sf}}. \quad (98)$$

Using the inverse-vector map of antisymmetric matrices introduced in the previous section, we write

$$\boldsymbol{\Omega}^{\text{sf}} = [\boldsymbol{\omega}^{\text{sf}}]_{\times} \quad \Big| \quad \boldsymbol{\Omega}^{\text{bf}} = [\boldsymbol{\omega}^{\text{bf}}]_{\times} \quad \Big| \quad \boldsymbol{\omega}^{\text{bf}} = \mathbf{F}^T\boldsymbol{\omega}^{\text{sf}}. \quad (99)$$

Then, the relative velocity of each mass is

$$\dot{\mathbf{r}}_k(t) = \dot{\mathbf{F}}\mathbf{r}_k^0 = \dot{\mathbf{F}}\mathbf{F}^T\mathbf{r}_k(t) = \boldsymbol{\Omega}^{\text{sf}}\mathbf{r}_k(t) = \boldsymbol{\omega}^{\text{sf}} \times \mathbf{r}_k(t), \quad (100)$$

or, equivalently, in body components,

$$\frac{d}{dt}\overrightarrow{CP_k}(t) = \overrightarrow{\mathbf{f}}_C(t)\boldsymbol{\Omega}^{\text{bf}}\mathbf{r}_k^0 = \overrightarrow{\mathbf{f}}_C(t)(\boldsymbol{\omega}^{\text{bf}} \times \mathbf{r}_k^0). \quad (101)$$

For our chosen Euler zxz parametrization $\mathbf{F}_{zxz}(\alpha, \beta, \gamma)$, α is the precession around the fixed lab z -axis, β is the tilt/libration angle between the lab z -axis and the body symmetry axis, and γ is the spin around the body symmetry axis. For the geometric unit axes, this is equivalently summarized by

$$\boldsymbol{\omega} = \dot{\alpha}\hat{\mathbf{e}}_z + \dot{\beta}\hat{\mathbf{N}} + \dot{\gamma}\hat{\mathbf{f}}_3 \quad \Big| \quad \hat{\mathbf{N}} = c_\gamma\hat{\mathbf{f}}_1 - s_\gamma\hat{\mathbf{f}}_2 \quad \Big| \quad \hat{\mathbf{e}}_z = s_\beta(s_\gamma\hat{\mathbf{f}}_1 + c_\gamma\hat{\mathbf{f}}_2) + c_\beta\hat{\mathbf{f}}_3. \quad (102)$$

We define the angular momentum based on the time-derivative of the three Euler angles:

$$\boldsymbol{\theta} := \begin{pmatrix} \alpha \\ \beta \\ \gamma \end{pmatrix} \quad \Big| \quad \boldsymbol{\omega}^{\text{sf}} = \mathbf{M}(\boldsymbol{\theta})\dot{\boldsymbol{\theta}} \quad \Big| \quad \boldsymbol{\omega}^{\text{bf}} = \mathbf{N}(\boldsymbol{\theta})\dot{\boldsymbol{\theta}}, \quad (103)$$

and we find these matrices to be (for a complete derivation, see **Appendix A.6**):

$$\mathbf{M} = \begin{pmatrix} 0 & c_\alpha & s_\alpha s_\beta \\ 0 & s_\alpha & -c_\alpha s_\beta \\ 1 & 0 & c_\beta \end{pmatrix} \quad \Big| \quad \mathbf{N} = \begin{pmatrix} s_\beta s_\gamma & c_\gamma & 0 \\ s_\beta c_\gamma & -s_\gamma & 0 \\ c_\beta & 0 & 1 \end{pmatrix}. \quad (104)$$

The body-frame form is the one we will use most often, because the inertia tensor of a rigid body is simplest in its principal-axis frame (see **Sec. 3.3**).

At $\beta \in \{0, \pi\}$, the map from Euler rates to angular velocity loses rank:

$$\boldsymbol{\omega}^{\text{bf}}|_{\beta=0} = \begin{pmatrix} c_\gamma\dot{\beta} \\ -s_\gamma\dot{\beta} \\ \dot{\alpha} + \dot{\gamma} \end{pmatrix} \quad \Big| \quad \boldsymbol{\omega}^{\text{bf}}|_{\beta=\pi} = \begin{pmatrix} c_\gamma\dot{\beta} \\ -s_\gamma\dot{\beta} \\ -\dot{\alpha} + \dot{\gamma} \end{pmatrix}, \quad (105)$$

making α and γ become redundant at $\beta \in \{0, \pi\}$. That is, while the physical orientation remains regular, any such parametrization based on Euler angles (or, more broadly, three degrees of freedom) becomes singular at $\beta \in \{0, \pi\}$. This failure of our coordinate system is called *the gimbal lock* and, in **Sec. 5.1.2**, we present an alternative method which avoids it.

3.3 Inertia tensor, angular momentum, and full metric

Let us denote the (antisymmetric) time-dependent position matrix of mass k as

$$\mathbf{X}_k(t) := [\mathbf{r}_k(t)]_{\times} \quad \Big| \quad \mathbf{X}_k^0 := [\mathbf{r}_k^0]_{\times} \quad \Rightarrow \quad \dot{\mathbf{r}}_k = \boldsymbol{\omega}^{\text{sf}} \times \mathbf{r}_k = -\mathbf{X}_k\boldsymbol{\omega}^{\text{sf}}, \quad (106)$$

where the first equality on the r.h.s. velocity equation is **Eq. 100**.

Angular momentum is defined in terms of the distance from the c.o.m. and the momentum vector $\mathbf{p}_k$:

$$\begin{aligned} \mathbf{L}_k &= \mathbf{r}_k \times \mathbf{p}_k = \mathbf{r}_k \times \dot{\mathbf{r}}_k m_k = \mathbf{r}_k \times (\boldsymbol{\omega}_k^{\text{sf}} \times \mathbf{r}_k) m_k \\ &= \mathbf{r}_k \times (-\mathbf{X}_k\boldsymbol{\omega}^{\text{sf}}) m_k = (\mathbf{X}_k\boldsymbol{\omega}^{\text{sf}}) \times \mathbf{r}_k m_k = m_k (\mathbf{X}_k^T \mathbf{X}_k) \boldsymbol{\omega}^{\text{sf}}, \end{aligned} \quad (107)$$

where in the last step we used the antisymmetry of $\mathbf{X}_k$ and the matrix associativity property.

By definition, the inertia tensor relates angular velocity to angular momentum in the following manner:

$$\mathbf{L}_k = \mathbf{I}_k \cdot \boldsymbol{\omega}^{\text{sf}} \quad \Rightarrow \quad \mathbf{I} = \sum_{k=1}^n m_k \mathbf{X}_k^T \mathbf{X}_k = \sum_{k=1}^n m_k \left[(\mathbf{r}_k^T \mathbf{r}_k) \mathbf{1} - \mathbf{r}_k \mathbf{r}_k^T \right] \quad \Rightarrow \quad \mathbf{I} = \mathbf{I}^T, \quad (108)$$

where we made use of **Eq. 62**.

In the body frame, the inertia tensor is constant and one can always chose $\mathbf{F}(t)$ such that the tensor is diagonalized in that frame. That is, choosing the body axes along the principal axes $\hat{\mathbf{f}}_i$ make them the eigenfunctions of the the principal moments I_i and thus

$$\mathbf{I}_0 \hat{\mathbf{f}}_i = I_i \hat{\mathbf{f}}_i \quad \Bigg| \quad \mathbf{I}_0 := \sum_{k=1}^n m_k (\mathbf{X}_k^0)^T \mathbf{X}_k^0 = \text{diag}(I_1, I_2, I_3) \quad \Bigg| \quad \mathbf{I}(t) = \mathbf{F}(t) \mathbf{I}_0 \mathbf{F}(t)^T. \quad (109)$$

If two or more principal moments are equal, the corresponding principal axes are not uniquely determined by the inertia tensor: e.g., if $I_1 = I_2$, any rotation of the axes $\mathbf{f}_1, \mathbf{f}_2$ in their common plane leaves $\mathbf{I}_0 = \text{diag}(I_1, I_1, I_3)$ unchanged. *This is not a coordinate singularity, but a freedom in choosing the body frame inside a physically degenerate eigenspace.*

The total kinetic energy's definition follows from $\mathbf{s}_k = \mathbf{c} + \mathbf{r}_k$ as:

$$T = \frac{1}{2} \sum_{k=1}^n m_k \dot{\mathbf{s}}_k^T \dot{\mathbf{s}}_k = \frac{1}{2} \sum_{k=1}^n m_k (\dot{\mathbf{c}} + \dot{\mathbf{r}}_k)^T (\dot{\mathbf{c}} + \dot{\mathbf{r}}_k) = \frac{1}{2} m \dot{\mathbf{c}}^T \dot{\mathbf{c}} + \frac{1}{2} \sum_{k=1}^n m_k \dot{\mathbf{r}}_k^T \dot{\mathbf{r}}_k \quad (110)$$

$$T_{\text{trans}} = \frac{1}{2} m \dot{\mathbf{c}}^T \dot{\mathbf{c}} \quad \Bigg| \quad T_{\text{rot}} = \frac{1}{2} \sum_{k=1}^n m_k \dot{\mathbf{r}}_k^T \dot{\mathbf{r}}_k \quad (111)$$

where the mixed terms $\dot{\mathbf{c}} \dot{\mathbf{r}}_k$ and $\dot{\mathbf{r}}_k \dot{\mathbf{c}}$ vanish as a consequence of **Thm. 6**, that is, after differentiating $\sum_k m_k \mathbf{r}_k = \mathbf{0}$. We denote the first term as the translational (c.o.m.) kinetic energy of the system.

For the full coordinates $\mathbf{q} := (\mathbf{c}, \boldsymbol{\theta})$, we thus obtain the block diagonal kinetic metric:

$$T = \frac{1}{2} g_{ij} \dot{q}^i \dot{q}^j \quad \Bigg| \quad \mathbf{g}(\mathbf{q}) = \begin{pmatrix} m \mathbf{1}_3 & \mathbf{0} \\ \mathbf{0} & \mathbf{g}_{\text{rot}}(\boldsymbol{\theta}) \end{pmatrix} \quad \Bigg| \quad \det \mathbf{g}(\mathbf{q}) = m^3 I_1 I_2 I_3 s_\beta^2. \quad (112)$$

$\mathbf{g}(\mathbf{q})$'s zero off-diagonal blocks are a statement about the kinetic energy only and do not mean that translation and rotation are dynamically decoupled.

This kinetic cross term $g_{c\theta}$ vanishes because $\mathbf{c}$ is the c.o.m. coordinate, which, following the steps in **Eq. 110**, causes the the rigid-body kinetic decomposition from **Eq. 111**, and not from any assumption that the potential separates.

A velocity-independent potential $V(\mathbf{c}, \boldsymbol{\theta}, \mathbf{S}, t)$ can still couple translation and rotation through the e.o.m.s (e.g. through $\partial^2 V / (\partial c^a \partial \theta^\mu)$ terms), but this is a potential coupling, not a kinetic cross-metric.

Thus, momentum-dependent spin-rotation terms (such as Einstein-de Haas couplings) should instead be added directly to the Hamiltonian and not absorbed into V or $\mathbf{g}$:

$$H = H_{\text{trans}} + H_{\text{rot}} + V(\mathbf{q}, \mathbf{S}, t) + H_{\text{md}}(\mathbf{q}, \mathbf{p}_\theta, \mathbf{S}, t). \quad (113)$$

Using **Eq. 106**, the rotational kinetic energy term can be elaborated as

$$T_{\text{rot}} = \frac{1}{2} \sum_{k=1}^n m_k \dot{\mathbf{r}}_k^T \dot{\mathbf{r}}_k = \frac{1}{2} (\boldsymbol{\omega}^{\text{sf}})^T \left(\sum_{k=1}^n m_k \mathbf{X}_k^T \mathbf{X}_k \right) \boldsymbol{\omega}^{\text{sf}} = \frac{1}{2} (\boldsymbol{\omega}^{\text{sf}})^T \mathbf{I}(t) \boldsymbol{\omega}^{\text{sf}} = \frac{1}{2} (\boldsymbol{\omega}^{\text{bf}})^T \mathbf{I}_0 \boldsymbol{\omega}^{\text{bf}}, \quad (114)$$

where we now see how the minus sign from **Eq. 106** cancels out. Substituting the angular-velocity form in **Eq. 103** into **Eq. 114** lets us define the rotationally kinetic metric:

$$T_{\text{rot}} = \frac{1}{2} \dot{\boldsymbol{\theta}}^T \mathbf{g}_{\text{rot}}(\boldsymbol{\theta}) \dot{\boldsymbol{\theta}} \quad \Bigg| \quad \mathbf{g}_{\text{rot}}(\boldsymbol{\theta}) := \mathbf{N}^T \mathbf{I}_0 \mathbf{N} = \mathbf{M}^T \mathbf{I}(t) \mathbf{M} \quad \Bigg| \quad \det \mathbf{g}_{\text{rot}}(\boldsymbol{\theta}) = I_1 I_2 I_3 s_\beta^2. \quad (115)$$

s_β^2 vanishes both determinants at $\beta \in \{0, \pi\}$, precisely where the Euler-angles parametrizations suffer from a coordinate singularity (it is not a physical loss of rotational degrees of freedom): the gimbal lock.

3.4 Equations of motion

Consider a velocity-independent potential $V(\mathbf{q}, t)$, where other terms such as spin may be later included. The Lagrangian, Hamiltonian, and the Euler–Lagrange equation of motion are then

$$\mathcal{L} = T - V = \frac{1}{2}g_{ij}\dot{q}^i\dot{q}^j - V \quad \left| \quad H = \frac{1}{2}g^{ij}p_i p_j + V \quad \right| \quad \frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}^i} \right) = \frac{\partial \mathcal{L}}{\partial q^i}. \quad (116)$$

From them, we obtain the momentum and its first and second time derivative

$$p_i = \frac{\partial \mathcal{L}}{\partial \dot{q}^i} = g_{ij}\dot{q}^j \quad \left| \quad \dot{p}_i = \frac{\partial \mathcal{L}}{\partial q^i} = \frac{1}{2} \frac{\partial g_{jk}}{\partial q^i} \dot{q}^j \dot{q}^k - \frac{\partial V}{\partial q^i} = \frac{1}{2} \frac{\partial g_{jk}}{\partial q^i} g^{jm} g^{kn} p_m p_n - \frac{\partial V}{\partial q^i}. \quad (117)$$

Equivalently, the second-order coordinate equation is given by

$$\begin{aligned} \ddot{q}^i &= \frac{d}{dt} (g^{ij} p_j) = \frac{\partial g^{ij}}{\partial q^k} \dot{q}^k p_j + g^{ij} \dot{p}_j = \frac{\partial g^{ij}}{\partial q^k} \dot{q}^k g_{jm} \dot{q}^m + g^{ij} \left(\frac{1}{2} \frac{\partial g_{km}}{\partial q^j} \dot{q}^k \dot{q}^m - \frac{\partial V}{\partial q^j} \right) \\ &= \frac{\partial g^{ij}}{\partial q^k} g^{kl} p_l p_j + g^{ij} \left(\frac{1}{2} \frac{\partial g_{kl}}{\partial q^j} g^{km} g^{ln} p_m p_n - \frac{\partial V}{\partial q^j} \right) \end{aligned} \quad (118)$$

$$\ddot{q}^i + \Gamma_{jk}^i \dot{q}^j \dot{q}^k = -g^{ij} \frac{\partial V}{\partial q^j} \quad \left| \quad \Gamma_{jk}^i = \frac{1}{2} g^{il} (\partial_j g_{lk} + \partial_k g_{lj} - \partial_l g_{jk}). \quad (119)$$

Because the kinetic metric is block diagonal, for the spatial equation, the left side of **Eq. 119** is simply

$$m\ddot{\mathbf{c}} = -\nabla_{\mathbf{c}} V. \quad (120)$$

The expression w.r.t. q^i is used in computational methods such as RK4 (see **Sec. 5.1.3**). With appropriate initial conditions, the coordinate velocity $\dot{\mathbf{q}}(t)$ can be iterated and solved for, leading to the evolution of the angular coordinates $\boldsymbol{\theta}(t)$ themselves.

*The rotational metric therefore converts the Euler-angle rates into canonical momenta, determining the rotational Hamiltonian **Eq. 129**. Consequently, independently of the particular interaction potential, it fixes the rotational angle–momentum relations and the kinematic contribution to $\dot{p}_\beta$ (in **Eq. 308**), and hence $\ddot{\beta}$ ’s nonlinear geometrical term (in **Eq. 309**).*

3.5 Axially symmetric cylindrical rotor

If we approximate the nanorotor’s body to be axially symmetric around the body axis $\hat{\mathbf{f}}_3$,

$$\mathbf{I}_0 = \text{diag}(I_\perp, I_\perp, I_\parallel) \quad \left| \quad I_\perp := I_1 = I_2 \quad \right| \quad I_\parallel := I_3, \quad (121)$$

which, for a homogeneous cylinder of radius R , length L and mass m , become (see **Appendix A.7**)

$$R = \sqrt{\frac{m}{L\pi\rho}} \quad \left| \quad I_\parallel = \frac{1}{2}mR^2 \quad \right| \quad I_\perp = \frac{1}{12}m(3R^2 + L^2) \quad \left| \quad \frac{I_\parallel}{I_\perp} = \frac{6R^2}{3R^2 + L^2}. \quad (122)$$

Thus $2R/L \gg 1$ gives a disk-like rotor with $I_\parallel/I_\perp \rightarrow 2$, while $2R/L \ll 1$ gives a long cylinder with $I_\parallel/I_\perp \rightarrow 0$.

Using **Eq. 115**, the axial symmetry $I_1 = I_2 = I_\perp$ eliminates all terms proportional to $I_1 - I_2$, making the rotational metric and its inverse depend on the Euler angles only through β :

$$\mathbf{g}_{rot} = \begin{pmatrix} I_\perp s_\beta^2 + I_\parallel c_\beta^2 & 0 & I_\parallel c_\beta \\ 0 & I_\perp & 0 \\ I_\parallel c_\beta & 0 & I_\parallel \end{pmatrix} \quad \left| \quad \mathbf{g}_{rot}^{-1} = \begin{pmatrix} \frac{1}{I_\perp s_\beta^2} & 0 & -\frac{c_\beta}{I_\perp s_\beta^2} \\ 0 & \frac{1}{I_\perp} & 0 \\ -\frac{c_\beta}{I_\perp s_\beta^2} & 0 & \frac{1}{I_\parallel} + \frac{c_\beta^2}{I_\perp s_\beta^2} \end{pmatrix} \quad (123)$$

$$\det \mathbf{g}_{rot} = I_\perp^2 I_\parallel s_\beta^2. \quad (124)$$

Restricting **Eq. 117** to the canonical momenta conjugate to the Euler angles gives

$$\mathbf{p}_\theta := \frac{\partial \mathcal{L}}{\partial \dot{\boldsymbol{\theta}}} = \frac{\partial T_{rot}}{\partial \dot{\boldsymbol{\theta}}} = \mathbf{g}_{rot} \dot{\boldsymbol{\theta}} \quad \Rightarrow \quad \dot{\boldsymbol{\theta}} = \mathbf{g}_{rot}^{-1} \mathbf{p}_\theta \quad (125)$$

$$\begin{aligned} p_\alpha &= (I_\perp s_\beta^2 + I_\parallel c_\beta^2) \dot{\alpha} + I_\parallel c_\beta \dot{\gamma} \quad \Big| \quad p_\beta = I_\perp \dot{\beta} \quad \Big| \quad p_\gamma = I_\parallel (\dot{\alpha} c_\beta + \dot{\gamma}), \\ &= I_\perp \dot{\alpha} s_\beta^2 + p_\gamma c_\beta \end{aligned} \quad (126)$$

which we solve for the generalized angular velocities (i.e., Euler-angle rates):

$$\dot{\alpha} = \frac{p_\alpha - p_\gamma c_\beta}{I_\perp s_\beta^2} \quad \Big| \quad \dot{\beta} = \frac{p_\beta}{I_\perp} \quad \Big| \quad \dot{\gamma} = \frac{p_\gamma}{I_\parallel} - \frac{c_\beta (p_\alpha - p_\gamma c_\beta)}{I_\perp s_\beta^2}. \quad (127)$$

Similarly, using **Eqs. 115, 117**, the kinetic energy is

$$\begin{aligned} T_{rot} &= \frac{1}{2} \dot{\boldsymbol{\theta}}^T \mathbf{g}_{rot} \dot{\boldsymbol{\theta}} = \frac{1}{2} (g_{rot})_{ij} \dot{\theta}^i \dot{\theta}^j = \frac{1}{2} [I_\perp (\dot{\alpha}^2 s_\beta^2 + \dot{\beta}^2) + I_\parallel (\dot{\alpha} c_\beta + \dot{\gamma})^2] \\ &= \frac{1}{2} \mathbf{p}_\theta^T \mathbf{g}_{rot}^{-1} \mathbf{p}_\theta = \frac{1}{2} (g_{rot})^{ij} p_i p_j = \frac{1}{2} \left[\frac{p_\beta^2}{I_\perp} + \frac{p_\gamma^2}{I_\parallel} + \frac{(p_\alpha - p_\gamma \cos \beta)^2}{I_\perp \sin^2 \beta} \right], \end{aligned} \quad (128)$$

which we use to obtain the rotational Hamiltonian which is defined through the Legendre transform

$$H_{rot} := \mathbf{p}_\theta^T \dot{\boldsymbol{\theta}} - T_{rot} = \mathbf{p}_\theta^T \mathbf{g}_{rot}^{-1} \mathbf{p}_\theta - \frac{1}{2} \mathbf{p}_\theta^T \mathbf{g}_{rot}^{-1} \mathbf{p}_\theta = \frac{1}{2} \mathbf{p}_\theta^T \mathbf{g}_{rot}^{-1} \mathbf{p}_\theta = T_{rot}. \quad (129)$$

For a free rotor, the Lagrangian only contains kinetic energy, making $H_{rot} = T_{rot}$ as physical energies, but not as functions: $T_{rot}(\boldsymbol{\theta}, \dot{\boldsymbol{\theta}})$ is written in terms of Euler-angle rates, while $H_{rot}(\boldsymbol{\theta}, \mathbf{p})$ is in terms of the conjugate momenta. Thus, during quantization by transformation of the canonical momenta into the operators associated with the cyclic Euler coordinates, it is the Hamiltonian form that we use.

If the initial rotation is purely around the body symmetry axis,

$$\dot{\alpha}(0) = 0 \quad \Big| \quad \dot{\beta}(0) = 0 \quad \Big| \quad \dot{\gamma}(0) = \omega_0 \quad \Big| \quad \beta(0) = \beta_0 \quad (130)$$

$$p_\alpha(0) = I_\parallel \omega_0 c_{\beta_0} \quad \Big| \quad p_\beta(0) = 0 \quad \Big| \quad p_\gamma(0) = I_\parallel \omega_0. \quad (131)$$

These initial values can be used at later times only for those momenta which are conserved. In particular, if the full Hamiltonian is independent of α and γ ,

$$\dot{p}_\alpha = -\frac{\partial H}{\partial \alpha} = 0 \quad \Big| \quad \dot{p}_\gamma = -\frac{\partial H}{\partial \gamma} = 0, \quad (132)$$

then they are cyclic coordinates:

$$p_\alpha(t) = p_\alpha(0) = I_\parallel \omega_0 c_{\beta_0} \quad \Big| \quad p_\gamma(t) = p_\gamma(0) = I_\parallel \omega_0. \quad (133)$$

Substitution into **Eq. 127** gives the kinematic precession and spin rates

$$\dot{\alpha} = \frac{I_\parallel \omega_0 \frac{c_{\beta_0} - c_\beta}{s_\beta^2}}{I_\perp} \quad \Big| \quad \dot{\beta} = 0 \quad \Big| \quad \dot{\gamma} = \frac{I_\parallel \omega_0 \frac{1 - c_{\beta_0} c_\beta}{s_\beta^2}}{I_\perp} + \omega_0 \left(1 - \frac{I_\parallel}{I_\perp} \right). \quad (134)$$

The angles α and γ themselves are not constants of motion; rather, their conjugate momenta are conserved. The factors c_{β_0} come from the initially fixed conserved momenta, whereas c_β and s_β are evaluated at the instantaneous angle $\beta = \beta(t)$. Thus β remains dynamical, while α and γ are reconstructed from the conserved momenta once $\beta(t)$ is known.

These two relations are useful as the magnetic potential primarily drives the libration angle β , while α and γ follow from the conserved momenta whenever they remain cyclic.

The momentum p_β is not conserved. Although $p_\beta(0) = 0$, the later value $p_\beta(t) = I_\perp \dot{\beta}(t)$ must be found from the β e.o.m.,

$$\ddot{\beta} = \frac{(p_\alpha - p_\gamma c_\beta)(p_\alpha c_\beta - p_\gamma)}{I_\perp^2 s_\beta^3} - \frac{1}{I_\perp} \frac{\partial V}{\partial \beta} = -\frac{I_\parallel^2 \omega_0^2 (c_{\beta_0} - c_\beta)(1 - c_{\beta_0} c_\beta)}{I_\perp^2 s_\beta^3} - \frac{1}{I_\perp} \frac{\partial V}{\partial \beta}. \quad (135)$$

3.6 Angular momentum and rotational symmetries

The angular momentum is most transparent when written in both frames:

$$\mathbf{L}^{\text{sf}} := \mathbf{I}(t)\boldsymbol{\omega}^{\text{sf}} = \mathbf{F}\mathbf{L}^{\text{bf}} \quad \Big| \quad \mathbf{L}^{\text{bf}} := \mathbf{I}_0\boldsymbol{\omega}^{\text{bf}} = \mathbf{F}^T\mathbf{L}^{\text{sf}}, \quad (136)$$

where in the set of second equalities we used $\mathbf{I}(t) = \mathbf{F}\mathbf{I}_0\mathbf{F}^T$ and $\boldsymbol{\omega}^{\text{bf}} = \mathbf{F}^T\boldsymbol{\omega}^{\text{sf}}$.

The canonical Euler momenta are not, in general, the components of $\mathbf{L}$. Rather, using **Eq. 103**,

$$p_\theta = \frac{\partial T_{\text{rot}}}{\partial \dot{\theta}} = \mathbf{g}_{\text{rot}}\dot{\theta} = \mathbf{N}^T\mathbf{L}^{\text{bf}} = \mathbf{M}^T\mathbf{L}^{\text{sf}}, \quad (137)$$

which, for the zxz parametrization, implies the two especially useful identities

$$p_\alpha = L_z^{\text{sf}} \quad \Big| \quad p_\gamma = L_3^{\text{bf}}, \quad (138)$$

meaning that, in zxz , α generates rotation about the space-fixed z -axis, while γ generates one about the body-fixed symmetry axis $\hat{\mathbf{f}}_3$, as per the parametrization's definition.

These identities explain why p_α and p_γ are conserved whenever the Hamiltonian is independent of α and γ , respectively: they are symmetry statements, not automatic consequences of using Euler angles.

If an external torque $\boldsymbol{\tau}^{\text{sf}}$ acts on the body, then

$$\begin{aligned} \boldsymbol{\tau}^{\text{sf}} &:= \dot{\mathbf{L}}^{\text{sf}} \\ &= \frac{d}{dt}(\mathbf{F}\mathbf{L}^{\text{bf}}) = \mathbf{F}\dot{\mathbf{L}}^{\text{bf}} + \mathbf{F}\dot{\mathbf{L}}^{\text{bf}} = \mathbf{F}(\dot{\mathbf{L}}^{\text{bf}} + \boldsymbol{\Omega}^{\text{bf}}\mathbf{L}^{\text{bf}}) \\ &= \mathbf{F}(\dot{\mathbf{L}}^{\text{bf}} + \boldsymbol{\omega}^{\text{bf}} \times \mathbf{L}^{\text{bf}}) = \mathbf{F}\boldsymbol{\tau}^{\text{bf}} \quad \Rightarrow \quad \boldsymbol{\tau}^{\text{bf}} := \mathbf{F}^T\boldsymbol{\tau}^{\text{sf}} \\ &= \dot{\mathbf{L}}^{\text{bf}} + \boldsymbol{\omega}^{\text{bf}} \times \mathbf{L}^{\text{bf}}, \end{aligned} \quad (139)$$

For zero external torque ($\boldsymbol{\tau}^{\text{sf}} = \boldsymbol{\tau}^{\text{bf}} = \mathbf{0}$), remembering that $\dot{\mathbf{I}}_0 = \mathbf{0}$, gives Euler's equations

$$\dot{\mathbf{L}}^{\text{bf}} = \mathbf{I}_0\dot{\boldsymbol{\omega}}^{\text{bf}} = -\boldsymbol{\omega}^{\text{bf}} \times \mathbf{L}^{\text{bf}} \quad (140)$$

$$I_1\dot{\omega}_1^{\text{bf}} = (I_2 - I_3)\omega_2^{\text{bf}}\omega_3^{\text{bf}} \quad \Big| \quad I_2\dot{\omega}_2^{\text{bf}} = (I_3 - I_1)\omega_3^{\text{bf}}\omega_1^{\text{bf}} \quad \Big| \quad I_3\dot{\omega}_3^{\text{bf}} = (I_1 - I_2)\omega_1^{\text{bf}}\omega_2^{\text{bf}}, \quad (141)$$

which, for an axially symmetric rotor, $I_1 = I_2$, gives:

$$\dot{\omega}_3^{\text{bf}} = 0, \quad (142)$$

making ω_3^{bf} and $\dot{\mathbf{L}}_3^{\text{bf}}$ constants of motion. If, however, we have spherical symmetry, $I_1 = I_2 = I_3$, then $\dot{\boldsymbol{\omega}}^{\text{bf}}$ and $\dot{\mathbf{L}}^{\text{bf}}$ become constants of motion. The spin around the symmetry axis is thus constant unless an external torque has a component along said axis.

Finally, one must distinguish frame invariance from conservation. Since $\mathbf{F}$ is orthogonal,

$$(\mathbf{L}^{\text{sf}})^T \mathbf{L}^{\text{sf}} = (\mathbf{L}^{\text{bf}})^T \mathbf{L}^{\text{bf}}, \quad (143)$$

an equality which is true at every instant, since both vectors represent the same geometric angular momentum. In contrast, conservation of the magnitude of angular momentum requires

$$\frac{d}{dt} \left[(\mathbf{L}^{\text{sf}})^2 \right] = 2\mathbf{L}^{\text{sf}} \cdot \boldsymbol{\tau}^{\text{sf}} = 0 \quad \Longleftrightarrow \quad \boldsymbol{\tau}^{\text{sf}} = \mathbf{0} \quad \text{or} \quad \boldsymbol{\tau}^{\text{sf}} \perp \mathbf{L}^{\text{sf}}. \quad (144)$$

Full vector conservation, however, requires the stronger condition $\boldsymbol{\tau}^{\text{sf}} = \mathbf{0}$.

In conclusion, the preceding derivations separate the model-independent kinetic geometry from the model-dependent interaction (set by the particular potential). The next section retains this structure while deriving the interaction from a quantum spin: its adiabatic eigenenergy becomes a classical branch potential, while the same moving-frame kinematics produces the spin-rotation term $-\hbar\boldsymbol{\omega}^{\text{s}} \cdot \mathbf{S}$.

4 Semiclassical spin–rotational dynamics of a cylindrical nanorotor

We specialize the framework of kinematics and mechanics of **Secs. 2, 3** to that of a cylindrical nanorotor which contains a fixed spin defect and is placed in a prescribed magnetic field.

While the c.o.m. and orientation remain classical (the angular-velocity maps, rotational metric and corresponding canonical kinetic Hamiltonian are inherited), we introduce an internal quantum spin placed as a defect in the ND. Its Hamiltonian depends on the field sampled by the moving defect expressed in its rotating frame, and its adiabatic eigenenergy supplies a branch-dependent potential for the classical mechanical motion.

We first construct the complete three-dimensional model, from which we the centered, aligned and one-dimensional limit is then obtained. The order is essential: an off-center defect, a transverse zero-field splitting or a non-axial field may remove the cyclic symmetries used in the reduced model; they therefore cannot be introduced after those symmetries have already been assumed.

4.1 Physical system and model hierarchy

4.1.1 Rotor, defect position and spin frame

Let the c.o.m. coordinates in the fixed laboratory f.o.r. and the body-to-laboratory r.m. (see **Eq. 20**) be

$$\mathbf{c}(t) = (c_x(t) \ c_y(t) \ c_z(t))^T \quad \Bigg| \quad \mathbf{F}(t) = \mathbf{F}_{zzz}(\alpha(t), \beta(t), \gamma(t)) \in SO(3). \quad (145)$$

The body frame $\vec{f}_C(t)$ is centered at the c.o.m. C, with its third body axis $\hat{f}_3(t)$ chosen along the cylinder symmetry axis. The radius, mass density and principal moments were fixed in **Eq. 122**.

The spin defect occupies a fixed point of the rigid body $\mathbf{d}_0$ in its constant body-frame coordinate, which, in laboratory coordinates is

$$\mathbf{r}_s(t) = \mathbf{c}(t) + \mathbf{F}(t)\mathbf{d}_0, \quad (146)$$

meaning that any formula in which the defect displacement depends only on β assumes a restricted geometry or planar motion; in the general three-dimensional case, $\mathbf{F}(t)\mathbf{d}_0$ depends on the full rotor orientation.

We encode the fixed relative orientation of the defect's principal axes and that of the cylinder by a constant r.m. $\mathbf{C}_s \in SO(3)$. The ND spin frame is thus centered at the defect, and its basis triple is

$$\overrightarrow{\mathbf{m}}_{r_s} = \overrightarrow{\mathbf{e}}_{r_s} \mathbf{F} \mathbf{C}_s \quad (147)$$

$$\boxed{\mathbf{F} = \mathbf{R}_{sf \leftarrow bf} \quad \Bigg| \quad \mathbf{C}_s = \mathbf{R}_{bf \leftarrow s} \quad \Bigg| \quad \mathbf{F} \mathbf{C}_s = \mathbf{R}_{sf \leftarrow s}}. \quad (148)$$

The spin quantization axis in the laboratory frame is consequently

$$\mathbf{n}_s^{\text{sf}} = \mathbf{F} \mathbf{C}_s (0 \ 0 \ 1)^T. \quad (149)$$

Let $\mathbf{B}^{\text{sf}}(\mathbf{r}, t)$ denote the magnetic field in the laboratory frame. In the spin frame, at the defect, it is:

$$\mathbf{B}^s := (B_1 \ B_2 \ B_3)^T = \mathbf{C}_s^T \mathbf{F}^T \mathbf{B}^{\text{sf}}(\mathbf{c} + \mathbf{F} \mathbf{d}_0, t) \quad \Bigg| \quad B_{\parallel} := B_3 \quad \Bigg| \quad B_{\perp}^2 := B_1^2 + B_2^2, \quad (150)$$

where we note that the transverse spin-frame component $B_{\perp}$ may arise either from transverse field components in the laboratory frame or from the projection of an otherwise axial laboratory field onto a misaligned spin frame. Thus, setting $B_{\perp}^{\text{sf}} = 0$ does not generally imply $B_{\perp} = 0$, unless $\mathbf{n}_s^{\text{sf}} \parallel \mathbf{B}^{\text{sf}}(\mathbf{r}_s, t)$.

Eqs. 146, 150 constitute the first direct application of the preceding frame construction.

4.1.2 Hierarchy of models

We develop the theory based on the following hierarchy:

1. The *parent semiclassical model* retains the three c.o.m. coordinates, the exact defect geometry of **Eq. 146**, the complete 3×3 spin Hamiltonian and the finite-volume diamagnetic energy.

2. The *working exact model* retains the complete 3×3 spin Hamiltonian and three-dimensional field, but replaces the finite-volume diamagnetic energy with its point-particle limit.
3. The *analytical effective model* eliminates the $|0\rangle$ spin level perturbatively and retains every term generated by the resulting second-order 2×2 Hamiltonian.
4. The *reference reduction* takes the centered, aligned, one-dimensional small-transverse-field limit [19].
5. The *linearized model* expands the reference reduction around a small initial libration angle and is used only to identify stability conditions and scaling laws.

Below, the word "exact" always infers the stated semiclassical assumptions: rigid body, prescribed classical magnetic field and adiabatic following of one non-degenerate internal spin branch.

4.2 Magnetic environment and interferometric protocol

4.2.1 Total field and local Maxwell constraints

First, the particle is levitated by the static confinement (trap) field which remains present throughout the interferometric sequence. The Stern–Gerlach field is subsequently turned on, and alternated between its three operating stages, before finally being turned off. Thus, every spin and diamagnetic energy must be evaluated using the total field

$$\mathbf{B}^{\text{sf}}(\mathbf{r}, t) = \mathbf{B}_{\text{tr}}(\mathbf{r}) + \mathbf{B}_{\text{SG}}(\mathbf{r}, t). \quad (151)$$

If confinement is instead represented by an effective potential, one sets $\mathbf{B}_{\text{tr}} = \mathbf{0}$ in the spin Hamiltonian and separately retains said potential: these are different physical models and should not be implicitly mixed.

In a current-free region (containing no magnetic material), within the magnetoquasistatic approximation of the slowly switched fields considered in this thesis, a local scalar magnetic potential is introduced:

$$\mathbf{B} = -\mu_0 \nabla \Phi \quad \Bigg| \quad \nabla \times \mathbf{B} = \mathbf{0} \quad \Bigg| \quad \nabla \cdot \mathbf{B} = 0 \Rightarrow \nabla^2 \Phi = 0, \quad (152)$$

where the first condition neglects the displacement-current correction and is therefore a local magnetoquasistatic statement. For a general local expansion

$$B_i(\mathbf{r}, t) = B_i(\mathbf{r}_0, t) + G_{ij}(t) \Delta r_j + \frac{1}{2} H_{ijk}(t) \Delta r_j \Delta r_k + \dots \quad \Bigg| \quad \Delta \mathbf{r} := \mathbf{r} - \mathbf{r}_0, \quad (153)$$

where the third and second parts of **Eq. 152**, respectively, require the linear gradient tensor to satisfy

$$\mathbf{G} = \mathbf{G}^T \quad \Bigg| \quad \text{Tr } \mathbf{G} = 0. \quad (154)$$

Thus, a longitudinal gradient necessarily generates a transverse field structure: restricting the c.o.m. to one axis does not make the complete field one-dimensional, particularly when the spin is off-center.

4.2.2 Static multipole confinement field

The three lowest non-vanishing multipole potentials employed in the magneto-gravitational trap of [10] are:

$$\begin{aligned} \text{linear quadrupole: } \Phi_{2,-2}(x, y, z) &= \left(\frac{a_2 y_0}{2\mu_0} \right) \left(\frac{r}{y_0} \right)^2 \left(\frac{1}{2} \sqrt{\frac{15}{\pi}} \frac{xy}{r^2} \right) = \left(\frac{\sqrt{15}}{4\sqrt{\pi}} \frac{a_2}{y_0 \mu_0} \right) xy \\ \text{hexapole: } \Phi_{3,1}(x, y, z) &= \left(\frac{a_3 y_0}{3\mu_0} \right) \left(\frac{r}{y_0} \right)^3 \left(\frac{1}{4} \sqrt{\frac{21}{2\pi}} \frac{x(4z^2 - x^2 - y^2)}{r^3} \right) = \left(\frac{\sqrt{42}}{24\sqrt{\pi}} \frac{a_3}{y_0^2 \mu_0} \right) x(4z^2 - x^2 - y^2) \\ \text{octopole: } \Phi_{4,-4}(x, y, z) &= \left(\frac{a_4 y_0}{4\mu_0} \right) \left(\frac{r}{y_0} \right)^4 \left(\frac{3}{4} \sqrt{\frac{35}{\pi}} \frac{xy(x^2 - y^2)}{r^4} \right) = \left(\frac{3\sqrt{35}}{16\sqrt{\pi}} \frac{a_4}{y_0^3 \mu_0} \right) xy(x^2 - y^2), \end{aligned} \quad (155)$$

where y_0 is trap length scale, i.e., the distance from the centre of the trap to the top&bottom pole pieces, μ_0 is the permeability of free space, $r = \sqrt{x^2 + y^2 + z^2}$ is the distance from the geometrical centre of the trap and a_2, a_3, a_4 its fitted field coefficients.

Within the three-mode approximation, the total magnetic field is

$$\vec{B}(x, y, z) \approx -\mu_0 \vec{\nabla} (\Phi_{2,-2}(x, y, z) + \Phi_{3,1}(x, y, z) + \Phi_{4,-4}(x, y, z)) \quad (156)$$

$$\boxed{\begin{aligned} B_{\text{tr},x} &= -b_2 y - b_3 (4z^2 - 3x^2 - y^2) - b_4 y (3x^2 - y^2) \\ B_{\text{tr},y} &= -b_2 x + 2b_3 xy - b_4 x (x^2 - 3y^2) \\ B_{\text{tr},z} &= -8b_3 xz, \end{aligned}} \quad (157)$$

where we defined (in units of T/m, T/m² and T/m³, respectively)⁶

$$b_2 := \frac{\sqrt{15} a_2}{4\sqrt{\pi} y_0} \quad \Bigg| \quad b_3 := \frac{\sqrt{42} a_3}{24\sqrt{\pi} y_0^2} \quad \Bigg| \quad b_4 := \frac{3\sqrt{35} a_4}{16\sqrt{\pi} y_0^3}. \quad (159)$$

After a 45° rotation in the linear quadrupole's x - y plane, its gradient is diagonal, so its difference from other familiar linear-quadrupole expressions may only be a coordinate convention.

On the $x=z=0$ line, the field becomes

$$\mathbf{B}_{\text{tr}}(0, y, 0) = (-b_2 y + b_3 y^2 + b_4 y^3) \hat{e}_x, \quad (160)$$

thus the trap's equilibrium field being generally transverse to its vertical y -axis and axial z -axis. This point becomes important to define the spin quantization direction once $\mathbf{B}_{\text{SG}}$ is introduced.

4.2.3 Stern–Gerlach field and smooth switching

The interferometric field is constructed from the three source-free profiles introduced in [20]⁷ and, in an idealized form, in [19]. Together with the respective scalar magnetic field, they may be written as

$$\mathbf{B}_{\text{SG}}(x, z, t) = -G(t)x\hat{e}_x + [B_c(t) + G(t)z]\hat{e}_z \quad \Rightarrow \quad \Phi_{\text{SG}}(x, z, t) = \frac{G(t)}{2\mu_0} (x^2 - z^2) - \frac{B_c(t)}{\mu_0} z, \quad (161)$$

with the three nominal operating profiles

$$\boxed{\begin{aligned} (B_c, G) &= (B_0, -\eta) &\Rightarrow & \mathbf{B}_{\text{SG}}^{(1)}(x, z) = \eta x \hat{e}_x + (B_0 - \eta z) \hat{e}_z \\ (B_c, G) &= (B_1, 0) &\Rightarrow & \mathbf{B}_{\text{SG}}^{(2)}(x, z) = B_1 \hat{e}_z \\ (B_c, G) &= (-B_0, \eta) &\Rightarrow & \mathbf{B}_{\text{SG}}^{(3)}(x, z) = -\eta x \hat{e}_x + (-B_0 + \eta z) \hat{e}_z \end{aligned}} \quad \text{for } B_0, B_1, \eta > 0. \quad (162)$$

For any spatially uniform $B_c(t)$ and $G(t)$, each profile satisfies the spatial Maxwell constraints:

$$\nabla \cdot \mathbf{B}_{\text{SG}} = -G + G = 0 \quad \Bigg| \quad \nabla \times \mathbf{B}_{\text{SG}} = \mathbf{0}. \quad (163)$$

The first profile generates the spin-dependent Stern–Gerlach momentum through its gradient, separating the two trajectories. The second allows the trajectories to continue to separate through the already acquired momentum, aside from the maintained static confinement field and the corresponding diamagnetic dynamics which remain present. Its nonzero homogeneous bias B_1 maintains a finite Zeeman splitting and a well-defined spin-quantization axis, thereby avoiding a low-field region: it suppresses unwanted spin mixing and Majorana transitions. The third profile reverses the gradient and is used, together with the microwave spin flip, to cancel the relative momentum and recombine the trajectories. Its duration and the spin-flip time are not fixed by that of the first stage: they must be ultimately chosen so that

$$\Delta z(t_{\text{closed}}) = 0 \quad \Bigg| \quad \Delta \dot{z}(t_{\text{closed}}) = 0, \quad (164)$$

and will therefore be computed from the complete e.o.m.s.

⁶As an independent check of the normalization and signs of **Eq. 157**, we perform a numerical consistency check. The parameters reported in [10] with $y_{\text{eq}} = -19 \mu\text{m}$, indeed reproduce the quoted equilibrium field for that trap:

$$y_0 = 75 \mu\text{m} \quad \Bigg| \quad a_2 = -1.3 \text{ T} \quad \Bigg| \quad a_3 = 0.018 \text{ T} \quad \Bigg| \quad a_4 = 0.72 \text{ T} \quad \Rightarrow \quad \|\mathbf{B}_{\text{tr}}(0, y_{\text{eq}}, 0)\| = 0.187 \text{ T} \simeq 0.2 \text{ T}. \quad (158)$$

⁷We relabel the original construction: $x_{\text{M}} \mapsto z$ as the superposition coordinate and $y_{\text{M}} \mapsto x$ as its transverse partner.

While the piecewise field above is sufficient for analytical limits, a physical field must vary continuously. We therefore introduce the smooth ramp function $R_{\text{sw}}(t; t_0)$ and the corresponding magnetic-profile a smooth top-hat switching function S_w :

$$R_{\text{sw}}(t; t_0) := \frac{1}{2} \{1 + \tanh[\delta_{\text{sw}}(t - t_0)]\} \quad (165)$$

$$\boxed{S_w(t; t_{\text{on}}, t_{\text{off}}) := R_{\text{sw}}(t; t_{\text{on}}) [1 - R_{\text{sw}}(t; t_{\text{off}})]} \\ = \frac{1}{2} \{ \tanh[\delta_{\text{sw}}(t - t_{\text{on}})] + 1 \} \times \frac{1}{2} \{ \tanh[\delta_{\text{sw}}(t_{\text{off}} - t)] + 1 \}, \quad (166)$$

where $\delta_{\text{sw}} = 10^3 \text{ s}^{-1}$ [20] controls the switching rate. For $\delta_{\text{sw}}(t_{\text{off}} - t_{\text{on}}) \gg 1$,

$$S_w \simeq \begin{cases} 0, & t < t_{\text{on}} \\ 1, & t_{\text{on}} < t < t_{\text{off}} \\ 0, & t > t_{\text{off}}. \end{cases} \quad (167)$$

We thus apply one such envelope to each of the three stages:

$$S_1(t) := S_w(t; t_{\text{SG, on}}, t_{12}) \quad | \quad S_2(t) := S_w(t; t_{12}, t_{23}) \quad | \quad S_3(t) := S_w(t; t_{23}, t_{\text{SG, off}}), \quad (168)$$

thus, adjacent windows share their switching centers and the same nominal rate. The smoothly switched Stern–Gerlach field is then given by

$$\mathbf{B}_{\text{SG}}(x, z, t) = S_1(t) \mathbf{B}_{\text{SG}}^{(1)}(x, z) + S_2(t) \mathbf{B}_{\text{SG}}^{(2)}(x, z) + S_3(t) \mathbf{B}_{\text{SG}}^{(3)}(x, z), \quad (169)$$

or, alternatively, extending the expressions in **Eq. 162** to the entire field,

$$\mathbf{B}_{\text{SG}}(x, z, t) = -G(t)x\hat{\mathbf{e}}_x + [B_c(t) + G(t)z]\hat{\mathbf{e}}_z \quad (170)$$

$$B_c(t) = B_0 S_1(t) + B_1 S_2(t) - B_0 S_3(t) \quad | \quad G(t) = -\eta S_1(t) + \eta S_3(t). \quad (171)$$

Because the envelopes depend only on time, the spatial Maxwell constraints in **Eq. 163** remain satisfied.

Importantly, at the crossover SG field transitions, the adjacent switching functions perform complementary cross-fades (i.e., a partition-of-unity construction):

$$S_1(t) + S_2(t) \simeq 1 \quad \text{around } t = t_{12} \quad | \quad S_2(t) + S_3(t) \simeq 1 \quad \text{around } t = t_{23}, \quad (172)$$

which avoid unwanted amplification or suppression of the field.

When two windows overlap, their magnetic fields physically superpose. Such an overlap is neither an error nor an automatic amplification artifact: it represents the field produced while two control settings are simultaneously active. The overlap must instead be selected so that the total field remains sufficiently far from zero and the transition remains adiabatic.

Since the field enters the e.o.m.'s through its time derivative, we compute that of the switching envelope,

$$\dot{S}_w(t; t_{\text{on}}, t_{\text{off}}) = \frac{\delta_{\text{sw}}}{4} \text{sech}^2[\delta_{\text{sw}}(t - t_{\text{on}})] \{1 + \tanh[\delta_{\text{sw}}(t_{\text{off}} - t)]\} \\ - \frac{\delta_{\text{sw}}}{4} \{1 + \tanh[\delta_{\text{sw}}(t - t_{\text{on}})]\} \text{sech}^2[\delta_{\text{sw}}(t_{\text{off}} - t)], \quad (173)$$

which consequently gives

$$\dot{B}_c(t) = B_0 \dot{S}_1(t) + B_1 \dot{S}_2(t) - B_0 \dot{S}_3(t) \quad | \quad \dot{G}(t) = -\eta \dot{S}_1(t) + \eta \dot{S}_3(t), \quad (174)$$

and the explicit field derivative is

$$\frac{\partial \mathbf{B}_{\text{SG}}}{\partial t} = -\dot{G}(t)x\hat{\mathbf{e}}_x + [\dot{B}_c(t) + \dot{G}(t)z]\hat{\mathbf{e}}_z. \quad (175)$$

There are three considerations about the field at this current stage.

Minimum-field and spin-adiabaticity condition. Adiabatic evolution means that a spin initially prepared in an instantaneous eigenstate of the local spin Hamiltonian remains in the corresponding eigenstate as the complete field changes, continuously following its instantaneous quantisation direction without undergoing an unwanted transition to another spin branch.

To achieve this, $\mathbf{B} \neq 0$ must be maintained, as to prevent the reduction of the Zeeman splitting between the relevant spin states, which will in turn lead to a poorly defined local quantisation axis.

Unfortunately, we are not yet fully equipped to address this point and reserve it to **Sec. 4.3.5**.

The c.o.m. position, orientation and Stern–Gerlach field then change negligibly during the pulse. In our description, the resulting branch bookkeeping is

$$m_s^{(+)}(t) = \begin{cases} +1, & t < t_{\text{flip}} \\ -1, & t > t_{\text{flip}} \end{cases} \quad \Bigg| \quad m_s^{(-)}(t) = -m_s^{(+)}(t). \quad (176)$$

To better understand this process, however, one must consider the transverse zero-field-splitting of the defect and how it enters the Hamiltonian. Again, we are unprepared to fully do so inside the appropriate context, so we again reserve this point to **Sec. 4.3.6**.

Induced and motional electric fields. The explicit time dependence in **Eq. 175** does not enter the conservative magnetic force directly. It contributes to $\dot{H}_s$ in the spin-adiabaticity condition and, through Faraday’s law, generates an electric field.

A vector potential which generates the Stern–Gerlach field of **Eq. 161** is

$$\mathbf{A}_{\text{SG}}(x, z, t) = x [B_c(t) + G(t)z] \hat{\mathbf{e}}_y, \quad \text{since} \quad \nabla \times \mathbf{A}_{\text{SG}} = -G(t)x \hat{\mathbf{e}}_x + [B_c(t) + G(t)z] \hat{\mathbf{e}}_z = \mathbf{B}_{\text{SG}}. \quad (177)$$

Faraday’s law therefore gives the induced electric field

$$\mathcal{E}_{\text{ind}} = -\frac{\partial \mathbf{A}_{\text{SG}}}{\partial t} - \nabla \phi = -x [\dot{B}_c(t) + \dot{G}(t)z] \hat{\mathbf{e}}_y - \nabla \phi, \quad (178)$$

where the first term is the non-conservative Faraday field fixed by R_{sw} and the electrostatic contribution $-\nabla \phi$ depends on the conductors, chip geometry and boundary conditions, and cannot be determined from the magnetic profile alone.

Together with **Eqs. 168, 174**, the bounds satisfied by the switching functions imply

$$0 \leq R_{\text{sw}} \leq 1 \quad \Bigg| \quad 0 \leq \dot{R}_{\text{sw}} \leq \frac{\delta_{\text{sw}}}{2} \quad \Rightarrow \quad \left| \dot{B}_c(t) \right| \leq \frac{\delta_{\text{sw}}}{2} (2B_0 + B_1) \quad \Bigg| \quad \left| \dot{G}(t) \right| \leq \delta_{\text{sw}} \eta. \quad (179)$$

For the maximum x and z coordinates reached during the system’s evolution,

$$x_{\text{max}} := \max_{\ell, t} |r_{s, \ell, x}(t)| \quad \Bigg| \quad z_{\text{max}} := \max_{\ell, t} |r_{s, \ell, z}(t)|, \quad (180)$$

where $\ell \in \{+, -\}$ labels the two interferometer arms, the Faraday component of **Eq. 178** then satisfies

$$\mathcal{E}_{\text{F}}^{\text{max}} \leq x_{\text{max}} \delta_{\text{sw}} \left[\frac{2B_0 + B_1}{2} + \eta z_{\text{max}} \right]. \quad (181)$$

A motional electric field is also introduced by the passage of the defect through $\mathbf{B}$. To first order in v_s/c (where c is the speed of light), the electric field in the instantaneous rest frame of the defect is

$$\mathcal{E}_s^{\text{rest}} = \mathcal{E}_{\text{ind}} + \mathbf{v}_s \times \mathbf{B}^{\text{sf}} + \mathcal{O}\left(\frac{v_s^2}{c^2}\right) \quad \Bigg| \quad \mathbf{v}_s := \dot{\mathbf{r}}_s = \dot{\mathbf{c}} + \dot{\mathbf{F}} \mathbf{d}_0, \quad (182)$$

where we see that $\mathbf{v}_s$ contains both c.o.m. translation and rigid-body rotation.

For an electrically neutral nanodiamond, there is no monopole force proportional to $\mathcal{E}$. Its leading mechanical response is electric polarization. Using an effective scalar electric polarizability⁸ α_e , the corresponding potential and acceleration are

$$V_{\text{pol}} = -\frac{\alpha_e}{2} \|\mathcal{E}_s^{\text{rest}}\|^2 \quad \Bigg| \quad \mathbf{a}_{\text{pol}} = \frac{\alpha_e}{2m} \nabla \|\mathcal{E}_s^{\text{rest}}\|^2 \quad \Bigg| \quad \alpha_e \simeq 3\varepsilon_0 \frac{m \varepsilon_r - 1}{\rho \varepsilon_r + 2}, \quad (183)$$

where ε_0 is the vacuum permittivity and $\varepsilon_r \simeq 5.7$ is the relative permittivity of diamond.

The NV spin is also electrically sensitive. We use the measured NV Stark coefficient $\kappa_{\mathcal{E}, \perp}$ to estimate $|\Delta \nu_{\text{Stark}}|$, then compare it with the Zeeman and zero-field-splitting frequencies. The larger ground-state

⁸ α_e was given for an equal-volume dielectric sphere, a sufficient order-of-magnitude bound, despite the precise cylindrical polarizability is tensorial. However, it differs only by geometrical factors of order unity.

Stark coefficient⁹ is approximately [21]

$$\kappa_{\mathcal{E},\perp} \simeq 17 \text{ Hz cm/V} = 0.17 \frac{\text{Hz}}{\text{V/m}}, \quad (184)$$

so that a conservative upper bound on the corresponding frequency shift is

$$|\Delta\nu_{\text{Stark}}| \lesssim \kappa_{\mathcal{E},\perp} \|\boldsymbol{\mathcal{E}}_s^{\text{rest}}\|. \quad (185)$$

For the nominal parameters and the conservative spatial bounds $\delta_{\text{sw}} = 10^3 \text{ s}^{-1}$, $B_0 = 10^{-2} \text{ T}$, $B_1 = 10^{-4} \text{ T}$, $\eta = 45 \text{ T/m}$, $x_{\text{max}} = 1 \mu\text{m}$, $z_{\text{max}} = 500 \mu\text{m}$, **Eq. 181** gives $\mathcal{E}_{\text{F}}^{\text{max}} \lesssim 3.3 \times 10^{-5} \text{ V/m}$. Furthermore, even for $v_s \lesssim 10^{-3} \text{ m/s}$, $B \lesssim 0.2 \text{ T}$, the motional contribution is $\|\mathbf{v}_s \times \mathbf{B}\| \lesssim 2 \times 10^{-4} \text{ V/m}$.

Taking the deliberately conservative combined bound $\mathcal{E}_{\text{eff}} = 10^{-3} \text{ V/m}$, a nanodiamond with $m = 10^{-17} \text{ kg}$, $\rho = 3.5 \times 10^3 \text{ kg/m}^3$ and $\varepsilon_r = 5.7$ has $\alpha_e \simeq 4.6 \times 10^{-32} \text{ C m}^2/\text{V}$. Picking a field-variation scale $\ell_{\mathcal{E}} = 100 \mu\text{m}$ to replace the spatial derivative in **Eq. 183** therefore gives $a_{\text{pol}} \sim \frac{\alpha_e}{2m} \frac{\mathcal{E}_{\text{eff}}^2}{\ell_{\mathcal{E}}} \lesssim 2.3 \times 10^{-17} \text{ m/s}^2$.

The intended Stern–Gerlach acceleration is instead of order $a_{\text{SG}} \sim \frac{\mu\eta}{m} \simeq 8.3 \times 10^{-5} \text{ m/s}^2$, giving

$$\frac{a_{\text{pol}}}{a_{\text{SG}}} \lesssim 3 \times 10^{-13}. \quad (186)$$

The same conservative electric-field bound gives $|\Delta\nu_{\text{Stark}}| \lesssim 1.7 \times 10^{-4} \text{ Hz}$, whereas the smallest nominal Zeeman scale is $|\Delta\nu_{\text{Zeeman}}| = \frac{\mu B_1}{h} \simeq 2.8 \times 10^6 \text{ Hz}$, giving

$$\frac{|\Delta\nu_{\text{Stark}}|}{|\Delta\nu_{\text{Zeeman}}|} \lesssim 0.6 \times 10^{-10}. \quad (187)$$

Both the electric polarization force and the NV Stark correction are therefore negligible at the accuracy retained in the present model. Likewise is the magnetoquasistatic retardation parameter

$$\frac{\delta_{\text{sw}} \ell_{\mathcal{E}}}{c} \sim 3 \times 10^{-10}, \quad (188)$$

meaning that the field is effectively updated everywhere ”instantaneously” on the mechanical timescale, allowing us to neglect effects such as propagation delays across the apparatus, phase differences caused by finite light speed, displacement-current corrections, etc.

So far we assumed the nanodiamond being electrically neutral, but the negative charge state of the internal NV defect does not by itself imply that the entire nanodiamond carries a net charge; nevertheless, an uncompensated particle charge Q_{ND} would experience a force $\mathbf{F}_Q$ that, whose neglecting requires

$$\mathbf{F}_Q = Q_{\text{ND}} (\boldsymbol{\mathcal{E}}_{\text{ind}} + \mathbf{v}_c \times \mathbf{B}^{\text{sf}}) \quad \left| \quad |Q_{\text{ND}}| \mathcal{E}_{\text{eff}} \ll F_{\text{SG}} = \mu \left| \frac{\partial \mathbf{B}}{\partial z} \right| \approx \mu \eta. \right. \quad (189)$$

Residual charge and apparatus-dependent electrostatic fields must therefore be measured or controlled experimentally rather than being absorbed into the neutral-particle model.

4.3 Internal spin Hamiltonian and adiabatic energy surfaces

4.3.1 Spin convention and full 3×3 Hamiltonian

The $\mathbf{d}_0$ defect is described by a spin triplet. We use the dimensionless spin-1 ($s=1$) matrices

$$S_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} \quad \left| \quad S_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 & -i & 0 \\ i & 0 & -i \\ 0 & i & 0 \end{pmatrix} \quad \right| \quad S_3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix}, \quad (190)$$

acting on the triplet Hilbert space $\mathcal{H}_{\mathbf{d}_0} \simeq \mathbb{C}^3$, here written in the standard S_3 -eigenbasis $\{|1\rangle, |0\rangle, |-1\rangle\}$.

Following **Sec. 2.2.3**, S_i are the Hermitian quantum generators associated with the spin-1 rep

$$\rho(\mathbf{L}_i) = -iS_i \quad \left| \quad S_i^\dagger = S_i \quad \right| \quad (-iS_i)^\dagger = -(-iS_i), \quad (191)$$

⁹The transverse Stark interaction has scale $H_{\text{Stark},\perp} \sim d_{\perp} \mathcal{E}_{\perp}$, hence $|\Delta\nu_{\text{Stark}}| = |\Delta E_{\text{Stark}}|/h \lesssim (d_{\perp}/h) |\mathcal{E}_{\perp}| = \kappa_{\mathcal{E},\perp} |\mathcal{E}_{\perp}|$, where $d_{\perp}$ is the transverse electric-dipole coupling and $|\mathcal{E}_{\perp}| \leq \|\boldsymbol{\mathcal{E}}_s^{\text{rest}}\|$.

making $-iS_i$ the anti-Hermitian Lie-algebra elements entering the unitary spin rotation, and S_i the Hermitian spin observables. The ordered triple $\mathbf{S} = (S_1, S_2, S_3)^T$ is a vector operator in the sense of **Thm. 5**:

$$U_S[\mathbf{R}(\hat{\mathbf{n}}, \varphi)] := U_S(\hat{\mathbf{n}}, \varphi) = \exp(-i\varphi n_i S_i) \quad \Bigg| \quad U_S[\mathbf{R}]^\dagger \mathbf{S} U_S[\mathbf{R}] = \mathbf{R} \mathbf{S} \quad \Bigg| \quad [S_i, S_j] = i\varepsilon_{ijk} S_k \quad (192)$$

$$\mathbf{R}_{\text{sf} \leftarrow \text{s}}(t) := \mathbf{F}(t) \mathbf{C}_s \quad \Bigg| \quad U_S^{\text{sf} \leftarrow \text{s}}(t) := U_S[\mathbf{R}_{\text{sf} \leftarrow \text{s}}(t)] = U_S[\mathbf{F}(t) \mathbf{C}_s]. \quad (193)$$

The triplet's squared magnitude is the operator scalar product, i.e. the quadratic Casimir,

$$\mathbf{S}^2 := \mathbf{S} \cdot \mathbf{S} = \sum_{i=1}^3 S_i^2 = s(s+1) \mathbf{1}_3 = 2\mathbf{1}_3, \quad (194)$$

where s is the total spin quantum number. The states $|s, m\rangle$ are labelled additionally by the magnetic quantum number m , defined through

$$S_3|s, m\rangle = m|s, m\rangle, \quad m = -s, -s+1, \dots, s. \quad (195)$$

To obtain the Casimir eigenvalue, we define the ladder operators $S_\pm := S_1 \pm iS_2$ and consider the highest-weight state $|s, s\rangle$, satisfying $S_+|s, s\rangle = 0$ and $S_3|s, s\rangle = s|s, s\rangle$:

$$\mathbf{S}^2 = S_- S_+ + S_3(S_3 + 1) \quad \Rightarrow \quad \mathbf{S}^2|s, s\rangle = s(s+1)|s, s\rangle. \quad (196)$$

Since $\mathbf{S}^2$ commutes with all S_i , and hence with $S_\pm$, applying the $S_\pm$ changes m but leaves s and the Casimir eigenvalue unchanged. Thus every state in the spin- s multiplet has eigenvalue $s(s+1)$; for the present spin-1 representation, which, for $s=1$, gives $\mathbf{S}^2 = 2\mathbf{1}_3$.

In the spin frame (s), using the ordered basis $\{|+1\rangle, |0\rangle, |-1\rangle\}$, the internal spin (S) Hamiltonian is

$$H_S^s = \mu \mathbf{S} \cdot \mathbf{B}^s + D \left(S_3^2 - \frac{2}{3} \mathbf{1} \right) + E \left(S_1^2 - S_2^2 \right) = \begin{pmatrix} \frac{D}{3} + \mu B_3 & \frac{\mu}{\sqrt{2}}(B_1 - iB_2) & E \\ \frac{\mu}{\sqrt{2}}(B_1 + iB_2) & -\frac{2D}{3} & \frac{\mu}{\sqrt{2}}(B_1 - iB_2) \\ E & \frac{\mu}{\sqrt{2}}(B_1 + iB_2) & \frac{D}{3} - \mu B_3 \end{pmatrix}, \quad (197)$$

where the axial splitting D separates $|0\rangle$ from the $|\pm 1\rangle$ manifold and the transverse splitting E selects two crystal axes perpendicular to the quantization one and couples $|+1\rangle$ to $|-1\rangle$ when the defect's axial symmetry is broken [11, 22] (see **Sec. 4.3.5**).

Throughout the derivation we use this Cartesian form. Writing the transverse couplings in terms of $B_1 \pm iB_2$ avoids introducing the azimuth $\phi_B = \text{atan2}(B_2, B_1)$, which is undefined when $B_\perp = \sqrt{B_1^2 + B_2^2} = 0$.

The physical spin angular momentum is $\hbar \mathbf{S}$. We choose

$$\mu := g_e \mu_B > 0, \quad (198)$$

so μ has units of energy per tesla, while D and E have units of energy. Whenever an angular frequency is compared with an SI energy, the required factor of $\hbar$ is restored explicitly.

We choose those transverse axes such that E is real. For an NV center [11],

$$D = \hbar(2\pi \times 2.87 \text{ GHz}). \quad (199)$$

4.3.2 Exact instantaneous branches and adiabatic reduction

For fixed c.o.m. coordinates, orientation and time, the spin-frame Hamiltonian in **Eq. 197** defines the instantaneous eigenproblem with the representation invariant λ_n eigenvalues:

$$H_S^s(\mathbf{B}^s)|n^s\rangle = \lambda_n(\mathbf{B}^s)|n^s\rangle \quad \Bigg| \quad \langle m^s | n^s \rangle = \delta_{mn}, \quad m, n \in \{1, 2, 3\}, \quad (200)$$

where for each fixed value of $\mathbf{B}^s$, the scalar λ_n is the internal spin energy of branch n .

The same physical Hamiltonian and instantaneous eigenstates in the laboratory representation are

$$H_S^{\text{sf}} = U_S^{\text{sf} \leftarrow \text{s}}(t) H_S^s(U_S^{\text{sf} \leftarrow \text{s}}(t))^\dagger \quad \Bigg| \quad |n^{\text{sf}}\rangle = U_S^{\text{sf} \leftarrow \text{s}}(t) |n^s\rangle. \quad (201)$$

From **Eq. 150**, we remind ourselves that changing the c.o.m. position or any Euler angle changes the field experienced by the defect and therefore alters the eigenenergy λ_n . The spin state is an independent

dynamical variable and must be propagated together with the mechanical coordinates, which in turn causes the mechanical force $\dot{p}_a^{\text{spin}}$ generated by the spin to then depend on the evolving spin state:

$$i\hbar \frac{d}{dt} |\psi_S^{\text{sf}}(t)\rangle = H_S^{\text{sf}}(\mathbf{c}, \alpha, \beta, \gamma, t) |\psi_S^{\text{sf}}(t)\rangle \quad \Rightarrow \quad \dot{p}_a^{\text{spin}} = - \left\langle \psi_S^{\text{sf}} \left| \frac{\partial H_S^{\text{sf}}}{\partial q^a} \right| \psi_S^{\text{sf}} \right\rangle, \quad q^a \in \{c_x, c_y, c_z, \alpha, \beta, \gamma\}. \quad (202)$$

Thus, without an additional approximation, one must solve simultaneously for the mechanical variables and for the three-component spin state $|\psi_S^{\text{sf}}(t)\rangle$: $q^a(t)$ changes $H_S(q^a(t), t)$, which in turn changes $|\psi_S^{\text{sf}}(t)\rangle$, which then itself changes $\dot{p}^a(t)$, which, of course, determines $q^a(t)$.

Moreover, the spin evolves on the fast $\hbar/\Delta_{\text{spin}}$ timescale, while the c.o.m. and rotor evolve much slower. A direct numerical solution would therefore resolve both the fast spin oscillations and the slow mechanical motion over the entire interferometer duration, making the coupled system numerically stiff and unnecessarily expensive if spin transitions are negligible: this is the job of the constant non-zero $\mathbf{B}$.

If the adiabaticity condition is satisfied and the spin is initially prepared on branch n , it remains in the corresponding instantaneous eigenstate up to an overall phase:

$$|\psi_S^{\text{sf}}(t)\rangle \simeq e^{i\vartheta_n(t)} |n^{\text{sf}}(\mathbf{c}(t), \alpha(t), \beta(t), \gamma(t), t)\rangle. \quad (203)$$

At this point, the spin state is no longer an independent unknown: once the mechanical configuration is known, the instantaneous eigenvector is fixed by the eigenproblem in **Eq. 200**. Only the branch label n remains to be specified between microwave pulses. Projecting the spin Hamiltonian onto this branch gives the instantaneous (evaluated at set $\mathbf{B}$) spin eigenenergy, which consequently acts as the adiabatic potential-energy surface:

$$\langle n^{\text{sf}} | H_S^{\text{sf}} | n^{\text{sf}} \rangle = \lambda_n[\mathbf{B}^s(\mathbf{c}, \alpha, \beta, \gamma, t)] := V_n^{(3)}(\mathbf{c}, \alpha, \beta, \gamma, t), \quad (204)$$

where the superscript (3) indicates that no spin level has been eliminated.

Solving the cubic characteristic equation $\det(H_S^s - \lambda_n \mathbf{1}_3) = 0$ would provide explicit analytical expressions for the three eigenvalues, but they are lengthy and offer little physical insight. We therefore retain the exact eigenproblem in **Eq. 200**; its eigenvalues will be evaluated numerically in **Sec. 5.1.3** when constructing the exact adiabatic branch potentials and spin gaps.

Therefore, in the mechanical Hamiltonian, the matrix-valued spin Hamiltonian is replaced by the scalar energy of the followed branch, which is reevaluated continuously from the local field and mechanical configuration (while the branch label n remains fixed during each adiabatic segment):

$$H_{\text{mech}} + H_S \quad \longrightarrow \quad H_{\text{mech}} + V_n^{(3)} \quad \Rightarrow \quad \dot{p}_a^{\text{spin}} = - \frac{\partial \lambda_n}{\partial q^a} = - \frac{\partial V_n^{(3)}}{\partial q^a}, \quad (205)$$

so the spin continues to generate both translational forces and rotational torques, despite its three-component state no longer being explicitly propagated.

While RK4 could integrate these equations, its timestep must resolve the faster spin oscillations. For the NV zero-field splitting, $D/\hbar \simeq 2\pi \times 2.87 \text{ GHz}$ (see **Eq. 199**), which corresponds to a timescale of $\sim 10^{-10} \text{ s}$, while the interferometer lasts $\sim 1 \text{ s}$, requiring tens to hundreds of billions of steps.

The adiabatic approximation uses the physical scale separation: the spin adjusts almost instantaneously to the slowly changing mechanical configuration and remains on branch n . We then use RK4 only for the slow mechanical equations from **Eq. 205** while checking separately that $\mathcal{A}_{mn} \ll 1$.

However, this adiabatic reduction is only valid if the changing Hamiltonian does not appreciably couple the instantaneous eigenstate $|n^{\text{sf}}\rangle$ to either of the other branches. For $m \neq n$, we therefore define

$$\mathcal{A}_{mn}(t) := \frac{\hbar \left| \langle m^{\text{sf}}(t) | \dot{H}_S^{\text{sf}}(t) | n^{\text{sf}}(t) \rangle \right|}{|\lambda_m(t) - \lambda_n(t)|^2} \quad \left| \quad \max_{m \neq n} \mathcal{A}_{mn}(t) \ll 1, \right. \quad (206)$$

which must be evaluated along each interferometer arm and throughout every smooth field transition.

The numerator measures the instantaneous coupling generated by the changing Hamiltonian, whereas the energy-gap denominator suppresses transitions between the two branches.

The derivative $\dot{H}_S^{\text{sf}}$ contains the explicit field switching, the motion of the defect through the inhomogeneous field and the rotation of its crystal axes. Consequently, the switching rate δ_{sw} is acceptable only

after this condition has been verified along the complete trajectory.

The microwave inversion is the deliberate exception: it connects two otherwise adiabatic segments.

The momentum-dependent spin-rotation correction is introduced separately in **Sec. 4.7**.

4.3.3 Feshbach reduction to the $|\pm 1\rangle$ manifold

The exact branch problem is the parent model, but an analytical expression is obtained by separating the $|\pm 1\rangle$ manifold from $|0\rangle$ through

$$|+1\rangle = (1 \ 0 \ 0)^T \quad |0\rangle = (0 \ 1 \ 0)^T \quad |-1\rangle = (0 \ 0 \ 1)^T \quad (207)$$

$$P := |+1\rangle\langle+1| + |-1\rangle\langle-1| = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad Q := \mathbf{1} - P = |0\rangle\langle 0| = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}. \quad (208)$$

We decompose the complete spin state into its components inside the retained and eliminated subspaces:

$$|\psi^s\rangle = P|\psi^s\rangle + Q|\psi^s\rangle =: |\psi_P^s\rangle + |\psi_Q^s\rangle, \quad (209)$$

where

$$\begin{aligned} |\psi_P^s\rangle &:= P|\psi^s\rangle & P|\psi_P^s\rangle &= |\psi_P^s\rangle & P|\psi_Q^s\rangle &= 0 \\ |\psi_Q^s\rangle &:= Q|\psi^s\rangle & Q|\psi_Q^s\rangle &= |\psi_Q^s\rangle & Q|\psi_P^s\rangle &= 0. \end{aligned} \quad (210)$$

Substituting **Eq. 209** into the complete eigenvalue equation gives

$$H_S^s (|\psi_P^s\rangle + |\psi_Q^s\rangle) = \lambda (|\psi_P^s\rangle + |\psi_Q^s\rangle) \Rightarrow \begin{cases} PH_S^s P|\psi_P^s\rangle + PH_S^s Q|\psi_Q^s\rangle = \lambda |\psi_P^s\rangle \\ QH_S^s P|\psi_P^s\rangle + QH_S^s Q|\psi_Q^s\rangle = \lambda |\psi_Q^s\rangle, \end{cases} \quad (211)$$

The second equation may be rearranged as

$$\begin{aligned} (\lambda Q - QH_S^s Q)|\psi_Q^s\rangle &= QH_S^s P|\psi_P^s\rangle \Rightarrow Q|\psi_Q^s\rangle = (\lambda Q - QH_S^s Q)^{-1} QH_S^s P|\psi_P^s\rangle \\ &= (\lambda - QH_S^s Q)^{-1} QH_S^s P|\psi_P^s\rangle, \end{aligned} \quad (212)$$

where we used the notation convention based on Q being the identity operator within the Q -subspace.

Substituting this result back into the first equation of **Eq. 211** gives

$$[PH_S^s P + PH_S^s Q (\lambda - QH_S^s Q)^{-1} QH_S^s P] |\psi_P^s\rangle = \lambda |\psi_P^s\rangle, \quad (213)$$

from which we define the exact energy-dependent Feshbach Hamiltonian [23, 24] (acting only on $|\psi_P^s\rangle$) as

$$\boxed{H_{S,\text{eff}}^s(\lambda) = PH_S^s P + PH_S^s Q (\lambda - QH_S^s Q)^{-1} QH_S^s P.} \quad (214)$$

The second term in **Eq. 214** describes the process in which the state leaves the $|\pm 1\rangle$ manifold, enters the eliminated $|0\rangle$ subspace through $QH_S^s P$, and subsequently returns through $PH_S^s Q$.

Using the ordered basis, the projectors separate the retained $|\pm 1\rangle$ manifold from the eliminated $|0\rangle$ state. The Hamiltonian restricted to the eliminated subspace is

$$QH_S^s Q = |0\rangle\langle 0|H_S^s|0\rangle\langle 0| = (\langle 0|H_S^s|0\rangle)|0\rangle\langle 0| = \langle 0|H_S^s|0\rangle Q = -\frac{2D}{3}Q \Rightarrow \lambda_0 = -\frac{2D}{3}, \quad (215)$$

where λ_0 is the energy of the eliminated branch before its virtual coupling to the $|\pm 1\rangle$ manifold. The transverse field coupling components consequently shift this branch only at second order:

$$\lambda_0 = -\frac{2D}{3} + \mathcal{O}\left(\frac{\mu^2 B_\perp^2}{D}\right). \quad (216)$$

The retained subspace restricted Hamiltonian (in the ordered basis $\{|+1\rangle, |-1\rangle\}$) is

$$\begin{aligned} PH_S^s P &= \frac{D}{3}P + \begin{pmatrix} \mu B_3 & E \\ E & -\mu B_3 \end{pmatrix} \equiv \frac{D}{3}\mathbf{1}_2 + \begin{pmatrix} \mu B_3 & E \\ E & -\mu B_3 \end{pmatrix} \Rightarrow \lambda_\pm = \frac{D}{3} \pm \sqrt{\mu^2 B_3^2 + E^2} \\ &= \frac{D}{3} + \mathcal{O}(\mu\|\mathbf{B}^s\|, |E|), \end{aligned} \quad (217)$$

where the two $\lambda_\pm$ eigenvalues are those of the retained branches before the virtual coupling through $|0\rangle$. Including the transverse coupling to $|0\rangle$ adds corrections of order

$$\lambda_\pm = \frac{D}{3} + \mathcal{O}\left(\mu\|\mathbf{B}^s\|, |E|, \frac{\mu^2 B_\perp^2}{D}\right). \quad (218)$$

For either of these two retained branches, the operator appearing in the resolvent is therefore

$$\lambda_{\pm} - QH_S^s Q = \left(\lambda_{\pm} + \frac{2D}{3} \right) Q = D + \mathcal{O} \left(\mu \|\mathbf{B}^s\|, |E|, \frac{\mu^2 B_{\perp}^2}{D} \right), \quad (219)$$

which, under

$$\frac{\mu \|\mathbf{B}^s\|}{D} \ll 1 \quad \left| \quad \frac{|E|}{D} \ll 1, \quad (220)$$

may be replaced by D^{-1} and obtain

$$H_{S,\text{eff}}^{(2),s} = \begin{pmatrix} \mu B_3 & \varepsilon^* \\ \varepsilon & -\mu B_3 \end{pmatrix} + \left(\frac{D}{3} + \frac{\mu^2 B_{\perp}^2}{2D} \right) \mathbf{1} \quad \boxed{\varepsilon = E + \frac{\mu^2}{2D}(B_1 + iB_2)^2}, \quad (221)$$

where ε is the effective complex transverse coupling between $|+1\rangle$ and $|-1\rangle$.

This is the effective Hamiltonian used in Ref. [19], written here in Cartesian components and complete to second order in the coupling between $|0\rangle$ and the $|\pm 1\rangle$ manifold; its difference from the 3×3 model is the neglected energy dependence of the resolvent and the resulting higher-order terms.

The two effective branch energies (i.e., eigenvalues of $H_{S,\text{eff}}^{(2),s}$) are

$$V_{\sigma}^{(2)} = \frac{D}{3} + \frac{\mu^2 B_{\perp}^2}{2D} + \sigma \sqrt{\mu^2 B_3^2 + |\varepsilon|^2}, \quad \sigma \in \{+1, -1\}, \quad (222)$$

where, since E is real in the chosen transverse axes,

$$|\varepsilon|^2 = E^2 + \frac{\mu^2 E}{D} (B_1^2 - B_2^2) + \frac{\mu^4}{4D^2} (B_1^2 + B_2^2)^2. \quad (223)$$

The Cartesian form remains regular at $B_{\perp} \neq 0$, thus distinguishing the transverse crystal axes.

The label $\sigma = \pm 1$ denotes an instantaneous energy branch, whereas $m_s = \pm 1$ labels a (fixed) bare eigenstate of S_3 . They may be identified only while the corresponding eigenvector remains close to $|m_s = \pm 1\rangle$, i.e., when the eigenstates are weak mixtures of the two states ($\varepsilon \approx 0$).

For an NV center, $D/\mu \simeq 0.10$ T.

The equilibrium field $\|\mathbf{B}_{\text{tr}}\| \simeq 0.187$ T obtained for **Eq. 158** does not satisfy the first of **Eq. 220** if that trap field acts uncompensated on the defect. In that model the full 3×3 Hamiltonian must be used; the analytical 2×2 reduction applies only when the total field seen by the spin remains well below D/μ , for example after the trap contribution is compensated or replaced by an effective mechanical confinement.

4.3.4 Successive analytical limits

Neglecting the virtual $|0\rangle$ corrections proportional to $B_{\perp}^2/D$ in **Eq. 222**, while retaining the transverse splitting, requires a stronger condition than the Feshbach approximation itself:

$$\Delta_{\perp} := \frac{\mu^2 B_{\perp}^2}{2D} \quad \left| \quad \Delta_E := \sqrt{\mu^2 B_3^2 + E^2} \quad \Rightarrow \quad \text{requirement: } \boxed{\Delta_{\perp} \ll \Delta_E}, \quad (224)$$

which simplifies the energies down to

$$V_{\sigma}^{(E)} = \frac{D}{3} + \sigma \sqrt{\mu^2 B_3^2 + E^2}. \quad (225)$$

While this ensures that the common shift terms (i.e., the term linear in $E\Delta_{\perp}$ and the quadratic one $\Delta_{\perp}^2$) are all small compared with the retained branch splitting, for forces and torques, however, the derivatives of the discarded terms must also be small. This is checked separately by comparing $\partial_{\xi} V_{\sigma}^{(2)}$ with $\partial_{\xi} V_{\sigma}^{(E)}$ for every relevant coordinate ξ .

The further limit $E = 0$ may be written either by following the ordering of the two energies or by following a specified bare state (see the above note):

$$V_{\sigma}^{(0)} = \frac{D}{3} + \sigma \mu |B_3| \quad \left| \quad V_{m_s}^{(0)} = \frac{D}{3} + m_s \mu B_3, \quad m_s = \pm 1. \quad (226)$$

4.3.5 Minimum-field and spin-adiabaticity condition

Excluding the deliberate microwave pulse, the field protocol is admissible iff the branches satisfy the adiabaticity condition from **Eq. 206**. For the branch $n_\ell(t)$ followed by arm¹⁰ $\ell \in \{+, -\}$, we define

$$\Delta_{\text{spin}}^{\min} := \min_{\ell, t} \min_{m \neq n_\ell(t)} |\lambda_m(t) - \lambda_{n_\ell(t)}| \quad \Bigg| \quad \mathcal{A}_{\max} := \max_{\ell, t} \max_{m \neq n_\ell(t)} \mathcal{A}_{mn_\ell}(t), \quad (227)$$

with the trajectory-wide requirements

$$\boxed{\Delta_{\text{spin}}^{\min} > 0 \quad \Bigg| \quad \mathcal{A}_{\max} \ll 1.} \quad (228)$$

These are the primary conditions for the protocol in [20, 19]: they involve the complete field of **Eq. 151**, evaluated at the actual defect position and orientation of each arm.

Their meaning is especially transparent in the effective $|\pm 1\rangle$ manifold, where, making use of **Eq. 218**,

$$\Delta_{+-,\ell}^{(2)}(t) = |\lambda_+ - \lambda_-| = 2\sqrt{\mu^2 B_{3,\ell}^2(t) + |\varepsilon_\ell(t)|^2}. \quad (229)$$

Analogously, in the bare axial limit,

$$\Delta_{+-,\ell}^{(0)} = 2\mu|B_{3,\ell}| \quad \Bigg| \quad \Delta_{+0,\ell}^{(0)} = |D + \mu B_{3,\ell}| \quad \Bigg| \quad \Delta_{-0,\ell}^{(0)} = |D - \mu B_{3,\ell}|. \quad (230)$$

Thus, D separates $|0\rangle$ from $|\pm 1\rangle$ at $B_3=0$, but it does not separate $|+1\rangle$ from $|-1\rangle$.

If $\varepsilon_\ell=0$, the two effective branches are degenerate at $B_{3,\ell}=0$.

If $\varepsilon_\ell \neq 0$, the crossing is replaced by an avoided crossing of size $2|\varepsilon_\ell|$, while the corresponding eigenvectors become mixtures of the two bare states. A nonzero E usually opens this gap, but it is not by itself sufficient: the transverse virtual contribution in **Eq. 221** may partially or completely cancel E for a particular field magnitude and transverse direction.

For the axial ($B_\perp \ll B_3$), bare-state protocol, the instantaneous adjacent-level Larmor scale is

$$\omega_{L,\ell}(t) := \frac{\mu|B_{3,\ell}(t)|}{\hbar}. \quad (231)$$

Choosing a required minimum value ω_L^{ad} then defines the operational field threshold

$$B_{3,\min} := \min_{\ell, t} |B_{3,\ell}(t)| > B_{\text{ad}} := \frac{\hbar\omega_L^{\text{ad}}}{\mu} = \frac{2m_e\omega_L^{\text{ad}}}{g_e|e|} > 0. \quad (232)$$

The estimate $\delta_{\text{sw}} \ll \omega_L^{\text{ad}}$ provides only a rough adiabaticity criterion when the switching changes the instantaneous spin eigenbasis on a timescale of order δ_{sw}^{-1} . A change in the eigenvalues alone does not induce transitions; only the off-diagonal couplings

$$\langle m^{\text{sf}}(t) | \dot{H}_S^{\text{sf}}(t) | n^{\text{sf}}(t) \rangle, \quad m \neq n, \quad (233)$$

can mix different instantaneous branches. So, $\delta_{\text{sw}} \ll \omega_L^{\text{ad}}$ does not replace the complete condition $\mathcal{A}_{mn} \ll 1$.

In the near-axial regime, $\|\mathbf{B}^{\text{s}}\| \simeq |B_3|$. In the bare axial model, the $|+1\rangle$ and $|-1\rangle$ states become degenerate when $B_3=0$. The homogeneous holding bias B_{hold} and the smooth cross-fades must therefore be chosen so that

$$\min_{\ell, t} |B_{3,\ell}(t)| > 0. \quad (234)$$

Outside the near-axial regime, however, a nonzero total field magnitude does not by itself guarantee a nonzero gap between the relevant spin branches and the exact gaps defined in **Eq. 227** must then be used.

4.3.6 Microwave spin inversion

At $t=t_{\text{flip}}$, the two interferometer arms must exchange spin branch [20, 19].

Let $|\chi_{+,\ell}^{\text{s}}\rangle$ and $|\chi_{-,\ell}^{\text{s}}\rangle$, with $\ell \in \{+, -\}$, denote the two exact instantaneous eigenstates connected by the pulse at the local field experienced by arm ℓ .

As previously stated, when **Eq. 220** holds, they may be approximated by the eigenvectors of **Eq. 221** and in the bare $E=0$ limit they reduce to $|+1\rangle$ and $|-1\rangle$.

¹⁰e.g. if $n_\ell \leftrightarrow \sigma_\ell(t) \in \{-1, +1\}$, then $\ell=+1$ follows $\sigma=+1$ before the pulse and after it, it follows $\sigma=-1$.

Over a microwave (MW) pulse that is short enough that it effectively leaves the instantaneous basis fixed, a resonant effective drive may be written as

$$H_{S,MW}^{\text{eff}}(t) = \frac{\hbar\Omega_R(t)}{2} \left[e^{-i\phi_{MW}} |\chi_+^s\rangle\langle\chi_-^s| + e^{i\phi_{MW}} |\chi_-^s\rangle\langle\chi_+^s| \right], \quad (235)$$

where Ω_R and ϕ_{MW} are the effective Rabi angular frequency and pulse phase. A π -pulse returns the Rabi rotation angle (we denote its duration by τ_π)

$$\Theta_{MW} := \int_{t_{\text{flip}} - \tau_\pi/2}^{t_{\text{flip}} + \tau_\pi/2} \Omega_R(t) dt = \pi \quad (236)$$

and interchanges the two states up to known phases:

$$|\chi_+^s\rangle \mapsto -ie^{i\phi_{MW}} |\chi_-^s\rangle \quad \Big| \quad |\chi_-^s\rangle \mapsto -ie^{-i\phi_{MW}} |\chi_+^s\rangle. \quad (237)$$

To see how this pulse interchanges the two states, define

$$\Sigma_{\phi_{MW}} := e^{-i\phi_{MW}} |\chi_+^s\rangle\langle\chi_-^s| + e^{i\phi_{MW}} |\chi_-^s\rangle\langle\chi_+^s| \quad \Rightarrow \quad \Sigma_{\phi_{MW}}^2 = |\chi_+^s\rangle\langle\chi_+^s| + |\chi_-^s\rangle\langle\chi_-^s| = \mathbf{1}_2, \quad (238)$$

where in the last step we used the orthonormality of the two instantaneous states.

For a fixed pulse phase, the microwave Hamiltonian and its time-evolution operator become:

$$\begin{aligned} H_{S,MW}^{\text{eff}}(t) = \frac{\hbar\Omega_R(t)}{2} \Sigma_{\phi_{MW}} \quad \Rightarrow \quad U_{MW} &= \exp \left[-\frac{i}{\hbar} \int H_{S,MW}^{\text{eff}}(t) dt \right] = \exp \left[-\frac{i\Theta_{MW}}{2} \Sigma_{\phi_{MW}} \right] \\ &= \cos \left(\frac{\Theta_{MW}}{2} \right) \mathbf{1}_2 - i \sin \left(\frac{\Theta_{MW}}{2} \right) \Sigma_{\phi_{MW}}, \end{aligned} \quad (239)$$

the later of which, for a π -pulse, equates to

$$U_{MW} = -i\Sigma_{\phi_{MW}}. \quad (240)$$

Acting on the two states gives

$$U_{MW} |\chi_+^s\rangle = -ie^{i\phi_{MW}} |\chi_-^s\rangle \quad \Big| \quad U_{MW} |\chi_-^s\rangle = -ie^{-i\phi_{MW}} |\chi_+^s\rangle. \quad (241)$$

Thus, the π -pulse exchanges the two instantaneous states, while attaching a known phase to each one.

Since the two arms generally occupy different positions and orientations at t_{flip} , their transition frequencies need not be identical. The corresponding transition angular frequency on arm ℓ is the energy separation between the two states in units of $\hbar$:

$$\omega_{+,\ell} := \frac{|\lambda_{+,\ell} - \lambda_{-,\ell}|}{\hbar}, \quad \ell \in \{+, -\} \quad \Big| \quad \delta_\ell := \omega_{MW} - \omega_{+,\ell}, \quad (242)$$

where $\lambda_{+,\ell}$ and $\lambda_{-,\ell}$ are the instantaneous spin eigenenergies (i.e., evaluated at the position and orientation of arm ℓ at $t=t_{\text{flip}}$). Thus, a resonant microwave drive for that arm satisfies

$$\delta_\ell = 0. \quad (243)$$

For a two-level transition (with some detuning δ_ℓ), the maximum population-transfer probability is

$$P_\ell^{\text{max}} = \frac{(\Omega_R^{\text{max}})^2}{(\Omega_R^{\text{max}})^2 + \delta_\ell^2}, \quad (244)$$

which dictates that a high-fidelity inversion on arm ℓ requires

$$|\delta_\ell| \ll \Omega_R^{\text{max}}. \quad (245)$$

Since the same pulse must invert the spin on both interferometer arms, this condition must hold for the arm with the largest detuning:

$$\boxed{\max_\ell |\delta_\ell| \ll \Omega_R^{\text{max}} \quad \Big| \quad \max \left(\Omega_R^{\text{max}}, \frac{1}{\tau_\pi} \right) \ll \Delta_{\text{leak}}^{\text{min}}} \quad (246)$$

where $\Delta_{\text{leak}}^{\text{min}}$ is the smallest angular-frequency detuning from any unwanted transition involving the third spin branch. If the first condition is not satisfied, a composite, broadband or multi-frequency pulse must be used rather than a single resonant π -pulse [25, 26].

Although these phases do not modify the classical branch forces, they must be retained for the final spin coherence. In the bare model, **Eq. 241** gives precisely the branch bookkeeping introduced in **Eq. 176**; for $E \neq 0$, it is the eigenvectors rather than the sign of one eigenvalue which are exchanged. Leakage into the third spin branch must remain negligible for the chosen pulse bandwidth and detuning.

The pulse is instantaneous on the mechanical timescale when

$$\omega_{\text{mech}}\tau_\pi \ll 1 \quad \Bigg| \quad \delta_{\text{sw}}\tau_\pi \ll 1, \quad (247)$$

where ω_{mech} is the classical evolution's largest translational or rotational angular frequency. This that the mechanical coordinates and the applied $\mathbf{B}_{\text{SG}}$ field change negligibly during the pulse.

Continuity of the mechanical momenta additionally requires the pulse-induced impulse to be negligible:

$$\Delta p_a^{\text{pulse}} := \int_{t_{\text{flip}} - \tau_\pi/2}^{t_{\text{flip}} + \tau_\pi/2} Q_a^{\text{pulse}}(t) dt \simeq 0, \quad q^a \in \{c_x, c_y, c_z, \alpha, \beta, \gamma\}, \quad (248)$$

where $Q_a^{\text{pulse}}(t)$ is the generalized force produced during the pulse, conjugate to the coordinate p_a .

This condition includes the integrated magnetic force and torque during the pulse, together with any microwave recoil or spin-to-rotor angular-momentum transfer. Both **Eqs. 247, 248** must hold for the mechanical coordinates and momenta to be considered continuous across the idealized branch swap.

4.4 Spin-dependent forces and torques

4.4.1 Exact generalized force

Once an adiabatic branch is fixed, both its translational force and its rotational torque follow from the same scalar energy.

Theorem 7. The Hellmann–Feynman theorem. *Let $H(\xi)$ be a differentiable Hermitian operator depending on a real parameter ξ , and let $|\psi_n(\xi)\rangle$ be a normalized, non-degenerate eigenstate.*

Then, the derivative of its eigenvalue is

$$\frac{d\mathcal{E}_n}{d\xi} = \left\langle \psi_n(\xi) \left| \frac{\partial H(\xi)}{\partial \xi} \right| \psi_n(\xi) \right\rangle. \quad (249)$$

Corollary 3. Application to the instantaneous spin branches. *Let $|n^s\rangle$ be a normalized, non-degenerate instantaneous eigenstate of H_S^s , with eigenenergy $V_n^{(3)}$. For any generalized mechanical coordinate $q^a \in \{c_x, c_y, c_z, \alpha, \beta, \gamma\}$, the Hellmann–Feynman theorem gives*

$$Q_{a,n}^{\text{spin}} := -\frac{\partial V_n^{(3)}}{\partial q^a} = -\mu \sum_{i=1}^3 \langle S_i \rangle_n \frac{\partial B_i}{\partial q^a}. \quad (250)$$

where we made use of the fact that the mechanical coordinates enter H_S^s through the spin-frame field components, while D and E remain constant.

For $q^a = c_j$, this is the j th Cartesian (translational) force component; for $q^a \in \{\alpha, \beta, \gamma\}$, it is the generalized (rotational) torque conjugate to that Euler angle. The two must be obtained from the same chosen approximation to the branch potential; otherwise, the resulting equations would not describe one consistent Hamiltonian model: e.g., if one one should calculates the force using the exact $V_n^{(3)}$, then the torque must also do the same and not using the approximate $V_\sigma^{(2)}$.

Refine the laboratory-frame field Jacobian

$$\mathbf{J}_B(\mathbf{r}, t) := \frac{\partial \mathbf{B}^{\text{sf}}}{\partial \mathbf{r}} \quad \Bigg| \quad (J_B)_{ij} := \frac{\partial B_i^{\text{sf}}}{\partial r_j}. \quad (251)$$

Using **Eq. 150**, the translational and rotational derivatives are

$$\frac{\partial \mathbf{B}^s}{\partial c_a} = \mathbf{C}_s^T \mathbf{F}^T \mathbf{J}_B(\mathbf{r}_s, t) \hat{\mathbf{e}}_a, \quad (252)$$

$$\boxed{\frac{\partial \mathbf{B}^s}{\partial q} = \mathbf{C}_s^T \frac{\partial \mathbf{F}^T}{\partial q} \mathbf{B}^{\text{sf}}(\mathbf{r}_s, t) + \mathbf{C}_s^T \mathbf{F}^T \mathbf{J}_B(\mathbf{r}_s, t) \frac{\partial \mathbf{F}}{\partial q} \mathbf{d}_0, \quad q \in \{\alpha, \beta, \gamma\}.} \quad (253)$$

The first angular term rotates the defect axes relative to the local field and the second moves the off-center defect through the field gradient, the later thus vanishing for $\mathbf{d}_0 = \mathbf{0}$.

Analogously, in the spin frame,

$$\dot{\mathbf{B}}^s = \mathbf{C}_s^T \dot{\mathbf{F}}^T \mathbf{B}^{sf}(\mathbf{r}_s, t) + \mathbf{C}_s^T \mathbf{F}^T \left[\frac{\partial \mathbf{B}^{sf}}{\partial t}(\mathbf{r}_s, t) + \mathbf{J}_B(\mathbf{r}_s, t) (\dot{\mathbf{c}} + \dot{\mathbf{F}} \mathbf{d}_0) \right]. \quad (254)$$

It contains, in one expression, the explicit field switching, the c.o.m. motion, the rotation of the spin frame and the motion of the off-center defect.

For the smooth protocol (i.e. when we use the switching function S_w), the explicit derivative is given by **Eq. 175** and, together with the time dependence of $U_S^{sf \leftarrow s}$, it determines $\dot{H}_S^{sf}$ in **Eq. 206**.

The equivalent rotating-frame description is given by $H_{S, \text{rot}}^s$ in **Sec. 4.7.1**.

4.4.2 Analytical force from the second-order Hamiltonian

For the effective potential (i.e., branch energies) in **Eq. 222**, define

$$\mathcal{R} := \sqrt{\mu^2 B_3^2 + |\varepsilon|^2} = \sqrt{\mu^2 B_3^2 + E^2 + \frac{\mu^2 E}{D} (B_1^2 - B_2^2) + \frac{\mu^4}{4D^2} B_\perp^4}. \quad (255)$$

Its derivative with respect to any generalized coordinate or external parameter ξ is

$$\left[\frac{\partial V_\sigma^{(2)}}{\partial \xi} = \frac{\mu^2}{D} \left(B_1 \frac{\partial B_1}{\partial \xi} + B_2 \frac{\partial B_2}{\partial \xi} \right) + \sigma \frac{\mu^2}{\mathcal{R}} \left[B_3 \frac{\partial B_3}{\partial \xi} + \left(\frac{\mu^2 B_\perp^2}{2D^2} + \frac{E}{D} \right) B_1 \frac{\partial B_1}{\partial \xi} + \left(\frac{\mu^2 B_\perp^2}{2D^2} - \frac{E}{D} \right) B_2 \frac{\partial B_2}{\partial \xi} \right] \right]. \quad (256)$$

This equation applies only on a non-degenerate effective branch, for which $\mathcal{R} > 0$. At $\mathcal{R} = 0$, the two effective eigenvalues are degenerate and the derivative of an individually ordered branch is not uniquely defined; the coupled spin dynamics must then be retained.

This Cartesian expression is the primary analytical result and remains regular when $B_\perp = 0$. For comparison with the phase notation, let

$$B_1 + iB_2 = B_\perp e^{i\phi_B} \quad \left| \quad \mathcal{Q} := \sqrt{B_\parallel^2 + \frac{\mu^2}{4D^2} B_\perp^4 + \frac{E^2}{\mu^2} + \frac{E}{D} B_\perp^2 \cos(2\phi_B)}, \quad (257)$$

where $\mathcal{R} = \mu \mathcal{Q}$ for $\mu > 0$. The spin-dependent translational acceleration is then given by

$$\mathbf{a}_\sigma^{(2)} = -\sigma \frac{\mu}{m} \frac{B_\parallel \nabla B_\parallel + \left[\frac{\mu^2}{2D^2} B_\perp^3 + \frac{E}{D} B_\perp \cos(2\phi_B) \right] \nabla B_\perp}{\mathcal{Q}} + \sigma \frac{\mu}{m} \frac{\frac{E}{D} B_\perp^2 \sin(2\phi_B) \nabla \phi_B}{\mathcal{Q}} - \frac{\mu^2}{mD} B_\perp \nabla B_\perp, \quad (258)$$

where the gradient is taken at fixed orientation and the $\nabla \phi_B$ term survives whenever the transverse-field direction varies in space.

Although the phase form more clearly displays the contributions' origins, it is undefined at $B_\perp = 0$, unlike the Cartesian derivative in **Eq. 256** which has no such coordinate singularity.

4.4.3 Reference-force limit and its domain

Applying the first analytical limit of **Sec. 4.3.4**, namely neglecting the virtual $|0\rangle$ corrections proportional to $B_\perp^2/D$ while retaining E , reduces $V_\sigma^{(2)}$ to $V_\sigma^{(E)}$ (as given in **Eq. 225**). The corresponding spin-dependent acceleration is therefore

$$\mathbf{a}_\sigma^{(E)} = -\frac{1}{m} \nabla V_\sigma^{(E)} = -\sigma \frac{\mu}{m} \frac{B_3 \nabla B_3}{\sqrt{B_3^2 + E^2/\mu^2}}. \quad (259)$$

For $E=0$ and $B_3 \neq 0$, the energy-ordered and bare-state forms are, respectively,

$$\mathbf{a}_\sigma^{(0)} = -\sigma \frac{\mu}{m} \text{sgn}(B_3) \nabla B_3 \quad \left| \quad \mathbf{a}_{m_s}^{(0)} = -\frac{m_s \mu}{m} \nabla B_3, \quad (260)$$

where the second is the force used in the bare-state reference model of [19].

The condition $\mu\|\mathbf{B}^s\|/D \ll 1$ controls the expansion of the energy, but it does not by itself bound the error in its derivative. The neglected quantities ∇B_1 , ∇B_2 , $\nabla B_\perp$ and $\nabla\phi_B$ remain relevant even when their energy contribution is small. Thus, agreement between the exact and approximate branch energies does not automatically imply agreement between their forces and torques. The three models must consequently be compared separately at the level of the energies and of their coordinate derivatives.

4.5 Diamagnetism, gravity and confinement

4.5.1 Finite-volume diamagnetic energy

Let $\mathcal{V}_0 = \pi R^2 L$ be the cylinder volume in the body coordinates $\mathbf{u}$, centered at the c.o.m., and let ρ be its constant mass density. For an isotropic mass susceptibility χ_ρ , the diamagnetic energy is

$$V_{\text{dia}} = -\frac{\chi_\rho \rho}{2\mu_0} \int_{\mathcal{V}_0} \|\mathbf{B}^{\text{sf}}(\mathbf{c} + \mathbf{F}\mathbf{u}, t)\|^2 d^3u, \quad (261)$$

which increases with the field magnitude, since $\chi_\rho < 0$ for diamond. Despite susceptibility being isotropic, an orientation-dependent energy may still be acquired by sampling different regions of an inhomogeneous field. We neglect the distortion of the prescribed field by the particle, which is consistent with the small magnitude of χ_ρ ¹¹. To display the point-particle approximation and its leading correction, define

$$\mathbf{M}_0 := \rho \int_{\mathcal{V}_0} \mathbf{u}\mathbf{u}^T d^3u = \text{diag}\left(\frac{mR^2}{4}, \frac{mR^2}{4}, \frac{mL^2}{12}\right), \quad (263)$$

where in the second step we used the property that centered body coordinates satisfy $\rho \int_{\mathcal{V}_0} \mathbf{u} d^3u = \mathbf{0}$.

Expanding $f(\mathbf{c} + \mathbf{F}\mathbf{u}, t)$ around the c.o.m. gives

$$V_{\text{dia}} = -\frac{\chi_\rho}{2\mu_0} \left[m \|\mathbf{B}^{\text{sf}}(\mathbf{c}, t)\|^2 + \frac{1}{2} \text{Tr} \left(\mathbf{M}_0 \mathbf{F}^T \left(\nabla \nabla \|\mathbf{B}^{\text{sf}}(\mathbf{c}, t)\|^2 \right) \mathbf{F} \right) + \dots \right]. \quad (264)$$

The second term is the leading finite-size orientation-dependent correction, which is suppressed by the squared ratio between the characteristic rotor dimensions $a_{\text{rot}} \sim \sqrt{M_0/m}$ and the characteristic length scale $\ell_{B^2} \sim \sqrt{\|\mathbf{B}^{\text{sf}}\|^2 / \nabla \nabla \|\mathbf{B}^{\text{sf}}\|^2}$ over which $\|\mathbf{B}^{\text{sf}}\|^2$ varies appreciably.

4.5.2 Point-particle limit and force sign

Neglecting the finite-size terms in **Eq. 264** gives the point-particle diamagnetic potential and the corresponding force and acceleration

$$V_{\text{dia}}^{(0)} = -\frac{\chi_\rho m}{2\mu_0} \|\mathbf{B}^{\text{sf}}(\mathbf{c}, t)\|^2 \quad \Rightarrow \quad m\mathbf{a}_{\text{dia}}^{(0)} = \mathbf{F}_{\text{dia}}^{(0)} = -\nabla_{\mathbf{c}} V_{\text{dia}}^{(0)} = \frac{\chi_\rho m}{2\mu_0} \nabla_{\mathbf{c}} \|\mathbf{B}^{\text{sf}}(\mathbf{c}, t)\|^2, \quad (265)$$

which, since $\chi_\rho < 0$ for diamond, the diamagnetic acceleration points towards weaker field.

Because the point-particle energy contains no finite-volume correction, it is independent of the body orientation and generates no torque.

4.5.3 Gravity, equilibrium and harmonic confinement

We choose the positive stationary/laboratory y -axis to point upwards, so the gravitational potential is

$$V_g = mgc_y \quad \left| \quad \frac{\partial V_{\text{tr}}}{\partial c_y} \right|_{c_0} = 0, \quad (266)$$

¹¹The total magnetic flux has an external, apparatus prescribed component and, as the diamond becomes weakly magnetized and produces a tiny additional field, an induced one. These relate to the field through the material's magnetization $\mathbf{M}$ and the magnetic field strength $\mathbf{H}$ as:

$$\mathbf{B} = \mathbf{B}_{\text{ext}} + \mathbf{B}_{\text{ind}} = \mu_0(\mathbf{H} + \mathbf{M}) \quad \left| \quad \mathbf{M} = \chi_v \mathbf{H} \quad \left| \quad \chi_v = \rho\chi_\rho \quad \Rightarrow \quad \text{for } |\chi_\rho| \ll 1 \Rightarrow |\mathbf{M}| \ll |\mathbf{H}| \Rightarrow \mathbf{B} \approx \mathbf{B}_{\text{ext}}. \right. \right. \quad (262)$$

and gravity becomes an irrelevant additive constant only after $c_y \rightarrow c_0$ has been fixed at trap equilibrium.

We first define the spin-independent diamagnetic and gravitational trapping potential

$$\frac{V_{\text{tr}}^{(0)}}{m} = -\frac{\chi_\rho}{2\mu_0} \|\mathbf{B}_{\text{tr}}\|^2 + gc_y. \quad (267)$$

Its equilibrium and matrix of squared small-oscillation frequencies satisfy

$$\left. \frac{\chi_\rho}{2\mu_0} \nabla \|\mathbf{B}_{\text{tr}}\|^2 \right|_{\mathbf{c}_0^{(0)}} - g\hat{\mathbf{e}}_y = \mathbf{0} \quad \left| \quad (\Omega_{\text{tr}}^{(0)2})_{ij} := \frac{1}{m} \frac{\partial^2 V_{\text{tr}}^{(0)}}{\partial c_i \partial c_j} \right|_{\mathbf{c}_0^{(0)}} = -\frac{\chi_\rho}{2\mu_0} \frac{\partial^2 B_{\text{tr}}^2}{\partial c_i \partial c_j} \Big|_{\mathbf{c}_0^{(0)}}. \quad (268)$$

If the static field also enters the internal spin Hamiltonian, then equilibrium is branch dependent. For a fixed prepared orientation $\mathbf{F}_0$, the corresponding potential is

$$V_{\text{tr},n}(\mathbf{c}; \mathbf{F}_0) := V_{\text{tr}}^{(0)}(\mathbf{c}) + V_n^{(3)}(\mathbf{c}, \mathbf{F}_0) \quad \left| \quad \nabla_{\mathbf{c}} V_{\text{tr},n} \Big|_{\mathbf{c}_{0,n}} = \mathbf{0} \quad \left| \quad (\Omega_{\text{tr},n}^2)_{ij} := \frac{1}{m} \frac{\partial^2 V_{\text{tr},n}}{\partial c_i \partial c_j} \right|_{\mathbf{c}_{0,n}}. \quad (269)$$

If the orientation is not externally fixed, the translational and rotational equilibrium conditions must instead be solved simultaneously.

Expanding the branch-dependent potential around $\mathbf{c}_{0,n}$, with $\delta\mathbf{c} := \mathbf{c} - \mathbf{c}_{0,n}$, gives

$$V_{\text{tr},n}(\mathbf{c}) \simeq V_{\text{tr},n}(\mathbf{c}_{0,n}) + \frac{m}{2} \delta\mathbf{c}^T \Omega_{\text{tr},n}^2 \delta\mathbf{c}. \quad (270)$$

A stable branch equilibrium therefore requires a positive definite $\Omega_{\text{tr},n}^2$, so that sufficiently small displacements increase the potential energy, thus producing a restoring force. Its eigenvalues are then positive and equal to the squared normal-mode trap frequencies.

The magnetic profile from **Eq. 160** along the $x=z=0$ line, now written as

$$\|\mathbf{B}_{\text{tr}}(0, y, 0)\|^2 = b(y)^2 \quad \left| \quad b(y) := -b_2 y + b_3 y^2 + b_4 y^3. \quad (271)\right.$$

reduces the vertical equilibrium condition (spin-branch energy omitted or included, respectively) to

$$\frac{\chi_\rho}{\mu_0} b(y_{\text{eq}}^{(0)}) b'(y_{\text{eq}}^{(0)}) = g \quad \left| \quad \frac{\chi_\rho}{\mu_0} b(y_{\text{eq},n}) b'(y_{\text{eq},n}) = g + \frac{1}{m} \frac{\partial V_n^{(3)}}{\partial y} \right|_{y_{\text{eq},n}}. \quad (272)$$

When the explicit trapping apparatus is omitted from the spin Hamiltonian, its local mechanical confinement is retained through the harmonic expansion around a stable equilibrium:

$$V_{\text{tr}}(\mathbf{c}) \simeq V_{\text{tr}}(\mathbf{c}_0) + \nabla V_{\text{tr}}|_{\mathbf{c}_0} \cdot \delta\mathbf{c} + \frac{1}{2} \delta\mathbf{c}^T \nabla \nabla V_{\text{tr}}|_{\mathbf{c}_0} \delta\mathbf{c} + \dots. \quad (273)$$

The constant term does not affect the dynamics and the linear one by definition vanishes at equilibrium. Picking the transverse axes along the principal directions of the local Hessian gives the effective confinement

$$V_{\text{conf}} = \frac{m}{2} \sum_{i \in \{x,y,z\}} \omega_i^2 (c_i - c_{i0})^2 \quad \left| \quad \omega_i^2 := \frac{1}{m} \frac{\partial^2 V_{\text{trap}}}{\partial c_i^2}, \quad (274)\right.$$

where the z term should ideally be null, as the z -direction has to allow the Stern–Gerlach force to freely separate and recombine the arms: the one-dimensional c.o.m. reduction $\mathbf{c}(t) = (c_{x0} \ c_{y0} \ z(t))^T$.

This reduction is justified only when the particle is initially prepared close to the transverse equilibrium and the confinement frequencies and remains so throughout the interferometric sequence.

4.6 Master semiclassical Hamiltonian

At leading scalar-potential order in the adiabatic reduction, where the spin contributes through the instantaneous branch energy $V_n^{(3)}$ of **Eq. 205**, and before retaining the momentum-dependent spin–rotation correction of **Sec. 4.7**, the branch Hamiltonian is

$$H_n = \frac{\mathbf{p}_c^T \mathbf{p}_c}{2m} + \frac{p_\beta^2}{2I_\perp} + \frac{p_\gamma^2}{2I_\parallel} + \frac{(p_\alpha - p_\gamma \cos \beta)^2}{2I_\perp \sin^2 \beta} + V_{\text{dia}}(\mathbf{c}, \mathbf{F}, t) + mgc_y + V_{\text{conf}}(\mathbf{c}, t) + V_n^{(3)}(\mathbf{c}, \alpha, \beta, \gamma, t).$$

(275)

Geometric corrections involving derivatives of the instantaneous spin eigenvectors are beyond the adiabatic order retained here and are therefore neglected.

If the explicit trapping field is included in $\mathbf{B}^{\text{sf}}$, its diamagnetic and spin effects are already contained in V_{dia} and $V_n^{(3)}$, so V_{conf} is omitted. The analytical model follows by replacing $V_n^{(3)}$ with $V_\sigma^{(2)}$; the centered, aligned and one-dimensional reference reduction is obtained in **Sec. 4.9**.

Eq. 275 makes the coupling explicit: translation changes the field sampled at the defect, while rotation changes both the defect position and the components of that field in the spin frame.

The same branch surface thus generates the c.o.m. force and the mechanical torque. Since **Eqs. 169, 171** give an explicitly time dependent applied field, H_n is not conserved during the ramps: energy is exchanged with the external source even in the absence of dissipation and conservation only occurs when the potentials are time independent.

4.7 Spin–rotation coupling and angular-momentum transfer

4.7.1 Rotating-frame Hamiltonian

To obtain the Schrödinger generator in the rotating spin rep of H_S^s (see **Eq. 197**), we transform the laboratory-frame state according to the unitary operators from **Eq. 193**:

$$|\psi^s\rangle := U_S^{s\leftarrow\text{sf}}(t)|\psi^{\text{sf}}\rangle = \left(U_S^{\text{sf}\leftarrow s}(t)\right)^\dagger |\psi^{\text{sf}}\rangle, \quad (276)$$

and going forward, in order to de-clutter our equations, we omit the obvious time-dependence.

Using **Eq. 201**, its Schrödinger equation becomes

$$\begin{aligned} i\hbar \frac{d}{dt} |\psi^s\rangle &= H_{S,\text{rot}}^s |\psi^s\rangle \quad \Rightarrow \quad H_{S,\text{rot}}^s := \left(U_S^{\text{sf}\leftarrow s}\right)^\dagger H_S^{\text{sf}} U_S^{\text{sf}\leftarrow s} - i\hbar \left(U_S^{\text{sf}\leftarrow s}\right)^\dagger \dot{U}_S^{\text{sf}\leftarrow s} \\ &= H_S^s - i\hbar \left(U_S^{\text{sf}\leftarrow s}\right)^\dagger \dot{U}_S^{\text{sf}\leftarrow s}. \end{aligned} \quad (277)$$

Using the Lie-algebra representation in **Eq. 191** together with the body-frame angular-velocity definitions in **Eqs. 98, 99**, we have

$$-i\hbar \left(U_S^{\text{sf}\leftarrow s}\right)^\dagger \dot{U}_S^{\text{sf}\leftarrow s} = -\hbar \boldsymbol{\omega}^s \cdot \mathbf{S} \quad \Bigg| \quad \boldsymbol{\omega}^s = \mathbf{C}_s^T \boldsymbol{\omega}^{\text{bf}}, \quad (278)$$

and therefore

$$\boxed{H_{S,\text{rot}}^s = H_S^s - \hbar \boldsymbol{\omega}^s \cdot \mathbf{S} = \mu \mathbf{S} \cdot \mathbf{B}^s + D \left(S_3^2 - \frac{2}{3} \mathbf{1}\right) + E \left(S_1^2 - S_2^2\right) - \hbar \boldsymbol{\omega}^s \cdot \mathbf{S}.} \quad (279)$$

For a direct comparison with Eqs. (23)–(24) of [19], we split the rotating-frame spin Hamiltonian as

$$\boxed{H_{S,\text{rot}}^s = H_{S,Z+\text{ZFS}}^s + H_{S,\text{sr}}^s \quad \Bigg| \quad H_{S,Z+\text{ZFS}}^s := H_S^s \quad \Bigg| \quad H_{S,\text{sr}}^s := -\hbar \boldsymbol{\omega}^s \cdot \mathbf{S}.} \quad (280)$$

The first term is the Zeeman plus zero-field-splitting Hamiltonian from **Eq. 197**; setting $E = 0$ gives Eq. (23) from [19] with the equivalent Cartesian (rather than phase) notation. Using the spin-1 matrices (**Eq. 190**), its spin–rotation part is

$$\boxed{H_{S,\text{sr}}^s = -\hbar \begin{pmatrix} \omega_3^s & \frac{\omega_1^s - i\omega_2^s}{\sqrt{2}} & 0 \\ \frac{\omega_1^s + i\omega_2^s}{\sqrt{2}} & 0 & \frac{\omega_1^s - i\omega_2^s}{\sqrt{2}} \\ 0 & \frac{\omega_1^s + i\omega_2^s}{\sqrt{2}} & -\omega_3^s \end{pmatrix} = -\hbar \begin{pmatrix} \frac{L_3^{\text{bf}}}{I_\parallel} & \frac{L_1^{\text{bf}} - iL_2^{\text{bf}}}{\sqrt{2}I_\perp} & 0 \\ \frac{L_1^{\text{bf}} + iL_2^{\text{bf}}}{\sqrt{2}I_\perp} & 0 & \frac{L_1^{\text{bf}} - iL_2^{\text{bf}}}{\sqrt{2}I_\perp} \\ 0 & \frac{L_1^{\text{bf}} + iL_2^{\text{bf}}}{\sqrt{2}I_\perp} & -\frac{L_3^{\text{bf}}}{I_\parallel} \end{pmatrix},} \quad (281)$$

which is Eq. (24) from [19] in the present notation. The second matrix we uses an aligned spin and body frame ($\mathbf{C}_s = \mathbf{1}_3$) and an axially symmetric rotor, for which angular momentum is given by

$$\boxed{L_1^{\text{bf}} = I_\perp \omega_1^s \quad \Bigg| \quad L_2^{\text{bf}} = I_\perp \omega_2^s \quad \Bigg| \quad L_3^{\text{bf}} = I_\parallel \omega_3^s.} \quad (282)$$

The final term of **Eq. 279** is the spin–rotation coupling and acts as a shift of the Zeeman energy vector from $\mu\mathbf{B}^s$ to $\mu\mathbf{B}^s - \hbar\boldsymbol{\omega}^s$. It is not an additional term to be appended to H_S^{sf} : the rotating defect axes in the laboratory representation and the inertial term in the spin-frame representation describe the same kinematics. Thus, the gap and adiabaticity conditions of **Sec. 4.3.5** are now evaluated using $H_{S,\text{rot}}^s$.

Combining the xxz angular-velocity relation $\boldsymbol{\omega}^{\text{bf}} = \mathbf{N}(\alpha, \beta, \gamma)\dot{\boldsymbol{\theta}}$ (**Eq. 104**) (where the explicit xxz matrix $\mathbf{N}$ given in **Eq. 104**) with the uncoupled symmetric-top momentum relation $\dot{\boldsymbol{\theta}} = \mathbf{g}_{\text{rot}}^{-1}(\beta)\mathbf{p}_\theta$ (**Eq. 127**) and transforming the angular velocity from the body frame to the spin frame through $\boldsymbol{\omega}^s = \mathbf{C}_s^T\boldsymbol{\omega}^{\text{bf}}$ from **Eq. 278** above gives, for the uncoupled rotor,

$$\boldsymbol{\omega}^s = \mathbf{C}_s^T \mathbf{N}(\alpha, \beta, \gamma)\dot{\boldsymbol{\theta}} = \mathbf{C}_s^T \mathbf{N}(\alpha, \beta, \gamma)\mathbf{g}_{\text{rot}}^{-1}(\beta)\mathbf{p}_\theta. \quad (283)$$

Unlike the first form (containing $\dot{\boldsymbol{\theta}}$), the canonical relation (containing $\mathbf{g}_{\text{rot}}^{-1}(\beta)\mathbf{p}_\theta$) belongs specifically to the uncoupled rotor and may therefore be used inside the spin–rotation energy only perturbatively.

If the spin–rotation contribution is self-consistently retained, let $V_{n,\text{rot}}$ denote the instantaneous branch eigenenergy of $H_{S,\text{rot}}^s$. The general canonical definition in **Eq. 117** then gives

$$\mathbf{p}_\theta = \frac{\partial \mathcal{L}_{\text{rot},n}}{\partial \dot{\boldsymbol{\theta}}} = \mathbf{g}_{\text{rot}}\dot{\boldsymbol{\theta}} - \frac{\partial V_{n,\text{rot}}}{\partial \dot{\boldsymbol{\theta}}} \quad \left| \quad \frac{\partial V_{n,\text{rot}}}{\partial \dot{\boldsymbol{\theta}}} = \left\langle n^s \left| \frac{\partial H_{S,\text{rot}}^s}{\partial \dot{\boldsymbol{\theta}}} \right| n^s \right\rangle = -\hbar \mathbf{N}^T \mathbf{C}_s \langle \mathbf{S} \rangle_n, \quad (284)$$

where the second equality follows from **Thm. 7** (the Hellmann–Feynman theorem) and

$$-\hbar\boldsymbol{\omega}^s \cdot \mathbf{S} = -\hbar\dot{\boldsymbol{\theta}}^T \mathbf{N}^T \mathbf{C}_s \mathbf{S}. \quad (285)$$

Consequently,

$$\boxed{\mathbf{p}_\theta = \mathbf{g}_{\text{rot}}\dot{\boldsymbol{\theta}} + \hbar \mathbf{N}^T \mathbf{C}_s \langle \mathbf{S} \rangle_n.} \quad (286)$$

Solving this relation for $\dot{\boldsymbol{\theta}}$ and substituting the result into the exact kinematic identity **Eq. 283** gives

$$\boxed{\dot{\boldsymbol{\theta}} = \mathbf{g}_{\text{rot}}^{-1}(\mathbf{p}_\theta - \hbar \mathbf{N}^T \mathbf{C}_s \langle \mathbf{S} \rangle_n) \quad \left| \quad \boldsymbol{\omega}^s = \mathbf{C}_s^T \mathbf{N} \mathbf{g}_{\text{rot}}^{-1}(\mathbf{p}_\theta - \hbar \mathbf{N}^T \mathbf{C}_s \langle \mathbf{S} \rangle_n)} \quad (287)$$

Thus, the kinematic relation between $\boldsymbol{\omega}^s$ and $\dot{\boldsymbol{\theta}}$ is unchanged; only that between $\dot{\boldsymbol{\theta}}$ and the canonical momentum $\mathbf{p}_\theta$ is shifted by the spin–rotation coupling. Because $\langle \mathbf{S} \rangle_n$ is evaluated in an eigenstate of $H_{S,\text{rot}}^s$, it may depend on $\boldsymbol{\omega}^s$, thus making the non-perturbative relation implicit and required to be self-consistently solved.

4.7.2 Scale of the rotational correction

For the initial conditions of pure symmetry-axis rotation (**Eq. 130**) and an aligned spin frame ($\mathbf{C}_s = \mathbf{1}_3$),

$$\dot{\boldsymbol{\theta}}(0) = \begin{pmatrix} 0 \\ 0 \\ \omega_0 \end{pmatrix} \quad \left| \quad \boldsymbol{\omega}^{\text{bf}}(0) = \mathbf{N}(\boldsymbol{\theta}(0))\dot{\boldsymbol{\theta}}(0) = \omega_0 \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \quad \left| \quad \boldsymbol{\omega}^s(0) = \mathbf{C}_s^T \boldsymbol{\omega}^{\text{bf}}(0) = \begin{pmatrix} 0 \\ 0 \\ \omega_0 \end{pmatrix}. \quad (288)$$

The initial spin–rotation term in **Eq. 279** is consequently

$$-\hbar\boldsymbol{\omega}^s(0) \cdot \mathbf{S} = -\hbar\omega_0 S_3 = \begin{pmatrix} -\hbar\omega_0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & +\hbar\omega_0 \end{pmatrix}. \quad (289)$$

The relative size of the spin–rotation correction $-\hbar\boldsymbol{\omega}^s \cdot \mathbf{S}$ in **Eq. 279** is estimated by comparing its characteristic energy $\hbar\|\boldsymbol{\omega}^s\|$ with the axial zero-field-splitting scale D and the Zeeman scale $\mu\|\mathbf{B}^s\|$:

$$\epsilon_{\omega D} := \frac{\hbar\|\boldsymbol{\omega}^s\|}{D} \quad \left| \quad \epsilon_{\omega B} := \frac{\hbar\|\boldsymbol{\omega}^s\|}{\mu\|\mathbf{B}^s\|}. \quad (290)$$

When both remain small, the spin–rotation correction to the scalar adiabatic branch potential $V_n^{(3)}$ may be neglected after its magnitude has been quantified.

However, more generally, $\epsilon_{\omega D} \ll 1$ and $\epsilon_{\omega B} \ll 1$ are not sufficient near a small avoided crossing, where the separation between two instantaneous branches may be much smaller than either of those characteristic scales (**Eq. 229** gives the minimum avoided-crossing gap $\Delta_{+-,\ell}^{(2)} = 2|\varepsilon_\ell|$). A sufficient condition is therefore

$$\hbar\|\boldsymbol{\omega}^s\| \ll \Delta_{\text{spin}}^{\min}, \quad (291)$$

ensuring no appreciable shift or mixing of neighbouring instantaneous spin branches from the the spin-rotation term. In the simplified axial bare-state model, the relevant unperturbed energy gaps are

$$2\mu|B_3| \quad \left| \quad \quad |D - \mu B_3| \quad \quad \right| \quad \quad |D + \mu B_3|, \quad (292)$$

thus providing the sufficient scale separations

$$\omega_0 \ll \frac{\mu|B_3|}{\hbar} \quad \left| \quad \quad \omega_0 \ll \frac{|D - \mu B_3|}{\hbar} \quad \quad \right| \quad \quad \omega_0 \ll \frac{|D + \mu B_3|}{\hbar}. \quad (293)$$

4.7.3 Einstein–de Haas interpretation

The additional term in the rotating-frame Hamiltonian of **Eq. 279**,

$$H_{S,\text{sr}}^s := -\hbar\boldsymbol{\omega}^s \cdot \mathbf{S}, \quad (294)$$

is the spin-rotation, or inertial, coupling. It arises because the spin is described in a frame rotating with $\boldsymbol{\omega}^s$ and couples its projection to the mechanical angular velocity, shifting the spin levels when it is diagonal in their instantaneous basis (**Eq. 289**), or, otherwise, mixing them.

The Einstein–de Haas effect is instead the mechanical response associated with a change of the internal spin angular momentum, which occurs whenever the spin state or its orientation changes. Defining

$$\langle \mathbf{J}_S^{\text{sf}} \rangle := \hbar \langle \mathbf{S}^{\text{sf}} \rangle, \quad (295)$$

the angular-momentum balance of the driven rotor–spin subsystem is

$$\boxed{\frac{d}{dt} [\mathbf{L}_{\text{mech}}^{\text{sf}} + \langle \mathbf{J}_S^{\text{sf}} \rangle] = \boldsymbol{\tau}_{\text{ext}}^{\text{sf}} \quad \Longleftrightarrow \quad \frac{d\mathbf{L}_{\text{mech}}^{\text{sf}}}{dt} = \boldsymbol{\tau}_{\text{ext}}^{\text{sf}} - \frac{d}{dt} \langle \mathbf{J}_S^{\text{sf}} \rangle = \boldsymbol{\tau}_{\text{ext}}^{\text{sf}} + \boldsymbol{\tau}_{\text{EdH}}^{\text{sf}},} \quad (296)$$

where $\boldsymbol{\tau}_{\text{ext}}^{\text{sf}}$ contains the torques from the magnetic field and, during a MW pulse, by the electromagnetic drive and its source. The Einstein–de Haas reaction torque is

$$\boldsymbol{\tau}_{\text{EdH}}^{\text{sf}} := -\frac{d}{dt} \langle \mathbf{J}_S^{\text{sf}} \rangle. \quad (297)$$

In the isolated limit $\boldsymbol{\tau}_{\text{ext}}^{\text{sf}} = \mathbf{0}$, its integrated action gives

$$\Delta \mathbf{L}_{\text{mech}}^{\text{sf}} = \int \boldsymbol{\tau}_{\text{EdH}}^{\text{sf}}(t) dt = -\Delta \langle \mathbf{J}_S^{\text{sf}} \rangle, \quad (298)$$

which is the usual Einstein–de Haas angular-momentum transfer [27, 28].

Thus, the spin-rotation term and the Einstein–de Haas effect are fundamentally distinct.

During the microwave pulse, energy and angular momentum are explicitly supplied. Writing $\Delta \mathbf{J}_{\text{drive}}^{\text{sf}}$ as the contribution of the electromagnetic field and its source, the balance becomes

$$\Delta \mathbf{L}_{\text{mech}}^{\text{sf}} + \Delta \langle \mathbf{J}_S^{\text{sf}} \rangle + \Delta \mathbf{J}_{\text{drive}}^{\text{sf}} = \mathbf{0}. \quad (299)$$

Consequently, the ideal state swap in **Eq. 241** does not require the mechanical angular momentum to acquire exactly the negative of the spin change.

The paper [19] uses the spin-rotation term to identify rotating-frame level shifts and resonances, imposes the corresponding upper bounds on ω_0 , and then omits it from the trajectory Hamiltonian used for the spatial and Euler-angle dynamics. We adopt that reduced model as the reference baseline, but treat the omission as a numerical approximation to be verified rather than an assumption.

For this verification, we begin with the angular velocity obtained from the uncoupled rotor (**Eq. 283**) and its corresponding perturbing operator

$$H_{S,\text{sr},0}^s(\boldsymbol{\theta}, \mathbf{p}_\theta) := -\hbar\boldsymbol{\omega}_0^s(\boldsymbol{\theta}, \mathbf{p}_\theta) \cdot \mathbf{S}. \quad (300)$$

For an unperturbed instantaneous branch $|n^s\rangle$ of H_S^s , the first-order shift of branch n 's spin energy caused by the spin-rotation coupling is

$$\boxed{\Delta V_{n,\text{sr}}^{(1)} := \langle n^s | H_{S,\text{sr},0}^s | n^s \rangle = -\hbar\boldsymbol{\omega}_0^s \cdot \langle \mathbf{S} \rangle_n.} \quad (301)$$

Equivalently, for the full three-level 3×3 spin Hamiltonian including spin-rotation,

$$(H_S^s + H_{S,\text{sr},0}^s) |n_{\text{sr}}^s\rangle = V_{n,\text{sr}}^{(3)} |n_{\text{sr}}^s\rangle, \quad (302)$$

then, on a non-degenerate branch,

$$V_{n,\text{sr}}^{(3)} = V_n^{(3)} + \langle n^s | H_{S,\text{sr},0}^s | n^s \rangle + \sum_{m \neq n} \frac{|\langle m^s | H_{S,\text{sr},0}^s | n^s \rangle|^2}{\lambda_m - \lambda_n} + \dots = V_n^{(3)} + \Delta V_{n,\text{sr}}^{(1)} + \mathcal{O}\left(\frac{\hbar^2 \|\boldsymbol{\omega}_0^s\|^2}{\Delta_n}\right), \quad (303)$$

where $\Delta_n := \min_{m \neq n} |\lambda_m - \lambda_n|$. The perturbatively spin-rotation-corrected branch Hamiltonian is therefore

$$H_n^{\text{sr}(1)} = \frac{\mathbf{p}_c^T \mathbf{p}_c}{2m} + \frac{1}{2} \mathbf{p}_\theta^T \mathbf{g}_{\text{rot}}^{-1} \mathbf{p}_\theta + V_{\text{dia}} + mgc_y + V_{\text{conf}} + V_n^{(3)} + \Delta V_{n,\text{sr}}^{(1)} = T_{\text{trans}} + T_{\text{rot}} + \mathcal{V}_n + \Delta V_{n,\text{sr}}^{(1)}, \quad (304)$$

where $\mathcal{V}_n$ is the total coordinate-dependent potential before the spin-rotation correction.

Its canonical equations are

$$\begin{aligned} \dot{\mathbf{c}} &= \frac{\mathbf{p}_c}{m} & \dot{\mathbf{p}}_c &= -\nabla_c (\mathcal{V}_n + \Delta V_{n,\text{sr}}^{(1)}) \\ \dot{\boldsymbol{\theta}} &= \mathbf{g}_{\text{rot}}^{-1} \mathbf{p}_\theta + \frac{\partial \Delta V_{n,\text{sr}}^{(1)}}{\partial \mathbf{p}_\theta} & \dot{\mathbf{p}}_\theta &= -\frac{\partial}{\partial \boldsymbol{\theta}} \left[\frac{1}{2} \mathbf{p}_\theta^T \mathbf{g}_{\text{rot}}^{-1} \mathbf{p}_\theta + \mathcal{V}_n + \Delta V_{n,\text{sr}}^{(1)} \right]. \end{aligned} \quad (305)$$

The momentum derivative in the third equation reproduces, to first order in the spin-rotation coupling, the shifted Euler-angle rates of **Eq. 287**.

The operator 2-norm spectral condition

$$\frac{\|H_{S,\text{sr},0}^s\|_2}{\Delta_n} = \frac{\hbar \|\boldsymbol{\omega}_0^s\|}{\Delta_n} \ll 1 \quad (306)$$

reads as the spin-rotation energy must be much smaller than the distance from branch n to the nearest other branch. Its trajectory-wide version is **Eq. 291** (compared to this branch-specific version). It controls the perturbation of the spin eigenvalues and eigenvectors, but does not alone guarantee negligible mechanical corrections. Omitting the spin-rotation term from the trajectory equations additionally requires its derivatives to be small at the level of the equations of motion:

$$\left| \frac{\partial \Delta V_{n,\text{sr}}^{(1)}}{\partial q^a} \right| \ll \left| \frac{\partial \mathcal{V}_n}{\partial q^a} \right| \quad \left| \frac{\partial \Delta V_{n,\text{sr}}^{(1)}}{\partial p_\mu} \right| \ll \left| \frac{\partial T_{\text{rot}}}{\partial p_\mu} \right|, \quad (307)$$

where vanishing denominators are handled through absolute tolerances and direct trajectory comparison.

In conclusion, the numerical analysis will use the following hierarchy. The reference-paper [19] and (principal scalar-branch trajectories) set $\Delta V_{n,\text{sr}}^{(1)} = 0$. A second run retains $\Delta V_{n,\text{sr}}^{(1)}$ and evaluates the changes in the spin gaps, forces, torques and trajectories. If **Eqs. 291, 307** hold and the resulting trajectory differences remain below the required tolerance, we accept the reduced scalar-potential model. If they fail however, the perturbative model is insufficient and we must instead solve the implicit self-consistent system defined by **Eqs. 286, 287**.

The Einstein-de Haas effect enters through the spin-rotation coupling $H_{S,\text{sr}}^s$, which shifts and can even mix the instantaneous spin branches. Firstly, we use it to constrain the allowed mechanical rotation through the spin-spectrum conditions, then retain its first-order correction as a trajectory validation model. If both the spectral and trajectory-level omission conditions are satisfied, then we consistently neglect this correction while retaining the externally generated Zeeman force and torque.

4.8 Coupled equations at leading adiabatic order and rotational symmetries

4.8.1 Hamilton equations

We derive the leading scalar-potential limit of **Eq. 305** by setting $\Delta V_{n,\text{sr}}^{(1)} = 0$. The resulting equations apply on each smooth adiabatic segment of the interferometer arms and retain the externally driven spin force and torque through $\mathcal{V}_n$. The spin-rotation-corrected trajectories generated **Eq. 305** are to be used in a numerical validity audit. At the microwave event, the followed branch is changed according to **Sec. 4.3.6**.

Thus, for the complete coordinate-dependent branch potential ($\mathcal{V}_n$), ignoring the first-order shift of branch n 's spin energy caused by the spin-rotation coupling ($\Delta V_{n,\text{sr}}^{(1)}$) reduces **Eqs. 305** to:

$$\boxed{\begin{array}{l|l|l} \dot{\mathbf{c}} = \frac{\mathbf{p}_c}{m} & \dot{\mathbf{p}}_c = -\nabla_{\mathbf{c}} \mathcal{V}_n & \\ \hline \dot{\alpha} = \frac{p_\alpha - p_\gamma \cos \beta}{I_\perp \sin^2 \beta} & \dot{\beta} = \frac{p_\beta}{I_\perp} & \dot{\gamma} = \frac{p_\gamma}{I_\parallel} - \frac{\cos \beta (p_\alpha - p_\gamma \cos \beta)}{I_\perp \sin^2 \beta} \\ \hline \dot{p}_\alpha = -\frac{\partial \mathcal{V}_n}{\partial \alpha} & \dot{p}_\beta = \frac{(p_\alpha - p_\gamma \cos \beta)(p_\alpha \cos \beta - p_\gamma)}{I_\perp \sin^3 \beta} - \frac{\partial \mathcal{V}_n}{\partial \beta} & \dot{p}_\gamma = -\frac{\partial \mathcal{V}_n}{\partial \gamma} \end{array}} \quad (308)$$

Since $p_\beta = I_\perp \dot{\beta}$, the corresponding nonlinear libration equation is

$$\boxed{\ddot{\beta} = \frac{(p_\alpha - p_\gamma \cos \beta)(p_\alpha \cos \beta - p_\gamma)}{I_\perp^2 \sin^3 \beta} - \frac{1}{I_\perp} \frac{\partial \mathcal{V}_n}{\partial \beta}} \quad (309)$$

The $\beta=0$ singularity is the gimbal lock; the physical orientation and angular velocity remain regular.

Eq. 309 retains the universal symmetric-top structure derived in **Eq. 135**: the first term is entirely fixed by the (inverse) rotational metric, whereas all specific-model dependencies enter through the generalized branch torque $-\partial_\beta \mathcal{V}_n$.

However, in the complete semiclassical model, β does not generally have a closed equation, as p_α and p_γ evolve according to $\dot{p}_\alpha = -\partial_\alpha \mathcal{V}_n$ and $\dot{p}_\gamma = -\partial_\gamma \mathcal{V}_n$. It is only when the complete branch potential is independent of both α and γ that these momenta are conserved, and **Eq. 135** is fully recovered.

Thus, the semiclassical extension retains the classical rotational kinetic structure, while its branch potential determines whether the classical symmetry reduction remains valid.

4.8.2 Symmetry audit

The leading-order adiabatic branch Hamiltonian has can be decomposed as

$$H_n = T_{\text{trans}} + T_{\text{rot}}(\beta, \mathbf{p}_\theta) + \mathcal{V}_n(\mathbf{c}, \alpha, \beta, \gamma, t) = \frac{\mathbf{p}_c^T \mathbf{p}_c}{2m} + \frac{p_\beta^2}{2I_\perp} + \frac{p_\gamma^2}{2I_\parallel} + \frac{(p_\alpha - p_\gamma \cos \beta)^2}{2I_\perp \sin^2 \beta} + \mathcal{V}_n, \quad (310)$$

and, when invariant under an Euler rotation, the corresponding momentum is conserved:

$$\dot{p}_\theta = -\frac{\partial H_n}{\partial \theta} \quad \Bigg| \quad p_\theta = \text{const.} \quad \Longleftrightarrow \quad \frac{\partial H_n}{\partial \theta} = 0. \quad (311)$$

These derivatives are evaluated only after the defect position, spin-frame orientation, total field and diamagnetic model have been specified. In particular:

1. a centered defect, aligned spin axis, bare $E=0$, point-particle diamagnetism and an axial field produce an orientation potential depending only on β , so both p_α and p_γ are conserved;
2. $\mathbf{r}_s = \mathbf{c} + \mathbf{F} \mathbf{d}_0$ depends on the complete orientation and may remove either cyclicity;
3. when $E \neq 0$ and $B_\perp \neq 0$, the term $E(B_1^2 - B_2^2)$ in **Eq. 223** resolves the two axis of the transverse defect and removes the γ cyclicity;
4. the explicit multipole field of **Eq. 157** is not axially symmetric about the interferometer z -axis, while the finite-volume diamagnetic energy also depends on $\mathbf{F}$ through **Eq. 264**: p_α and p_γ are no longer generally conserved, thus breaking the laboratory- z rotational symmetry, while the finite-volume diamagnetic term produces orientation-dependent torques.

Thus, saying that no force acts "along" α or γ is insufficient: their momenta are conserved only when the corresponding continuous transformations leave the complete Hamiltonian unchanged (i.e., **Eq. 311**).

4.8.3 Controlled one-dimensional translation

We now assume that the particle is initially prepared near the transverse equilibrium and that, with negligible transverse momenta, the confinement frequencies keep the defect at

$$c_x(t) \simeq c_{x0} \quad \Big| \quad c_y(t) \simeq c_{y0} \quad \Big| \quad z(t) := c_z(t) \quad \Rightarrow \quad \mathbf{r}_s = (c_{x0} \ c_{y0} \ z)^T + \mathbf{F}\mathbf{d}_0, \quad (312)$$

allowing for off-center defects to sample all three field components despite the c.o.m. having only one active coordinate. Under this reduction, gravity and the transverse harmonic confinement have no z derivative under this reduction, leaving the only translational equation,

$$m\ddot{z} = -\frac{\partial}{\partial z} (V_{\text{dia}} + V_n^{(3)}). \quad (313)$$

The spin contribution is still evaluated from the complete three-dimensional field at $\mathbf{r}_s$ through **Eq. 252**.

4.9 Gyroscopic baseline and recovery of the reference model

4.9.1 Trajectory hierarchy and full branch acceleration

For the full three-dimensional adiabatic branch model of $H_n^{sr(1)}$ (see **Eq. 304**), the complete c.o.m. acceleration at leading scalar-potential order (i.e., excluding $\Delta V_{n,\text{sr}}^{(1)}$) is

$$\boxed{\ddot{\mathbf{c}}_n^{(3)} = -\frac{1}{m} \nabla_{\mathbf{c}} \mathcal{V}_n}, \quad (314)$$

which, in the point-particle diamagnetic limit (i.e. ignoring the field variation across the particle volume),

$$\boxed{\ddot{\mathbf{c}}_n^{(3)} = \frac{\chi_\rho}{2\mu_0} \nabla_{\mathbf{c}} \left\| \mathbf{B}^{\text{sf}}(\mathbf{c}, t) \right\|^2 - g\hat{\mathbf{e}}_y - \frac{1}{m} \nabla_{\mathbf{c}} V_{\text{conf}} - \frac{\mu}{m} \sum_{i=1}^3 \langle S_i \rangle_n \nabla_{\mathbf{c}} B_i}, \quad (315)$$

where the spin-frame field and its derivatives are evaluated at the exact defect position $\mathbf{r}_s = \mathbf{c} + \mathbf{F}\mathbf{d}_0$. If the explicit trap field is retained in $\mathbf{B}^{\text{sf}}$, V_{conf} is omitted (as stated below **Eq. 275**).

At first spin-rotation order, the translational acceleration is

$$\boxed{\ddot{\mathbf{c}}_n^{\text{sr}(1)} = \ddot{\mathbf{c}}_n^{(3)} - \frac{1}{m} \nabla_{\mathbf{c}} \Delta V_{n,\text{sr}}^{(1)}}, \quad (316)$$

while the full rotational dynamics is the canonical first-order system in **Eq. 305**. Because ω_0^s depends on $\mathbf{p}_\theta$ through **Eq. 288**, the correction $\Delta V_{n,\text{sr}}^{(1)}$ of **Eq. 301** is momentum dependent and the spin-rotation-retained rotor must therefore be integrated using the canonical system of **Eq. 305** (i.e., using the canonical variables $(\boldsymbol{\theta}, \mathbf{p}_\theta)$), rather than forced into the coordinate-only libration equation **Eq. 309**.

The following subsections form a set of successive reductions of these equations:

1. the working (full 3×3) adiabatic model uses $V_n^{(3)}$ and omits $\Delta V_{n,\text{sr}}^{(1)}$ only after the numerical audit;
2. the analytical model replaces $V_n^{(3)}$ with $V_\sigma^{(2)}$;
3. the reference model replaces $V_\sigma^{(2)}$ with $V_{m_s}^{(0)}$.

Thus, the reduced equations below do not replace the full acceleration in **Eq. 314**, but simply provide controlled analytical limits of it.

4.9.2 Centered and aligned axial model

We now recover the reference trajectory model by setting $\Delta V_{n,\text{sr}}^{(1)} = 0$ in **Eq. 304**, while retaining the externally generated spin force and torque through the branch energy $V_n^{(3)}$ contained in $\mathcal{V}_n$. Their generalized-force form follows from **Cor. 3**. The spin-rotation correction is omitted only after both the spectral condition of **Eq. 291** and the trajectory-level conditions of **Eq. 307** have been verified.

After translating the transverse equilibrium to the origin ($c_x = c_x^{eq}$ and $c_y = c_y^{eq}$), the centered, aligned and one-dimensional assumptions are

$$\mathbf{d}_0 = \mathbf{0} \quad \Bigg| \quad \mathbf{C}_s = \mathbf{1}_3 \quad \Bigg| \quad \mathbf{c}(t) = z(t)\hat{\mathbf{e}}_z \quad \Bigg| \quad \mathbf{B}^{\text{sf}}(z, t) = B_z(z, t)\hat{\mathbf{e}}_z, \quad (317)$$

where the smoothly switched axial field is

$$B_z(z, t) = B_c(t) + G(t)z \quad \Bigg| \quad G(t) = \partial_z B_z. \quad (318)$$

The final assumption describes the field on the constrained trajectory and requires the explicit transverse trap contribution either to be omitted from the spin Hamiltonian, compensated locally or replaced by the effective confinement of **Eq. 274**.

Using the zxx r.m. (**Eq. 20**) we see that an axial laboratory field still retains a transverse spin-frame projection if the rotor is tilted:

$$\mathbf{B}^s = \mathbf{F}^T B_z \hat{\mathbf{e}}_z = B_z \begin{pmatrix} \sin \beta \sin \gamma \\ \sin \beta \cos \gamma \\ \cos \beta \end{pmatrix} \quad \Bigg| \quad B_{\parallel} = B_z \cos \beta \quad \Bigg| \quad B_{\perp}^2 = B_z^2 \sin^2 \beta. \quad (319)$$

For $E=0$, the complete second-order branch potential (**Eq. 222**) becomes

$$V_{\sigma}^{(2)}(z, \beta, t) = \frac{D}{3} + \frac{\mu^2 B_z^2}{2D} \sin^2 \beta + \sigma \sqrt{\mu^2 B_z^2 \cos^2 \beta + \frac{\mu^4 B_z^4}{4D^2} \sin^4 \beta}, \quad (320)$$

which is independent of α and γ as the $E=0$ defect is axially symmetric. Thus, **Eq. 255** becomes

$$\mathcal{R} := \sqrt{\mu^2 B_z^2 + |\varepsilon|^2} = \sqrt{\mu^2 B_z^2 \cos^2 \beta + \frac{\mu^4 B_z^4}{4D^2} \sin^4 \beta}, \quad (321)$$

and, following **Eq. 256**, the derivatives required by the reduced equations (**Eqs. 325. 331**) are

$$\begin{aligned} \frac{\partial V_{\sigma}^{(2)}}{\partial B_z} &= \frac{\mu^2 B_z}{D} \sin^2 \beta + \sigma \frac{\mu^2 B_z \cos^2 \beta + \frac{\mu^4 B_z^3}{2D^2} \sin^4 \beta}{\mathcal{R}} \\ \frac{\partial V_{\sigma}^{(2)}}{\partial \beta} &= \frac{\mu^2 B_z^2}{D} \sin \beta \cos \beta + \sigma \frac{-\mu^2 B_z^2 \sin \beta \cos \beta + \frac{\mu^4 B_z^4}{2D^2} \sin^3 \beta \cos \beta}{\mathcal{R}}. \end{aligned} \quad (322)$$

We reiterate the non-degeneracy qualification stated after **Eq. 256**. These expressions apply only on a non-degenerate effective branch, for which $\mathcal{R} > 0$. At $\mathcal{R}=0$, the two effective eigenvalues are degenerate and the derivative of an individually ordered branch is not uniquely defined; the coupled spin dynamics must then be retained.

4.9.3 Spatial equation and reference limit

For the centered and axially constrained full 3×3 branch model, **Eq. 314** gives

$$\ddot{z}_n^{(3)} = -\frac{G(t)}{m} \frac{\partial V_n^{(3)}}{\partial B_z} + \frac{\chi_{\rho}}{\mu_0} B_z(z, t) G(t), \quad (323)$$

where the exact derivative is evaluated through **Thm. 7** (the Hellmann–Feynman theorem) At first spin-rotation order,

$$\ddot{z}_n^{\text{sr}(1)} = \ddot{z}_n^{(3)} - \frac{G(t)}{m} \frac{\partial \Delta V_{n, \text{sr}}^{(1)}}{\partial B_z}. \quad (324)$$

The second-order and bare-state equations below are successive reductions of **Eqs. 323, 324**.

Within the point-particle diamagnetic approximation, the effective-branch spin force $Q_{z, \sigma}^{\text{spin}} := -\partial_z V_{\sigma}^{(2)}$, defined analogously to **Eq. 250**, and the diamagnetic force $F_{\text{dia}, z}^{(0)} := -\partial_z V_{\text{dia}}^{(0)}$ (see **Eq. 265**) give

$$\ddot{z}_{\sigma} = -\frac{G(t)}{m} \frac{\partial V_{\sigma}^{(2)}}{\partial B_z} + \frac{\chi_{\rho}}{\mu_0} B_z(z, t) G(t). \quad (325)$$

This retains every term in the second-order branch energy while omitting the perturbative spin-rotation correction $\Delta V_{n,\text{sr}}^{(1)}$.

The bare-state reference equation follows by setting $E = 0$ and neglecting the second-order terms generated by virtual transverse-field coupling through $|0\rangle$ (i.e., those proportional to $\mu^2 B_\perp^2/(2D)$ and $\mu^2(B_1 + iB_2)^2/(2D)$)

$$V_{m_s}^{(0)} = \frac{D}{3} + m_s \mu B_z \cos \beta, \quad m_s = \pm 1, \quad (326)$$

which gives

$$\ddot{z}_{m_s} = -\frac{m_s \mu G(t)}{m} \cos \beta + \frac{\chi_\rho}{\mu_0} B_z(z, t) G(t). \quad (327)$$

During the homogeneous stage $G=0$, both translational force terms vanish, while the finite holding bias B_{hold} continues to determine the spin gap and the orientation-dependent energy. Since $\chi_\rho < 0$ for diamond, the diamagnetic acceleration points towards weaker field. At fixed field and orientation, the bare spin acceleration scales as m^{-1} , whereas the diamagnetic acceleration is mass independent.

4.9.4 Nonlinear reduced rotational equations

In the centered ($\mathbf{d}_0 = 0$) bare ($E = 0$) axial ($c = z\hat{e}_z$ and $\mathbf{B}^{\text{sf}} = B_z(z, t)\hat{e}_z$) model, the branch potential depends on the orientation only through β (see **Eq. 320**), so α and γ remain cyclic despite the applied field explicitly depending on time. The gyroscopic initial conditions of **Eqs. 130, 131** give

$$p_\alpha = I_\parallel \omega_0 \cos \beta_0 \quad \Bigg| \quad p_\gamma = I_\parallel \omega_0, \quad (328)$$

and, by revisiting **Eq. 122**,

$$\kappa := \frac{2R}{L} \quad \Bigg| \quad r_I := \frac{I_\parallel}{I_\perp} = \frac{6R^2}{3R^2 + L^2} = \frac{6\kappa^2}{3\kappa^2 + 4}, \quad (329)$$

the angular rates become

$$\dot{\alpha} = r_I \omega_0 \frac{\cos \beta_0 - \cos \beta}{\sin^2 \beta} \quad \Bigg| \quad \dot{\beta} = \frac{p_\beta}{I_\perp} \quad \Bigg| \quad \dot{\gamma} = \omega_0 - \dot{\alpha} \cos \beta. \quad (330)$$

In particular, $\dot{\beta}(0) = 0$ and $\dot{\gamma}(0) = \omega_0$.

For the centered branch potential $V_n(z, \beta, t)$ retained at leading scalar-potential order, the nonlinear libration equation is

$$\ddot{\beta}_n = -r_I^2 \omega_0^2 \frac{(\cos \beta_0 - \cos \beta_n)(1 - \cos \beta_0 \cos \beta_n)}{\sin^3 \beta_n} - \frac{1}{I_\perp} \frac{\partial V_n}{\partial \beta_n}. \quad (331)$$

Taking $V_n = V_n^{(3)}$ gives the full 3×3 adiabatic branch equation, while $V_n = V_n^{(2)}$ gives its complete second-order analytical reduction. Both retain the externally generated branch torque through $-\partial_\beta V_n$ but omit the momentum-dependent spin-rotation correction under **Eq. 291**.

In the bare $E=0$ reference limit, the nonlinear libration equation used in the centered reference model of [19], with the external Zeeman torque retained explicitly is

$$\ddot{\beta}_{m_s} = -r_I^2 \omega_0^2 \frac{(\cos \beta_0 - \cos \beta_{m_s})(1 - \cos \beta_0 \cos \beta_{m_s})}{\sin^3 \beta_{m_s}} + \frac{m_s \mu B_z(z, t)}{I_\perp} \sin \beta_{m_s}. \quad (332)$$

4.9.5 Small-angle dynamics and stability

Let the tilt angle libration centre β_0 be small and have a weak time-dependence given by

$$\beta_n(t) = \beta_0 + \delta\beta_n(t), \quad |\delta\beta_n| \ll \beta_0 \ll 1. \quad (333)$$

Linearizing **Eq. 331** about β_0 gives

$$\delta\ddot{\beta}_n + \Omega_n^2(t) \delta\beta_n = -\frac{1}{I_\perp} \frac{\partial V_n}{\partial \beta} \Bigg|_{\beta_0} \quad \Bigg| \quad \Omega_n^2(t) := r_I^2 \omega_0^2 + \frac{1}{I_\perp} \frac{\partial^2 V_n}{\partial \beta^2} \Bigg|_{\beta_0}. \quad (334)$$

The first derivative shifts the libration centre, whereas the second changes its local curvature. If the branch potential varies slowly on the timescale Ω_n^{-1} , the instantaneous equilibrium is

$$\delta\bar{\beta}_n(t) = -\frac{\partial_\beta V_n|_{\beta_0}}{I_\perp \Omega_n^2(t)}, \quad \Omega_n^2(t) > 0, \quad (335)$$

where the second condition is the local linear-stability requirement.

For $V_{m_s}^{(0)} = m_s \mu B_z \cos \beta + D/3$, **Eq. 334** becomes

$$\delta\ddot{\beta}_{m_s} + \left[r_I^2 \omega_0^2 - \frac{m_s \mu B_z(t)}{I_\perp} \right] \delta\beta_{m_s} = \frac{m_s \mu B_z(t)}{I_\perp} \beta_0. \quad (336)$$

For $B_z > 0$, the $m_s = +1$ branch is stable only if

$$r_I^2 \omega_0^2 > \frac{\mu B_z}{I_\perp}, \quad (337)$$

while changing the sign of B_z swaps which branch receives this destabilizing curvature. The $m_s = 0$ branch receives no magnetic curvature.

A branch-independent strong-gyroscopic regime is therefore

$$\boxed{r_I \omega_0 \gg \sqrt{\frac{\mu |B_z|}{I_\perp}}}, \quad (338)$$

so the effective gyroscopic scale depends on $r_I \omega_0$, not just ω_0 alone.

The equilibrium shift within the linearized bare model and its strong-rotation limit are

$$\delta\bar{\beta}_{m_s} = \frac{m_s \mu B_z \beta_0 / I_\perp}{r_I^2 \omega_0^2 - m_s \mu B_z / I_\perp} \simeq \frac{m_s \mu B_z}{I_\perp r_I^2 \omega_0^2} \beta_0 \quad \Rightarrow \quad \delta\bar{\beta}_+ - \delta\bar{\beta}_- \simeq \frac{2\mu B_z}{I_\perp r_I^2 \omega_0^2} \beta_0. \quad (339)$$

The spin-dependent displacement of the libration centre is therefore suppressed as ω_0^{-2} . The same expansion of **Eq. 330** gives

$$\dot{\alpha}_n \simeq r_I \omega_0 \frac{\delta\beta_n}{\beta_0} \quad \left| \quad \dot{\gamma}_n \simeq \omega_0 - r_I \omega_0 \frac{\delta\beta_n}{\beta_0} \quad \Rightarrow \quad \delta\dot{\gamma} \simeq -\delta\dot{\alpha}. \quad (340)$$

The individual γ trajectory retains the common baseline ω_0 ; only the difference between the two arms is opposite to the precession mismatch.

Finally, defining

$$B_z^{\max} := \max_{\ell, t} |B_z(z_\ell(t), t)|, \quad (341)$$

the strong-gyroscopic condition of **Eq. 338** and the spin-rotation perturbativity condition of **Eq. 291** combine into the axial operating window

$$\boxed{\frac{1}{r_I} \sqrt{\frac{\mu B_z^{\max}}{I_\perp}} \ll \omega_0 \ll \frac{\Delta_{\text{spin}}^{\min}}{\hbar}}. \quad (342)$$

The lower bound is the trajectory-wide version of the mechanical stability condition derived from **Eqs. 336, 337**: it ensures that the gyroscopic restoring curvature dominates the largest destabilizing magnetic curvature encountered by either arm.

The upper bound follows from **Eq. 291**, with $\Delta_{\text{spin}}^{\min}$ defined by **Eq. 227**: it ensures that the spin-rotation term does not appreciably shift or mix neighbouring instantaneous spin branches.

Both extrema are evaluated along the complete trajectories throughout the smooth field ramps. If we keep an uncompensated non-axial trap field, the lower bound must then be replaced by the exact branch-curvature condition obtained from $\Omega_n^2(t)$ (see **Eq. 334**), and the full spin-gap and adiabaticity conditions of **Eqs. 291, 206** must be verified.

4.9.6 Shape and mass scaling

The inertia ratio in **Eq. 329** can be specified for a long cylinder and a disk-like cylinder, respectively

$$\kappa \rightarrow 0 \quad \Rightarrow \quad r_I \rightarrow 0 \quad \left| \quad \kappa \rightarrow \infty \quad \Rightarrow \quad r_I \rightarrow 2. \quad (343)$$

Since the gyroscopic restoring scale is $r_I \omega_0$, for the same stabilization (i.e., same resistance to the externally driven magnetic torque), a long cylinder rotating about its symmetry axis requires a substantially larger ω_0 . This stability (or lack thereof) in the present libration problem not imply that torque-free rotation about the symmetry axis of a long cylinder is unstable; that axis is a principal one with an extremal moment of inertia and therefore supports stable, free rotation.

At fixed shape and density,

$$R, L \propto m^{1/3} \quad \Rightarrow \quad I_{\parallel}, I_{\perp} \propto m^{5/3}, \quad (344)$$

so the bare spin acceleration scales as m^{-1} , while the magnetic angular acceleration scales as $I_{\perp}^{-1} \propto m^{-5/3}$ at fixed field and orientation. For fixed L , the radius and aspect ratio change through **Eq. 122**.

4.9.7 Off-center defect: exact geometry and planar limit

The field is evaluated at $\mathbf{r}_s = \mathbf{c} + \mathbf{F}\mathbf{d}_0$, and the resulting force and torque are given by **Eqs. 250, 253**. Because $\mathbf{F}\mathbf{d}_0$ depends on the complete orientation, the generic model requires all three momentum equations in **Eq. 308**.

The planar construction of [19] is recovered only after imposing the laboratory y - z -plane ansatz

$$\mathbf{n}_s^{\text{sf}} = \sin \beta \hat{\mathbf{e}}_y + \cos \beta \hat{\mathbf{e}}_z \quad \Big| \quad \mathbf{d}^{\text{sf}} = d [\sin(\beta + \alpha') \hat{\mathbf{e}}_y + \cos(\beta + \alpha') \hat{\mathbf{e}}_z], \quad (345)$$

where α' is the fixed geometric angle between the spin axis and the defect-displacement vector within the y - z plane, and should not be confused with the Euler angle α .

For an axial field, the corresponding bare-state potential is

$$V_{m_s}^{\text{pl}} = m_s \mu B_z [z + d \cos(\beta + \alpha'), t] \cos \beta, \quad (346)$$

and its generalized torque is therefore

$$Q_{\beta, m_s}^{\text{pl}} := -\frac{\partial V_{m_s}^{\text{pl}}}{\partial \beta} = m_s \mu \left[B_z(\mathbf{r}_s, t) \sin \beta + d \frac{\partial B_z}{\partial z} \cos \beta \sin(\beta + \alpha') \right]. \quad (347)$$

The sign of the gradient term follows from the vector convention in **Eq. 345**; reversing the definition of α' changes it accordingly. For $\beta \ll 1$, $|dG| \ll |B_z|$ and neglected terms of order $dG\beta$,

$$Q_{\beta, m_s}^{\text{pl}} \simeq m_s \mu [B_z \beta + dG \sin \alpha']. \quad (348)$$

Thus, γ dependence is suppressed by this restricted comparison model, while the complete off-center dynamics must instead use **Eq. 146**.

Moreover, since $c_x = 0$ does not imply $r_{s,x} = 0$, a divergence-free Stern–Gerlach field gives

$$B_x(\mathbf{r}_s, t) = -G(t)r_{s,x}, \quad (349)$$

meaning that off-center defects sample the transverse field even for axial c.o.m. trajectories.

4.9.8 Nonzero transverse splitting

For the centered ($\mathbf{d}_0 = \mathbf{0}$), aligned ($\mathbf{C}_s = \mathbf{1}_3$) and axially constrained ($\mathbf{c} = z\hat{\mathbf{e}}_z$ and $\mathbf{B}^{\text{sf}} = B_z(z, t)\hat{\mathbf{e}}_z$) model, **Eq. 319** gives

$$B_1^2 - B_2^2 = -B_z^2 \sin^2 \beta \cos(2\gamma). \quad (350)$$

Substituting the centered axial field components of **Eq. 319** into the general second-order branch energy of **Eqs. 222, 223** gives

$$V_{\sigma}^{(2)} = \frac{D}{3} + \frac{\mu^2 B_z^2}{2D} \sin^2 \beta + \sigma \left[\mu^2 B_z^2 \cos^2 \beta + E^2 + \frac{\mu^4 B_z^4}{4D^2} \sin^4 \beta - \frac{\mu^2 E B_z^2}{D} \sin^2 \beta \cos(2\gamma) \right]^{1/2}, \quad (351)$$

where the sign of the final term follows from the adopted xxz convention and transverse defect axes. A constant rotation of those axes shifts the origin of γ , but does not restore the continuous symmetry.

Consequently, for $E \neq 0$ and $B_\perp \neq 0$, the complete potential depends on γ and its canonical momentum is (generally) not conserved,

$$\dot{p}_\gamma = -\frac{\partial V_\sigma^{(2)}}{\partial \gamma}, \quad (352)$$

which is despite the derivative possibly vanishing at isolated orientations. The laboratory axial symmetry still leaves α cyclic and therefore conserves p_α .

By contrast, the E -split reduced branch potential

$$V_\sigma^{(E)} = \frac{D}{3} + \sigma \sqrt{\mu^2 B_z^2 \cos^2 \beta + E^2} \quad (353)$$

is independent of γ , since we have already discarded the $B_\perp^2/D$ coupling (through which E resolves the transverse-field direction). The complete and reduced $E \neq 0$ models therefore possess different rotational symmetries and must be treated as distinct approximations.

We have obtained the three nested systems required before the numerical analysis: the full 3×3 adiabatic branch model, the complete second-order analytical model and the centered bare-state reference reduction. Their numerical implementation will determine the actual size of every approximation made above.

5 Numerical calibration and classical interferometric dynamics

The previous section established the complete semiclassical Hamiltonian, the exact and reduced spin branches, the corresponding forces and torques, and the conditions under which the centered one-dimensional model is recovered. We now implement the model numerically, fix the static trap, reproduce the centre-of-mass trajectory of the reduced reference model, replace its piecewise controls by smooth ones and then determine what survives in the complete three-dimensional dynamics.

The calculation follows the physical order of the experiment. The magneto-gravitational trap is first reconstructed and its exact spin spectrum is then evaluated, because a mechanically stable trap need not provide a usable spin environment. Only after that decision is made do we retime the Stern–Gerlach sequence, propagate the two conditional trajectories and audit the reductions introduced in **Sec. 4**.

All translational and rotational variables remain classical. Accordingly, the final statements concern conditional trajectories, stability and classical phase-space closure.

5.1 Numerical construction and declared operating point

5.1.1 Nominal physical inputs

Universal and material constants, nominal design choices and preparation conditions are fixed before the calculation. The radius and moments of inertia are derived from them. The multipole coefficients are fitted by the trap inverse problem. The final switching times are optimized outputs.

The fixed and derived physical inputs used throughout the nominal calculation can be found in **Tabs. 8–10** from **Appendix B.1**. The values are stated explicitly so that the numerical results do not depend on hidden defaults.

The numerical resolution, quadrature orders, finite-difference steps and Monte Carlo seed are implementation settings rather than physical parameters. They are collected in **Tabs. 11, 12** from **Appendix B.2**, together with the convergence checks that justify them. The uncertainty widths used only in the robustness ensemble are declared separately in **Eqs. 389, 390**. Together, these tables and equations exhaust the editable input cell of the notebook; fitted trap coefficients, continued event times and closure quantities are outputs rather than preset values.

5.1.2 Symbolic model and propagated states

The numerical notebook mirrors the hierarchy of **Sec. 4.1.2**:

$$\mathcal{M}_3 : V_n^{(3)} \quad \Big| \quad \mathcal{M}_2 : V_\sigma^{(2)} \quad \Big| \quad \mathcal{M}_E : V_\sigma^{(E)} \quad \Big| \quad \mathcal{M}_{\text{ref}} : V_{m_s}^{(0)}, \quad (354)$$

where $\mathcal{M}_3$ (3×3) is the numerical authority, the complete 2×2 , E -reduced and bare-state models are used to measure approximation errors, while finite-volume diamagnetism and first-order spin–rotation coupling are introduced one at a time in dedicated checks.

The main numerical propagation represents the rotor orientation by a unit quaternion rather than by the Euler angles directly. This is a numerical choice, not a change of physical model. The zxz Euler coordinates derived in **Sec. 3** become singular when $\beta \rightarrow 0$, despite the physical orientation and angular velocity remaining well defined. Since the nominal rotor is initialized close to this configuration, directly propagating (α, β, γ) would introduce unnecessarily ill-conditioned factors proportional to $1/\sin \beta$.

The propagated state is therefore

$$\mathbf{y}_Q = \left(\mathbf{c}, \mathbf{p}_c, \mathbf{q}, \mathbf{L}^{\text{bf}} \right)^T \quad \Big| \quad \mathbf{q} = (q_0, q_1, q_2, q_3)^T \quad \Big| \quad \mathbf{q}^T \mathbf{q} = 1, \quad (355)$$

where $\mathbf{q}$ determines the same body-to-laboratory production r.m. $\mathbf{F} = \mathbf{F}(\mathbf{q})$ used throughout **Sec. 4**. Its

evolution is

$$\dot{\mathbf{q}} = \frac{1}{2}\mathbf{q} \otimes (0, \boldsymbol{\omega}^{\text{bf}}) \quad \left| \quad \dot{\mathbf{L}}^{\text{bf}} = \boldsymbol{\tau}^{\text{bf}} - \boldsymbol{\omega}^{\text{bf}} \times \mathbf{L}^{\text{bf}}, \right. \quad (356)$$

with $\boldsymbol{\omega}^{\text{bf}} = \mathbf{I}_0^{-1}\mathbf{L}^{\text{bf}}$ at the leading mechanical order retained here. Writing the scalar-first quaternion as $\mathbf{q} = (q_0, \mathbf{q}_v)^T$, the product appearing above is

$$\mathbf{q} \otimes (0, \boldsymbol{\omega}^{\text{bf}}) = \left(-\mathbf{q}_v \cdot \boldsymbol{\omega}^{\text{bf}}, q_0 \boldsymbol{\omega}^{\text{bf}} + \mathbf{q}_v \times \boldsymbol{\omega}^{\text{bf}} \right)^T \quad \left| \quad \boldsymbol{\omega}^{\text{bf}} = \mathbf{I}_0^{-1}\mathbf{L}^{\text{bf}}. \right. \quad (357)$$

The quaternion is renormalized after each accepted RK4 step to remove accumulated floating-point drift.

Euler angles remain useful as they have a direct physical interpretation and are therefore reconstructed from $\mathbf{F}(\mathbf{q})$ for plotting. In addition, selected trajectories are independently propagated using the canonical xxz Hamilton equations to compare the two implementations through the physical r.m., angular momentum, c.o.m. trajectory and final closure, rather than through the raw Euler coordinates alone:

$$\|\mathbf{F}_Q - \mathbf{F}_E\|_F \ll 1 \quad \left| \quad \|\mathbf{L}_Q^{\text{sf}} - \mathbf{L}_E^{\text{sf}}\| \ll L_{\text{scale}} \quad \left| \quad \mathbf{F}_Q^T \mathbf{F}_Q \simeq \mathbf{1}_3 \quad \left| \quad \det \mathbf{F}_Q \simeq 1. \right. \right. \quad (358)$$

Thus, the quaternion implementation supplies the nonsingular main trajectory, while the Euler implementation provides an independent verification and the angles used in the physical interpretation.

5.1.3 Branch tracking, integration and events

At every force evaluation, including the intermediate Runge–Kutta stages, the field is evaluated at the actual defect position $\mathbf{r}_s = \mathbf{c} + \mathbf{F}\mathbf{d}_0$ and transformed according to **Eq. 150**. The exact Hamiltonian of **Eq. 197** is diagonalized and the new basis is assigned by eigenvector continuity. If the squared overlap between the previously assigned eigenvector u_i^{old} and the newly computed eigenvector u_j^{new} is

$$O_{ij} := \left| \left\langle u_i^{\text{old}} \middle| u_j^{\text{new}} \right\rangle \right|^2, \quad (359)$$

then the accepted permutation (i.e., the one which maximizes the total overlap between the two bases) is

$$\pi_* := \operatorname{argmax}_{\pi \in S_3} \sum_{i=1}^3 O_{i, \pi(i)}. \quad (360)$$

The assigned eigenvectors are phase-aligned before the next step: we multiply the new one by a phase factor so its overlap with the previous one is real and positive, preventing artificial sign/phase jumps. Eigenvalue ordering alone is not used, since it can exchange the physical branches near an avoided crossing¹².

For a step h , the classical fourth-order Runge–Kutta update is

$$\begin{aligned} \mathbf{k}_1 &= \mathbf{f}(t_j, \mathbf{y}_j) & \left| \quad \mathbf{k}_2 &= \mathbf{f}\left(t_j + \frac{h}{2}, \mathbf{y}_j + \frac{h}{2}\mathbf{k}_1\right) \\ \mathbf{k}_3 &= \mathbf{f}\left(t_j + \frac{h}{2}, \mathbf{y}_j + \frac{h}{2}\mathbf{k}_2\right) & \left| \quad \mathbf{k}_4 &= \mathbf{f}(t_j + h, \mathbf{y}_j + h\mathbf{k}_3), \end{aligned} \quad (361)$$

$$\mathbf{y}_{j+1} = \mathbf{y}_j + \frac{h}{6}(\mathbf{k}_1 + 2\mathbf{k}_2 + 2\mathbf{k}_3 + \mathbf{k}_4). \quad (362)$$

All switching centres and the microwave event lie exactly on the integration grid. At $t = t_{\text{flip}}$, the two tracked branch labels are exchanged while the mechanical coordinates and momenta remain continuous. The force and torque at every stage are obtained from the same branch energy through **Eq. 250**.

The nominal reference calculation (reduced one-dimensional reference model) uses $h = 20 \mu\text{s}$, while the nominal exact calculation (3×3 model propagated with quaternions) is also repeated with $h = 10 \mu\text{s}$ and $5 \mu\text{s}$ to verify convergence. The high-frequency design scans use a timestep that decreases with the imposed spin rate, i.e., to resolve the increasingly rapid rotational motion: a step can resolve the slow c.o.m. motion while still aliasing the 10 kHz rotational dynamics.

¹²When two energy levels approach, their eigenvectors can exchange character. Sorting only by energy keeps calling one "lowest" and the other "middle," despite the followed physical state having moved onto the other energy branch.

5.1.4 Verification

The principal verification results are collected in **Tab. 1**. The symbolic and compiled forms agree to numerical precision, the magnetic fields satisfy their local Maxwell constraints and the exact forces and torques agree with independent finite differences of the branch energy.

Table 1: Numerical verification summary. Relative errors are stated explicitly; dimensional residuals are reported with their units.

Check	Largest discrepancy	Result
SymPy versus numerical rotation, fields and spin Hamiltonian	1.8×10^{-12}	pass
$\nabla \cdot \mathbf{B}$ and $\nabla \times \mathbf{B}$	1.9×10^{-13} T/m and 0	pass
Hellmann–Feynman translational force	9.5×10^{-11}	pass
Hellmann–Feynman physical torque	1.8×10^{-9}	pass
Trap frequencies: Hessian versus RK4	1.77×10^{-3}	pass
RK4 error ratio $E_h/E_{h/2}$	15.14	fourth order
Minimum exact branch overlap across the tested steps	> 0.9999	pass
Maximum driven work–energy residual	2.63×10^4 Hz	pass

The ratio $E_h/E_{h/2} = 15.14$, where $E_h := \|\mathbf{y}_h(t_f) - \mathbf{y}_{h/2}(t_f)\|$ and $E_{h/2} := \|\mathbf{y}_{h/2}(t_f) - \mathbf{y}_{h/4}(t_f)\|$, is close to the fourth-order expectation $2^4 = 16$, confirming the intended RK4 error scaling.

The last two rows of **Tab. 1** test different aspects of the calculation. The branch overlap > 0.9999 confirms continuous numerical tracking of the same instantaneous eigenbranch, whereas physical adiabaticity is checked separately through $\mathcal{A}_{mn}$. The work–energy residual of 2.63×10^4 Hz checks the consistency of the driven dynamics; it does not test interferometric closure.

The RK4 ratio in the sixth row confirms fourth-order convergence, but the rotational variables provide the stricter practical timestep test. Reducing h from $20 \mu\text{s}$ to $5 \mu\text{s}$ changes the maximum separation by only 0.017 nm , while the invariant orientation mismatch changes by 0.063 mrad . The finer step is therefore used for the quoted final rotational residuals, with the detailed refinement results collected in **Appendix B**.

5.2 Static trap calibration and confinement model

5.2.1 Inverse reconstruction and mechanical stability

Following the multipole reconstruction of [10], the static field is fixed before the Stern–Gerlach sequence. With the specific potential $u_{\text{tr}}^{(0)} = V_{\text{tr}}^{(0)}/m$ from **Eq. 267**, the unknown vector is

$$\boldsymbol{\xi}_{\text{tr}} := (a_2, a_3, a_4, y_{\text{eq}})^T \quad (363)$$

and is determined from the equilibrium condition and the three target frequencies of **Eq. 268**:

$$\partial_y u_{\text{tr}}^{(0)} = 0 \quad \Big| \quad \partial_x^2 u_{\text{tr}}^{(0)} = \omega_x^2 \quad \Big| \quad \partial_y^2 u_{\text{tr}}^{(0)} = \omega_y^2 \quad \Big| \quad \partial_z^2 u_{\text{tr}}^{(0)} = \omega_z^2, \quad (364)$$

all evaluated at $(0, y_{\text{eq}}, 0)^T$. A scaled multi-start solve returns four mathematical roots. Two lie at $y_{\text{eq}} \simeq -230 \mu\text{m}$, well outside the local region set by $y_0 = 75 \mu\text{m}$, while the remaining two are related by the overall field-sign symmetry. We adopt the local member consistent with the sign convention of **Sec. 4**.

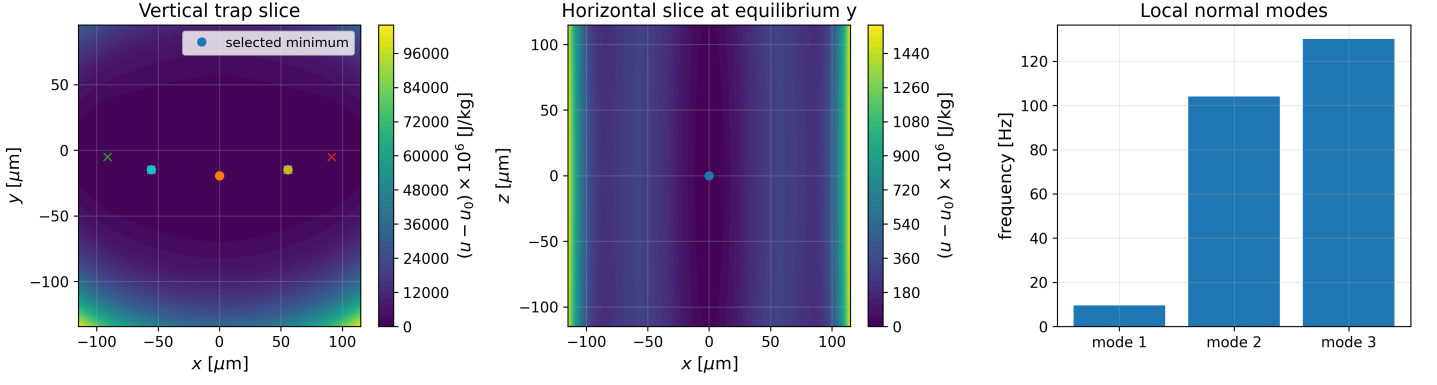


Figure 4: Mechanical reconstruction of the static multipole trap. The left panel shows the vertical specific-potential slice relative to the selected minimum together with the stationary points found in the search region. The centre panel shows the horizontal slice at $y = y_{\text{eq}}$, and the right panel gives the three local normal-mode frequencies.

Table 2: Selected local multipole-trap solution. The fitted coefficients and equilibrium are separated from the derived mechanical diagnostics to avoid compressing the rows.

a_2 [T]	a_3 [T]	a_4 [T]	y_{eq} [μm]	$ B_{\text{eq}} $ [T]
-1.28031	0.0180607	0.743612	-19.3679	0.188442

f_x [Hz]	f_y [Hz]	f_z [Hz]	$\lambda_{\text{min}}(\Omega_{\text{tr}}^2)$ [s^{-2}]	Δu_{esc} [J/kg]
104	130	9.6	3.63833×10^3	8.0549×10^{-5}

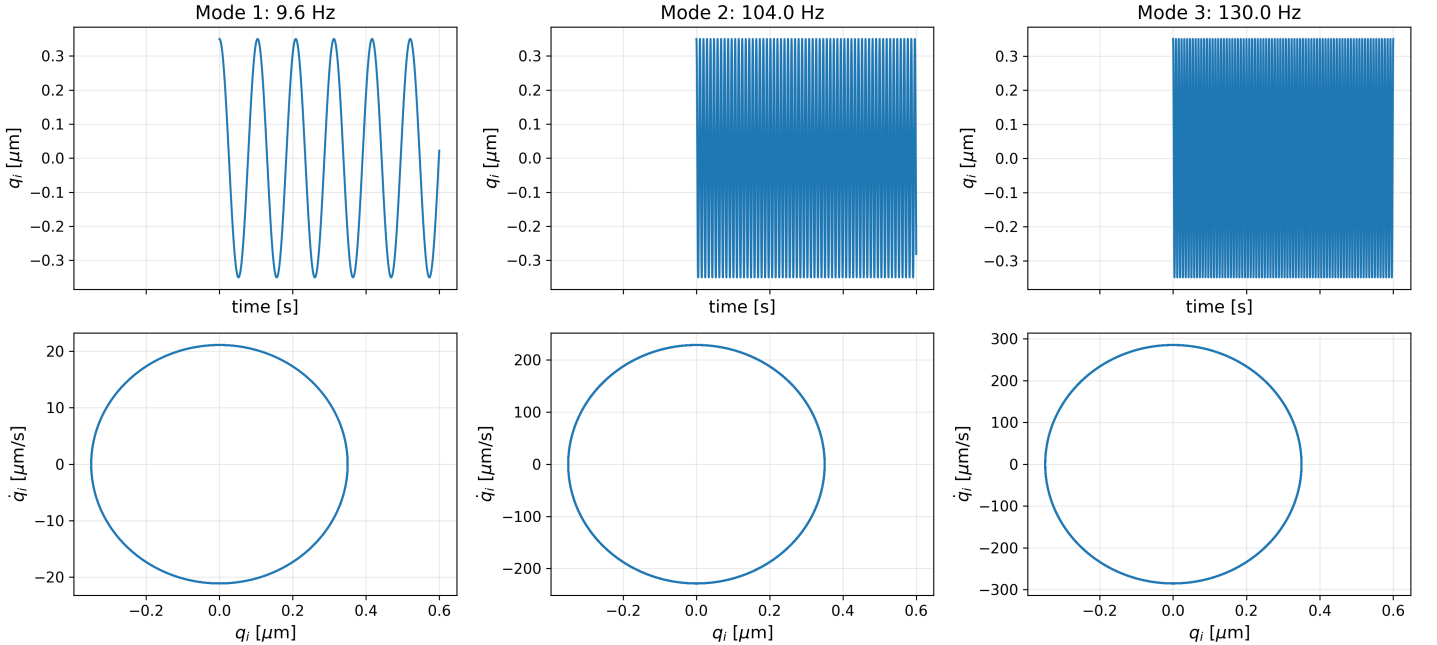


Figure 5: Small-amplitude validation of the three local trap modes. The upper row shows the normal coordinates at the Hessian frequencies, while the lower row shows the corresponding phase-space trajectories.

The local Hessian is positive, and the frequencies recovered from the trap-only RK4 trajectories agree with the Hessian values 104 Hz, 130 Hz and 9.6 Hz to within a maximum relative discrepancy of 0.18% (see **Tab. 1**). The corresponding small-amplitude trajectories remain single-frequency and closed, with no

visible intermode transfer or secular distortion; the numerical validation is shown in **Fig. 24**. The lowest saddle of the truncated three-mode model lies near $(\pm 91.28, -5.15, 0) \mu\text{m}$, giving $\Delta V_{\text{esc}} = 8.05 \times 10^{-22} \text{ J}$ for the nominal particle, or 58 K in temperature units. This is vastly larger than the illustrative microkelvin preparation scale, so *thermally activated escape is negligible within the model*. At the illustrative preparation temperature $T = 1 \mu\text{K}$, the weak 9.6 Hz mode dominates the equilibrium position width, while the transverse modes remain much more tightly localized; the corresponding ensemble is shown in **Fig. 25**. This ensemble characterizes the reconstructed magnetic trap only and is not the distribution propagated through the final guide, whose axial frequency is zero.

Two qualifications are important. First, the saddle lies at a transverse coordinate only slightly larger than y_0 ; the quoted barrier is therefore a global estimate of the truncated multipole field, not a calibrated apparatus depth: the local equilibrium and Hessian are the more reliable outputs. Second, the 9.6 Hz weak mode has a period of only 0.104 s. If this trap remained active during a 1.3 s interferometer, the nominal splitting axis would execute about thirteen oscillation periods rather than undergo a free middle stage.

The inverse system determines a local multipole solution which reproduces the Hsu equilibrium and trap frequencies. It does not search over arbitrary magnet geometries or optimize the coefficients for maximum stability. Among the mathematical roots of the stated inverse problem, the retained one is selected because it lies in the intended local region and follows the adopted field-sign convention.

The trap-only trajectories remain single-frequency and their phase-space curves remain closed at the sampled sub-micrometre amplitude. No visible intermode transfer or secular distortion occurs, confirming that the Hessian eigenvectors provide an accurate local dynamical basis in the preparation region.

5.2.2 Exact spin gate

Mechanical stability does not imply spin compatibility. At the defect, the equilibrium field is predominantly transverse to the nominal spin axis and has magnitude 0.1885 T. Consequently,

$$\frac{\mu |\mathbf{B}_{\text{eq}}|}{D} = 1.839, \quad (365)$$

so the condition of **Eq. 220** is not satisfied. The complete 3×3 spectrum must therefore be used to decide whether the trap can remain active.

The left panel of **Fig. 6** shows that all three exact branches remain well separated in energy, with a smallest nearest gap of 4.05 GHz. This large gap suppresses unintended transitions during a slow preparation. The middle panel, however, shows that the corresponding eigenstates are not the bare $|+1\rangle$, $|0\rangle$ and $|-1\rangle$ states required by the intended Stern–Gerlach interpretation. The transverse trap field strongly mixes the bare basis, so the numerical $+$ and $-$ symbols denote tracked eigenbranches rather than pure spin projections. One force-carrying mixed branch has bare-state probabilities (0.316, 0.368, 0.316), while the dark antisymmetric branch is approximately $(|+1\rangle - |-1\rangle)/\sqrt{2}$. The numerical $+$ and $-$ registry labels may interchange in this overlap tie without changing the physical branch content.

This pattern follows directly from a predominantly transverse Zeeman coupling: S_1 couples $|0\rangle$ to the symmetric $|+1\rangle + |-1\rangle$ combination, while the antisymmetric combination is dark. The dark branch consequently carries almost no spin force or torque, whereas the other two branches are strongly mixed and force-carrying. Thus, an exact eigenstate can be well isolated spectrally and still be unusable as the bare spin label assumed by the intended interferometer.

Thus, the trap produces spectrally stable spin eigenstates, but not the nearly pure $|\pm 1\rangle$ states required for the intended interferometric protocol. This rules out retaining the uncompensated multipole field during the Stern–Gerlach sequence, although the trap remains useful as a mechanical preparation and calibration benchmark.

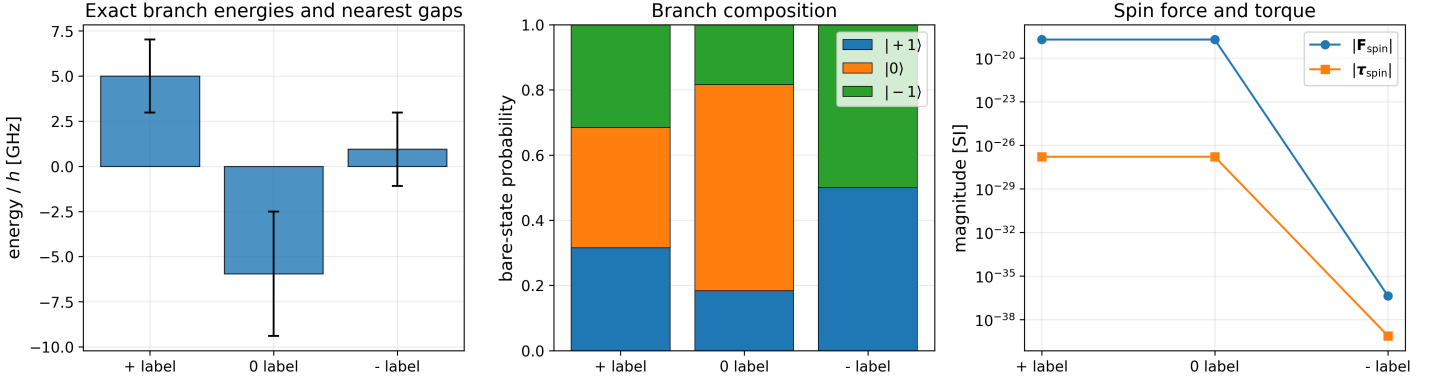


Figure 6: Exact trap–spin compatibility test at the selected equilibrium. The left panel shows the exact branch energies as bars measured from zero, while the black vertical intervals are centred on each branch energy and have total length equal to the corresponding nearest spectral gap; they therefore represent level separation rather than uncertainty. The centre panel resolves each eigenstate into its bare-state probabilities in the basis $\{|+1\rangle, |0\rangle, |-1\rangle\}$. The right panel shows the corresponding spin-force and spin-torque magnitudes on a logarithmic scale.

In the principal interferometric model, the particle is instead assumed to be transferred to, or retained by, an effective transverse guide with frequencies (104, 130, 0) Hz. Physically, this requires the multipole field to be switched off, compensated at the defect or replaced by a separate SG-compatible confinement mechanism; it cannot simply be omitted from the spin Hamiltonian while remaining present. The Stern–Gerlach field then defines the relevant spin eigenstates and restores the intended branches interpretation.

The imposed axial rotation instead gyroscopically suppresses libration and rotational mismatch. It does not cancel the transverse trap field or restore pure $|\pm 1\rangle$ states. Keeping the explicit trap would therefore require field compensation or reorientation and a redesigned full 3×3 protocol.

The two force-carrying branches experience $|\mathbf{F}_{\text{spin}}| = 1.89 \times 10^{-19}$ N and $|\boldsymbol{\tau}_{\text{spin}}| = 1.63 \times 10^{-27}$ N m. This force is about 2.3×10^2 times the nominal Stern–Gerlach scale $\mu\eta = 8.35 \times 10^{-22}$ N. Their vertical equilibria shift by approximately ± 28.3 nm. Thus, the trap would prepare branch-dependent mechanical equilibria before the intended splitting begins.

The physical failure is consequently stronger than the failure of the 2×2 expansion. Preparing bare $|\pm 1\rangle$ states would not prepare stationary internal states of the trap, while preparing the exact mixed branches would already imprint large branch-dependent forces, torques and equilibrium displacements.

The explicit trap is therefore rejected for the principal interferometric sequence, despite being an acceptable local mechanical preparation benchmark.

The working interferometer instead uses the gravity-compensated confinement of **Eq. 274**,

$$\frac{\boldsymbol{\omega}_{\text{conf}}}{2\pi} = (104, 130, 0) \text{ Hz} \quad \Bigg| \quad \mathbf{c}_0 = (0, 14.699 \mu\text{m}, 0)^T, \quad (366)$$

where the displaced harmonic centre makes $\mathbf{c} = \mathbf{0}$ an equilibrium after gravity is included. The magnetic trap is then set to zero in the spin Hamiltonian (i.e., a different physical configuration, and not a partial removal of the field from one term of the Hamiltonian). Its experimental realization is not specified by the present calculation and remains an apparatus requirement.

5.3 Reference benchmark and smooth protocol design

5.3.1 Piecewise reference trajectory

The centered, aligned model of **Secs. 4.9.3, 4.9.4** is first propagated with the parameters of [19]:

$$B_0 = 10^{-2} \text{ T} \quad \Bigg| \quad B_1 = 10^{-4} \text{ T} \quad \Bigg| \quad \eta = 45 \text{ T/m}, \quad (367)$$

$$t_{12} = 0.482 \text{ s} \quad \Bigg| \quad t_{23} = 0.514 \text{ s} \quad \Bigg| \quad t_{\text{flip}} = 0.8022 \text{ s} \quad \Bigg| \quad t_f = 1.320 \text{ s}. \quad (368)$$

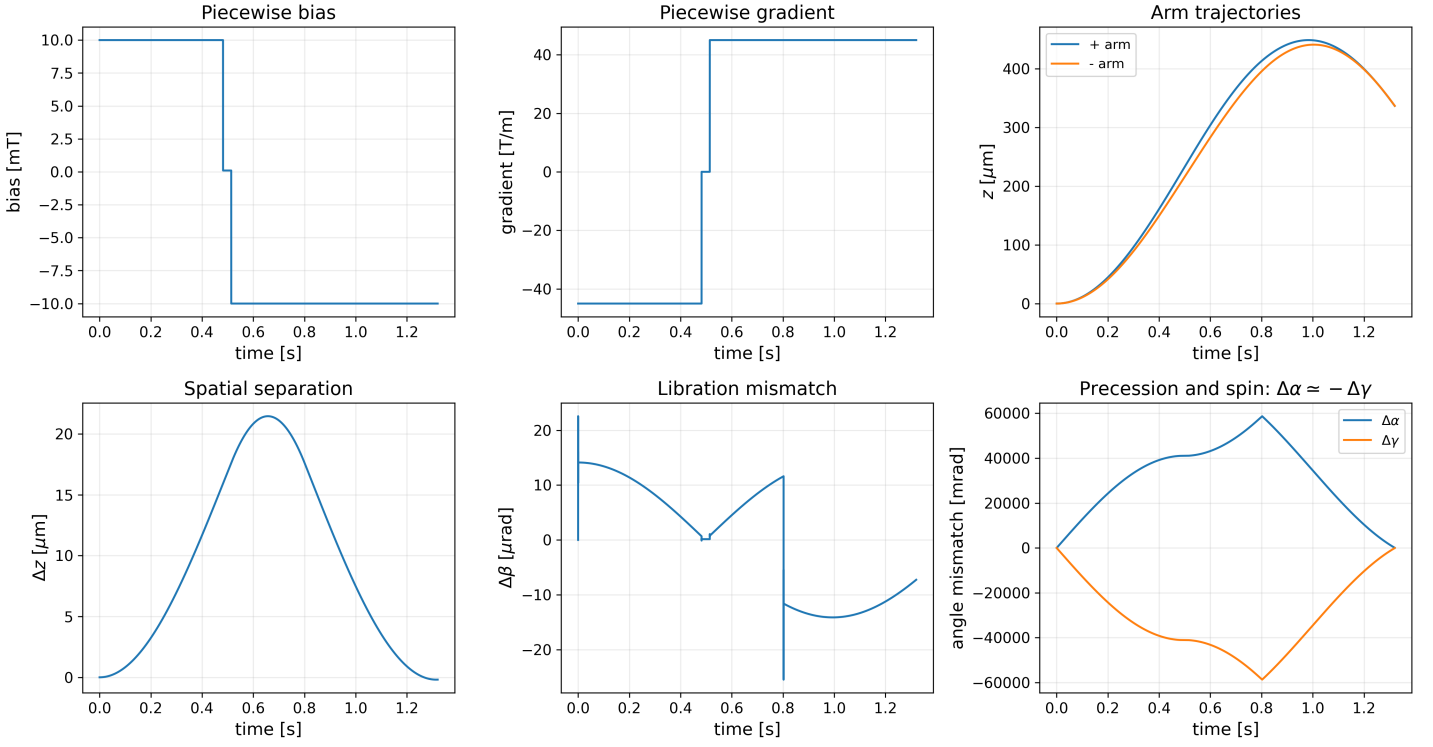


Figure 7: Reproduction of the piecewise reference model. The top row shows the bias, gradient and two arm trajectories. The bottom row shows the spatial separation, libration mismatch and the approximately antisymmetric precession and spin mismatches $\Delta\alpha \simeq -\Delta\gamma$.

For $m = 10^{-17}$ kg, the maximum separation is $21.455 \mu\text{m}$.¹³ At the quoted closing time, the residuals are $\Delta z_f = -189.8 \text{ nm}$ and $\Delta v_{z,f} = 0.977 \mu\text{m/s}$. The final libration mismatch is $-7.27 \mu\text{rad}$ and the invariant orientation mismatch is 0.530 mrad . The nearly 0.45 mm common excursion is already present in the reduced benchmark. It is therefore inherited from the diamagnetic orbit and is not generated by the exact spin treatment, the off-centre defect or the three-dimensional propagation.

The benchmark therefore reproduces the separation scale, the rapid gyroscopic oscillations and the approximate $\Delta\alpha \simeq -\Delta\gamma$ structure. It does not exactly reproduce the stated closure at the finite-precision times quoted in the paper. Since the reference calculation is reduced and the event times are published only to a few significant figures, these values are used as a timing seed rather than treated as an exact numerical target.

The apparent jumps in $\Delta\beta$ at the beginning of the sequence and at $t_{\text{flip}} = 0.8022 \text{ s}$ are not discontinuities of the rotor orientation. The piecewise field turn-on and the ideal instantaneous spin inversion abruptly change the magnetic torque and the corresponding libration equilibrium, so $\dot{\beta}$ changes discontinuously while β and $\dot{\beta}$ remain continuous. The resulting high-frequency libration oscillations are not resolved on the full-duration plotting scale and are visually compressed into nearly vertical features. They should therefore be interpreted as rapidly excited transients rather than as numerical instabilities or instantaneous angular displacements.

5.3.2 Smooth switching and timing continuation

Next, the piecewise field is replaced by the smooth windows of **Eqs. 168, 171** with $\delta_{\text{sw}} = 10^3 \text{ s}^{-1}$.

Keeping the original event times with the smooth switching yields $\Delta z_f = -155.4 \text{ nm}$ and $\Delta v_{z,f} = 0.832 \mu\text{m/s}$, compared with -189.8 nm and $0.977 \mu\text{m/s}$ for the piecewise field. The maximum separation changes only slightly, from $21.455 \mu\text{m}$ to $21.480 \mu\text{m}$, whereas the final closure residuals remain appreciable.

The four stage durations are first adjusted in the reference model and then continued into the complete 2×2 , three-dimensional rigid-body model. Only spatial closure is imposed because the available timing

¹³This value agrees with the $m = 10^{-17}$ kg curve in Fig. 6(c) of [19]. The nearby prose associates approximately $40 \mu\text{m}$ with the same mass, but that scale corresponds to the lighter 5×10^{-18} kg curve.

parameters are chosen primarily to recombine the two c.o.m. trajectories in position and momentum. Including rotational closure in the same objective would require arbitrary relative weights between quantities with different units and could improve the angles only by degrading the spatial recombination.

The rotational mismatch is therefore left unconstrained and reported afterwards as an independent physical limitation of the protocol. The final controls used by the exact calculation are

$$t_{12} = 0.483085062 \text{ s} \quad | \quad t_{23} = 0.515090178 \text{ s} \quad | \quad t_{\text{flip}} = 0.801531322 \text{ s}. \quad (369)$$

$$t_{\text{off}} = 1.315634715 \text{ s} \quad | \quad t_f = 1.323634715 \text{ s}. \quad (370)$$

The final time is set 8 ms after turn-off, allowing the smooth tail to decay before closure is checked. The additional digits are retained because the six-decimal values do not reproduce the quoted sub-nanometre closure.

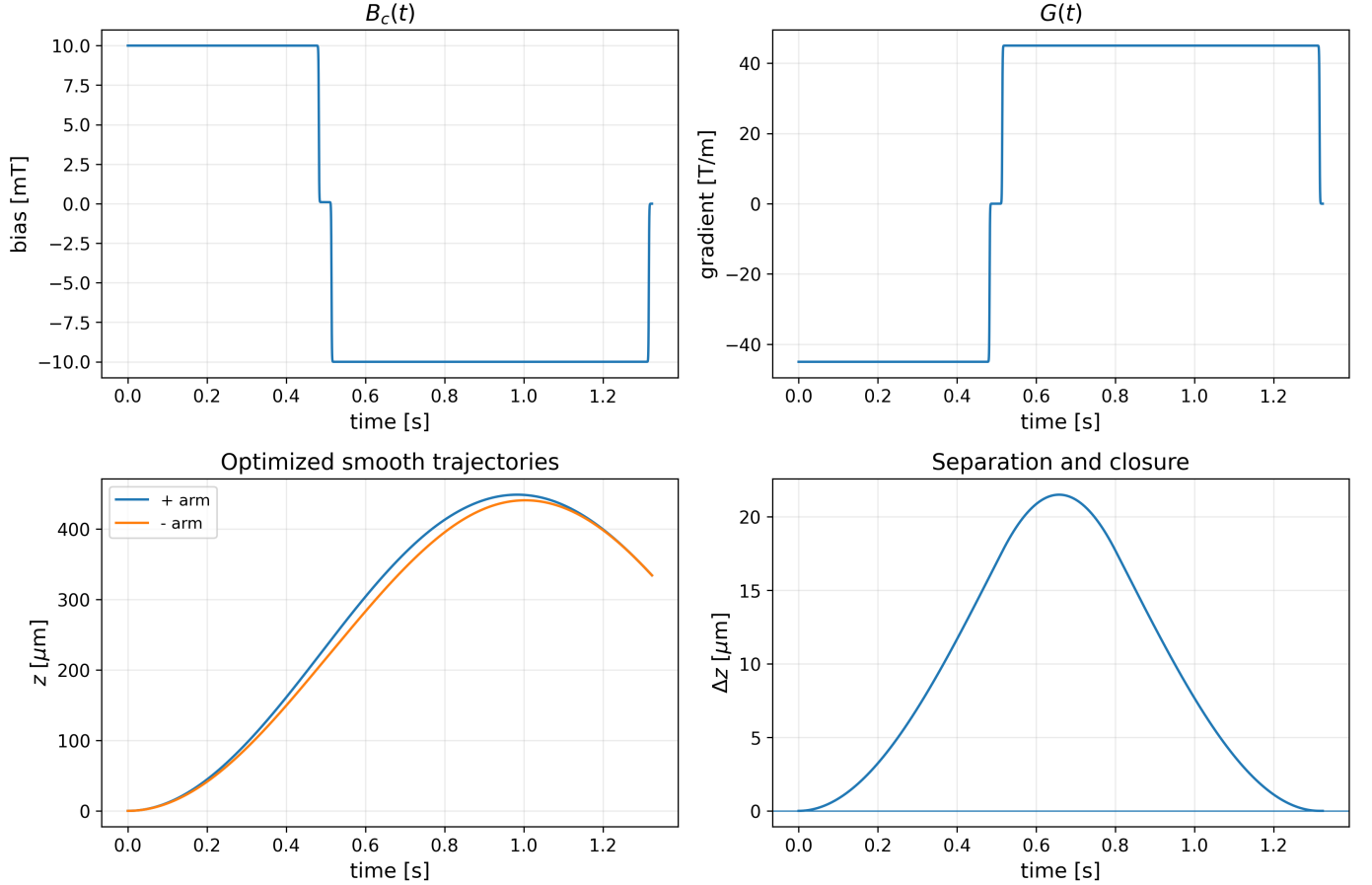


Figure 8: Retimed smooth reference protocol. The upper panels show the continuous bias and gradient; the lower panels show the two trajectories and their separation. This calculation fixes the timing seed used for continuation into the complete models.

In the retimed reference model, the smooth trajectory reaches $21.505 \mu\text{m}$ and closes its axial position and velocity to numerical precision. However, the smooth change from the positive holding bias to the negative third-stage bias necessarily passes through $B_3 \simeq 0$. For $E=0$, the $|\pm 1\rangle$ branches then become degenerate, so a unique adiabatic branch is not defined at the crossing. In the nominal exact model, $E/h=10 \text{ MHz}$ opens an avoided crossing of size $2E/h=20 \text{ MHz}$ and restores spectral isolation.

The transverse strain term is therefore essential to the smooth protocol rather than a small cosmetic correction. It keeps the exact branch spectrally isolated while its bare $|\pm 1\rangle$ composition becomes temporarily mixed near the avoided crossing. The tracked object is thus the continuous instantaneous eigenstate, or equivalently its projector, rather than a fixed bare-spin label. Both the force assignment and the microwave branch exchange must follow this instantaneous branch structure.

The exact smooth sequence is therefore not merely a differentiable version of the paper’s bare-state protocol. In the nominal trajectory, the near-zero-field passage does not itself perform the later branch

exchange: the followed branch recovers its original dominant bare-state character before t_{flip} , and the explicit microwave event then transfers the trajectory to the opposite exact branch.

At the converged $5\text{ }\mu\text{s}$ step, the complete 2×2 model with the optimized times gives

$$\begin{array}{l|l} \Delta z_{\text{max}} = 21.4358\text{ }\mu\text{m} & \|\Delta \mathbf{c}_f\| = 0.681\text{ nm} \\ \frac{|\Delta p_{z,f}|}{m} = 0.00534\text{ }\mu\text{m/s} & \Delta \Theta_f = 2.089\text{ mrad.} \end{array} \quad (371)$$

Here, Δz_{max} is the largest arm separation, $\|\Delta \mathbf{c}_f\|$ is the final c.o.m. mismatch, $|\Delta p_{z,f}|/m$ is the final axial velocity mismatch and $\Delta \Theta_f$ is the invariant final orientation mismatch. The first three quantities show that the spatial loop is well closed, whereas the nonzero $\Delta \Theta_f$ records the remaining rotational Humpty–Dumpty error. These optimized times are then passed unchanged to the exact 3×3 model, so any later difference measures the spin-model correction rather than a separate retuning.

A second basis issue appears when the field is finally turned off. As $\mathbf{B} \rightarrow \mathbf{0}$ with $E \neq 0$, the tracked branches approach strain eigenstates containing nearly equal $|+1\rangle$ and $|-1\rangle$ components. A practical readout therefore requires a final bias field or pulse sequence which maps this strain basis onto the desired measurement basis. This mapping does not change the classical forces already calculated, but it is required before a later spin-contrast result can be interpreted experimentally.

5.3.3 Microwave event

The mechanical propagation treats the microwave operation as an ideal instantaneous exchange of the two exact branches. Its spectral admissibility is then checked at the positions and orientations of both arms. The quantities in **Tab. 3** assume an effective target-transition Rabi frequency $\Omega_R/(2\pi) = 1\text{ MHz}$.

Table 3: Exact arm-dependent microwave diagnostic at $t = t_{\text{flip}}$. The required field uses $f_R = (\mu/h)B_{\text{mw}}|\langle \chi_+ | S_1 | \chi_- \rangle|$, with B_{mw} defined as the resonant co-rotating field amplitude. A linearly polarized peak amplitude would be twice as large under the usual rotating-wave convention.

Arm	$\omega_{+-}/(2\pi)$ [MHz]	$ \langle \chi_+ S_1 \chi_- \rangle $	B_{mw} for 1 MHz [mT]	nearest gap to third branch [GHz]
+	482.178	9.40×10^{-4}	38.0	2.629
–	437.890	6.80×10^{-4}	52.5	2.651

Tab. 3 exposes two distinct obstacles to a direct branch swap. First, the two target frequencies differ by 44.29 MHz, so one narrowband tone cannot be resonant with both arms. Second, the direct S_1 matrix elements are below 10^{-3} because S_1 connects states differing by one unit of m_s , whereas the target branches lie mainly in the $|\pm 1\rangle$ manifold and couple only through their small $|0\rangle$ admixtures. An effective 1 MHz direct transition would therefore require microwave fields of tens of millitesla.

The large separation from the third branch does not remove this weak-coupling problem, but it leaves more than 2.15 GHz of one-photon detuning from the nearest unwanted transition. The ideal branch exchange should therefore be interpreted as an abstract control operation which may be realizable through a two-tone, Raman, composite or sequential pulse involving the third level, rather than through a single direct $|+1\rangle \leftrightarrow |-1\rangle$ tone.

For a sub-microsecond effective swap, the mechanical motion remains negligible on the pulse timescale. The estimated impulses are below $4.2 \times 10^{-28}\text{ kg m/s}$ and $8.2 \times 10^{-34}\text{ J s}$, while a $0.5\text{ }\mu\text{s}$ pulse produces about 0.2 nm of common displacement but only 26 pm of arm-dependent displacement. These bounds must be recalculated if a realistic multilevel pulse is appreciably longer.

5.4 Nominal exact three-dimensional dynamics

The nominal trajectory uses the physical inputs of **Tabs. 8–10**, the optimized controls of **Eqs. 369–370** and the exact tracked branch model $\mathcal{M}_3$. The two arms begin from the same mechanical state and differ only in the followed branch.

5.4.1 Spatial and rotational trajectories

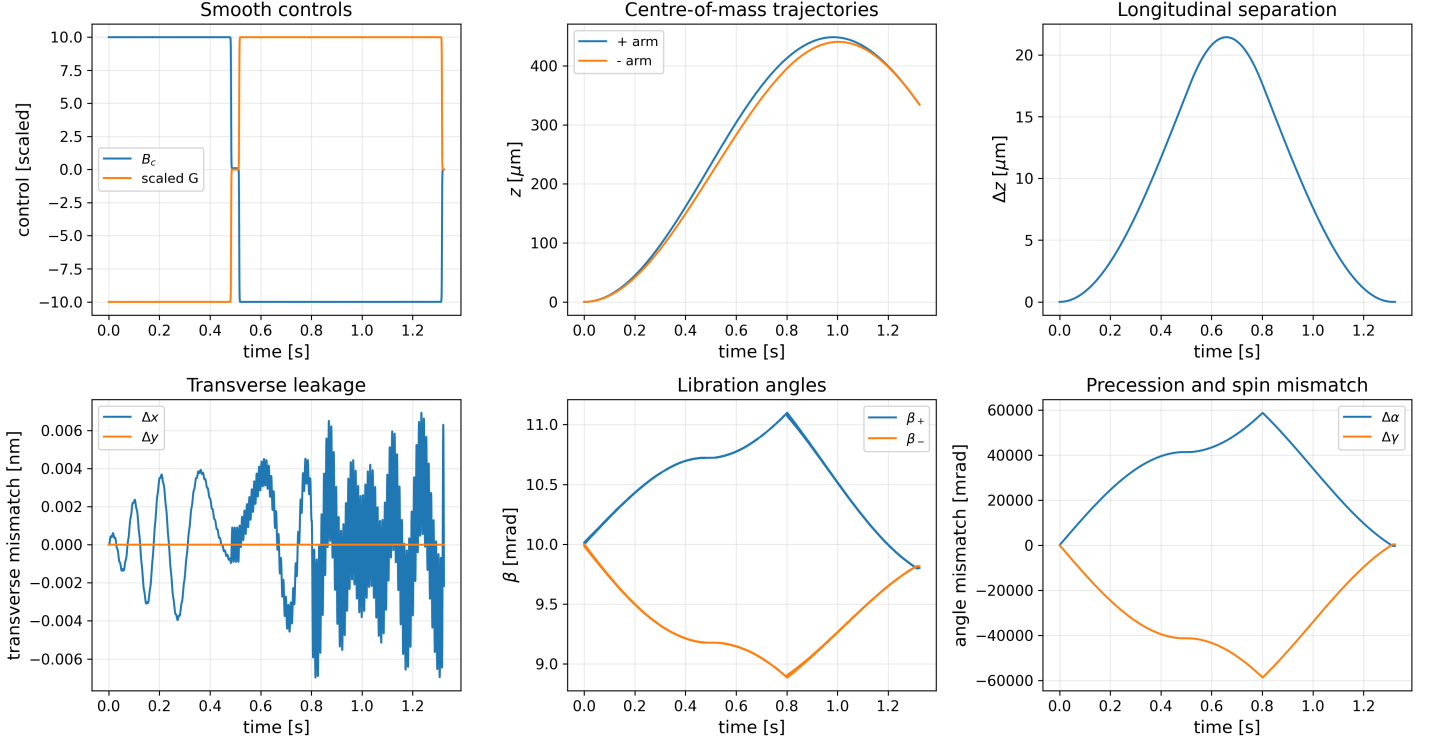


Figure 9: Nominal exact 3×3 trajectory. The panels show the smooth controls, c.o.m. paths, longitudinal separation, transverse leakage, libration angles and Euler-angle differences retained for comparison with the reference model.

Fig. 9 separates the large common motion from the useful branch-dependent displacement. The mean coordinate $\bar{z}$ reaches $444.45 \mu\text{m}$ and ends at $334.35 \mu\text{m}$, while the maximum relative separation is

$$\max_t |\Delta c_z(t)| = 21.4358 \mu\text{m}. \quad (372)$$

The common motion is mainly diamagnetic. In the locally axial field, the point-particle acceleration is

$$a_{\text{dia},z} \simeq \frac{\chi_\rho}{\mu_0} B_z G, \quad (373)$$

which is independent of mass because χ_ρ is the susceptibility per unit mass. During the first stage, $B_z = B_0 - \eta z$ turns this term into harmonic motion about the field zero,

$$\ddot{z}_{\text{dia}} = \omega_{\text{dia}}^2 (Z_0 - z) \quad \left| \quad Z_0 = \frac{B_0}{\eta} = 222.22 \mu\text{m} \quad \right| \quad \frac{\omega_{\text{dia}}}{2\pi} = \frac{\eta}{2\pi} \sqrt{\frac{-\chi_\rho}{\mu_0}} = 0.503 \text{ Hz}. \quad (374)$$

Here $Z_0 = 222.22 \mu\text{m}$ is the field zero and the centre of the diamagnetic harmonic motion, rather than its turning point. For uninterrupted evolution from $z = 0$ with zero initial velocity, the solution is $z(t) = Z_0[1 - \cos(\omega_{\text{dia}} t)]$, so the first turning point occurs at $z = 2Z_0 = 444.44 \mu\text{m}$. Although the actual protocol contains a short homogeneous interval and smooth switching, its first and third gradient stages generate the same approximate diamagnetic restoring law. The observed maximum $\bar{z} = 444.45 \mu\text{m}$ is therefore consistent with this common diamagnetic orbit rather than with the much smaller branch separation.

The relative displacement is set primarily by the spin force. Neglecting the smaller rotational and differential-diamagnetic corrections gives

$$\Delta a_{\text{spin}} \simeq \frac{2\mu\eta}{m} = 1.67 \times 10^{-4} \text{ m/s}^2 \quad \Rightarrow \quad \frac{1}{2}\Delta a_{\text{spin}} t_{12}^2 \simeq 19.5 \mu\text{m}. \quad (375)$$

The acquired relative velocity, $\Delta v_{12} \simeq 80.7 \mu\text{m/s}$, adds another $2.58 \mu\text{m}$ during the 32.0 ms holding interval. This simple estimate reproduces the observed separation scale, with the smooth ramps, rotational projection and differential diamagnetic sampling providing the remaining correction.

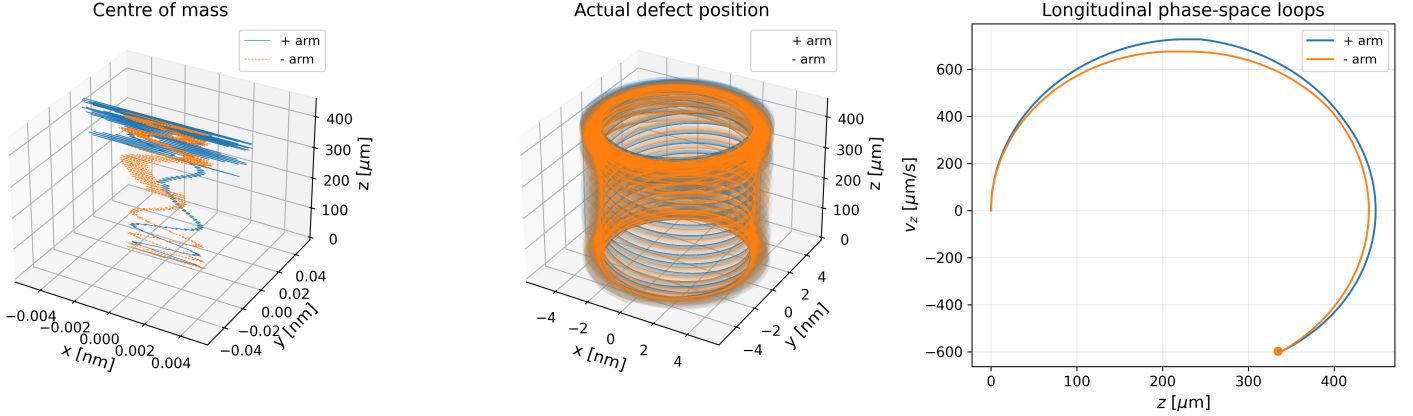


Figure 10: Three-dimensional motion and longitudinal phase space. The c.o.m. trajectories exhibit only picometre-scale transverse differential motion. The dense cylindrical traces in the centre panel are the paths swept out by the rapidly rotating off-centre defect, rather than spatial distributions. The right panel shows the two absolute longitudinal phase-space trajectories, with markers denoting their final states; interferometric closure concerns their relative displacement rather than closure of either absolute curve.

The sequence therefore closes the relative translational phase space rather than returning the particle to its launch point. Sub-nanometre arm closure is the necessary classical centre condition for recombination, but the apparatus must still support the 0.45 mm common excursion and maintain the assumed field profile over that distance. Compensating the common diamagnetic drift would reduce this aperture requirement without directly suppressing the spin-dependent separation.

The transverse differential motion is extremely small:

$$\frac{\max_t \|\Delta \mathbf{c}_\perp(t)\|}{\max_t |\Delta c_z(t)|} = 3.25 \times 10^{-7}, \quad (376)$$

corresponding to a maximum x separation of 6.97 pm and no y separation in the chosen symmetric geometry. The exact three-dimensional and constrained z -only trajectories differ by the same 6.97 pm, validating the one-dimensional translational reduction for this aligned, effectively guided operating point. This conclusion does not extend automatically to misaligned fields or arbitrary spin-frame orientations.

The defect traces a nanometre-scale orbit around each c.o.m., yet its maximum longitudinal separation is $21.4359 \mu\text{m}$ and is therefore indistinguishable from the c.o.m. result at the plotted scale. The off-centre geometry has little effect on the useful spatial splitting, but it remains relevant to the final defect closure and torque decomposition.

The libration remains bounded around the imposed 10 mrad tilt. The raw α and γ differences accumulate to tens of radians and nearly cancel near closure, but their individual values depend on the Euler chart. The physical orientation mismatch is therefore evaluated below through the invariant relative rotation.

5.4.2 Exact spin spectrum and mechanical budget

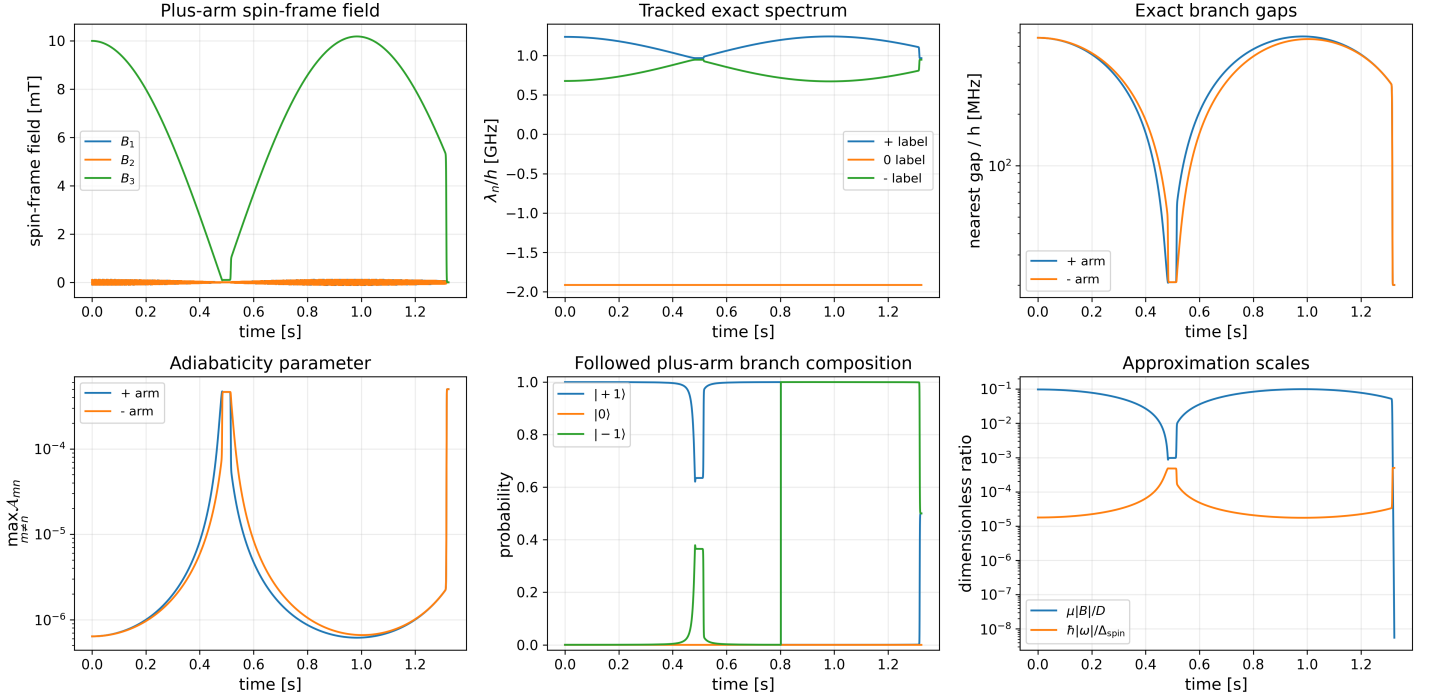


Figure 11: Exact spin diagnostics along the two arms. The figure shows the spin-frame field, tracked spectrum, nearest gaps, adiabaticity parameter, followed-branch composition and the dimensionless expansion scales $\mu|\mathbf{B}|/D$ and $\hbar|\boldsymbol{\omega}|/\Delta_{\text{spin}}$.

Fig. 11 checks spectral isolation, physical adiabaticity and numerical branch continuity separately. The spin-frame field remains almost purely longitudinal, with B_1 and B_2 negligible on the plotted scale. The nontrivial branch mixing is therefore localized to the small-gap passages rather than being driven by a sustained transverse field.

The minimum exact gap is

$$\frac{\Delta_{\text{spin}}^{\min}}{\hbar} = 20.0 \text{ MHz} \simeq \frac{2E}{\hbar}, \quad (377)$$

while $\mathcal{A}_{\max} = 5.0 \times 10^{-4}$ confirms adiabatic following and the minimum overlap > 0.9999 confirms unambiguous numerical branch assignment. Because E directly sets the avoided-crossing gap, reducing the strain splitting at fixed switching rate would weaken the adiabatic margin and must be reassessed explicitly.

Near the mid-sequence field minimum, the followed state temporarily mixes the two bare $|\pm 1\rangle$ components while remaining on a unique exact branch. The ideal microwave event at $t = t_{\text{flip}}$ then transfers the trajectory to the opposite exact branch. During the final turn-off, the exact gap again approaches the strain scale and the followed branch approaches an equal $|+1\rangle - |-1\rangle$ mixture. The turn-off is therefore a second adiabatic bottleneck, and its ramp must satisfy the same gap-controlled adiabatic requirement as the mid-sequence passage. The spectrum consequently confirms that preparation, branch exchange and readout must be defined in the instantaneous eigenbasis.

The largest 2×2 expansion parameter is $\mu|\mathbf{B}|/D = 0.0994$, while the spin-rotation scale is $\hbar|\boldsymbol{\omega}|/\Delta_{\text{spin}} = 5.0 \times 10^{-4}$. These values support the reduced model for efficient translational calculations and show that spin-rotation is spectrally perturbative, but neither replaces the direct force and torque audit below. The exact spectrum remains necessary for trap compatibility, microwave addressability and the accumulated rotational phase.

The two approximation measures are most restrictive in different field regions. Near the field minimum, $\mu|\mathbf{B}|/D$ decreases and the Zeeman expansion improves, whereas the adiabaticity and spin-rotation ratios increase because the exact gap is smallest. Low field therefore improves the energetic 2×2 expansion while tightening the branch-following and rotational margins.

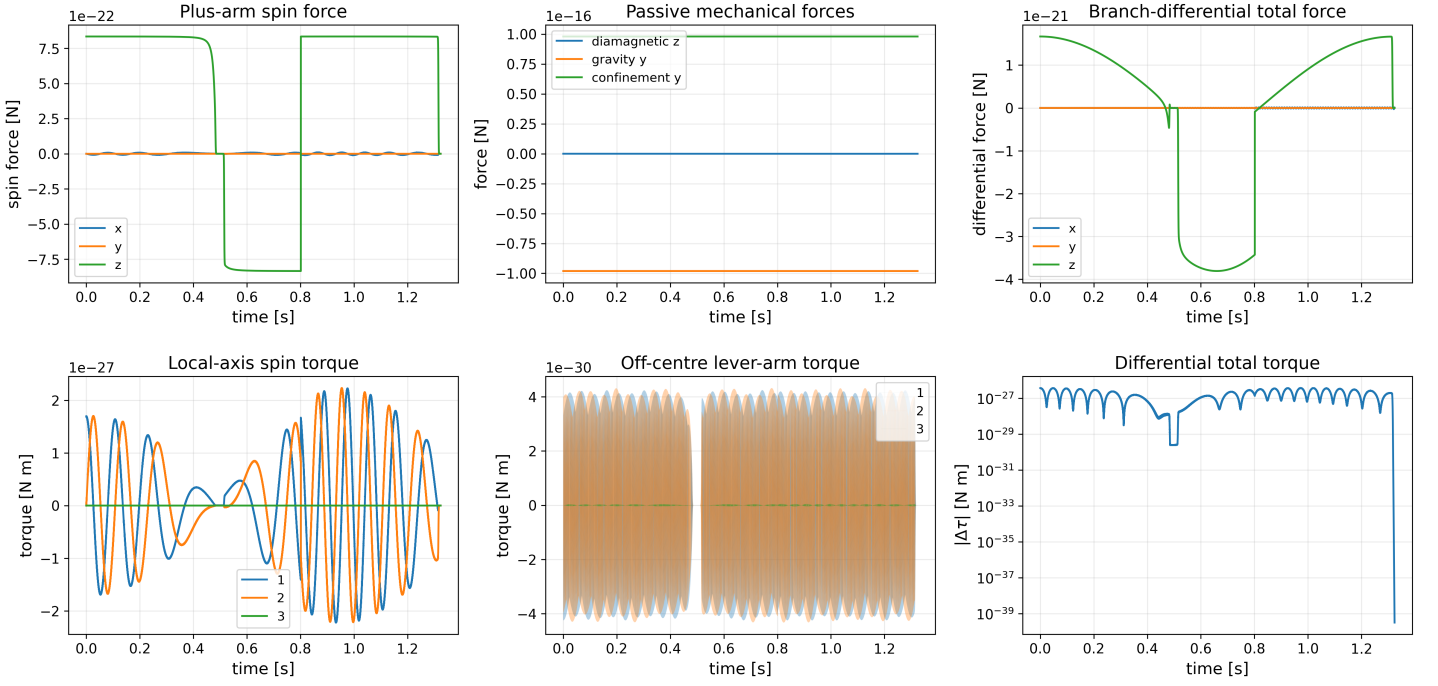


Figure 12: Force and torque budget. The top row separates the plus-arm spin force, passive mechanical forces and branch-differential total force. The lower row resolves the local-axis and off-centre spin torques and their differential total.

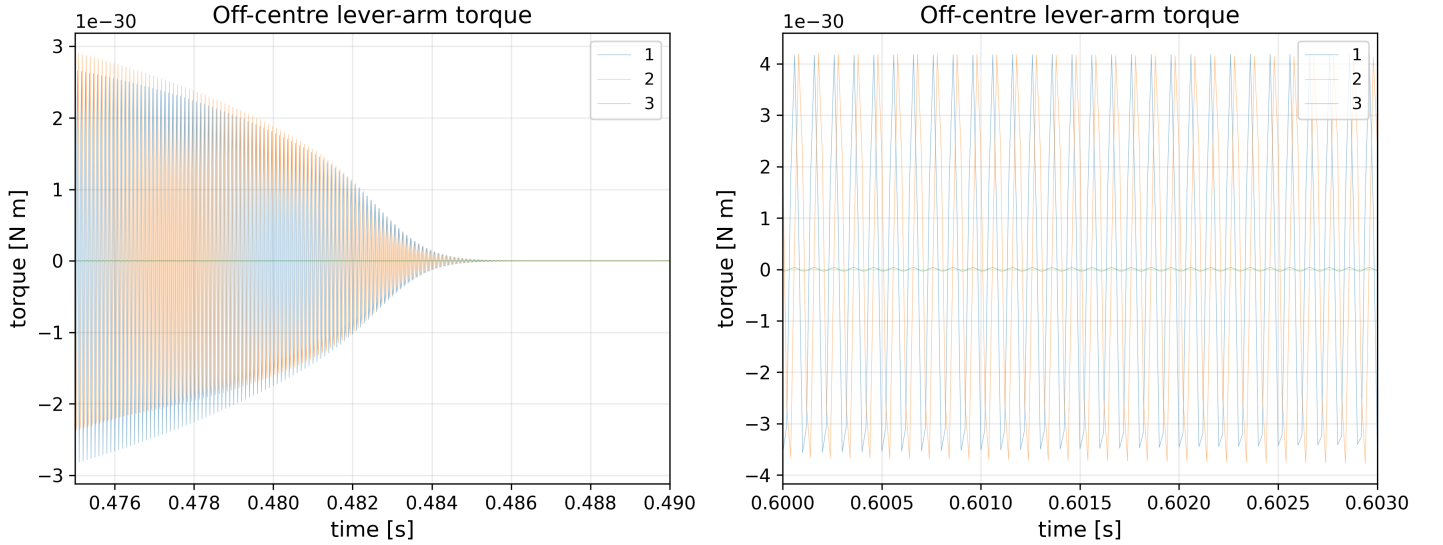


Figure 13: Zoom of the plus-arm off-centre lever-arm torque from **Fig. 12**. The left panel resolves its suppression during the transition to the zero-gradient holding stage around t_{12} . The right panel shows the approximately constant-amplitude oscillation during the third stage. Components 1 and 2 form the dominant phase-shifted transverse torque, while component 3 remains strongly suppressed.

The maximum spin force on either arm is 8.34×10^{-22} N, giving a branch-differential contribution of 1.67×10^{-21} N. Gravity and the displaced harmonic guide each contribute approximately 9.81×10^{-17} N in the y direction and cancel by construction. The remaining passive force is dominated by the longitudinal diamagnetic term, which reaches 2.26×10^{-20} N but is largely common-mode.

After the arms separate, they sample different values of B^2 and acquire a differential passive force of order 2.14×10^{-21} N, comparable to the differential spin force. Spatial closure is therefore produced by a cancellation between the spin impulse and branch-dependent sampling of a nominally spin-independent

curvature. Their different gradient scalings,

$$F_{\text{spin}} \propto G \quad \left| \quad \frac{\partial F_{\text{dia},z}}{\partial z} = \frac{m\chi_\rho}{\mu_0} G^2, \right. \quad (378)$$

mean that a gradient error changes the balance between them rather than uniformly stretching the trajectory. This directly anticipates the strong closure sensitivity to η found in the Monte Carlo analysis.

The differential longitudinal force also displays the impulse sequence which constructs the interferometer. The first positive lobe opens the arms, the negative middle lobe brakes and reverses their relative motion and the final positive lobe removes the returning relative velocity near recombination.

The local-axis torque reaches 2.23×10^{-27} N m, over five hundred times the 4.33×10^{-30} N m off-centre lever-arm contribution. The dominant rotational effect is therefore the tendency of the branch magnetic moment to align with the local field, rather than the displacement of the defect through the gradient.

Its body-frame components lie almost entirely in the transverse 1–2 plane. The local-axis torque therefore drives libration and precession rather than directly changing the imposed spin about the symmetry axis.

The full-duration curve compresses the rapid off-centre oscillations into a slowly varying envelope. The rapid oscillation follows the imposed axial rotation: the off-centre lever arm rotates at approximately $\omega_0/(2\pi) = 10$ kHz relative to the slowly varying magnetic force, causing the transverse torque components to oscillate at the same scale. The left panel of **Fig. 13** shows this envelope collapsing around t_{12} as the gradient is switched to zero. During the constant-gradient interval in the right panel, its amplitude instead remains approximately constant. Components 1 and 2 are phase shifted by the rapid axial rotation, while component 3 remains small because the nearly axial force produces a torque mainly in the transverse plane. The slight inequality between the sampled positive and negative extrema does not indicate secular growth and is consistent with the phase shift and slowly varying background.

Reducing $|\mathbf{d}_0|$ cannot solve the rotational closure problem. Improved alignment, stronger gyroscopic stabilization, a more favourable inertia ratio or explicit torque compensation are more effective controls.

5.4.3 Classical closure and stability

The raw Euler-angle differences are replaced by the invariant relative rotation

$$\mathbf{R}_{\text{rel}} := \mathbf{F}_-^T \mathbf{F}_+ \quad \left| \quad \delta\phi := \text{Log}(\mathbf{R}_{\text{rel}})^\vee \quad \left| \quad \Delta\Theta := \|\delta\phi\|, \right. \quad (379)$$

where $\mathbf{R}_{\text{rel}}$ is the relative rotation matrix between the two interferometer arms and $\delta\phi$ is the principal rotation vector carrying one final orientation into the other. The complete classical closure residual is

$$\mathbf{C}_f := (\Delta\mathbf{c}_f, \Delta\mathbf{p}_{c,f}, \delta\phi_f, \Delta\mathbf{L}_f^{\text{sf}}, \Delta\mathbf{r}_{s,f}). \quad (380)$$

At the plotted production step $h = 20 \mu\text{s}$, the relative rotation vector is

$$\delta\phi_f = (-2.052, 2.655, -0.004)^T \text{ mrad} \quad \left| \quad \Delta\Theta_f = 3.356 \text{ mrad}. \right. \quad (381)$$

The quoted closure values use the converged $5 \mu\text{s}$ refinement,

$$\|\Delta\mathbf{c}_f\| = 0.673 \text{ nm} \quad \left| \quad \frac{\|\Delta\mathbf{p}_{c,f}\|}{m} = 0.00670 \mu\text{m/s} \quad \left| \quad \|\Delta\mathbf{r}_{s,f}\| = 0.681 \text{ nm}, \right. \quad (382)$$

$$\Delta\Theta_f = 3.419 \text{ mrad} \quad \left| \quad \|\Delta\mathbf{L}_f^{\text{sf}}\| = 9.76 \times 10^{-30} \text{ J s} = 0.342\% L_0. \right. \quad (383)$$

The final c.o.m. mismatch is only 3.1×10^{-5} of the maximum arm separation, so the translational loop is extremely well closed on its classical scale. Because the event times were optimized in the complete 2×2 model and then transferred unchanged to the exact model, the remaining sub-nanometre offset measures model transfer rather than a fundamental closure floor. Exact retiming could reduce it further, although improving one closure component may worsen another.

The rotational result is qualitatively different: the invariant mismatch reaches about 20 mrad during the sequence and returns only to 3.419 mrad. The plus-arm orientation Hessian remains positive,

$$\min_t \Omega_+^2(t) = 8.45 \times 10^9 \text{ s}^{-2} > 0 \quad \left| \quad \frac{\sqrt{\min_t \Omega_+^2}}{2\pi} = 14.6 \text{ kHz}. \right. \quad (384)$$

This excludes a local exponential libration instability on the sampled arm, while direct propagation also shows bounded motion on the minus arm. It is not a formal two-arm Hessian proof because the minus-arm curvature was not scanned independently.

The 14.6 kHz local scale corresponds to roughly 1.9×10^4 libration cycles over the interferometer, so a small branch-dependent torque or frequency difference can accumulate a coherent milliradian phase without causing instability. Local stability and rotational closure are therefore distinct: both arms remain bounded, but they do not retrace the same rotational phase. A later wavefunction calculation must compare these translational and rotational residuals with their respective quantum widths before any contrast can be inferred.

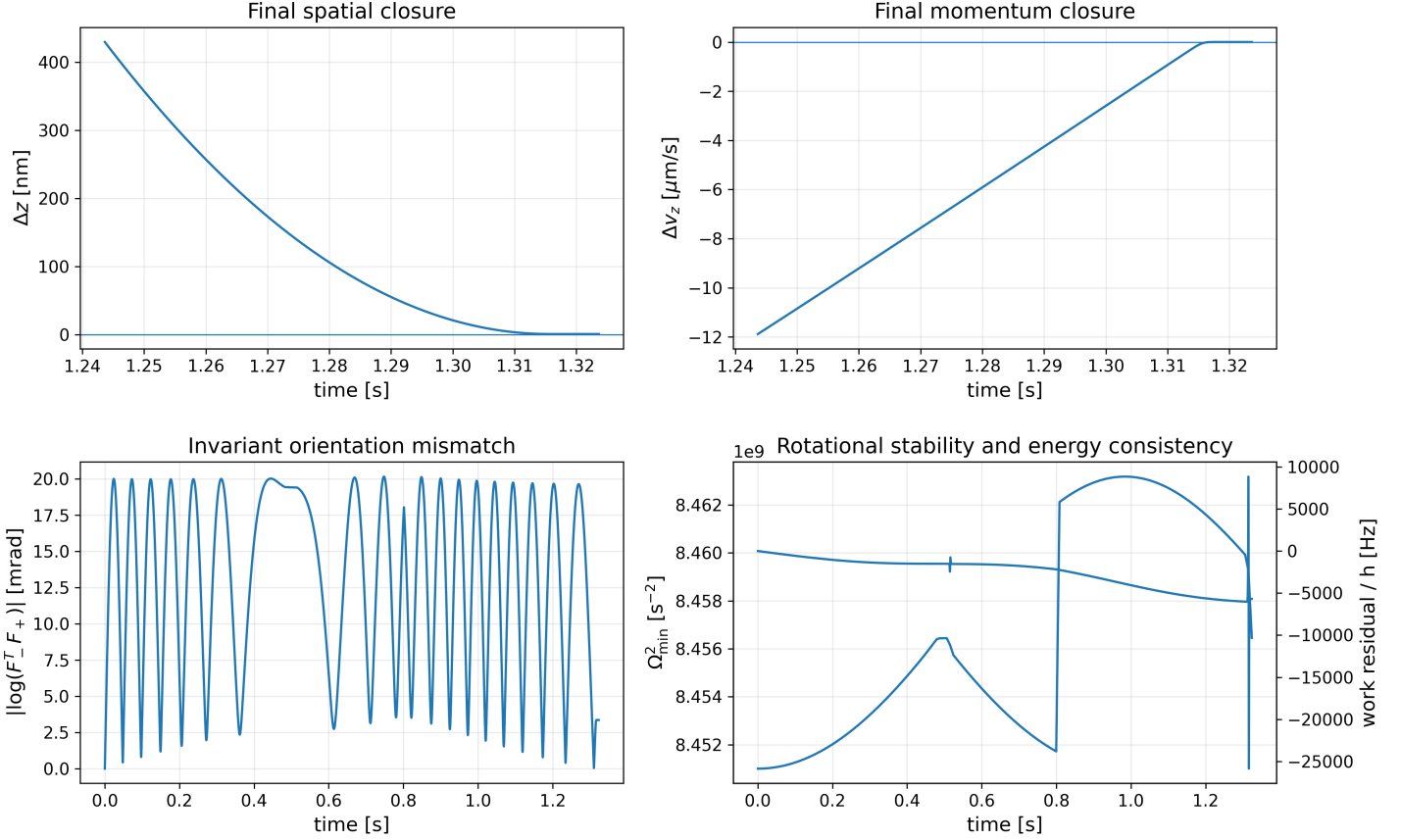


Figure 14: Final closure and stability. The upper panels magnify the spatial and momentum residuals near the end of the sequence. The lower panels show the invariant orientation mismatch, sampled plus-arm rotational curvature and driven work–energy residual.

The endpoint magnification shows that the axial position and velocity residuals are minimized at the same final time. The optimization therefore closes the translational phase space rather than producing only a spatial crossing with appreciable residual momentum. The invariant orientation mismatch, by contrast, passes through several small values at other times but remains nonzero at translational closure. Rotational recombination is therefore an additional phase-matching condition rather than a consequence of translational closure or positive local curvature.

5.5 Approximation audit and physical operating region

5.5.1 Spin and mechanical reductions

The reduced branches are evaluated on the exact trajectories under identical fields and orientations. Energy, force and torque are compared separately because a small potential error can still produce a significant derivative and accumulated phase error.

Table 4: Worst local spin-model error on either exact arm.

Model	$\max \delta V /h$	maximum relative force error	maximum relative torque error
Complete 2×2	181 Hz	1.19×10^{-4}	1.02×10^{-2}
E -reduced	1.83 kHz	1.20×10^{-3}	1.03×10^{-1}
Bare state	10.0 MHz	6.33	46.4

The complete 2×2 model reproduces the exact branch energy and translational force extremely well, but its 1% local torque error accumulates over the rapid libration cycles. It predicts $\Delta\Theta_f = 2.089$ mrad instead of 3.419 mrad, giving a 1.33 mrad or 39% final orientation error. The model is therefore adequate for timing continuation and ensemble scans, but the exact 3×3 branch is required for rotational closure. The E -reduced model already loses ten-percent torque accuracy, while the bare model fails at the avoided crossing and cannot represent the smooth $E \neq 0$ protocol.

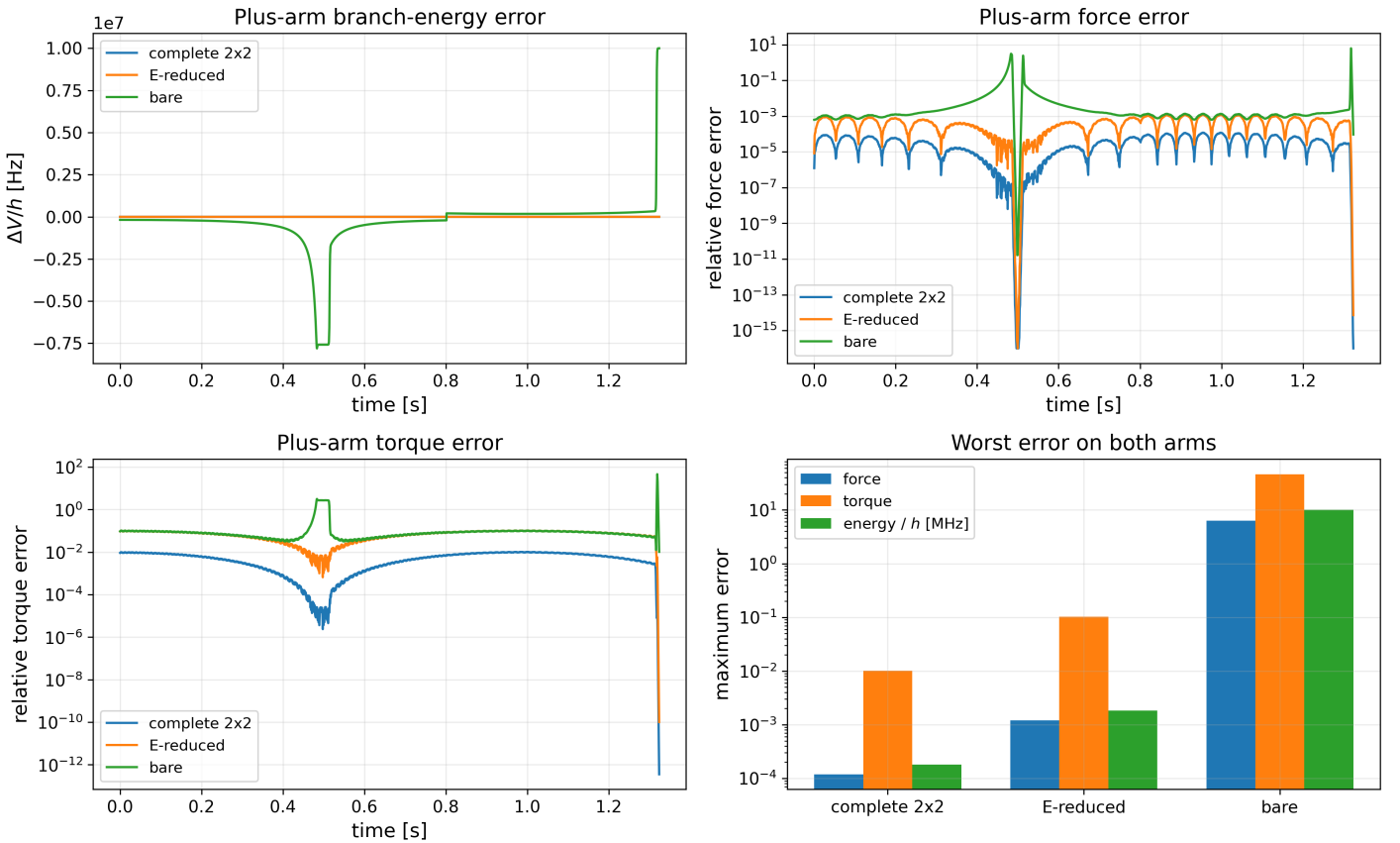


Figure 15: Error hierarchy of the complete 2×2 , E -reduced and bare models relative to the exact 3×3 branch. The time traces show the plus-arm energy, force and torque errors, while the final panel reports the worst error on either arm.

The largest discrepancies are localized near the mid-sequence small-gap passage and the final turn-off, where the exact eigenvectors vary most rapidly. Torque is more sensitive than force because it probes angular derivatives and small transverse admixtures which have little effect on the branch energy but accumulate over many libration cycles.

The exact three-dimensional and exact z -only trajectories differ by only 6.97 pm, confirming that geometry reduction is negligible at the accepted operating point. The finite-volume diamagnetic correction changes the energy by 6.07×10^{-8} relatively and generates only 1.28×10^{-33} N m of torque. For the linear field used here, the finite-volume second-moment term is independent of the c.o.m. coordinate, so its additional translational force vanishes analytically and the measured 2.66×10^{-16} relative force difference is numerical roundoff. Point-particle diamagnetism is therefore fully adequate for the nominal trajectory.

The first-order spin–rotation energy reaches 9.99 kHz, only 5.0×10^{-4} of the minimum exact gap. This makes spin–rotation spectrally perturbative and unlikely to mix neighbouring branches, but it does not bound the accumulated mechanical orientation shift below a sub-milliradian target. A definitive rotational correction would require propagating the momentum-dependent branch, so spin–rotation remains controlled but not fully included.

5.5.2 Gyroscopic window

The reference model is used for a fast design scan over the aspect ratio $\kappa = 2R/L$ and initial rotation frequency. The timestep is reduced with the local gyroscopic frequency, while the exact nominal point remains the final authority. At the nominal aspect ratio, the two characteristic scales are

$$\frac{\omega_{0,\min}}{2\pi} \simeq 270 \text{ Hz} \quad \left| \quad \frac{\Delta_{\text{spin}}^{\min}}{h} \simeq 20 \text{ MHz}. \quad (385)$$

The first marks the onset of positive local curvature, while the second is a spectral scale rather than a hard upper endpoint; perturbativity requires $\omega_0/(2\pi) \ll \Delta_{\text{spin}}^{\min}/h$. The nominal 10 kHz rotation lies about a factor 2×10^3 below the spin scale and is safely perturbative.

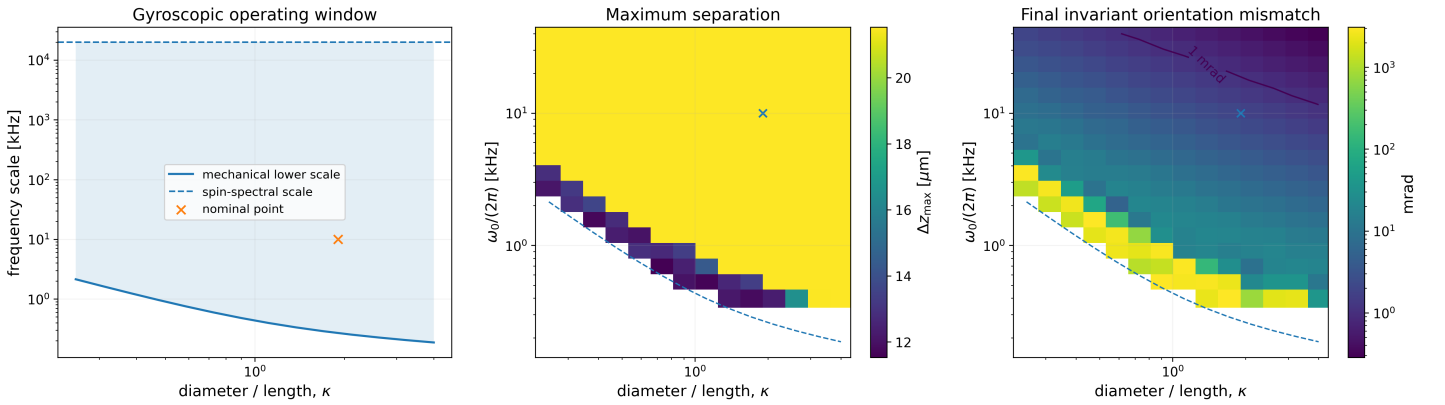


Figure 16: Gyroscopic design region. The left panel compares the mechanical lower scale from **Eq. 337** with the spin-spectral scale from **Eq. 342**. The centre and right panels show the reference-model maximum separation and final invariant orientation mismatch over the same (κ, ω_0) scan. The mechanical lower boundary and nominal operating point are indicated, while the 1 mrad contour marks the illustrative rotational-closure target.

For every pair (κ, ω_0) , the rotor dimensions and moments of inertia are recalculated and the complete reference trajectory is propagated. The mechanical condition is tested through the minimum curvature encountered by either arm, while the spectral condition compares the imposed rotation with the smallest relevant spin gap. Within the plotted range, the spectral bound is inactive, so the visible feasibility boundary is set by the onset of positive rotational curvature. The other panels then determine whether points which are locally stable also provide adequate spatial separation and rotational closure.

Local stability alone is not sufficient for rotational recombination. Near the 270 Hz limit, the rotor may remain formally stable while accumulating orientation errors of hundreds of milliradians. At the nominal aspect ratio, the first sampled point below $\Delta\Theta_f = 1$ mrad occurs at a rotation frequency of order 20 kHz. This is a practical closure scale for the fixed control sequence, not an additional stability threshold.

The separation map is smoother than the orientation map, confirming that the imposed rotation mainly controls the rotor rather than generating the spatial splitting. Points near the stability boundary remain fragile because larger angular excursions can modify the projected spin force. Disk-like geometries increase gyroscopic stiffness as $I_{\parallel}/I_{\perp} \rightarrow 2$, but do not ensure rotational closure.

5.5.3 Mass and geometry scaling

Two inequivalent mass families are compared: fixed κ , for which all dimensions scale together, and fixed $L = 100 \text{ nm}$, for which increasing mass also makes the cylinder progressively disk-like. The controls are not retimed at each mass, so the closure panels test how well the nominal sequence transfers rather than giving a separately optimized bound.

Across $5 \times 10^{-18} \leq m \leq 10^{-15} \text{ kg}$,

$$\Delta z_{\max} : 43.07 \mu\text{m} \longrightarrow 0.215 \mu\text{m}, \quad (386)$$

with a fitted scaling indistinguishable from $\Delta z_{\max} \propto m^{-1}$. The spin force is essentially mass independent, so the branch-dependent acceleration and resulting separation scale as $1/m$. Increasing the mass therefore improves rotational rigidity but directly reduces the useful spatial splitting unless ηt^2 is increased proportionally.

The rotational response falls more rapidly and depends on how the geometry is varied. For fixed shape,

$$R, L \propto m^{1/3} \quad \left| \quad I_{\perp}, I_{\parallel} \propto m^{5/3}, \quad (387)$$

so the calculated orientation mismatch follows $m^{-1.68}$, close to the expected inertial suppression $m^{-5/3}$. For fixed length,

$$R \propto m^{1/2} \quad \left| \quad I_{\parallel} \propto m^2, \quad (388)$$

and the simultaneous evolution towards a disk-like shape steepens the calculated mismatch to approximately $m^{-2.01}$. This is why the two families have nearly identical spatial curves but substantially different rotational ones.

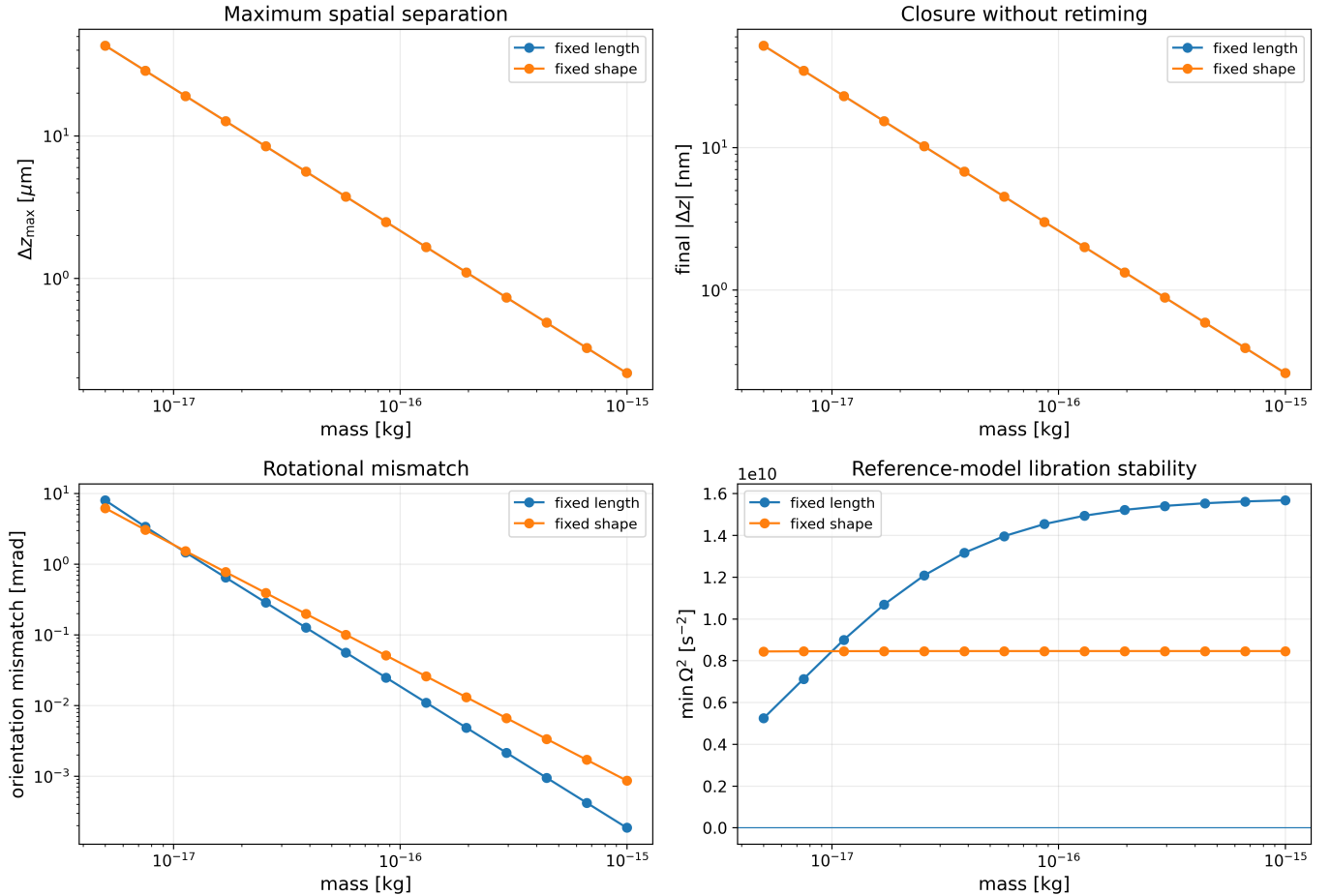


Figure 17: Reference-model mass scaling for fixed-shape and fixed-length families. The panels show maximum separation, closure without retiming, invariant orientation mismatch and minimum libration curvature.

Because the controls are not retimed at each mass, the spatial and rotational closure residuals are non-monotonic even though the broad separation envelope decreases with mass.

The design trade-off is therefore not simply "larger is better". Increasing mass suppresses rotation but penalizes separation, whereas a larger spin rate or more disk-like inertia ratio can improve rotational closure without the same direct $1/m$ spatial cost. Increasing η to recover separation is also constrained because the diamagnetic curvature grows as η^2 , while increasing the duration tightens coherence and control requirements. The operating point must therefore be selected from the simultaneous spatial, rotational and apparatus constraints.

5.6 Robustness and final assessment

5.6.1 Sensitivity and Monte Carlo ensemble

The local sensitivity calculation first ranks the preparation and control errors. The seven retained inputs are then sampled as independent zero-mean Gaussian deviations around the nominal point, with one-standard-deviation widths

$$\sigma_{z_0} = 20 \text{ nm} \quad \left| \quad \sigma_{v_{z0}} = 0.2 \mu\text{m/s} \quad \left| \quad \sigma_{\beta_0} = 10 \mu\text{rad} \quad \left| \quad \sigma_{\dot{\beta}_0} = 5 \text{ mrad/s}, \quad (389)$$

$$\frac{\sigma_{\omega_0}}{2\pi} = 5 \text{ Hz} \quad \left| \quad \sigma_{t_{\text{flip}}} = 2 \mu\text{s} \quad \left| \quad \frac{\sigma_{\eta}}{\eta} = 2 \times 10^{-4}. \quad (390)$$

These are illustrative technical tolerances rather than measured posterior distributions. They define a reproducible robustness test and should be replaced by experimental uncertainty distributions when available.

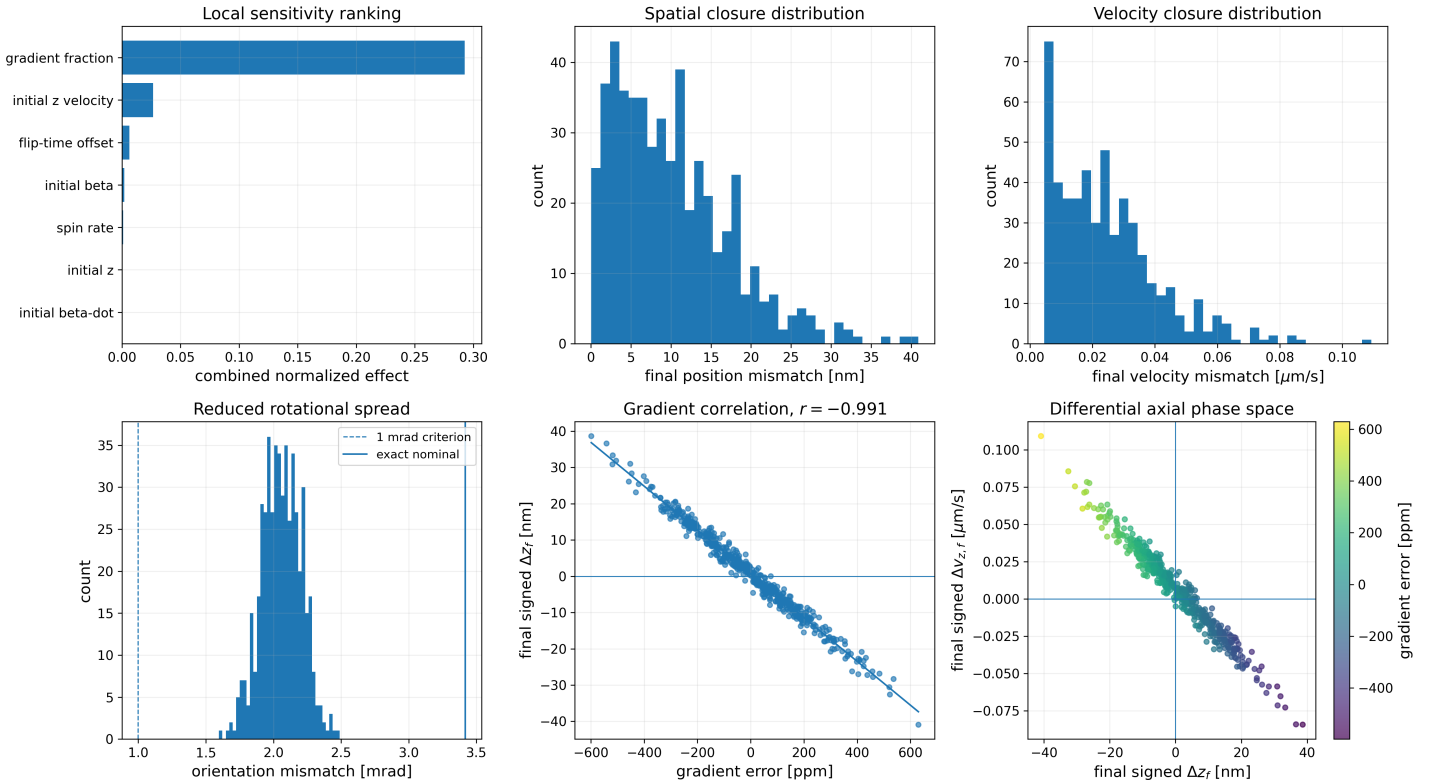


Figure 18: Robustness of the smooth protocol. The top row shows the local sensitivity ranking and the signed spatial and velocity closure distributions. The lower row shows the reduced-model orientation distribution with the 1 mrad criterion and exact nominal marker, the dominant gradient correlation and the final differential axial phase space.

The 512 samples are propagated with the complete 2×2 model at a validated $20 \mu\text{s}$ step. Its nominal timestep shift in orientation is about 0.03 mrad , smaller than the sampled technical spread, while its

translational force agrees with the exact model to better than 1.2×10^{-4} . The ensemble therefore resolves the dominant technical sensitivities efficiently, but its rotational distribution describes fluctuations around the reduced-model result; the exact nominal orientation remains the absolute reference.

The gradient amplitude is the dominant local sensitivity. A one-standard-deviation error of 200 ppm changes the signed final axial residual by 12.1 nm, the final axial velocity by $0.0272 \mu\text{m/s}$ and the reduced-model orientation by 0.144 mrad, but changes the maximum separation by only 1.12 nm. Maximum separation is set by the gross integrated force, whereas closure is the small remainder of several larger impulses whose spin and diamagnetic contributions scale differently with η . A gradient error therefore detunes their cancellation rather than simply rescaling the trajectory.

The correlation coefficient between the gradient error and final signed Δz_f is -0.991 . This nearly linear relation identifies gradient calibration or feed-forward correction as the most direct route to improved translational closure. The next important spatial input is the initial axial velocity, which changes the common trajectory and hence the diamagnetic curvature sampled by the two arms. The final phase-space distribution also shows correlated position and velocity residuals, consistent with a common gradient-dominated perturbation.

Table 5: Monte Carlo output distributions. The intervals are the central 68% and 95% ranges of the 512 complete- 2×2 trajectories.

Output	Median	central 68%	central 95%
Maximum separation [μm]	21.4359	[21.4340, 21.4377]	[21.4322, 21.4397]
Final position mismatch [nm]	8.87	[2.86, 17.44]	[0.57, 27.27]
Final velocity mismatch [$\mu\text{m/s}$]	0.0221	[0.00810, 0.0392]	[0.00473, 0.0634]
Orientation mismatch [mrad]	2.06	[1.91, 2.21]	[1.76, 2.32]
Angular-momentum mismatch [J s]	5.87×10^{-30}	$[5.47, 6.33] \times 10^{-30}$	$[5.03, 6.64] \times 10^{-30}$
Defect-position mismatch [nm]	8.87	[2.85, 17.43]	[0.56, 27.27]
Minimum spin gap [MHz]	20.0	20.0	20.0

The closure entries in **Tab. 5** are magnitudes and are therefore positive and mildly skewed even when their signed residuals are nearly Gaussian. In particular, the signed Δz_f retains the almost linear dependence on the gradient error shown in **Fig. 18**.

No sample violates the illustrative tolerances of 100 nm in position, $1 \mu\text{m/s}$ in velocity or 1 MHz in minimum spin gap. Zero observed failures do not imply zero probability: the rule of three gives an approximate one-sided 95% upper bound of $3/512 \simeq 0.59\%$ for each zero-count criterion. By contrast, no sample satisfies the illustrative 1 mrad orientation limit. The reduced ensemble is centred near 2.06 mrad, while the exact nominal result is 3.42 mrad, so the systematic spin-model correction exceeds the sampled rotational spread. The ensemble therefore identifies the dominant technical sensitivities, but cannot replace the exact rotational result or determine a quantum success probability.

5.6.2 Final operating point

The physical inputs have already been declared in **Tabs. 8–10**. The final table therefore collects only the selected controls and principal outputs.

The reconstructed multipole trap is mechanically stable but spin-incompatible: its transverse field strongly mixes the intended bare states and produces branch forces more than two orders of magnitude above the nominal SG scale. The principal protocol must therefore use a separate SG-compatible confinement mechanism rather than retaining the uncompensated trap field.

With that confinement, the smooth sequence produces $21.44 \mu\text{m}$ of relative separation and sub-nanometre exact c.o.m. closure. The spatial splitting follows the differential spin acceleration, whereas the $444.45 \mu\text{m}$ common excursion is the first diamagnetic turning point and sets an independent apparatus requirement.

Translational closure is robust to the sampled preparation errors but remains most sensitive to gradient calibration because the spin and diamagnetic contributions scale differently with η .

The 20 MHz strain-protected avoided crossing makes the near-zero-field passage adiabatic, but it also means that the followed states are not fixed bare $|\pm 1\rangle$ states near the crossing or final turn-off. Together with the weak direct microwave matrix element and the arm-dependent transition frequencies, this requires a genuine multilevel preparation, exchange and readout sequence beyond the ideal mechanical event.

The aligned translational reduction is exceptionally accurate, whereas rotational closure remains the limiting classical result. A 1% local torque error in the complete 2×2 model becomes a 39% error in the final orientation, and the exact propagation leaves a converged 3.42 mrad mismatch despite bounded libration. A larger spin rate or more disk-like inertia ratio can improve this residual without the direct $1/m$ separation penalty of increasing the mass, although each introduces its own preparation or fabrication constraint.

Table 6: Fact-checked nominal classical result. Converged closure values use the $5 \mu\text{s}$ exact refinement.

Quantity	Value	Physical interpretation
Selected confinement during SG	effective (104, 130, 0) Hz guide	explicit multipole field rejected by the spin gate
First two transition times	$t_{12} = 0.483085062 \text{ s}$, $t_{23} = 0.515090178 \text{ s}$	continued timing solution
Flip and turn-off times	$t_{\text{flip}} = 0.801531322 \text{ s}$, $t_{\text{off}} = 1.315634715 \text{ s}$	closes the continued 2×2 spatial trajectory
Final evaluation time	1.323634715 s	includes the smooth turn-off tail
Maximum common c.o.m. excursion	$444.45 \mu\text{m}$	apparatus aperture and field-linearity requirement
Maximum arm separation	$21.4358 \mu\text{m}$	useful spin-conditioned splitting
Final c.o.m. mismatch	0.673 nm	converged relative spatial closure
Final velocity mismatch	$0.00670 \mu\text{m/s}$	converged momentum closure divided by mass
Final defect mismatch	0.681 nm	includes the off-centre rotor geometry
Final orientation mismatch	3.419 mrad	dominant classical Humpty–Dumpty residual
Final angular-momentum mismatch	$9.76 \times 10^{-30} \text{ J s}$	0.342% of the initial mechanical angular momentum
Minimum exact spin gap	20.0 MHz	strain-protected avoided crossing
Maximum adiabaticity parameter	5.0×10^{-4}	adiabatic at the nominal E and switching rate
Maximum transverse/longitudinal separation	3.25×10^{-7}	validates the aligned one-dimensional translation
Complete 2×2 force error	1.19×10^{-4}	adequate for timing and ensemble scans
Direct single-tone branch swap	not established	two-arm detuning and suppressed matrix element require multilevel control

6 Quantum wave-packet dynamics and interferometric spin contrast

The preceding chapter determines two spin-conditioned semiclassical arms, labelled by $\ell \in \{+, -\}$. The label ℓ identifies the interferometer path and remains fixed throughout the protocol, whereas $n_\ell(t)$ denotes the instantaneous adiabatic spin branch followed on that path. The microwave π -pulse exchanges the two followed branches at $t = t_{\text{flip}}$ without changing the mechanical coordinates or momenta.

For the centered, aligned and axially constrained calculation considered in this chapter, the mechanical configuration and its conjugate momenta are

$$\mathbf{q} = (z \quad \alpha \quad \beta \quad \gamma)^T \quad \Bigg| \quad \mathbf{p} = (p_z \quad p_\alpha \quad p_\beta \quad p_\gamma)^T \quad \Bigg| \quad \boldsymbol{\xi} := \begin{pmatrix} \mathbf{q} \\ \mathbf{p} \end{pmatrix}. \quad (391)$$

Thus, $\boldsymbol{\xi}$ is the eight-dimensional canonical mechanical phase-space vector. Its first four components are the axial position and three Euler angles, while the final four are their corresponding canonical momenta.

The Hamiltonian followed by arm ℓ is

$$H_\ell(\boldsymbol{\xi}, t) := H_{n_\ell(t)}(\boldsymbol{\xi}, t), \quad (392)$$

where H_n is the adiabatic branch Hamiltonian derived in **Eq. 275** and reduced to the coordinates of **Eq. 391**. The classical packet centre $\bar{\boldsymbol{\xi}}_\ell(t) = (\bar{\mathbf{q}}_\ell, \bar{\mathbf{p}}_\ell)^T$ is the trajectory obtained from Hamilton's equations,

$$\dot{\bar{\mathbf{q}}}_\ell = \left. \frac{\partial H_\ell}{\partial \mathbf{p}} \right|_{\bar{\boldsymbol{\xi}}_\ell} \quad \Bigg| \quad \dot{\bar{\mathbf{p}}}_\ell = - \left. \frac{\partial H_\ell}{\partial \mathbf{q}} \right|_{\bar{\boldsymbol{\xi}}_\ell}. \quad (393)$$

The local curvature of the same Hamiltonian is described by its phase-space Hessian,

$$(K_\ell)_{ab}(t) := \left. \frac{\partial^2 H_\ell}{\partial \xi_a \partial \xi_b} \right|_{\boldsymbol{\xi} = \bar{\boldsymbol{\xi}}_\ell(t)}, \quad a, b \in \{1, \dots, 8\}. \quad (394)$$

The matrix $\mathbf{K}_\ell(t)$ therefore contains the local translational curvature, rotational curvature and all translation-rotation couplings sampled along arm ℓ .

We now place a mechanical quantum state $|\psi_\ell(t)\rangle$ on each classical trajectory. The packet is centred on $\bar{\boldsymbol{\xi}}_\ell(t)$ when

$$\langle \psi_\ell(t) | \hat{\boldsymbol{\xi}} | \psi_\ell(t) \rangle = \bar{\boldsymbol{\xi}}_\ell(t) \quad \Bigg| \quad \delta \hat{\boldsymbol{\xi}}_\ell := \hat{\boldsymbol{\xi}} - \bar{\boldsymbol{\xi}}_\ell(t). \quad (395)$$

Its quantum widths and correlations are collected in the covariance matrix

$$(\Sigma_\ell)_{ab}(t) := \frac{1}{2} \langle \psi_\ell(t) | \delta \hat{\xi}_{\ell,a} \delta \hat{\xi}_{\ell,b} + \delta \hat{\xi}_{\ell,b} \delta \hat{\xi}_{\ell,a} | \psi_\ell(t) \rangle. \quad (396)$$

The diagonal entries of Σ_ℓ are the coordinate and momentum variances, while its off-diagonal entries describe correlations between the mechanical degrees of freedom.

At t_f , the mechanical overlap appears in the off-diagonal spin coherence as

$$\mathcal{M}(t_f) = e^{i\Delta\phi(t_f)} \langle \psi_-(t_f) | \psi_+(t_f) \rangle \quad \Bigg| \quad C := |\mathcal{M}(t_f)| = |\langle \psi_-(t_f) | \psi_+(t_f) \rangle|. \quad (397)$$

Here, $\Delta\phi(t_f)$ is the relative phase accumulated by the two arms, including the branch and microwave-pulse phases, while $C \in [0, 1]$ is the interferometric spin contrast. The phase changes the final interference fringe, whereas the magnitude of the mechanical overlap determines its visibility.

The calculation performed in this chapter can therefore be summarized as

$$\boxed{\underbrace{\{H_\ell, \bar{\boldsymbol{\xi}}_\ell, \mathbf{K}_\ell\}}_{\text{semiclassical input}} \longrightarrow \underbrace{\{|\psi_\ell\rangle, \Sigma_\ell\}}_{\text{quantum propagation}} \longrightarrow \underbrace{\mathcal{M}(t_f)}_{\text{spin coherence}} \longrightarrow \underbrace{C = |\mathcal{M}(t_f)|}_{\text{contrast}}}. \quad (398)$$

The wavefunction calculation is therefore the next layer of the same model hierarchy. The trajectories obtained in the preceding chapter become the centres of the two conditional quantum packets, and the Hessians evaluated along those trajectories determine the evolution of their widths, squeezing and correlations. The final question is no longer only whether the two classical centres close, but whether the complete quantum packets centred on them substantially overlap.

6.1 Gaussian quantization about the semiclassical arms

In the canonical ordering of **Eq. 391**, the symplectic matrix is

$$\mathbf{J} = \begin{pmatrix} \mathbf{0} & \mathbf{1}_4 \\ -\mathbf{1}_4 & \mathbf{0} \end{pmatrix}. \quad (399)$$

For arm ℓ , write the mechanical phase-space operator as

$$\hat{\xi} = \bar{\xi}_\ell(t) + \delta\hat{\xi}_\ell(t). \quad (400)$$

Expanding the corresponding branch Hamiltonian of **Eq. 392** around the classical centre gives

$$\hat{H}_\ell(t) = H_\ell(\bar{\xi}_\ell(t), t) + \nabla_\xi H_\ell|_{\bar{\xi}_\ell} \cdot \delta\hat{\xi}_\ell + \frac{1}{2} \delta\hat{\xi}_\ell^T \mathbf{K}_\ell(t) \delta\hat{\xi}_\ell + \mathcal{O}(\delta\hat{\xi}_\ell^3). \quad (401)$$

The packet centre satisfies Hamilton's equation,

$$\dot{\bar{\xi}}_\ell = \mathbf{J} \nabla_\xi H_\ell(\bar{\xi}_\ell, t), \quad (402)$$

so the linear term disappears in the displaced frame. Neglecting the cubic remainder leaves a quadratic generator. An initially Gaussian state then remains Gaussian, and its tangent matrix and covariance obey

$$\boxed{\dot{\mathbf{S}}_\ell = \mathbf{J} \mathbf{K}_\ell \mathbf{S}_\ell, \quad \mathbf{S}_\ell(0) = \mathbf{1}_{2n_m}, \quad \Sigma_\ell(t) = \mathbf{S}_\ell(t) \Sigma_0 \mathbf{S}_\ell^T(t)}, \quad (403)$$

where

$$\Sigma_{\ell,ab} = \frac{1}{2} \langle \delta\hat{\xi}_{\ell,a} \delta\hat{\xi}_{\ell,b} + \delta\hat{\xi}_{\ell,b} \delta\hat{\xi}_{\ell,a} \rangle. \quad (404)$$

Equation 403 is the central bridge between the two chapters: the same variational flow used to test classical stability now propagates the quantum packet widths and correlations.

6.1.1 Prepared covariance and rotor correlations

Let

$$r_I := \frac{I_\parallel}{I_\perp} \quad \Bigg| \quad \Omega_{\text{lib}} \simeq r_I \omega_0 \quad (405)$$

be the inertia ratio and the strong-gyroscopic libration frequency. If the axial preparation trap has frequency ω_z , ground-state preparation gives

$$\sigma_z^2 = \frac{\hbar}{2m\omega_z} \quad \Bigg| \quad \sigma_{p_z}^2 = \frac{\hbar m \omega_z}{2} \quad \Bigg| \quad \sigma_\beta^2 = \frac{\hbar}{2I_\perp \Omega_{\text{lib}}} \quad \Bigg| \quad \sigma_{p_\beta}^2 = \frac{\hbar I_\perp \Omega_{\text{lib}}}{2}. \quad (406)$$

Following the reference construction [19], p_α and p_γ are Gaussian packets centred on

$$\bar{p}_\alpha = I_\parallel \omega_0 \cos \beta_0, \quad \bar{p}_\gamma = I_\parallel \omega_0, \quad (407)$$

with standard deviations σ_{p_α} and σ_{p_γ} . A product minimum-uncertainty packet is a useful benchmark, but the symmetric-top kinetic energy makes the local libration minimum depend on the cyclic-momentum fluctuations. Writing

$$p'_\alpha := p_\alpha - \bar{p}_\alpha, \quad p'_\gamma := p_\gamma - \bar{p}_\gamma, \quad (408)$$

the quadratic minimum shifts by

$$\delta\bar{\beta}_0 = a_\alpha p'_\alpha + a_\gamma p'_\gamma \quad \Bigg| \quad a_\alpha = -\frac{1}{I_\parallel \omega_0 \sin \beta_0} \quad \Bigg| \quad a_\gamma = \frac{\cos \beta_0}{I_\parallel \omega_0 \sin \beta_0}. \quad (409)$$

This is a coherent preparation correlation, not an incoherent average over different classical equilibria. It is included through the symplectic shear

$$\Sigma_0^{(\text{corr})} = \mathbf{T}_{\text{prep}} \Sigma_0^{(\text{prod})} \mathbf{T}_{\text{prep}}^T, \quad \mathbf{T}_{\text{prep}} = \begin{pmatrix} \mathbf{1}_4 & \mathbf{G} \\ \mathbf{0}_4 & \mathbf{1}_4 \end{pmatrix}, \quad \mathbf{G}^T = \mathbf{G}, \quad (410)$$

where the only nonzero entries are

$$G_{\alpha\beta} = G_{\beta\alpha} = a_\alpha, \quad G_{\beta\gamma} = G_{\gamma\beta} = a_\gamma. \quad (411)$$

The symmetry of $\mathbf{G}$ makes $\mathbf{T}_{\text{prep}}$ symplectic, preserving the initial Gaussian purity.

6.2 Reference coherent-state limit and thermal libration

The reference paper describes the stabilized libration as a driven harmonic mode. This limit is recovered by taking a common, constant curvature Ω_{lib}^2 on both arms and retaining their difference only through the branch-dependent generalized torque. With

$$x := \beta - \beta_0 \quad \left| \quad \tau_{\beta,\ell}(t) := - \frac{\partial V_{n_\ell(t)}}{\partial \beta} \right|_{z=\bar{z}_\ell(t), \beta=\beta_0}, \quad (412)$$

the conditional libration Hamiltonian is

$$\hat{H}_{\beta,\ell}(t) = \frac{\hat{p}_\beta^2}{2I_\perp} + \frac{1}{2}I_\perp\Omega_{\text{lib}}^2\hat{x}^2 - \tau_{\beta,\ell}(t)\hat{x} + \epsilon_\ell(t), \quad (413)$$

where ϵ_ℓ is independent of x and changes only the interferometric phase.

An initial coherent state remains coherent, with final amplitude difference

$$\Delta\kappa_f = \frac{ie^{-i\Omega_{\text{lib}}t_f}}{\sqrt{2\hbar I_\perp \Omega_{\text{lib}}}} \int_0^{t_f} e^{i\Omega_{\text{lib}}t} \Delta\tau_\beta(t) dt, \quad \Delta\tau_\beta := \tau_{\beta,+} - \tau_{\beta,-}. \quad (414)$$

Equivalently, in terms of the final libration phase-space mismatch,

$$\Delta\kappa_f = \sqrt{\frac{I_\perp \Omega_{\text{lib}}}{2\hbar}} \Delta\beta_f + \frac{i\Delta p_{\beta,f}}{\sqrt{2\hbar I_\perp \Omega_{\text{lib}}}}. \quad (415)$$

The corresponding zero-temperature contrast is

$$C_\beta^{(\text{coh})} = \exp\left[-\frac{1}{2}|\Delta\kappa_f|^2\right] = \exp\left[-\frac{I_\perp \Omega_{\text{lib}} \Delta\beta_f^2}{4\hbar} - \frac{\Delta p_{\beta,f}^2}{4\hbar I_\perp \Omega_{\text{lib}}}\right]. \quad (416)$$

Equation 414 identifies the part of the torque history which survives at recombination. Defining

$$\tilde{\tau}_\beta(\Omega_{\text{lib}}; t_f) := \int_0^{t_f} e^{i\Omega_{\text{lib}}t} \Delta\tau_\beta(t) dt \quad \Rightarrow \quad -\ln C_\beta^{(\text{coh})} = \frac{|\tilde{\tau}_\beta(\Omega_{\text{lib}}; t_f)|^2}{4\hbar I_\perp \Omega_{\text{lib}}}. \quad (417)$$

Therefore,

$$\tilde{\tau}_\beta(\Omega_{\text{lib}}; t_f) = 0 \quad \Longleftrightarrow \quad \Delta\beta_f = \Delta p_{\beta,f} = 0 \quad (418)$$

within the equal-curvature approximation. The real and imaginary parts impose the two libration phase-space closure conditions. This filter description explains the dependence on ω_0 and T_{lib} shown below, while the complete covariance propagation retains the time-dependent curvature and omitted mode correlations.

For a thermal libration state, the exact Bose occupation is

$$\bar{n}_{\text{lib}} = \frac{1}{\exp(\hbar\Omega_{\text{lib}}/k_B T_{\text{lib}}) - 1}. \quad (419)$$

The thermal displacement contrast is then

$$C_\beta^{(\text{th})} = \exp\left[-\frac{1}{2}(2\bar{n}_{\text{lib}} + 1)|\Delta\kappa_f|^2\right]. \quad (420)$$

This is the exact form whose high-temperature limit reproduces the occupation factor used in the reference paper. It applies to the equal-curvature displacement model; if the two arms finish with different squeezing, the full covariance or a direct thermal wavefunction propagation is required.

6.3 Mechanical overlap and invariant rotational comparison

After recombination, the mechanical degrees of freedom are traced out. For an initial mechanical density operator $\rho_{\text{m},0}$, the off-diagonal spin coherence is

$$\mathcal{M}(t_f) = e^{i\phi_{\text{ctrl}}} \text{Tr}_{\text{m}} \left[U_+(t_f, 0) \rho_{\text{m},0} U_-^\dagger(t_f, 0) \right], \quad (421)$$

where U_ℓ propagates the mechanical state along arm ℓ and ϕ_{ctrl} contains the controlled spin and microwave phases. For a pure initial state,

$$\mathcal{M}(t_f) = e^{i\phi_{\text{ctrl}}} \langle \psi_-(t_f) | \psi_+(t_f) \rangle. \quad (422)$$

The observable contrast is

$$C := |\mathcal{M}(t_f)|. \quad (423)$$

A final Ramsey pulse of phase ϕ_R gives

$$P_{\pm} = \frac{1}{2} [1 \pm C \cos(\Phi + \phi_R)], \quad \Phi := \arg \mathcal{M}. \quad (424)$$

The present calculation concerns the modulus C ; the branch, microwave and geometric phases affect Φ but not the overlap modulus.

6.3.1 Common canonical chart on $SO(3)$

The final rotational packets cannot be compared by treating $\Delta\alpha$, $\Delta\beta$ and $\Delta\gamma$ as Cartesian displacements. Their invariant mean-orientation mismatch is the principal logarithm of the relative rotation,

$$\mathbf{R}_{\text{rel}} := \mathbf{F}_{-}^T(t_f) \mathbf{F}_{+}(t_f), \quad \Delta\boldsymbol{\varphi}_f := \text{Log}(\mathbf{R}_{\text{rel}})^{\vee}. \quad (425)$$

Its norm is the principal relative angle.

The conjugate momenta and covariances must follow the same coordinate map. Choose $\mathbf{F}_{-}(t_f)$ as the origin of a local logarithmic chart and let $\boldsymbol{\theta} = (\alpha, \beta, \gamma)^T$. With the right Jacobian $\mathbf{J}_r(\boldsymbol{\varphi})$ of $SO(3)$ and the body-frame Euler-rate matrix $\mathbf{N}_{zzz}(\boldsymbol{\theta})$ from **Eq. 104**, the Euler-to-logarithm Jacobian reads

$$\mathbf{A} = \frac{\partial \boldsymbol{\varphi}}{\partial \boldsymbol{\theta}} = \mathbf{J}_r^{-1}(\boldsymbol{\varphi}) \mathbf{N}_{zzz}(\boldsymbol{\theta}). \quad (426)$$

The canonical momentum in the common chart follows from the cotangent lift:

$$\boxed{\boldsymbol{\pi} = \mathbf{A}^{-T} \mathbf{p}_{\theta} = \mathbf{J}_r^T(\boldsymbol{\varphi}) \mathbf{L}^{\text{bf}}}, \quad (427)$$

where $\mathbf{p}_{\theta} = (p_{\alpha}, p_{\beta}, p_{\gamma})^T$. Equation 427 is essential: rotating $\mathbf{L}^{\text{bf}}$ as an ordinary vector is not the canonical transformation associated with the logarithmic coordinate.

To first order, the rotational fluctuations transform as

$$\begin{pmatrix} \delta \boldsymbol{\varphi} \\ \delta \boldsymbol{\pi} \end{pmatrix} = \mathbf{T}_{\text{rot}} \begin{pmatrix} \delta \boldsymbol{\theta} \\ \delta \mathbf{p}_{\theta} \end{pmatrix} \quad \left| \quad \mathbf{T}_{\text{rot}} = \begin{pmatrix} \mathbf{A} & \mathbf{0}_3 \\ \mathbf{B} & \mathbf{A}^{-T} \end{pmatrix} \quad \left| \quad (\mathbf{B})_{:j} := \frac{\partial}{\partial \theta_j} \left(\mathbf{A}^{-T} \mathbf{p}_{\theta} \right) \right|_{\mathbf{p}_{\theta}}. \quad (428)$$

The lower-left block contains the coordinate dependence of the transformed momentum. As a cotangent lift, $\mathbf{T}_{\text{rot}}$ is symplectic. Both final means and covariances are mapped through this transformation before the overlap is calculated.

6.3.2 Gaussian contrast and reference factorization

Let tildes denote final means and covariances expressed in the common translational-rotational chart. For two pure n_m -mode Gaussian states, their exact overlap modulus is

$$\boxed{C_G = \frac{\hbar^{n_m/2}}{\det(\tilde{\boldsymbol{\Sigma}}_+ + \tilde{\boldsymbol{\Sigma}}_-)^{1/4}} \exp \left[-\frac{1}{4} \Delta \tilde{\boldsymbol{\xi}}_f^T (\tilde{\boldsymbol{\Sigma}}_+ + \tilde{\boldsymbol{\Sigma}}_-)^{-1} \Delta \tilde{\boldsymbol{\xi}}_f \right]}, \quad (429)$$

where

$$\Delta \tilde{\boldsymbol{\xi}}_f := \tilde{\boldsymbol{\xi}}_+(t_f) - \tilde{\boldsymbol{\xi}}_-(t_f). \quad (430)$$

The exponent separates into a displacement and a packet-shape contribution,

$$\boxed{-\ln C_G = \Gamma_{\text{disp}} + \Gamma_{\text{shape}}} \quad (431)$$

$$\Gamma_{\text{disp}} := \frac{1}{4} \Delta \tilde{\boldsymbol{\xi}}_f^T (\tilde{\boldsymbol{\Sigma}}_+ + \tilde{\boldsymbol{\Sigma}}_-)^{-1} \Delta \tilde{\boldsymbol{\xi}}_f \quad \left| \quad \Gamma_{\text{shape}} := \frac{1}{4} \ln \left[\frac{\det(\tilde{\boldsymbol{\Sigma}}_+ + \tilde{\boldsymbol{\Sigma}}_-)}{\hbar^{2n_m}} \right]. \quad (432)$$

The first term measures the final phase-space separation in units of the combined quantum width; the second measures differential squeezing and correlations. Unit contrast requires coincident means and identical covariances. Exact position closure alone is therefore insufficient.

For comparison with the reference calculation, independent equal-width packets give

$$C_{\text{fact}} = C_z C_{\alpha\gamma} C_\beta \quad (433)$$

$$-\ln C_z = \frac{\Delta z_f^2}{8\sigma_z^2} + \frac{\sigma_z^2 \Delta p_{z,f}^2}{2\hbar^2} \quad \left| \quad -\ln C_{\alpha\gamma} = \frac{\Delta \alpha_f^2 \sigma_{p_\alpha}^2 + \Delta \gamma_f^2 \sigma_{p_\gamma}^2}{2\hbar^2} + \frac{\Delta p_{\alpha,f}^2}{8\sigma_{p_\alpha}^2} + \frac{\Delta p_{\gamma,f}^2}{8\sigma_{p_\gamma}^2} \quad \right| \quad -\ln C_\beta = \frac{1}{2} |\Delta \kappa_f|^2. \quad (434)$$

In the centered axial reference model, p_α and p_γ are conserved and the corresponding momentum-mismatch terms vanish. The general Gaussian expression additionally retains the time-dependent covariance, mode correlations and differential squeezing.

6.4 Numerical wave-packet results

The numerical calculation retains the one-loop Stern–Gerlach geometry and gyroscopic mechanism of [19], but uses the smooth field, event-corrected classical trajectories and full reduced covariance obtained from the preceding chapter. The nominal rotor and preparation are

$$\begin{aligned} m &= 10^{-17} \text{ kg}, & L &= 100 \text{ nm}, & R &= 95.365 \text{ nm}, & r_I &= 1.463573, \\ \beta_0 &= 10^{-2} \text{ rad}, & \frac{\omega_0}{2\pi} &= 10 \text{ kHz}, & \frac{\Omega_{\text{lib}}}{2\pi} &= 14.6357 \text{ kHz}. \end{aligned} \quad (435)$$

The axial packet is prepared in the 9.6 Hz ground state, giving

$$\sigma_z = 2.9566 \times 10^{-10} \text{ m}, \quad \sigma_\beta = 1.3585 \times 10^{-4} \text{ rad}, \quad \sigma_{p_\alpha} = \sigma_{p_\gamma} = 10\hbar. \quad (436)$$

6.4.1 Classical handoff and packet centres

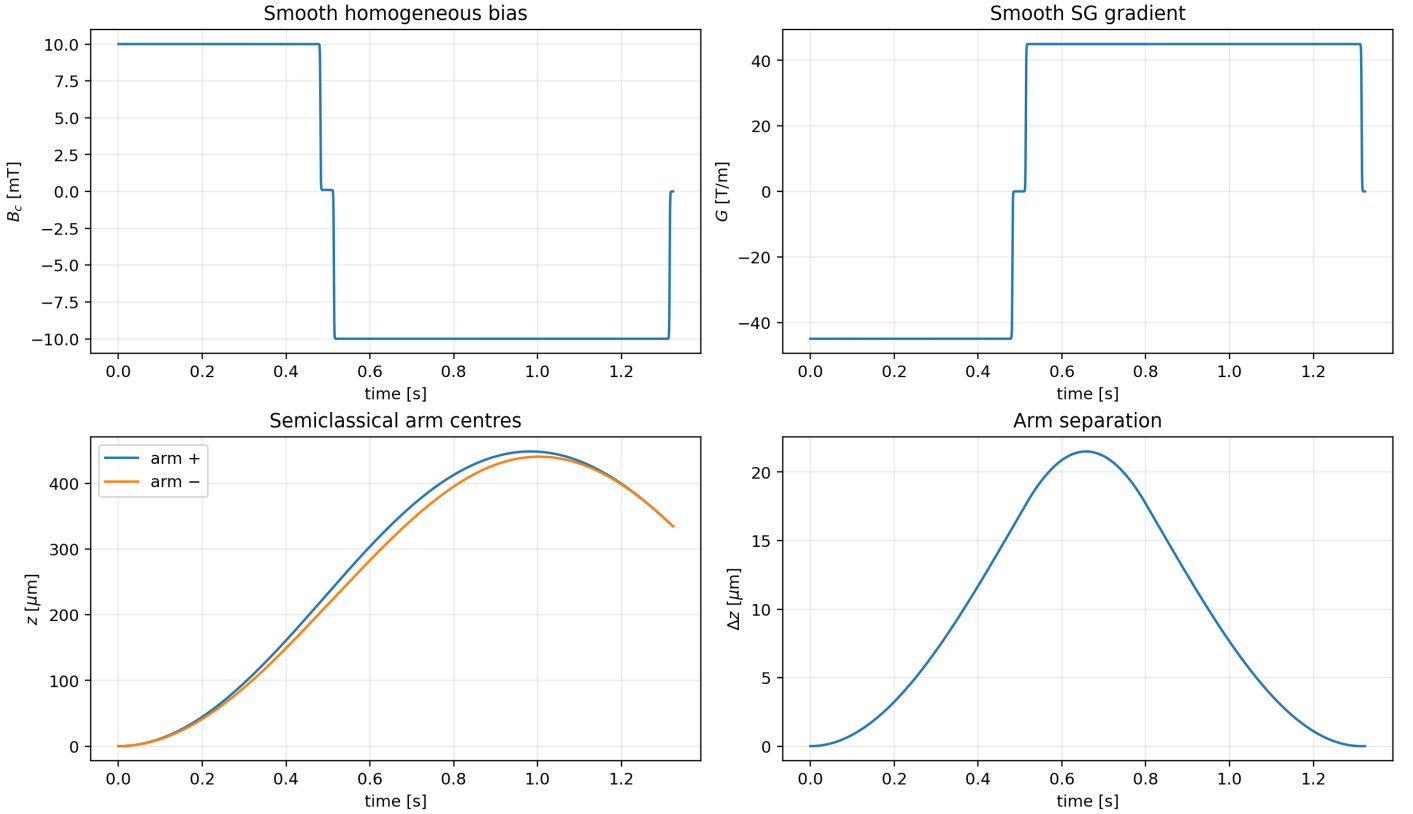


Figure 19: Classical input to the wave-packet calculation. The upper panels show the smooth homogeneous bias $B_c(t)$ and gradient $G(t)$. The lower panels show the two semiclassical c.o.m. centres and their separation. These curves are not recomputed by an independent quantum model: they are the packet centres inherited directly from the preceding chapter.

The microwave inversion is placed exactly on the integration grid and the branch is kept fixed throughout every complete RK4 step. Reoptimizing the relevant times for the smooth field gives

$$\begin{aligned} t_{\text{flip}} &= 0.8006448473 \text{ s}, & t_{\text{off}} &= 1.3152251478 \text{ s}, & t_f &= 1.3232251478 \text{ s}, \\ \Delta z_{\text{max}} &= 21.504698 \mu\text{m}, & \Delta z_f &= 3.79 \times 10^{-18} \text{ m}, & \Delta p_{z,f} &= -1.74 \times 10^{-33} \text{ kg m/s}. \end{aligned} \quad (437)$$

The packet centres therefore inherit a $21.5 \mu\text{m}$ one-loop trajectory which closes in axial phase space to numerical precision.

Figure 19 makes the chapter boundary explicit. The control sequence and the arm centres are classical inputs; the new calculation concerns the finite widths around those centres and their final overlap. The approximately common axial excursion is much larger than the branch separation, while the latter grows to $21.5 \mu\text{m}$ and then returns to zero.

6.4.2 Invariant Gaussian contrast

The raw endpoint differences in the Euler chart are

$$\Delta\alpha_f = 4.23829 \times 10^{-2} \text{ rad}, \quad \Delta\beta_f = 1.92853 \times 10^{-5} \text{ rad}, \quad \Delta\gamma_f = -4.23749 \times 10^{-2} \text{ rad}. \quad (438)$$

The nearly opposite α and γ differences largely cancel in the physical orientation. Equation 425 gives

$$\|\Delta\varphi_f\| = 4.24247 \times 10^{-4} \text{ rad} = 0.424247 \text{ mrad}. \quad (439)$$

The difference between the raw Euler coordinates and the invariant rotation is numerically decisive.

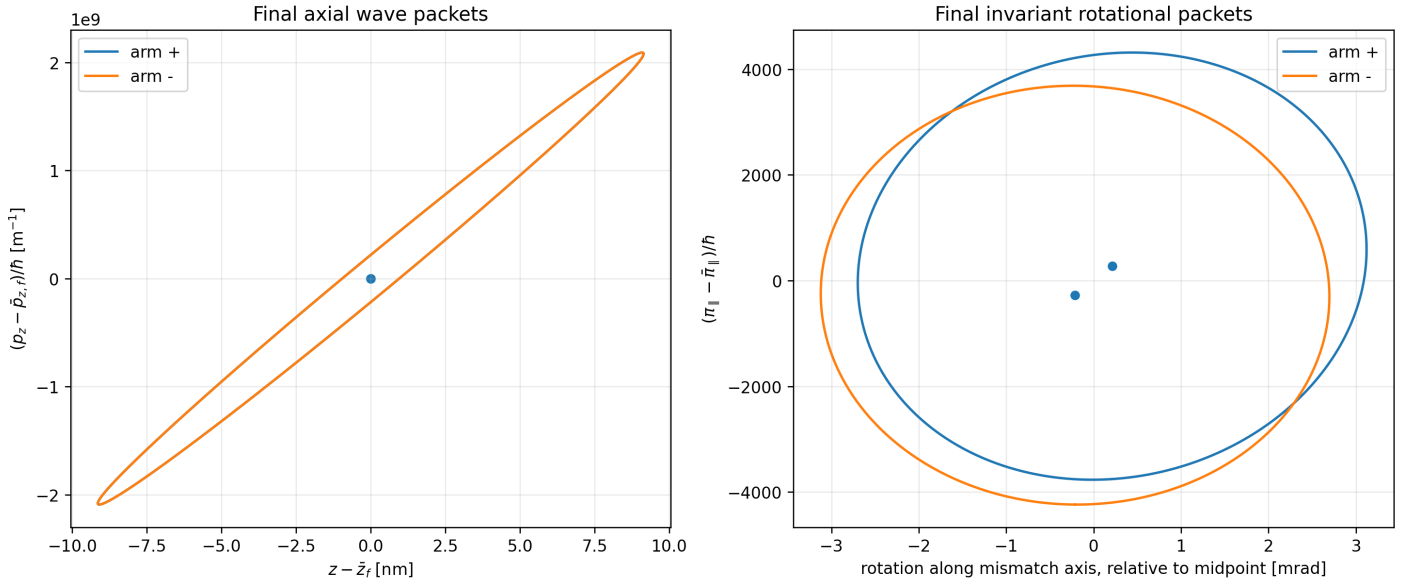


Figure 20: Final wave packets in axial and invariant rotational phase space. The ellipses are two-standard-deviation contours after removing the irrelevant common offsets. The axial packets are centred on the same point to the displayed precision, whereas the rotational packets retain a small relative displacement and a modest shape difference in the common canonical chart.

The left panel of **Fig. 20** confirms that the spatial closure of the centres is also excellent relative to the axial packet width. The right panel shows why the contrast is nevertheless below unity: after the correct cotangent lift, the two rotational packets do not have identical centres or covariances.

The resulting hierarchy is

Calculation	Contrast	Content retained
Reference-paper lower bound	0.823329	Conservative coherent-state amplitude bound
Trajectory-resolved coherent reference	0.835111	Actual smooth-protocol torque history in the equal-curvature model
Factorized endpoint packets	0.832668	Actual final means with independent fixed-width packets
Invariant Gaussian, product preparation	0.819292	Propagated covariance and common canonical $SO(3)$ chart
Invariant Gaussian, correlated preparation	0.816803	Momentum-dependent prepared libration centre from Eq. 409

Table 7: Nested intrinsic-contrast calculations at the nominal operating point. The reference expression is a conservative lower bound, so the rows need not form a strictly monotonic approximation sequence.

The production value is

$$C_G^{(\text{corr})} = 0.8168035 \pm 9.5 \times 10^{-6}. \quad (440)$$

Its exponent is

$$-\ln C_G^{(\text{corr})} = \underbrace{0.193320}_{\Gamma_{\text{disp}}} + \underbrace{0.0090367}_{\Gamma_{\text{shape}}} = 0.202357. \quad (441)$$

Thus, approximately 95.5% of the intrinsic contrast exponent comes from the final covariance-weighted phase-space displacement and 4.5% from differential squeezing and correlations.

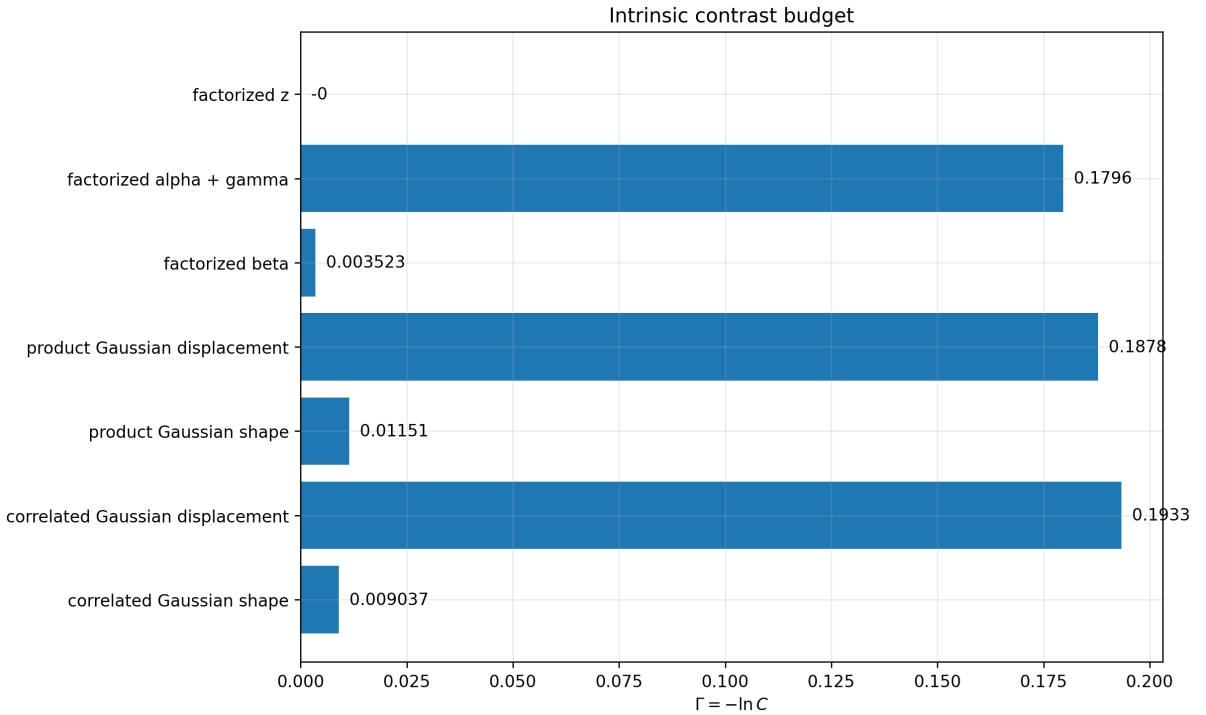


Figure 21: Intrinsic contrast budget. The factorized rows separate the axial, cyclic-angle and libration contributions. The product- and correlated-Gaussian rows then separate the complete result into the displacement term Γ_{disp} and the packet-shape term Γ_{shape} of **Eqs. 431–432**.

Figure 21 identifies the physical priority directly. The axial contribution is negligible, and the isolated β contribution is small. The dominant loss is associated with the complete rotational phase-space displacement, especially the cyclic directions once they are represented in the common canonical chart. Further improvement should therefore target invariant rotational recombination rather than the already negligible axial residual.

6.4.3 Direct nonlinear libration wavefunction

The Gaussian calculation is checked independently for the libration coordinate without attempting a global propagation on $\mathbb{R}^3 \times SO(3)$. In the centered axial model, α and γ are cyclic. We therefore select one exact angular-momentum sector,

$$p_\alpha = \hbar \mathbf{m}, \quad p_\gamma = \hbar \mathbf{k}, \quad \mathbf{m}, \mathbf{k} \in \mathbb{Z}, \quad (442)$$

using the integer pair nearest the mean values in **Eq. 407**. The corresponding rotor wavefunction is

$$\Psi_\ell(\alpha, \beta, \gamma, t) = \frac{e^{i\mathbf{m}\alpha} e^{i\mathbf{k}\gamma}}{2\pi} f_\ell(\beta, t), \quad \int_0^\pi \sin \beta |f_\ell(\beta, t)|^2 d\beta = 1. \quad (443)$$

The transformation

$$u_\ell(\beta, t) := \sqrt{\sin \beta} f_\ell(\beta, t) \quad (444)$$

converts the Haar-weighted norm into $\int_0^\pi |u_\ell|^2 d\beta = 1$. The exact fixed-sector equation is then

$$i\hbar \frac{\partial u_\ell}{\partial t} = \left[-\frac{\hbar^2}{2I_\perp} \frac{\partial^2}{\partial \beta^2} + U_{\mathbf{m}\mathbf{k},\ell}(\beta, t) \right] u_\ell \quad (445)$$

$$U_{\mathbf{m}\mathbf{k},\ell}(\beta, t) = \frac{\hbar^2 \mathbf{k}^2}{2I_\parallel} + \frac{\hbar^2 (\mathbf{m} - \mathbf{k} \cos \beta)^2}{2I_\perp \sin^2 \beta} + V_{n_\ell(t)}(\bar{z}_\ell(t), \beta, t) - \frac{\hbar^2}{8I_\perp} (1 + \csc^2 \beta). \quad (446)$$

The first two terms are the exact symmetric-top kinetic energy in the selected $(\mathbf{m}, \mathbf{k})$ sector. The third is the branch potential evaluated on the previously calculated c.o.m. centre $\bar{z}_\ell(t)$, and the last arises from the transformation to the flat wavefunction u_ℓ . Regularity requires $u_\ell(0, t) = u_\ell(\pi, t) = 0$.

The time-independent part of **Eq. 445** is diagonalized once, after which the branch-dependent drive is propagated by a unitary midpoint Strang step in the retained low-energy basis. The final fixed-sector contrast is

$$C_\beta^{(1D)} = \left| \int_0^\pi u_-^*(\beta, t_f) u_+(\beta, t_f) d\beta \right|. \quad (447)$$

Grid, modal and time-step refinement give

$$C_\beta^{(1D)} = 0.9989966 \pm 1.1 \times 10^{-6}. \quad (448)$$

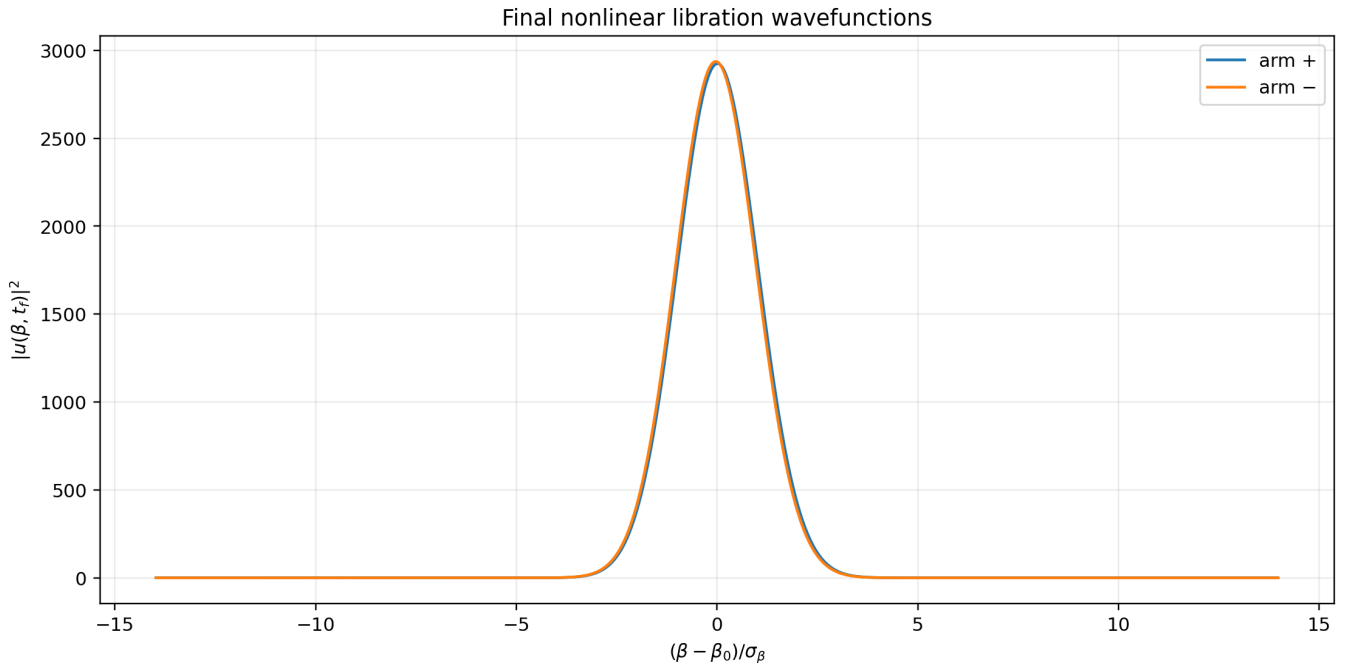


Figure 22: Final nonlinear fixed-sector libration densities. The coordinate is measured relative to the prepared angle in units of the ground-state width σ_β . The two branch-conditioned densities remain almost coincident, in agreement with the overlap in **Eq. 448**. This benchmark tests the nonlinear β wavefunction in the central populated sector; it does not include the full sum over $(\mathbf{m}, \mathbf{k})$ or the multimode correlations.

The near coincidence in **Fig. 22** confirms that non-Gaussian deformation of the isolated central libration sector is not responsible for the total $C_G \simeq 0.817$ result. The larger loss appears only after the complete rotational phase space and its correlations are included.

6.4.4 Thermal tolerance and covariance-weighted timing

For the nominal rotor,

$$\frac{\hbar\Omega_{\text{lib}}}{k_B} = 0.7024 \mu\text{K}. \quad (449)$$

The exact Bose factor in **Eq. 419** therefore saturates to the zero-temperature value below this scale. Using the actual torque history in the equal-curvature model gives approximately

$$C_\beta^{(\text{th})} \simeq 0.9982, \quad 0.9832, \quad 0.8445 \quad \text{for} \quad T_{\text{lib}} = 1, \quad 10, \quad 100 \mu\text{K}, \quad (450)$$

respectively. These are libration-only reference values, not the complete multimode contrast of **Eq. 440**.

The classically closed timing was also subjected to a local, covariance-weighted refinement. Let

$$\boldsymbol{\vartheta} := (t_{\text{flip}} \quad t_{\text{off}})^T \quad \Bigg| \quad \mathbf{d}(\boldsymbol{\vartheta}) := \Delta\tilde{\boldsymbol{\xi}}_f \quad (451)$$

be the two adjusted times and the final common-chart phase-space mismatch. Around the spatially closed solution,

$$\mathbf{d}(\boldsymbol{\vartheta} + \delta\boldsymbol{\vartheta}) \simeq \mathbf{d} + \mathbf{D}_\vartheta \delta\boldsymbol{\vartheta} \quad \Bigg| \quad \mathbf{D}_\vartheta := \frac{\partial \mathbf{d}}{\partial \boldsymbol{\vartheta}}. \quad (452)$$

Holding the combined covariance fixed locally, define

$$\mathbf{W} := (\tilde{\boldsymbol{\Sigma}}_+ + \tilde{\boldsymbol{\Sigma}}_-)^{-1}. \quad (453)$$

Minimizing the displacement exponent $\Gamma_{\text{disp}} = \mathbf{d}^T \mathbf{W} \mathbf{d} / 4$ from **Eq. 432** gives the local weighted least-squares update

$$\boxed{\delta\boldsymbol{\vartheta} = - \left(\mathbf{D}_\vartheta^T \mathbf{W} \mathbf{D}_\vartheta \right)^{-1} \mathbf{D}_\vartheta^T \mathbf{W} \mathbf{d}}. \quad (454)$$

For the nominal protocol, the resulting shift is

$$\Delta t_{\text{flip}} = -1.5122 \times 10^{-7} \text{ s}, \quad \Delta t_{\text{off}} = 1.9926 \times 10^{-8} \text{ s}, \quad (455)$$

which changes the contrast on the common numerical grid from 0.8169547 to 0.8171063. The candidate accepts the small axial residual

$$\Delta z_f = 1.60 \times 10^{-11} \text{ m} \simeq 0.054 \sigma_z, \quad \Delta p_{z,f} = 5.96 \times 10^{-30} \text{ kg m/s}. \quad (456)$$

This is a local correction rather than a new global optimization of the magnetic protocol. Its small gain shows that the spatially closed solution is already close to the quantum-width-weighted optimum for the chosen preparation.

The left panel of **Fig. 23** shows the expected gyroscopic suppression of the torque-filter response: increasing ω_0 moves the system into a broader high-contrast region. The narrow vertical structure is the phase sensitivity of the Fourier component in **Eq. 417**. The scan changes only the reference oscillator frequency; each new operating point would still require the complete classical stability, gap and adiabaticity checks. The right panel shows that every controlled refinement remains on the same approximately 0.82 contrast scale established by the reference paper.

The Gaussian result was checked by halving the maximum RK4 step, monitoring the symplectic tangent flow and comparing the isolated libration prediction with the direct wavefunction propagation. The uncertainty in **Eq. 440** is the conservative Richardson estimate from the two finest Gaussian grids.

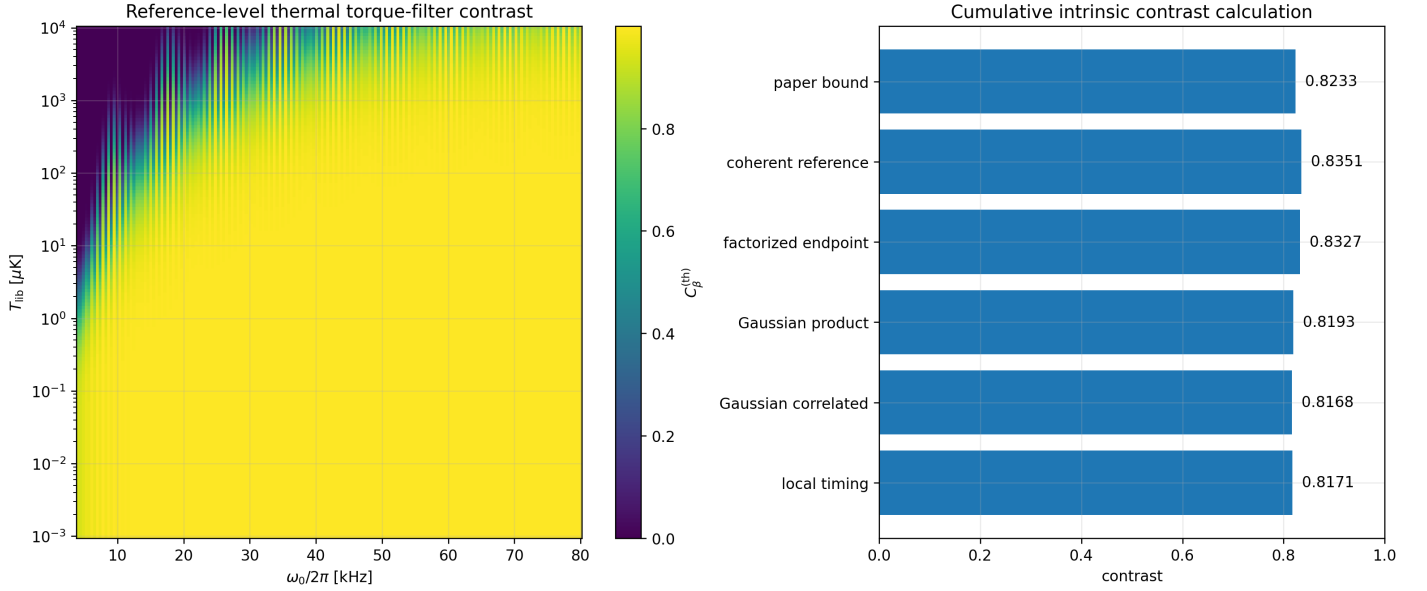


Figure 23: Thermal and timing extensions. The left panel evaluates the reference-level thermal torque-filter contrast from **Eq. 420** as a function of the gyrosopic frequency and libration temperature. The right panel shows the cumulative zero-temperature contrast hierarchy, ending with the small local timing refinement.

6.5 Scope and conclusion

The calculation above is the intrinsic mechanical contrast of the centered, aligned and axially constrained quantum extension of the reference configuration. It includes the quantum spread along the superposition axis, all three local rotational canonical pairs, the momentum-dependent prepared libration centre, time-dependent covariance evolution and the invariant cotangent-lift comparison of the final orientations. It does not include environmental decoherence, current noise or spin dephasing, which would multiply the mechanical coherence by additional factors.

The same overlap theory extends directly to the complete classical model. The reduced eight-component trajectory and Hessian are replaced by the full twelve-component data for translation and rotation. Their tangent matrices are propagated through **Eq. 403**. The same rotational cotangent lift is then applied, and the contrast is evaluated with **Eq. 429**. The exact spin gaps, adiabaticity and off-centre forces and torques remain those already verified in the classical chapter.

For the nominal smooth one-loop protocol, the central result is

$$\boxed{C_G^{(\text{corr})} \simeq 0.817}. \quad (457)$$

This remains near the 0.82 scale of the reference paper, but now follows from the smooth semiclassical trajectories, translational packet, propagated multimode covariance and invariant $SO(3)$ geometry. Exact spatial closure is necessary but insufficient: the remaining contrast is set mainly by rotational phase-space recombination in the common chart.

7 Discussion

The purpose of this thesis was not to construct a new QGEM protocol, but to test whether the single-nanorotor dynamics required by the proposed one-loop scheme remain self-consistent once translation, rotation and the embedded spin are treated together.

The results give a qualified positive answer. A sufficiently large mechanical angular momentum stabilises the two $m_s = \pm 1$ branches and preserves the largest Stern–Gerlach separation, but neither a nonsingular numerical representation nor bounded rotational motion guarantees interferometric recombination.

The central distinction is therefore between *stability*, which prevents the rotor from departing uncontrollably from the operating configuration, and *closure*, which requires the two arms to return to the same complete mechanical state.

7.1 Interpretation of the results

Without an imposed rotation, the aligned configuration favours the $|0\rangle, |-1\rangle$ pair because the $|+1\rangle$ branch is rotationally unstable near $\beta = 0$ [19]. This avoids the unstable branch, but also uses only one magnetically responsive arm and reduces the attainable separation. The gyroscopic proposal addresses precisely this trade-off: the imposed angular momentum makes both $m_s = \pm 1$ branches dynamically usable without weakening the leading differential spin force. The contribution of the present thesis is therefore not a search over unrelated stabilisation mechanisms, but a test of what the proposed mechanism actually solves once the rigid-body, spin and control dynamics are made mutually consistent.

The complete classical calculation shows that it solves the original instability in its intended sense. The exact 3×3 trajectories reach a maximum separation of $21.4358 \mu\text{m}$, retain a minimum spin gap of 20.0 MHz and close to 0.673 nm in c.o.m. position and $0.00670 \mu\text{m/s}$ in velocity. The sampled plus-arm curvature remains positive and the minus-arm motion remains bounded. Nevertheless, the final invariant orientation mismatch is 3.419 mrad and the angular momenta do not close exactly.

These statements are not contradictory. Positive curvature excludes a local exponential libration instability, whereas the final mismatch records the coherent phase accumulated from small branch-dependent torque and frequency differences over many rapid rotational cycles. Gyroscopic rotation therefore converts an unstable branch into a bounded control problem; it does not automatically make the two rotational histories identical.

The quaternion implementation is important to this conclusion because it separates a coordinate failure from a physical one. The production trajectories propagate the mean orientation with a unit quaternion, so the xxz singularity at $\beta = 0$ cannot generate the observed mismatch. Euler angles are reconstructed for interpretation and independently reproduce the same rotation matrices and angular momenta away from their singular chart. The residual is evaluated through the logarithm of the relative rotation in $SO(3)$.

Thus, the previous gimbal-lock difficulty has been removed at the classical numerical level, while the milliradian Humpty–Dumpty residual remains as an invariant property of the two trajectories.

The spatial results also separate useful differential motion from a much larger common orbit. The $21.44 \mu\text{m}$ branch separation is accompanied by a $444.45 \mu\text{m}$ common diamagnetic excursion, so good relative closure does not imply a small apparatus aperture or a locally sampled field.

Moreover, the final closure is unusually sensitive to the gradient because the spin force scales as η , whereas the differential diamagnetic curvature scales as η^2 . The near-linear Monte Carlo correlation between gradient error and final axial mismatch is therefore a physical consequence of the force balance, not merely a feature of the optimizer.

7.2 From classical closure to quantum contrast

The wave-packet calculation makes the distinction between stability and closure quantitative. For the centred, aligned and axially constrained quantum reduction, the event-corrected packet centres close in axial phase space to numerical precision and have an invariant final orientation mismatch of 0.424247 mrad . After propagating the complete reduced covariance and comparing both packets in one common canonical

chart, the correlated Gaussian contrast is

$$C_G^{(\text{corr})} = 0.8168035 \pm 9.5 \times 10^{-6}. \quad (458)$$

This result belongs to the separately retimed reduced model and must not be combined directly with the 3.419 mrad residual of the complete off-centre trajectory. Its role is to determine how the remaining mechanical record enters the overlap and to validate the invariant comparison method before it is extended to the parent model.

The geometric treatment is decisive here as well. The local Gaussian dynamics use canonical Euler fluctuations only inside the narrow patch occupied by each packet, while the final orientations, momenta and covariances are mapped to a common logarithmic chart by the cotangent lift. Large and nearly opposite differences in α and γ therefore do not masquerade as a large physical rotation, and the conjugate momenta are not transformed as ordinary vectors. The quantum calculation does not promote the four quaternion components to independent Gaussian variables; the quaternions describe the nonsingular classical centres, while three canonical tangent-space coordinates describe the local quantum fluctuations.

Approximately 95.5% of the contrast exponent comes from the final covariance-weighted phase-space displacement and only 4.5% from differential packet shape. The direct nonlinear propagation of the central libration sector gives $C_\beta^{(1D)} = 0.9989966$, showing that non-Gaussian deformation of the isolated β mode is not the source of the total loss. The dominant limitation lies instead in the complete rotational phase space, especially the cyclic directions, their conjugate momenta and their preparation correlations. The very small gain obtained by retuning only t_{flip} and t_{off} further indicates that substantial improvement requires additional control freedom rather than finer adjustment of the same two event times.

7.3 Assumptions and limitations

The conclusions remain conditional on the model hierarchy. The complete 2×2 branch reproduces the translational force to better than 1.2×10^{-4} and is therefore suitable for timing and robustness scans, but its percent-level torque error produces a 39% error in the final orientation mismatch. The exact 3×3 spectrum is consequently required whenever rotational closure is the observable of interest. Conversely, the quantum result propagates the reduced four-mode covariance rather than the full six-mode covariance of the complete trajectory. Equation 458 is therefore an intrinsic benchmark, not yet the quantum contrast of the complete parent model.

Two experimental assumptions are equally important. First, the reconstructed multipole trap is mechanically stable but spin-incompatible, so the accepted operating point assumes a separate gravity-compensated guide whose physical implementation is not specified. Second, the microwave operation is an ideal instantaneous branch exchange. The two arms have transition frequencies separated by 44.29 MHz and very small direct matrix elements, while the smooth zero-field passages and final turn-off place the spin in strain-defined instantaneous eigenstates. A realistic protocol must therefore provide multilevel branch exchange and an explicit mapping between the instantaneous eigenbasis and the final readout basis.

The local Gaussian treatment also neglects cubic wave-packet dynamics, Berry and Born–Huang corrections, environmental decoherence, spin dephasing and current noise. Together with the rigid, homogeneous-cylinder approximation and prescribed classical fields, these exclusions mean that the thesis establishes a necessary closed-system single-interferometer result rather than the visibility of a complete two-particle QGEM experiment.

7.4 Priorities for further work

The first priority is to propagate the full 12×12 tangent matrices and six-mode covariances around the verified complete 3×3 trajectories, using quaternions for the mean orientations and three canonical tangent directions for the rotational fluctuations. This would place the classical and quantum residuals in one calculation and determine the intrinsic contrast of the parent interferometer.

The next step is a joint control problem rather than another isolated correction. The confinement, Stern–Gerlach waveform, strain-protected passages, microwave exchange and final basis mapping should be designed together and optimized directly for the invariant covariance-weighted mismatch, while retaining

the required separation, spin gap, adiabaticity and field aperture. The gradient sensitivity identified here should be included explicitly in that optimization. Only after these closed-system requirements are satisfied is it meaningful to add environmental decoherence and the two-particle gravitational interaction.

Within the present assumptions, gyroscopic stabilisation has therefore achieved its central purpose: it makes the full-separation $|+1\rangle, |-1\rangle$ protocol dynamically admissible. The remaining problem is narrower but nontrivial—to erase the complete rotational record left by two stable, yet still distinguishable, nanorotor trajectories.

8 Conclusion

This thesis investigated whether a massive spin-embedded cylindrical nanorotor can be separated and recombined in a one-loop Stern–Gerlach interferometer without leaving an observable record of the followed spin branch in its mechanical state. This is a necessary mechanical precursor to QGEM rather than a calculation of the gravity-mediated entangling phase itself. The answer is conditional but clear: gyroscopic rotation makes the $m_s = \pm 1$ protocol dynamically possible and preserves the largest available Stern–Gerlach separation, yet stable motion and excellent c.o.m. closure do not by themselves guarantee a large mechanical overlap. The interferometer must close translation, orientation and angular momentum, together with the widths and correlations of the corresponding quantum wave packets.

8.1 Scientific result

The thesis first developed a consistent description of rotations, moving frames and rigid-body dynamics and then used it to construct the complete spin-conditioned nanorotor Hamiltonian. The later numerical treatment preserves the distinction between physical orientation and its coordinates. The main classical trajectories are propagated with unit quaternions, thereby avoiding the zxz Euler singularity near $\beta = 0$, and are independently checked against the canonical Euler equations through the rotation matrix and angular momentum. Final rotational closure is evaluated through the logarithm of the relative rotation in $SO(3)$ rather than through raw angle differences. The gimbal-lock problem is therefore removed without changing the physical model or hiding a genuine rotational mismatch.

The complete semiclassical parent model retains three-dimensional translation, the off-centre defect geometry, the general spin frame, the full 3×3 internal Hamiltonian, diamagnetism, gravity and confinement. Reduced 2×2 , centred, aligned and one-dimensional models are derived only afterwards and are compared against the same exact trajectories. This hierarchy shows why an approximation cannot be judged from its branch energy alone. At the nominal point the complete 2×2 model reproduces the translational force to better than 1.2×10^{-4} , but its percent-level torque error changes the final orientation by 39%. The reduced model is therefore adequate for timing and robustness scans, whereas the exact spin spectrum is required for rotational closure.

The same consistency test rejects the reconstructed multipole trap from the interferometric sequence. Although mechanically stable, its predominantly transverse field strongly mixes the intended bare states and produces branch-dependent forces more than two orders of magnitude above the nominal Stern–Gerlach scale. The principal calculation consequently assumes a separate gravity-compensated guide. This is not a minor implementation detail: levitation, spin preparation and interferometric control must be compatible within the same physical field configuration.

With the accepted guide and smoothly switched controls, the complete exact trajectory produces

$$\Delta z_{\max} = 21.4358 \mu\text{m} \quad \Bigg| \quad \|\Delta \mathbf{c}_f\| = 0.673 \text{ nm} \quad \Bigg| \quad \frac{\|\Delta \mathbf{p}_{c,f}\|}{m} = 0.00670 \mu\text{m/s}. \quad (459)$$

The minimum exact spin gap is 20.0 MHz and the maximum adiabaticity parameter is 5.0×10^{-4} . The sampled plus-arm rotational curvature remains positive and the minus-arm trajectory remains bounded, confirming gyroscopic stabilisation. Nevertheless,

$$\Delta \Theta_f = 3.419 \text{ mrad} \quad \Bigg| \quad \|\Delta \mathbf{L}_f^{\text{sf}}\| = 9.76 \times 10^{-30} \text{ J s} = 0.342\% L_0. \quad (460)$$

The classical rotor is therefore stable and translationally recombined, but not fully closed in rotational phase space. The $444.45 \mu\text{m}$ common diamagnetic excursion also remains an independent field-linearity and apparatus-aperture requirement, despite the much smaller relative residual.

The quantum calculation then converts the remaining mechanical record into an intrinsic wave-packet overlap. The global rotor configuration space is identified as $SO(3)$, while the narrow fluctuations around each semiclassical arm are propagated as a local Gaussian state. At the endpoint, both rotational means and covariances are mapped into one common logarithmic chart by the canonical cotangent lift. This removes the coordinate dependence of the raw Euler residuals and preserves the correct transformation of their conjugate momenta. For the separately retimed, centred, aligned and axially constrained reduction,

the production result is

$$C_G^{(\text{corr})} = 0.8168035 \pm 9.5 \times 10^{-6}. \quad (461)$$

Approximately 95.5% of the contrast exponent comes from the final covariance-weighted phase-space displacement and 4.5% from differential squeezing and correlations. A direct nonlinear propagation of the central libration sector gives $C_\beta^{(\text{1D})} = 0.9989966$, showing that the principal loss is not non-Gaussian deformation of the isolated β mode, but incomplete recombination of the complete rotational phase space. The negligible gain from a local two-time quantum retuning indicates that substantial improvement is unlikely to follow from small adjustments of only the flip and turn-off times.

The classical and quantum numbers above deliberately belong to different levels of the hierarchy. The 3.419 mrad residual is obtained from the complete off-centre 3×3 trajectory, whereas **Eq. 461** belongs to the reduced trajectory whose invariant endpoint mismatch is 0.424247 mrad. The contrast is therefore not the exact quantum prediction of the complete parent model. Its significance is that it validates the covariance propagation and invariant $SO(3)$ overlap calculation, reproduces the approximately 0.82 scale of the reference proposal and identifies the rotational canonical directions which dominate the intrinsic loss.

8.2 Answer to the research question

The central result can be stated as

$$\begin{array}{l} \text{large spatial separation and bounded rotation are achievable,} \\ \text{but quantum closure remains a full rotational phase-space problem.} \end{array} \quad (462)$$

The imposed mechanical angular momentum removes the instability which otherwise favours the $|0\rangle, |-1\rangle$ pair and permits the full-force $|+1\rangle, |-1\rangle$ protocol. It does not, however, force the two branches to acquire identical precession phases, final angular momenta or covariance matrices. A coordinate-regular and dynamically stable nanorotor can still carry which-path information.

The thesis therefore provides neither an unqualified feasibility claim nor a no-go result. Within the prescribed-field, adiabatic-branch and ideal-exchange model, the operating point is mechanically promising: a $21.4 \mu\text{m}$ superposition is generated, the exact spin branch remains isolated, the c.o.m. loop closes accurately and the reduced intrinsic contrast remains finite and of order 0.82. At the same time, the rotational mismatch is the dominant intrinsic limitation. The assumed branch exchange is also not realised by one narrowband 1 MHz tone, because the two arms differ in transition frequency by 44.29 MHz and have strongly suppressed direct matrix elements. The smooth zero-field passages require the control to follow the instantaneous strain-protected branches, while the final turn-off requires an explicit mapping from the strain basis to the chosen readout basis. These are specific control requirements rather than unexplained sources of contrast loss.

The result should not be interpreted as the visibility of a complete two-mass QGEM experiment. The calculation excludes environmental decoherence, current noise, spin dephasing and the gravity-mediated phase, and it does not provide the engineering design of the assumed guide. It also propagates a local Gaussian state rather than a global wavefunction on $L^2(\mathbb{R}^3 \times SO(3))$. Its value is to isolate the closed-system mechanical contribution and to show where the remaining distinguishability resides before those additional effects are introduced.

8.3 Final perspective

The work advances the reference proposal by turning its chain of assumptions into a testable hierarchy. It connects rigid-rotor geometry to a complete spin-conditioned Hamiltonian, uses quaternions to remove the classical coordinate singularity, tests reduced branches against exact forces and torques, distinguishes mechanically stable from spin-compatible confinement, replaces discontinuous controls by a strain-protected smooth protocol, evaluates closure invariantly on $SO(3)$ and converts the final phase-space record into a quantum overlap.

The immediate next step is the full six-mode quantum calculation: the 12×12 tangent matrices and covariances should be propagated around the verified complete 3×3 trajectories, using quaternions for

the mean orientations and three canonical tangent directions for the rotational fluctuations. The branch exchange, final basis mapping and confinement should then be incorporated into one driven control model and optimized directly for invariant covariance-weighted closure, with gradient sensitivity and apparatus aperture included as constraints. Only after these tasks are completed should environmental decoherence and the two-particle gravitational interaction be added.

The final conclusion is therefore that gyroscopic stabilisation is necessary and effective, but not by itself sufficient to erase rotational which-path information. It changes the problem from an unstable dynamics into a bounded, coordinate-independent and quantitatively diagnosable mismatch. This thesis identifies that mismatch, determines its intrinsic consequence in the reduced quantum model and provides the framework required to bring the complete interferometer to quantum closure.

A Supporting derivations

A.1 Proof of Thm. 1: The matrix handedness determines its reflectivity.

Proof Let $\mathbf{R} \in O(3)$ and $\mathbf{r}_i$ be its columns. Since the latter form an orthonormal basis, it suffices to project $\mathbf{r}_i \times \mathbf{r}_j$ onto each $\mathbf{r}_k$ and make use of the column permutation $(1, 2, 3) \mapsto (i, j, k)$:

$$(\mathbf{r}_i \times \mathbf{r}_j) \cdot \mathbf{r}_k = \det(\mathbf{r}_i \ \mathbf{r}_j \ \mathbf{r}_k) = \varepsilon_{ijk} \det(\mathbf{R}).$$

Expanding the cross product in the orthonormal basis $\{\mathbf{r}_k\}$ gives

$$\mathbf{r}_i \times \mathbf{r}_j = \sum_{k=1}^3 [(\mathbf{r}_i \times \mathbf{r}_j) \cdot \mathbf{r}_k] \mathbf{r}_k = \det(\mathbf{R}) \sum_{k=1}^3 \varepsilon_{ijk} \mathbf{r}_k.$$

Next, while it is easy to fix the $\det(\mathbf{R})$ to ± 1 (since $\mathbf{R}$ is orthogonal),

$$\det(\mathbf{R}) = \mathbf{r}_1 \cdot (\mathbf{r}_2 \times \mathbf{r}_3) = \pm \sum_k \varepsilon_{23k} \mathbf{r}_1 \cdot \mathbf{r}_k = \pm \varepsilon_{231} = \pm 1,$$

in order to make the connection to reflections, we look at the ε term.

In the convention that for p.r.m., $\det(\mathbf{R}) = 1$, the columns obey the same cross-product relations as the standard basis, while an imp.r.m. has $\det(\mathbf{R}) = -1$, introducing an overall minus sign:

$$\text{p.r.m.: } \mathbf{r}_i \times \mathbf{r}_j = \sum_{k=1}^3 \varepsilon_{ijk} \mathbf{r}_k \quad \Bigg| \quad \text{imp.r.m.: } \mathbf{r}_i \times \mathbf{r}_j = \sum_{k=1}^3 -\varepsilon_{ijk} \mathbf{r}_k.$$

Hence, a p.r.m. changes the orientation of the frame in space, but preserves its handedness and an imp.r.m. reverses the handedness of the orthonormal frame, which is the geometric effect of a reflection. $\square$

A.2 Proof of Thm. 2: Euler-angle coverage of $SO(3)$

Proof Many other parametrizations are extremely similar in their proof, so we will focus on the Tait–Bryan zyx and the Euler zxz ones, which we re-write below for easier reference.

$$R_{zyx} = \mathbf{R}_z(\alpha) \mathbf{R}_y(\beta) \mathbf{R}_x(\gamma) = \begin{pmatrix} c_\alpha c_\beta - s_\alpha c_\gamma + s_\beta s_\gamma c_\alpha & s_\alpha s_\gamma + s_\beta c_\alpha c_\gamma \\ s_\alpha c_\beta & s_\alpha s_\beta s_\gamma + c_\alpha c_\gamma & s_\alpha s_\beta c_\gamma - s_\gamma c_\alpha \\ -s_\beta & s_\gamma c_\beta & c_\beta c_\gamma \end{pmatrix} \quad \text{Tait–Bryan} \quad (463)$$

$$R_{zxz} = \mathbf{R}_z(\alpha) \mathbf{R}_x(\beta) \mathbf{R}_z(\gamma) = \begin{pmatrix} c_\alpha c_\gamma - s_\alpha c_\beta s_\gamma - c_\alpha s_\gamma - s_\alpha c_\beta c_\gamma & s_\alpha s_\beta \\ s_\alpha c_\gamma + c_\alpha c_\beta s_\gamma - s_\alpha s_\gamma + c_\alpha c_\beta c_\gamma & -c_\alpha s_\beta \\ s_\beta s_\gamma & s_\beta c_\gamma & c_\beta \end{pmatrix} \quad \text{Euler.} \quad (464)$$

Thus, our goal is to show that each parametrization defines a surjective map onto $SO(3)$ (every p.r.m. can be produced by at least one choice of α, β, γ):

$$\forall \mathbf{R} \in SO(3), \quad \exists (\alpha, \beta, \gamma) \quad \text{such that} \quad \mathbf{R} = \mathbf{R}_z(\alpha) \mathbf{R}_y(\beta) \mathbf{R}_x(\gamma).$$

$$\forall \mathbf{R} \in SO(3), \quad \exists (\alpha, \beta, \gamma) \quad \text{such that} \quad \mathbf{R} = \mathbf{R}_z(\alpha) \mathbf{R}_x(\beta) \mathbf{R}_z(\gamma).$$

First, we remember that $SO(3)$ is closed under matrix product, therefore $\mathbf{R} \in SO(3)$. It remains to prove the converse: every element of $SO(3)$ is obtained by a suitable choice of angles.

The core idea is simple: assume arbitrary $\mathbf{R} = (R_{ij}) \in SO(3)$, from which we build the $\{\alpha, \beta, \gamma\}$ parameters to initially fit it, with the goal to reconstruct the original matrix with no extra parameters. This is still general, as long as they are not forcefully restricted.

However, a cleaner, equivalent method is to simply "undo" the proposed rotations and show that nothing is left. We do so by probing (while still keeping $\mathbf{R}$ general) if

$$zyx\text{-parametrization: } \mathbf{S} := \mathbf{R}_z(-\alpha) \mathbf{R} \mathbf{R}_x(-\gamma) \quad \longrightarrow \quad \mathbf{S} = \mathbf{R}_y(\beta)$$

$$zxz\text{-parametrization: } \mathbf{S} := \mathbf{R}_z(-\alpha)\mathbf{R}\mathbf{R}_z(-\gamma) \longrightarrow \mathbf{S} = \mathbf{R}_x(\beta).$$

We use $\text{atan2}(y, x)$ (assuming $(x, y) \neq (0, 0)$) for the angle ϕ satisfying

$$\cos \phi = \frac{x}{\sqrt{x^2 + y^2}} \quad \Big| \quad \sin \phi = \frac{y}{\sqrt{x^2 + y^2}}.$$

Finally, a useful step when determining α and γ is to remember from **Lem. 2** that the norm of any row or column of any $\mathbf{R} \in SO(3)$ is unitary.

The Tait–Bryan zyx -parametrization.

If $|R_{31}| < 1$, we can set

$$\beta = \arcsin(-R_{31}) \quad \Big| \quad -\frac{\pi}{2} < \beta < \frac{\pi}{2} \quad \Rightarrow \quad c_\beta > 0.$$

Since the first column and third row of $\mathbf{R}$ have norm 1,

$$R_{11}^2 + R_{21}^2 = R_{32}^2 + R_{33}^2 = 1 - R_{31}^2 = c_\beta^2.$$

Thus we may equivalently define

$$\begin{aligned} \alpha = \text{atan2}(R_{21}, R_{11}) \quad \Big| \quad \gamma = \text{atan2}(R_{32}, R_{33}) \\ R_{11} = c_\alpha c_\beta \quad R_{21} = s_\alpha c_\beta \quad R_{31} = -s_\beta \quad R_{32} = c_\beta s_\gamma \quad R_{33} = c_\beta c_\gamma, \end{aligned}$$

where the first line was obtained by noting that $(\mathbf{R}_{11}, \mathbf{R}_{21}) = c_\beta(c_\alpha, s_\alpha)$, thus pointing in the direction with angle α , and analogously for (R_{32}, R_{33}) .

From this, the first column and the third row of $\mathbf{S}$ are set in place by the choice of α and γ , leading to

$$\mathbf{S} = \begin{pmatrix} c_\beta & a & b \\ 0 & d & e \\ -s_\beta & 0 & c_\beta \end{pmatrix}.$$

We again remind ourselves that, since $\mathbf{S} \in SO(3)$, its rows are orthonormal and all rows and columns have the unitary norm. We thus obtain:

$$a = 0 \quad \Big| \quad d = \pm 1 \quad \Big| \quad 0 = -c_\beta s_\beta + b c_\beta \Rightarrow b = s_\beta \quad \Big| \quad 0 = e c_\beta \Rightarrow e = 0,$$

but since $\det \mathbf{S} = 1$, this forces $d = 1$:

$$\mathbf{S} = \begin{pmatrix} c_\beta & 0 & s_\beta \\ 0 & 1 & 0 \\ -s_\beta & 0 & c_\beta \end{pmatrix} = \mathbf{R}_y(\beta).$$

By the stripped-equation equivalence,

$$\mathbf{R} = \mathbf{R}_z(\alpha)\mathbf{R}_y(\beta)\mathbf{R}_x(\gamma).$$

We now have to treat the singular cases.

If $R_{31} = -1$, then the first column and third row force

$$\mathbf{R} = \begin{pmatrix} 0 & a & b \\ 0 & c & d \\ -1 & 0 & 0 \end{pmatrix} \quad \Big| \quad \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} -s_\alpha & c_\alpha \\ c_\alpha & s_\alpha \end{pmatrix}.$$

where the r.h.s. was obtained by noting that the 2×2 block is orthogonal with determinant -1 . The parameter setting that leads to this result is

$$\beta = \frac{\pi}{2} \quad \Big| \quad \gamma = 0 \quad \Big| \quad \alpha = \text{atan2}(R_{23}, R_{13}).$$

$$\mathbf{R} = \mathbf{R}_z(\alpha)\mathbf{R}_y(\pi/2)\mathbf{R}_x(0).$$

If $R_{31} = 1$, then similarly

$$\mathbf{R} = \begin{pmatrix} 0 & a & b \\ 0 & c & d \\ 1 & 0 & 0 \end{pmatrix} \quad \Big| \quad \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} -s_\alpha & -c_\alpha \\ c_\alpha & -s_\alpha \end{pmatrix}.$$

where the r.h.s. was obtained by noting that the 2×2 block is orthogonal with determinant 1. The parameter setting that leads to this result is

$$\beta = -\frac{\pi}{2} \quad \Big| \quad \gamma = 0 \quad \Big| \quad \alpha = \text{atan2}(-R_{12}, R_{22}).$$

$$\mathbf{R} = \mathbf{R}_z(\alpha)\mathbf{R}_y(-\pi/2)\mathbf{R}_x(0).$$

Therefore the zyx Tait–Bryan parametrization is surjective onto $SO(3)$.

The Euler zxz -parametrization.

If $|R_{33}| < 1$, we can set

$$\beta = \arccos(R_{33}) \quad \Big| \quad 0 < \beta < \pi \quad \Rightarrow \quad s_\beta > 0.$$

Since the third column and third row of $\mathbf{R}$ have norm 1,

$$R_{13}^2 + R_{23}^2 = R_{31}^2 + R_{32}^2 = 1 - R_{33}^2 = s_\beta^2.$$

Thus we may equivalently define

$$\begin{aligned} \alpha = \text{atan2}(R_{13}, -R_{23}) \quad & \Big| \quad \gamma = \text{atan2}(R_{31}, R_{32}) \\ R_{13} = s_\alpha s_\beta \quad R_{23} = -c_\alpha s_\beta \quad R_{31} = s_\beta s_\gamma \quad R_{32} = s_\beta c_\gamma \quad R_{33} = c_\beta, \end{aligned}$$

where the first line was obtained by noting that $(-R_{23}, R_{13}) = s_\beta(c_\alpha, s_\alpha)$, thus pointing in the direction with angle α , and analogously for (R_{32}, R_{31}) .

From this, the third column and the third row of $\mathbf{S}$ are set in place by the choice of α and γ , leading to $\mathbf{S}$ having the form

$$\mathbf{S} = \begin{pmatrix} a & b & 0 \\ d & e & -s_\beta \\ 0 & s_\beta & c_\beta \end{pmatrix}.$$

We again remind ourselves that, since $\mathbf{S} \in SO(3)$, its rows are orthonormal and all rows and columns have unit norm. We thus obtain:

$$0 = bs_\beta \Rightarrow b = 0 \quad \Big| \quad a = \pm 1 \quad \Big| \quad 0 = ad \Rightarrow d = 0 \quad \Big| \quad 0 = s_\beta(e - c_\beta) \Rightarrow e = c_\beta,$$

but since $\det \mathbf{S} = 1$, this forces $a = 1$. Hence

$$\mathbf{S} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & c_\beta & -s_\beta \\ 0 & s_\beta & c_\beta \end{pmatrix} = \mathbf{R}_x(\beta).$$

By the stripped-equation equivalence,

$$\mathbf{R} = \mathbf{R}_z(\alpha)\mathbf{R}_x(\beta)\mathbf{R}_z(\gamma).$$

We now have to treat the singular cases.

If $R_{33} = 1$, then the third column and third row force

$$\mathbf{R} = \begin{pmatrix} a & b & 0 \\ c & d & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad \Big| \quad \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} c_\alpha & -s_\alpha \\ s_\alpha & c_\alpha \end{pmatrix}.$$

where the r.h.s. was obtained by noting that the 2×2 block is orthogonal with determinant 1. The parameter setting that leads to this result is

$$\beta = 0 \quad \Big| \quad \gamma = 0 \quad \Big| \quad \alpha = \text{atan2}(R_{21}, R_{11}).$$

$$\mathbf{R} = \mathbf{R}_z(\alpha)\mathbf{R}_x(0)\mathbf{R}_z(0).$$

If $R_{33} = -1$, then similarly

$$\mathbf{R} = \begin{pmatrix} a & b & 0 \\ c & d & 0 \\ 0 & 0 & -1 \end{pmatrix} \quad \Big| \quad \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} c_\alpha & s_\alpha \\ s_\alpha & -c_\alpha \end{pmatrix}.$$

where the r.h.s. was obtained by noting that the 2×2 block is orthogonal with determinant -1 . The parameter setting that leads to this result is

$$\beta = \pi \quad \Big| \quad \gamma = 0 \quad \Big| \quad \alpha = \text{atan2}(R_{21}, R_{11}).$$

$$\mathbf{R} = \mathbf{R}_z(\alpha)\mathbf{R}_x(\pi)\mathbf{R}_z(0).$$

Therefore the zxz Euler parametrization is surjective onto $SO(3)$.

Thus every p.r.m. admits both a zyx Tait–Bryan factorization and a zxz Euler-angle factorization. The angle choices are not globally unique, so the statement is surjectivity, not bijectivity. $\square$

A.3 Proof of Thm. 3: Euler's axis theorem for proper rotations

Proof Let $\mathbf{R} \in SO(3)$. We start with proving the existence of the invariant eigenvector and its eigenvalue for both the theorem and its extension.

Suppose we are interested in showing matrix $\mathbf{R}$ has a non-trivial eigenvector $\mathbf{n}$ with eigenvalue λ :

$$\mathbf{R}\mathbf{n} = \lambda\mathbf{n} \iff (\mathbf{R} - \lambda\mathbf{1})\mathbf{n} = 0. \quad (465)$$

If the bracket term is invertible,

$$\mathbf{n} = \mathbf{1}\mathbf{n} = ((\mathbf{R} - \lambda\mathbf{1})^{-1}(\mathbf{R} - \lambda\mathbf{1}))\mathbf{n} = (\mathbf{R} - \lambda\mathbf{1})^{-1}((\mathbf{R} - \lambda\mathbf{1})\mathbf{n}) = 0. \quad (466)$$

Therefore, the existence of a non-trivial eigenvalue requires and is satisfied by the non-invertibility of the bracket term in the second part of **Eq. 465**.

In the case of a p.r.m., $\lambda = +1$ and for an imp.r.m., $\lambda = -1$:

$$\det(\mathbf{R}) = \pm 1 \Rightarrow \det(\mathbf{R})^{-1} = \det(\mathbf{R}^{-1}) = \pm 1.$$

We give here an outline for a proof on determinant multiplicativity (i.e. $\det(\mathbf{AB}) = \det(\mathbf{A})\det(\mathbf{B})$): if either $\mathbf{A}$ or $\mathbf{B}$ are not invertible, their determinant is null, satisfying the above, but if they both aren't, we can always factorize an invertible matrix into a product of elementary matrices:

$$\mathbf{A} = \prod_i^n \mathbf{E}_i \text{ with } \det(\mathbf{E}_i) = \pm c \text{ for } c \geq 0, \quad (467)$$

where $\mathbf{E}_i$ is an elementary matrix. Those, in essence, are matrices that behave as row operations on the matrix they act on (with a possible multiplicative factor c). Because of that, the determinant of this product can be directly broken down into a product:

$$\det(\mathbf{E}_i\mathbf{A}) = \det(\mathbf{E}_i)\det(\mathbf{A}), \quad (468)$$

which directly leads to our result:

$$\begin{aligned} \det(\mathbf{AB}) &= \det\left(\prod_i^m \mathbf{E}_i^A \prod_i^n \mathbf{E}_i^B\right) = \prod_i^m \det(\mathbf{E}_i^A) \prod_i^n \det(\mathbf{E}_i^B) \\ &= \det\left(\prod_i^m \mathbf{E}_i^A\right) \det\left(\prod_i^n \mathbf{E}_i^B\right) = \det(\mathbf{A})\det(\mathbf{B}). \end{aligned} \quad (469)$$

This same methodology can be used to prove that:

$$\det(\mathbf{A}^{-1}) = \det(\mathbf{A})^{-1}. \quad (470)$$

In the case of a p.r.m., $\lambda = +1$ and for an imp.r.m., $\lambda = -1$. This gives

$$\begin{aligned} \det(\mathbf{R} \mp \mathbf{1}) &= \det((\mathbf{R} \mp \mathbf{1})^T) = \det(\mathbf{R}^T \mp \mathbf{1}) = \det(\mathbf{R}^{-1} \mp \mathbf{1}) = \det(\mathbf{R}^{-1}(\mathbf{1} \mp \mathbf{R})) \\ &= \det(\mp \mathbf{R}^{-1}(\mathbf{R} \mp \mathbf{1})) = \mp \det(\mathbf{R}^{-1})\det(\mathbf{R} \mp \mathbf{1}) = -\det(\mathbf{R} \mp \mathbf{1}) \Rightarrow \det(\mathbf{R} \mp \mathbf{1}) = 0, \end{aligned} \quad (471)$$

thus proving the existence of the eigenvector $\mathbf{n}$ and respective ± 1 eigenvalues. **This implies $\mathbf{n}$ is the r.ax. of $\mathbf{R}$ and leads to the $(\mathbf{n}, \varphi)$ parametrization of $\mathbf{R}$.**

Now suppose $\mathbf{R} \neq \mathbf{1}$ and it has two linearly independent invariant vectors $\mathbf{u}$ and $\mathbf{v}$. Since $\mathbf{R}$ is proper, **Lem. 1** gives

$$\mathbf{R}(\mathbf{u} \times \mathbf{v}) = (\mathbf{R}\mathbf{u}) \times (\mathbf{R}\mathbf{v}) = \mathbf{u} \times \mathbf{v}. \quad (472)$$

Thus, $\mathbf{R}$ fixes the basis $\{\mathbf{u}, \mathbf{v}, \mathbf{u} \times \mathbf{v}\}$ and must be $\mathbf{1}$. Therefore, every non-identity p.r.m. has only one invariant axis, up to sign.

For the improper case, where we use the notation $\mathbf{w} = \mathbf{u} \times \mathbf{v}$, we get

$$(\mathbf{R}\mathbf{u}) \times (\mathbf{R}\mathbf{v}) = (-\mathbf{u}) \times (-\mathbf{v}) = \mathbf{w}. \quad (473)$$

But the cross product under an imp.r.m. obeys

$$(\mathbf{R}\mathbf{u}) \times (\mathbf{R}\mathbf{v}) = \det(\mathbf{R})\mathbf{R}(\mathbf{u} \times \mathbf{v}) = -\mathbf{R}\mathbf{w} \Rightarrow \mathbf{R}\mathbf{w} = -\mathbf{w} \Rightarrow \mathbf{R} = -\mathbf{1}, \quad (474)$$

since here $\mathbf{R}$ fixes the basis $\{\mathbf{u}, \mathbf{v}, \mathbf{w}\}$. $\square$

A.4 Equivalent Taylor proof of Thm. 5: Unitary rotations of vector operators

Proof The main text proves **Thm. 5** by solving the differential equation for the rotated vector operator. The same result can be obtained directly from the Taylor expansion of the adjoint action using the commutator relation from **Eq. 65**, thus explicitly showing where the sine and cosine series come from. Let

$$\begin{aligned} \hat{G} = \hat{\mathbf{n}} \cdot \hat{\mathbf{L}} \quad \Big| \quad \hat{\mathbf{F}}(\varphi) = e^{\frac{i}{\hbar}\varphi\hat{G}}\hat{\mathbf{V}}e^{-\frac{i}{\hbar}\varphi\hat{G}} \quad \Rightarrow \quad \hat{\mathbf{F}}(\varphi) &= \sum_{m=0}^{\infty} \frac{\varphi^m}{m!} \left. \frac{d^m \hat{\mathbf{F}}}{d\varphi^m} \right|_{\varphi=0} \\ \Rightarrow \quad \left. \frac{d\hat{\mathbf{F}}_i}{d\varphi} \right|_{\varphi=0} &= \frac{i}{\hbar} [\hat{G}, \hat{V}_i] = \frac{i}{\hbar} \hat{n}_j [\hat{L}_j, \hat{V}_i] = -\hat{n}_j \varepsilon_{jik} \hat{V}_k \\ &= \hat{n}_j \varepsilon_{ijk} \hat{V}_k = (\hat{\mathbf{n}} \times \hat{\mathbf{V}})_i. \end{aligned} \quad (475)$$

Thus, we can write the first and second derivatives in full operator form:

$$\left. \frac{d\hat{\mathbf{F}}}{d\varphi} \right|_{\varphi=0} = \hat{\mathbf{n}} \times \hat{\mathbf{V}} \Rightarrow \left. \frac{d^2 \hat{\mathbf{F}}}{d\varphi^2} \right|_{\varphi=0} = \hat{\mathbf{n}} \times (\hat{\mathbf{n}} \times \hat{\mathbf{V}}) = \hat{\mathbf{n}}(\hat{\mathbf{n}} \cdot \hat{\mathbf{V}}) - \hat{\mathbf{V}}. \quad (476)$$

The higher derivatives then alternate:

$$\left. \frac{d^{2k+1} \hat{\mathbf{F}}}{d\varphi^{2k+1}} \right|_{\varphi=0} = (-1)^k \hat{\mathbf{n}} \times \hat{\mathbf{V}}, \quad k \geq 0 \quad \Big| \quad \left. \frac{d^{2k} \hat{\mathbf{F}}}{d\varphi^{2k}} \right|_{\varphi=0} = (-1)^k [\hat{\mathbf{V}} - \hat{\mathbf{n}}(\hat{\mathbf{n}} \cdot \hat{\mathbf{V}})], \quad k \geq 1. \quad (477)$$

Substituting these derivatives into the Taylor series and using the Taylor series of sine and cosine gives

$$\begin{aligned} \hat{\mathbf{F}}(\varphi) &= \hat{\mathbf{V}} + [\hat{\mathbf{V}} - \hat{\mathbf{n}}(\hat{\mathbf{n}} \cdot \hat{\mathbf{V}})] \sum_{k=1}^{\infty} \frac{(-1)^k \varphi^{2k}}{(2k)!} + (\hat{\mathbf{n}} \times \hat{\mathbf{V}}) \sum_{k=0}^{\infty} \frac{(-1)^k \varphi^{2k+1}}{(2k+1)!} \\ &= \hat{\mathbf{V}} + [\hat{\mathbf{V}} - \hat{\mathbf{n}}(\hat{\mathbf{n}} \cdot \hat{\mathbf{V}})] (\cos(\varphi) - 1) + (\hat{\mathbf{n}} \times \hat{\mathbf{V}}) \sin(\varphi) \\ &= \hat{\mathbf{V}} \cos(\varphi) + (\hat{\mathbf{n}} \times \hat{\mathbf{V}}) \sin(\varphi) + (1 - \cos(\varphi))(\hat{\mathbf{n}} \cdot \hat{\mathbf{V}})\hat{\mathbf{n}}, \end{aligned} \quad (478)$$

which is exactly Rodrigues' formula acting on the vector operator:

$$\hat{\mathbf{F}}(\varphi) = \mathbf{R}(\hat{\mathbf{n}}, \varphi) \hat{\mathbf{V}}. \quad (479)$$

The sign of the sine term is tied to the convention $\hat{U}^\dagger \hat{\mathbf{V}} \hat{U}$ used in the main text. If the operator order is reversed to $\hat{U} \hat{\mathbf{V}} \hat{U}^\dagger$, the sine term changes sign, corresponding to the inverse rotation. $\square$

A.5 Proof of Thm. 6: Rigidity through the Gram matrix

Proof Let $\mathbf{x}_k(t)$ be the absolute position of the k -th point mass. We write:

$$m = \sum_{j=1}^n m_j \quad \Big| \quad \mu_j = \frac{m_j}{m} \quad \Big| \quad \mathbf{c}(t) = \sum_{j=1}^n \mu_j \mathbf{x}_j(t) \quad \Big| \quad \mathbf{r}_k(t) = \mathbf{x}_k(t) - \mathbf{c}(t). \quad (480)$$

Since the masses are fixed, the weights μ_j are also time-independent.

A point-mass system is rigid iff all mutual distances

$$D_{kl}(t) := \|\mathbf{x}_k(t) - \mathbf{x}_l(t)\|^2 = \|\mathbf{r}_k - \mathbf{r}_l\|^2 = \mathbf{r}_k^2 + \mathbf{r}_l^2 - 2\mathbf{r}_k \cdot \mathbf{r}_l \quad (481)$$

are time-independent. In the second step we used that translations cancel in differences. Therefore, if all scalar products $\mathbf{r}_k \cdot \mathbf{r}_l$ are time-independent, then all D_{kl} are time-independent, and the system is rigid.

Conversely, suppose the system is rigid, so that all D_{kl} are time-independent. Because

$$\sum_{i=1}^n \mu_i \mathbf{r}_i = \mathbf{0}, \quad (482)$$

we obtain

$$\sum_{i=1}^n \mu_i D_{ki} = \sum_{i=1}^n \mu_i \|\mathbf{r}_k - \mathbf{r}_i\|^2 = \mathbf{r}_k^2 + \sum_{i=1}^n \mu_i \mathbf{r}_i^2 - 2\mathbf{r}_k \cdot \sum_{i=1}^n \mu_i \mathbf{r}_i = \mathbf{r}_k^2 + \sum_{i=1}^n \mu_i \mathbf{r}_i^2 \quad (483)$$

$$\sum_{i,j=1}^n \mu_i \mu_j D_{ij} = \sum_{i,j=1}^n \mu_i \mu_j \|\mathbf{r}_i - \mathbf{r}_j\|^2 = 2 \sum_{i=1}^n \mu_i \mathbf{r}_i^2. \quad (484)$$

Hence

$$\mathbf{r}_k^2 = \sum_{i=1}^n \mu_i D_{ki} - \frac{1}{2} \sum_{i,j=1}^n \mu_i \mu_j D_{ij}. \quad (485)$$

Substituting this into $\mathbf{r}_k \cdot \mathbf{r}_l$ to obtain:

$$\mathbf{r}_k \cdot \mathbf{r}_l = \frac{1}{2} (\mathbf{r}_k^2 + \mathbf{r}_l^2 - D_{kl}) = \frac{1}{2} \left(\sum_{i=1}^n \mu_i D_{ki} + \sum_{i=1}^n \mu_i D_{li} - D_{kl} - \sum_{i,j=1}^n \mu_i \mu_j D_{ij} \right). \quad (486)$$

The r.h.s. only depends on the fixed masses and on the mutual squared distances D_{ij} . Thus, if the system is rigid, every entry $\mathbf{r}_k \cdot \mathbf{r}_l$ of the c.o.m.-relative Gram matrix is time-independent: *the system is rigid iff the c.o.m.-relative Gram matrix is time-independent.* $\square$

A.6 Angular-velocity matrices in the zxz parametrization

Proof

$$\Omega = \dot{\mathbf{F}} \mathbf{F}^T$$

$$\begin{aligned} &= [\dot{\mathbf{R}}_z(\alpha) \mathbf{R}_x(\beta) \mathbf{R}_z(\gamma) + \mathbf{R}_z(\alpha) \dot{\mathbf{R}}_x(\beta) \mathbf{R}_z(\gamma) + \mathbf{R}_z(\alpha) \mathbf{R}_x(\beta) \dot{\mathbf{R}}_z(\gamma)] [\mathbf{R}_z(\gamma)^T \mathbf{R}_y(\beta)^T \mathbf{R}_z(\alpha)^T] \\ &= [[\dot{\alpha} \mathbf{L}_z \mathbf{R}_z(\alpha)] \mathbf{R}_x(\beta) \mathbf{R}_z(\gamma) + \mathbf{R}_z(\alpha) [\dot{\beta} \mathbf{L}_x \mathbf{R}_x(\beta)] \mathbf{R}_z(\gamma) + \mathbf{R}_z(\alpha) \mathbf{R}_x(\beta) [\dot{\gamma} \mathbf{L}_z \mathbf{R}_z(\gamma)]] [\mathbf{R}_z(\gamma)^T \mathbf{R}_x(\beta)^T \mathbf{R}_z(\alpha)^T] \\ &= \dot{\alpha} \mathbf{L}_z + \dot{\beta} [\mathbf{R}_z(\alpha) \mathbf{L}_x \mathbf{R}_z(\alpha)^T] + \dot{\gamma} [\mathbf{R}_z(\alpha) \mathbf{R}_x(\beta) \mathbf{L}_z [\mathbf{R}_z(\alpha) \mathbf{R}_x(\beta)]^T] \\ &= \dot{\alpha} \mathbf{L}_z + \dot{\beta} \det(\mathbf{R}_z(\alpha)) \sum_{i=1}^3 \left[\mathbf{R}_z(\alpha) \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \right]_i \mathbf{L}_i + \dot{\gamma} \det(\mathbf{R}_z(\alpha) \mathbf{R}_x(\beta)) \sum_{i=1}^3 \left[(\mathbf{R}_z(\alpha) \mathbf{R}_x(\beta)) \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \right]_i \mathbf{L}_i \\ &= \dot{\alpha} \mathbf{L}_z + \dot{\beta} (\mathbf{L}_x \mathbf{L}_y \mathbf{L}_z) \begin{pmatrix} c_\alpha \\ s_\alpha \\ 0 \end{pmatrix} + \dot{\gamma} (\mathbf{L}_x \mathbf{L}_y \mathbf{L}_z) \begin{pmatrix} s_\alpha s_\beta \\ -c_\alpha s_\beta \\ c_\beta \end{pmatrix} \\ &= (\dot{\beta} c_\alpha + \dot{\gamma} s_\alpha s_\beta) \mathbf{L}_x + (\dot{\beta} s_\alpha - \dot{\gamma} c_\alpha s_\beta) \mathbf{L}_y + (\dot{\alpha} + \dot{\gamma} c_\beta) \mathbf{L}_z, \end{aligned}$$

where in the fifth step we used **Lem. 3** with $\mathbf{L}_y$ and $\mathbf{L}_y$, respectively, as $\mathbf{A}$, and $\mathbf{R}_z(\alpha)$ and $\mathbf{R}_z(\alpha) \mathbf{R}_y(\beta)$, respectively, as $\mathbf{B}$; the rest follows. By making use of the first part of the same theorem:

$$\Omega = \sum_{i=1}^3 \omega_i \mathbf{L}_i \Rightarrow \boldsymbol{\omega} = \begin{pmatrix} \omega_1 \\ \omega_2 \\ \omega_3 \end{pmatrix} = \begin{pmatrix} 0 & c_\alpha & s_\alpha s_\beta \\ 0 & s_\alpha & -c_\alpha s_\beta \\ 1 & 0 & c_\beta \end{pmatrix} \begin{pmatrix} \dot{\alpha} \\ \dot{\beta} \\ \dot{\gamma} \end{pmatrix}, \quad (487)$$

and by trivial calculation of $\mathbf{N}$ we finish proving the lemma. $\square$

A.7 Cylindrical symmetry: inertia tensor

Proof Consider a homogeneous cylinder of radius R , length L and mass m . Let the cylinder symmetry axis coincide with the body axis $\hat{\mathbf{f}}_3$, and introduce cylindrical coordinates

$$\left. \begin{array}{l} x = r \cos \phi \\ y = r \sin \phi \end{array} \right| \quad \left. \begin{array}{l} dm = \rho r dr d\phi dz \\ \rho = \frac{m}{\pi R^2 L} \end{array} \right| \quad (488)$$

The moment of inertia around the symmetry axis is

$$I_{\parallel} = \int_V (x^2 + y^2) dm = \rho \int_{-L/2}^{L/2} dz \int_0^{2\pi} d\phi \int_0^R r^3 dr = \rho L (2\pi) \frac{R^4}{4} = \frac{1}{2} m R^2. \quad (489)$$

The moment of inertia around a transverse principal axis $\hat{\mathbf{f}}_1$ is

$$I_{\perp} = \int_V (y^2 + z^2) dm = \rho \int_{-L/2}^{L/2} dz \int_0^{2\pi} d\phi \int_0^R (r^2 \sin^2 \phi + z^2) r dr = \frac{1}{4} m R^2 + \frac{1}{12} m L^2 = \frac{1}{12} m (3R^2 + L^2) \quad (490)$$

$$\frac{I_{\parallel}}{I_{\perp}} = \frac{\frac{1}{2} m R^2}{\frac{1}{12} m (3R^2 + L^2)} = \frac{6R^2}{3R^2 + L^2}. \quad (491)$$

□

B Supplementary numerical verification and preparation diagnostics

This appendix collects numerical checks which are necessary for reproducibility but would interrupt the results-first structure of **Sec. 5**.

B.1 Constants and base parameters

Table 8: Universal constants and nominal material and spin parameters used by the calculation.

Quantity	Symbol	Value
Planck constant	h	$6.62607015 \times 10^{-34} \text{ J s}$
Reduced Planck constant	$\hbar$	$1.05457181765 \times 10^{-34} \text{ J s}$
Vacuum permeability	μ_0	$1.25663706127 \times 10^{-6} \text{ H/m}$
Gravitational acceleration	g	9.80665 m/s^2
Boltzmann constant	k_B	$1.380649 \times 10^{-23} \text{ J/K}$
Diamond mass density	ρ	$3.50 \times 10^3 \text{ kg/m}^3$
Diamond mass susceptibility	χ_ρ	$-6.20 \times 10^{-9} \text{ m}^3/\text{kg}$
Spin magnetic coefficient	μ/h	28.0 GHz/T
Axial zero-field splitting	D/h	2.87 GHz
Transverse zero-field splitting	E/h	10.0 MHz

The value of D is the nominal NV axial splitting, whereas E is defect dependent and can vary with strain. The choice $E/h = 10 \text{ MHz}$ is therefore an operating-point assumption rather than a universal diamond constant; the gap and adiabaticity conclusions below scale with it.

Table 9: Nominal rotor, defect and preparation inputs. The radius and principal moments are derived from the specified mass, density and length.

Quantity	Symbol	Value
Particle mass	m	$1.00 \times 10^{-17} \text{ kg}$
Cylinder length	L	100 nm
Cylinder radius	R	95.365 nm
Diameter-to-length ratio	$\kappa = 2R/L$	1.90731
Perpendicular moment of inertia	$I_{\perp}$	$3.10698 \times 10^{-32} \text{ kg m}^2$
Parallel moment of inertia	$I_{\parallel}$	$4.54728 \times 10^{-32} \text{ kg m}^2$
Inertia ratio	$I_{\parallel}/I_{\perp}$	1.46357
Defect position in the body frame	$\mathbf{d}_0$	$(0, -5.00, 8.66) \text{ nm}$
Spin-to-body orientation	$\mathbf{C}_s$	$\mathbf{1}_3$
Initial Euler angles	$(\alpha_0, \beta_0, \gamma_0)$	$(0, 0.0100, 0) \text{ rad}$
Initial body angular velocity	$\boldsymbol{\omega}^{\text{bf}}(0)$	$(0, 0, 2\pi \times 10.0 \text{ kHz})$
Initial c.o.m. state	$\mathbf{c}(0), \dot{\mathbf{c}}(0)$	$\mathbf{0}, \mathbf{0}$

Table 10: Trap, confinement and Stern–Gerlach inputs. The multipole coefficients and final event times are fitted outputs reported later.

Quantity	Symbol	Value	Role
Multipole length scale	y_0	$75 \mu\text{m}$	trap fit
Target trap frequencies	(f_x, f_y, f_z)	$(104, 130, 9.6) \text{ Hz}$	trap fit
High and holding biases	B_0, B_1	10.0 mT, 0.100 mT	SG control
Gradient magnitude	η	45.0 T/m	SG control
Switching rate	δ_{sw}	10^3 s^{-1}	cross-fades
Nominal turn-on centre	t_{on}	-8.0 ms	initial ramp
Early timing seed	$(t_{12}, t_{23}, t_{\text{flip}})$	$(0.482, 0.514, 0.8022) \text{ s}$	continuation
Turn-off and evaluation seed	(t_{off}, t_f)	$(1.320, 1.328) \text{ s}$	smooth tail
Effective confinement	$\boldsymbol{\omega}_{\text{conf}}/(2\pi)$	$(104, 130, 0) \text{ Hz}$	after trap rejection
Harmonic centre	$\mathbf{c}_0$	$(0, 14.699, 0) \mu\text{m}$	gravity compensation
Magnetic trap during SG	$\mathbf{B}_{\text{tr}}$	$\mathbf{0}$	separate model

B.2 Numerical settings

The numerical values in **Tab. 11** are implementation choices rather than independent physical parameters. Their adequacy is established by the symbolic, finite-difference and timestep tests reported in **Sec. 5**.

Table 11: Numerical settings used by the nominal calculations.

Setting	Value	Purpose
Trap coefficient seed	$(-1.3, 0.018, 0.72)$ T	starting guess only
Trap-equilibrium seed	$-19 \mu\text{m}$	starting guess only
Trap inverse random starts	40, seed 7	search for distinct nonlinear roots
Stationary-point random starts	250, seed 8	basin and saddle search
Trap inverse algorithm	scaled trust-region least squares	inverse multipole fit
Stationary-point algorithm	modified Powell hybrid	basin and saddle search
Nominal reference and exact RK4 step	$20 \mu\text{s}$	production trajectories and figures
Exact refinement steps	40, 20, 10, $5 \mu\text{s}$	convergence of final observables
Minimum accepted branch overlap	0.50	failure guard; actual minimum > 0.9999
Absolute gap floor	$h \times 1 \text{ kHz}$	numerical degeneracy guard
Cartesian finite-difference step	10^{-10} m	force and field-derivative checks
Angular finite-difference step	10^{-7} rad	torque checks
Cylinder quadrature	$4 \times 4 \times 8$	radial, axial and azimuthal points
Monte Carlo sample number	512	focused robustness ensemble
Random seed for the ensemble	7	reproducibility of the Gaussian draws

Table 12: Secondary diagnostic, scan and ensemble settings.

Setting	Value	Purpose
Illustrative preparation temperature	$1\ \mu\text{K}$	scale of the magnetic-trap ensemble
Illustrative preparation draws	2500	Fig. 25
Complete- 2×2 timing-optimizer step	$50\ \mu\text{s}$	optimizes only final z and p_z ; rotational outputs are reevaluated
Converged complete- 2×2 diagnostic step	$5\ \mu\text{s}$	final reduced-model orientation and closure
Monte Carlo trajectory step	$20\ \mu\text{s}$	validated ensemble propagation
Microwave diagnostic rate	$\Omega_R/(2\pi) = 1\ \text{MHz}$	bandwidth and impulse estimate
Implied ideal π -pulse duration	$0.5\ \mu\text{s}$	diagnostic only, not a pulse design
Gyroscopic scan	$0.25 \leq \kappa \leq 4, 1\text{--}50\ \text{kHz}$	14×18 logarithmic grid
Gyroscopic-scan step	$1.29\text{--}20\ \mu\text{s}$	reduced with the local rotational scale
Mass scan	$5 \times 10^{-18}\text{--}10^{-15}\ \text{kg}$	14 logarithmic points in each geometry family

The $50\ \mu\text{s}$ timing solve is acceptable for its slow translational objective, but it aliases the $10\ \text{kHz}$ rotation and its raw angular diagnostic is discarded. The fixed event times are therefore reevaluated at $5\ \mu\text{s}$ before any reduced-model rotational number is quoted. The corrected gyroscopic and mass scans similarly use converged steps; unconverged coarse-scan files are not part of the final data package.

The overlap threshold is deliberately loose because it is a failure guard, not a claimed accuracy. The observed overlaps are orders of magnitude closer to unity. Likewise, the $1\ \text{kHz}$ gap floor only detects a numerical branch degeneracy; the physical minimum gap of the accepted trajectory is $20\ \text{MHz}$.

B.3 Trap roots and normal-mode propagation

The multi-start inverse solve of **Eq. 364** returns the four roots shown in **Tab. 13**. The roots at $y_{\text{eq}} \simeq -230\ \mu\text{m}$ are outside the local region set by $y_0 = 75\ \mu\text{m}$. Each remaining pair differs only by the overall sign of the magnetic field and therefore generates the same spin-independent diamagnetic potential.

Table 13: Distinct roots of the static inverse problem.

$a_2\ [\text{T}]$	$a_3\ [\text{T}]$	$a_4\ [\text{T}]$	$y_{\text{eq}}\ [\mu\text{m}]$	$ B_{\text{eq}} \ [\text{T}]$	minimum Hessian $[\text{s}^{-2}]$	interpretation
0.547852	0.0180607	0.0625269	-230.336	0.188442	3638.33	outside local trap region
-0.547852	-0.0180607	-0.0625269	-230.336	0.188442	3638.33	sign-related outer root
-1.28031	0.0180607	0.743612	-19.3679	0.188442	3638.33	selected local root
1.28031	-0.0180607	-0.743612	-19.3679	0.188442	3638.33	sign-related local root

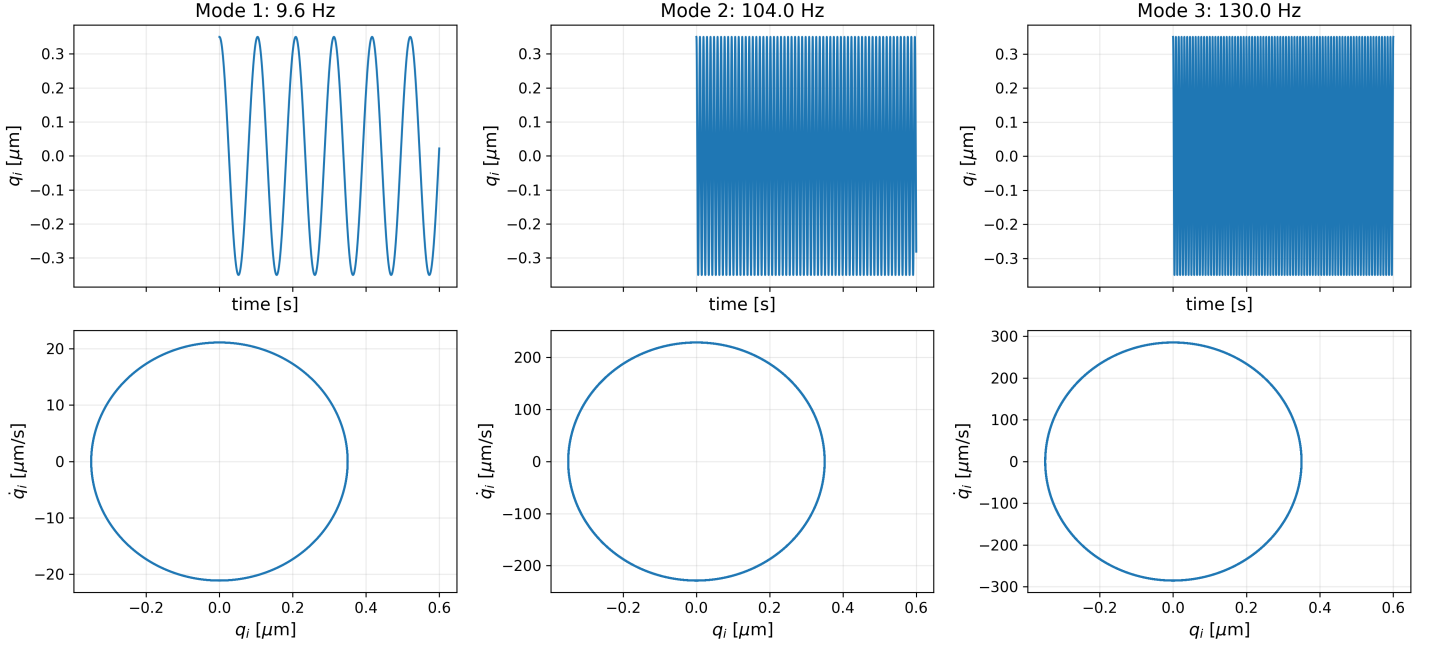


Figure 24: Trap-only propagation of the three Hessian eigenmodes. The upper row shows their normal coordinates and the lower row their phase-space loops.

Table 14: Hessian frequencies compared with the frequencies extracted from RK4 trajectories.

Mode	Hessian prediction [Hz]	RK4 trajectory [Hz]	relative difference
1	9.6000	9.5830	1.77×10^{-3}
2	104.000	103.955	4.34×10^{-4}
3	130.000	129.996	3.33×10^{-5}

The trap-only trajectories remain single-frequency and their phase-space curves remain closed at the sampled sub-micrometre amplitude. No visible intermode transfer or secular distortion occurs, confirming that the Hessian eigenvectors provide an accurate local dynamical basis in the preparation region.

B.4 Classical preparation ensemble

For the illustrative translational preparation temperature $T = 1 \mu\text{K}$, each normal mode is sampled according to

$$q_i \sim \mathcal{N}\left(0, \frac{k_B T}{m \omega_i^2}\right) \quad \Bigg| \quad p_i \sim \mathcal{N}(0, m k_B T). \quad (492)$$

The corresponding one-standard-deviation position widths are 19.48 nm, 1.80 nm and 1.44 nm for the 9.6, 104 and 130 Hz modes, respectively; all three velocity widths are $1.175 \mu\text{m/s}$.

The sampled ensemble is approximately Gaussian and centred on the selected equilibrium. The weak 9.6 Hz mode carries the largest positional width, consistent with $\langle q_i^2 \rangle = k_B T / (m \omega_i^2)$, and its phase-space cloud shows no appreciable position–velocity correlation. The slow longitudinal mode therefore sets the dominant thermal preparation uncertainty, while the transverse coordinates remain much more tightly localized.

This distribution characterizes the scale of the reconstructed magnetic preparation trap. It is not the ensemble propagated through the final effective-guide protocol, whose axial frequency is zero. The robustness study of **Sec. 5.6.1** therefore uses the explicitly declared technical widths of **Eqs. 389, 390** rather than assuming a common thermal equilibrium for every degree of freedom.

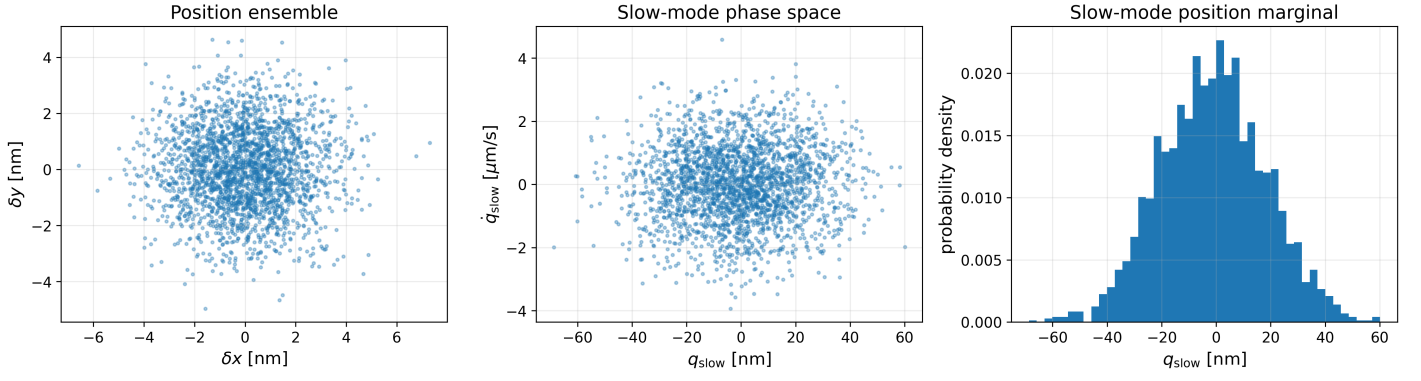


Figure 25: Illustrative $1\,\mu\text{K}$ classical preparation ensemble in normal-mode coordinates. The panels show the spatial projection, weak-mode phase space and weak-mode position marginal.

B.5 Exact timestep refinement

Table 15: Exact 3×3 trajectory under timestep refinement.

h [μs]	Δz_{max} [μm]	$\ \Delta \mathbf{c}_f\ $ [nm]	$\ \Delta \mathbf{v}_f\ $ [$\mu\text{m/s}$]	$\Delta \Theta_f$ [mrad]	$\ \Delta \mathbf{L}_f\ $ [J s]	minimum overlap
40	21.43447	0.2118	0.01517	2.6669	7.62×10^{-30}	0.999912
20	21.43581	0.6519	0.006583	3.3555	9.59×10^{-30}	0.999978
10	21.43580	0.6724	0.006698	3.4214	9.76×10^{-30}	0.999995
5	21.43580	0.6731	0.006699	3.4186	9.76×10^{-30}	0.999999

The $40\,\mu\text{s}$ result does not yet resolve the rotational motion. The 20, 10 and $5\,\mu\text{s}$ rows agree at the level required for the conclusions of Chapter 4. The production figures use $20\,\mu\text{s}$ because it resolves the trajectory while keeping the exact diagnostics and repeated audits tractable.

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